

ANESTHESIOLOGIST'S Manual of Surgical Procedures

SIXTH
EDITION

Richard A. Jaffe
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SURGICAL CONSIDERATIONS

INDUCTION

MAINTENANCE

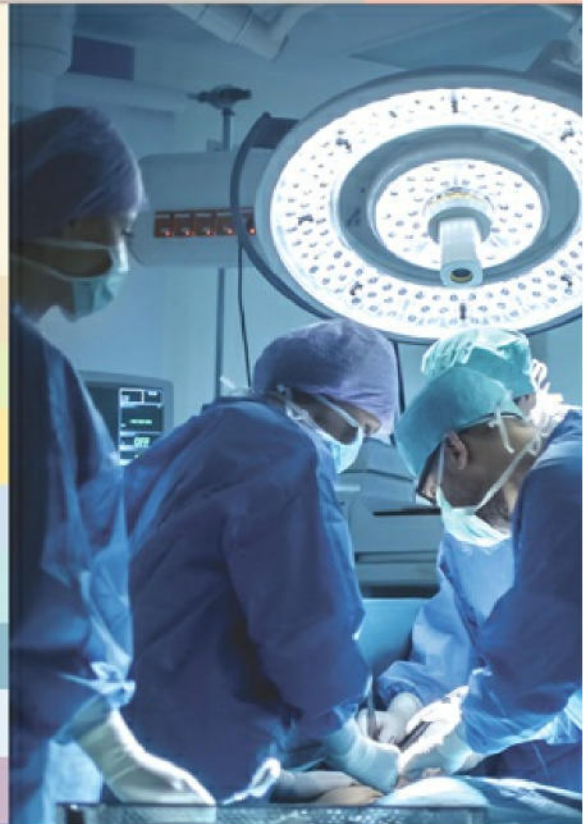
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Wolters Kluwer

Anesthesiologist's Manual of Surgical Procedures

Sixth Edition

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PREFACE

The goals of this sixth edition remain unchanged from those of the first—that is, to provide an easily accessible source of clinically relevant information about a wide variety of common and not-so-common surgical and nonsurgical procedures requiring anesthesia. As with the first edition, this edition does not pretend to be either a textbook of anesthesia or a textbook of surgery. Indeed, in the formulation of any anesthetic plan, there is no substitute for experience and sound clinical judgment.

Those familiar with the preceding editions will notice that the format and organization of this edition remain largely unchanged. Along with the addition of new procedures, every existing procedure was carefully reviewed and revised to reflect current practice. We continue to expand the use of color to improve readability of the tables and clarity of the figures. New to this edition is an emphasis on common sense enhanced recovery considerations that we believe can be applied perioperatively to most patients undergoing most surgical procedures. Also new to this edition is a chapter reviewing the basic principles and anatomy of common regional anesthetic techniques, an appendix reviewing individual components of enhanced recovery (ERAS) protocols, including fluid management and multimodal pain management, an appendix describing the intraoperative use of Transnasal Humidified Rapid-Insufflation Ventilatory Exchange (THRIVE), and a collection of interesting facts and figures printed on the inside cover pages.

Once again, we have made extensive use of abbreviations, medical symbols, and telegraphic sentence structure to present a large body of information in a condensed format. Although we realized that it may be aesthetically more pleasing to read: “hypoxia or hypercapnia can lead to the development of tachycardia and hypertension”, it takes up a lot less space to write: $\downarrow\text{PaO}_2$ or $\uparrow\text{PaCO}_2 \rightarrow \uparrow\text{HR} + \uparrow\text{BP}$.

As always the editorial team welcomes comments and suggestions from our readers (e-mail: anesthesiologistsmanual.Jaffe@stanford.edu), and we remain hopeful that this edition will become a useful resource in their training and subsequent clinical practice.

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This edition is built on the foundation of its predecessors and thus includes material from prior editions. We remain grateful to our many former contributors, some of whose words have survived the test of time and continue to live on in the current edition.

We would like to thank our families for their support, patience, and understanding; our colleagues on both sides of the blood–brain barrier for the time and effort they have generously invested in this educational project; our readers, whose many kind comments have been very much appreciated; Donna J. Allison, PhD, for the difficult job of proofreading under adverse circumstances; our Acquisitions Editor, Keith Donnellan, and our Developmental Editor, Rebeca Barroso, for their patient and enthusiastic support of this project; and finally, our teachers, mentors, and trainees, who have made anesthesiology such an exciting and fulfilling profession for us all.

CONTENTS

Contributing Authors

Preface

Acknowledgments

1.0 NEUROSURGERY

1.1 Intracranial Neurosurgery

SURGEONS Gary K. Steinberg, Robert L. Dodd, Navjot Chaudhary, Lawrence M. Shuer, Steven D. Chang

ANESTHESIOLOGISTS Richard A. Jaffe, Mark A. Burbridge

Craniotomy for Intracranial Aneurysms

Craniotomy for Cerebral Embolectomy

Craniotomy for Intracranial Vascular Malformations

Craniotomy for Extracranial-Intracranial Revascularization (EC-IC Bypass)

Craniotomy for Tumor

Considerations for Awake Craniotomy

Craniotomy for Skull Tumor

Craniotomy for Trauma

Microvascular Decompression of Cranial Nerve

Bifrontal Craniotomy for CSF Leak

Transoral or Transnasal Approach to the Cervicomedullary Junction and Odontoid

Transsphenoidal Resection of Pituitary Tumor

Ventricular Shunt Procedures

Craniocervical Decompression (Chiari Malformation)

Stereotactic Neurosurgery

1.2 Functional Neurosurgery

SURGEONS Jaimie M. Henderson, Lawrence M. Shuer

ANESTHESIOLOGIST Richard A. Jaffe

Functional Neurosurgery—Introduction: The Surgical Treatment of Pain, Movement Disorders, and Epilepsy

Stereotactic Procedures: Deep Brain Stimulation, Laser Interstitial Thermal Therapy, and MR-Guided Focused

Ultrasound
Surgical Analgesics: Spinal Cord Stimulation, Intrathecal Pumps,
and Cortical Stimulation
Vagus Nerve Stimulation
Epilepsy Surgery
Surgery for Spasticity
Percutaneous Procedures for Trigeminal Neuralgia

1.3 Spinal Neurosurgery

SURGEONS Anand Veeravagu, Lawrence M. Shuer

ANESTHESIOLOGISTS Karl Zheng, C. Philip Larson Jr., Martin S. Angst

Anterior Fusion/Fixation of the Upper Cervical (C1-C2) Spine
Posterior Fusion/Fixation of the Upper Cervical Spine
Anterior Fusion/Fixation of the Mid and Lower Cervical Spine
Posterior Fusion/Fixation of the Mid and Lower Cervical Spine
Anterior Cervicothoracic Spine Surgery
Anterior Thoracic Spine Surgery
Posterior Thoracic Spine Surgery
Anterior Lumbar/Lumbosacral Spine Surgery
Posterior Lumbar Spine Surgery
Posterior Lumbar Fusion and Instrumentation
Combined Anterior and Posterior Instrumentation of the Thoracic
and Lumbar Spine

1.4 Carotid Endarterectomy

SURGEONS Gary K. Steinberg, Robert L. Dodd

ANESTHESIOLOGIST Richard A. Jaffe

Carotid Endarterectomy

2.0 OPHTHALMIC SURGERY

SURGEONS Peter H. Tang, Darius M. Moshfeghi

ANESTHESIOLOGISTS Richard A. Jaffe, Lindsey Ralls

Cataract Extraction with Intraocular Lens Insertion
Corneal Transplantation
Trabeculectomy
Ectropion Repair

Entropion Repair
Ptosis Repair
Eyelid Reconstruction
Pterygium Excision
Repair of Ruptured or Lacerated Globe
Dacryocystorhinostomy
Enucleation
Orbitotomy—Anterior and Lateral
Retinal Surgery

3.0 OTOLARYNGOLOGY—HEAD AND NECK SURGERY

SURGEONS Michael J. Kaplan, Edward J. Damrose, Robert K. Jackler, Robson Capasso, Peter H. Hwang, Sam P. Most, Donald M. Sesso, Jerome E. Hester

ANESTHESIOLOGISTS Vladimir Nekhendzy, Tiffany Cheng

Introduction—Surgeon's Perspective
Introduction—Anesthesiologist's Perspective
Anesthetic Management
Laryngoscopy/Bronchoscopy/Esophagoscopy
Tracheotomy/Tracheostomy and Cricothyroidotomy
Intubation for Epiglottitis
Zenker Diverticulectomy (Open Approach)
Laryngeal Framework Surgery (Thyroplasty, Arytenoid Adduction, Injection Laryngoplasty)
Tracheal and Cricotracheal Resection
Laryngectomy: Partial and Total
Tonsillectomy and/or Adenoidectomy
Glossectomy
Maxillectomy and Orbit Exenteration
Composite Resection with Marginal or Segmental Mandibulectomy (and Neck Dissection)
Neck Dissection
Lymph Node Biopsy
Parotidectomy: Superficial, Total, or Radical
Submandibular Gland Excision
Endoscopic Sinus Surgery
External Sinus Surgery
Nasal Surgery (Rhinoplasty, Septoplasty, Septorhinoplasty)

Facial Plastic Surgery
Otology and Neurotology
Neurotologic Skull Base Surgery
Reconstructive Surgery for Sleep-Disordered Breathing

4.0 DENTAL SURGERY

SURGEON Stephen A. Schendel

ANESTHESIOLOGIST Richard A. Jaffe

Temporomandibular Joint Arthroscopy/Arthrotomy/Arthroplasty
Oral Surgery
Restorative Dentistry

5.0 THORACIC SURGERY

SURGEON Natalie Shaubie Lui

ANESTHESIOLOGIST Vivek Kulkarni

Introduction: Surgeon's Perspective
Video-Assisted Thoracic Surgery
Lung Resections
Chest Wall Resection
Repair of Pectus Excavatum or Carinatum
Thoracoplasty
Drainage of Empyema
Tracheal Resection
Tracheobronchoplasty
Excision of Blebs or Bullae
Lung Volume Reduction Surgery
Bronchopulmonary Lavage
Endobronchial Ultrasound-Guided Transbronchial Needle
Aspiration
Bronchoscopy—Flexible and Rigid
Airway Laser Surgery
Thymectomy
Excision of Other Mediastinal Tumors
Mediastinoscopy
Lung Transplant

6.0 CARDIOVASCULAR SURGERY

6.1 Cardiac Surgery

SURGEON R. Scott Mitchell

ANESTHESIOLOGISTS Linda E. Foppiano, Daryl Oakes, Albert Tsai

Cardiopulmonary Bypass
Coronary Artery Bypass Graft Surgery
Left Ventricular Aneurysmectomy
Aortic Valve Replacement
Minimally Invasive Aortic Valve Procedures
Mitral Valve Repair or Replacement
Tricuspid Valve Repair
Septal Myectomy/Myotomy
Pacemaker Insertion
Pericardiectomy

6.2 Minimally Invasive Cardiac Surgery

SURGEON James I. Fann

ANESTHESIOLOGISTS Albert Tsai, Daryl Oakes

Off-Pump and Minimally Invasive Coronary Artery Bypass
Grafting
Port-Access and TECAB Coronary Revascularization
Limited Thoracotomy and Port-Access Approaches to Valve
Surgery
Transcatheter Valve Replacement Procedures

6.3 Vascular Surgery

SURGEONS James I. Fann, R. Scott Mitchell, Clayton A. Kaiser,
Michael D. Dake

ANESTHESIOLOGISTS Jessica L. Brodt, Albert Tsai, Daryl Oakes

Carotid Endarterectomy (Vascular)
Repair of Thoracic Aortic Aneurysms
Endovascular Stent Grafting of Aortic Aneurysms
Repair of Acute Aortic Dissections
Repair of Aneurysms of the Thoracoabdominal Aorta
Surgery of the Abdominal Aorta
Infrainguinal Arterial Bypass
Arterial Embolectomy
Lumbar Sympathectomy

Venous Surgery—Thrombectomy or Vein Excision
Surgery for Portal Hypertension
Arteriovenous Access for Hemodialysis
Permanent Vascular Access
Venous Surgery—Vein Stripping and Perforator Ligation
Varicose Vein Stripping and Ablation

6.4 Heart/Lung Transplantation

SURGEON Hari R. Mallidi

ANESTHESIOLOGISTS Albert Tsai, Daryl Oakes, Linda E. Foppiano

Surgery for Heart Transplantation

Surgery for the Lung and Heart/Lung Transplantation

7.0 GENERAL SURGERY

7.1 Esophageal Surgery

SURGEONS Jeffrey A. Norton, Mark F. Berry

ANESTHESIOLOGIST Vivek Kulkarni

Esophagostomy

Esophageal Diverticulectomy

Management of Esophageal Perforation

Esophagomyotomy

Esophagogastric Fundoplasty

Esophagectomy

7.2 Stomach Surgery

SURGEONS Dan Eisenberg, Sherry M. Wren, Maureen M. Tedesco

ANESTHESIOLOGISTS Jerry Ingrande, Clifford A. Schmiesing, Brendan Carvalho

Gastric Resections

Gastric or Duodenal Perforation

Open Operations for Morbid Obesity

Gastrostomy Placement

7.3 Intestinal Surgery

SURGEON Jeffrey A. Norton

ANESTHESIOLOGIST Clifford A. Schmiesing

Duodenotomy
Open Appendectomy
Excision of Meckel Diverticulum
Enterostomy
Continent Ileostomy Pouch (Kock)
Small-Bowel Resection with Anastomosis
Enterolysis
Closure of Enteric Fistulae

7.4 Colorectal Surgery

SURGEON Andrew A. Shelton

ANESTHESIOLOGIST Clifford A. Schmiesing

Enhanced Recovery After Surgery (ERAS)
Laparoscopic Colorectal Surgery
Total Proctocolectomy
Segmental (Partial) Colectomy
Stoma Closure or Peristomal Hernia Repair
Operations for Rectal Prolapse
Rectal Surgery
Anorectal Surgery
Surgery for Anal Fistulae
Hemorrhoidectomy/Stapled Hemorrhoidopexy
Operations for Fecal Incontinence

7.5 Hepatic Surgery

SURGEON Samuel K. S. So

ANESTHESIOLOGIST Hendrikus J. M. Lemmens

Hepatic Resection
Hepatorrhaphy

7.6 Biliary Tract Surgery

SURGEON C. Andrew Bonham

ANESTHESIOLOGIST Hendrikus J. M. Lemmens

Open Cholecystectomy and Common Bile Duct Exploration
Biliary Drainage Procedures
Excision of Bile Duct Tumor
Choledochal Cyst Excision or Anastomosis
Anesthetic Considerations for Biliary Tract Surgery

7.7 Laparoscopic General Surgery

SURGEONS Sherry M. Wren, Anthony J. Carden, Maureen M. Tedesco, Ruth M. Fanning

ANESTHESIOLOGISTS Clifford A. Schmiesing, Jerry Ingrande, Brendan Carvalho

Laparoscopic Repair of Perforated Peptic Ulcer

Laparoscopic Esophageal Fundoplication

Laparoscopic Heller Myotomy ± Antireflux Procedure

Laparoscopic Cholecystectomy ± Common Duct Exploration

Laparoscopic Splenectomy

Laparoscopic Adrenalectomy

Laparoscopic and Robotic Bowel Resection

Laparoscopic Appendectomy

Laparoscopic Inguinal Hernia Repair

Laparoscopic Bariatric Surgery

Anesthesia for Laparoscopy in Pregnancy

7.8 Pancreatic Surgery

SURGEON Jeffrey A. Norton

ANESTHESIOLOGIST Martin S. Angst

Operative Drainage for Pancreatitis

Drainage of Pancreatic Pseudocyst

Longitudinal Pancreaticojejunostomy

Pancreatectomy

Whipple Resection

7.9 Peritoneal Surgery

SURGEON Jeffrey A. Norton

ANESTHESIOLOGIST Martin S. Angst

Exploratory or Staging Laparotomy

Splenectomy

Excision of Intra-abdominal and Retroperitoneal Tumors

Drainage of Subphrenic Abscess

Inguinal Herniorrhaphy

Femoral Herniorrhaphy

Repair of Incisional Hernia

Repair of Abdominal Dehiscence

7.10 Breast Surgery

SURGEONS Irene L. Wapnir, Deshka S. Foster, Arash Momeni

ANESTHESIOLOGIST Clifford A. Schmiesing

Breast Biopsy

Lymph Node Surgery

Mastectomy ± Reconstruction

7.11 Endocrine Surgery

SURGEONS Hadiza S. Kazaure, Dana T. Lin

ANESTHESIOLOGISTS Javier Lorenzo, Frederick G. Mihm

Excision of Thyroglossal Duct Cyst

Thyroidectomy

Parathyroidectomy

Adrenalectomy

7.12 Liver/Kidney/Pancreas Transplantation

SURGEON Marc L. Melcher

ANESTHESIOLOGISTS Timothy P. Angelotti, Hendrikus J. M. Lemmens

Kidney Transplantation: Cadaveric and Live Donor

Cadaveric Kidney/Pancreas Transplantation

Live-Donor Nephrectomy: Laparoscopic and Open

Kidney Transplant Nephrectomy

Liver Transplantation

Living-Donor Liver Transplantation

Multiorgan Procurement

7.13 Trauma Surgery

SURGEONS Lisa M. Knowlton, Aussama K. Nassar, Timothy Browder

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Initial Assessment and Airway Management for Trauma Surgery

Emergency Tube Thoracostomy

Emergency Department Thoracotomy

Exploratory Surgery for Neck Trauma

Anesthetic Considerations for Neck Trauma Surgery

Chest Trauma: Pericardial Window, Release of Tamponade, and

Repair of Cardiac Laceration
Chest Trauma: Repair of Great Vessels
Chest Trauma: Pneumonectomy, Lobectomy, and Repair of
Tracheobronchial Injury
Abdominal Trauma: Damage Control
Abdominal Trauma: Hepatic and Splenic Injuries
Abdominal Trauma: Vascular Injuries
Pediatric Trauma

8.0 OBSTETRIC/GYNECOLOGIC SURGERY

8.1 Gynecologic Oncology

SURGEONS Zhenhua Du, Nelson N. H. Teng

ANESTHESIOLOGIST Clifford A. Schmiesing

Staging Laparotomy for Ovarian, Fallopian Tube, and Primary
Peritoneal Cancer

Interval, Secondary Cytoreductive, and Second-Look
Assessment Laparotomy

Radical Vulvectomy

Conization of the Cervix

Anesthetic Considerations for Laser Therapy to the Vulva,
Vagina, and Cervix

Suction Curettage for Gestational Trophoblastic Disease

Pelvic Exenteration

Exploratory Laparotomy or Laparoscopy and Hysterectomy/BSO
for Uterine Cancer

Radical Hysterectomy

Robotic-Assisted Laparoscopy and Surgery in Gynecologic
Oncology

Interstitial Perineal Implants

8.2 Gynecology/Infertility Surgery

SURGEONS Deirdre Lum, Amin A. Milki, Jonathan Kort, Lynn M.
Westphal

ANESTHESIOLOGIST Clifford A. Schmiesing

Dilatation and Curettage (D&C)

Therapeutic Abortion, Dilatation, and Evacuation (D&E)

Hysteroscopy

Pelvic Laparotomy
Transvaginal Oocyte Retrieval
Infertility Operations/Assisted Reproductive Technologies
Hysterectomy—Vaginal, Abdominal, Laparoscopic, or Robotic
Operations for Pelvic Organ Prolapse: Anterior and Posterior
Colporrhaphy, Enterocoele Repair, Sacrospinous Ligament
Vaginal Vault Suspension, Uterosacral Ligament Vaginal Vault
Suspension, and Sacrocolpopexy
Operations for Stress Urinary Incontinence

8.3 Obstetric Surgery

SURGEONS Samantha Corey Do, Yasser Y. El-Sayed

ANESTHESIOLOGISTS Brendan Carvalho, Lee Coleman

Cesarean Section—Lower Segment and Classic
Medical and Surgical Management of Postpartum Hemorrhage
Repair of Uterine Rupture
Postpartum Tubal Ligation
Repair of Vaginal/Cervical Lacerations
Cervical Cerclage—Elective and Emergent
Removal of Retained Placenta
Management of Uterine Inversion

8.4 Laparoscopic Procedures with and Without Robotic Assistance for Gynecologic Surgery

SURGEONS Camran R. Nezhat, Stacy Young, Megan Kennedy
Burns, Lucia DiFrancesco, Azadeh A. Nezhat

ANESTHESIOLOGIST Clifford A. Schmiesing

Laparoscopic Surgery for Endometriosis
Laparoscopic Surgery for Ectopic Pregnancy or Adnexal Mass
Laparoscopic Myomectomy
Laparoscopic Hysterectomy
Laparoscopic Surgery for Vaginal Vault Suspension

9.0 UROLOGY

SURGEON Harcharan S. Gill

ANESTHESIOLOGISTS Periklis Panousis, Roya Saffary

Diagnostic Transurethral (Endoscopic) Procedures

Therapeutic Transurethral Procedures (Except TURP)
Transurethral Resection of the Prostate (TURP)
Open Prostate Operations
Nephrectomy
Operations on the Renal Pelvis and Upper Ureter
Cystectomy
Open Bladder Operations (Other than Cystectomy)
Inguinal Operations
Penile Operations
Scrotal Operations
Perineal Operations
Vaginal Operations
Special Considerations for Robotic-Assisted Laparoscopic
Procedures

10.0 ORTHOPEDIC SURGERY

10.1 Hand Surgery

SURGEONS Paige M. Fox, Catherine Curtin, Vincent R. Hentz

ANESTHESIOLOGIST Lindsey Vokach-Brodsky

Darrach Procedure

Dorsal Stabilization and Extensor Synovectomy of the
Rheumatoid Wrist

Metacarpophalangeal and Interphalangeal Joint Arthroplasty and
Arthrodesis

Arthrodesis of the Wrist

Total Wrist Replacement

Thumb Carpometacarpal Joint Fusion/Arthroplasty/Stabilization

Excision of Ganglion of the Wrist

Palmar and Digital Fasciectomy

Repair of Lacerated Tendons/Nerves

Wrist Arthroscopy/Repair of Triangular Fibrocartilage Complex
Tears

Carpal Tunnel Release

Fixation of Fractures and Dislocations of the Wrist and Hand

Digit and Hand Replantation

10.2 Shoulder/Arm Surgery

SURGEONS Amy L. Ladd, Emilie V. Cheung

ANESTHESIOLOGIST Lindsey Vokach-Brodsky

Surgeon's Perspective: Regional Anesthesia in Shoulder Surgery

Arthroscopic Shoulder Surgery

Surgery for Acromial Impingement, Rotator Cuff Tears, and Acromioclavicular Joint Arthritis

Surgery for Shoulder Instability

Glenohumeral Shoulder Arthroplasty

Shoulder Girdle Procedures

Brachial Plexus Surgery

Arm Surgery

10.3 Spine Surgery

SURGEONS Ivan Cheng, Alexander H. Arzeno

ANESTHESIOLOGISTS Karl Zheng, C. Philip Larson Jr.

Minimally Invasive Posterior Lumbar Discectomy (Microdiscectomy)

Minimally Invasive Anterior Lumbar Interbody Fusion Through a Transpsoas Approach

Spinal Reconstruction and Fusion: Thoracic and Thoracolumbar Spine

Spinal Reconstruction and Fusion: Lumbosacral Spine

Spinal Reconstruction and Fusion: Cervical Spine

10.4 Hip, Pelvis, and Upper Leg Surgery

SURGEONS James I. Huddleston III, Stuart B. Goodman, Michael J. Bellino

ANESTHESIOLOGISTS Frederick G. Mihm, Christopher J. Painter, Pedro P. Tanaka

Open Reduction and Internal Fixation (ORIF) of the Pelvis or Acetabulum

Closed Reduction and External Fixation of the Pelvis

Open Reduction and Internal Fixation (ORIF) of Acetabulum Fractures

Osteotomy and Bone Graft Augmentation of the Pelvis

Arthrodesis of the Sacroiliac Joint

Amputations About the Hip and Pelvis: Disarticulation of the Hip and Hindquarter Amputation

Arthroplasty of the Hip
Arthrodesis of the Hip
Synovectomy of the Hip
Open Reduction and Internal Fixation (ORIF) of Proximal Femoral Fractures (Femoral Neck, Intertrochanteric, Subtrochanteric Fractures)
Open Reduction and Internal Fixation (ORIF) of Distal Femur Fractures
Open Reduction and Internal Fixation (ORIF) of the Femoral Shaft with Plate
Intramedullary Nailing of Femoral Shaft
Repair of Nonunion/Malunion of Proximal Third of the Femur, Proximal Femoral Osteotomy for Osteoarthritis
Closed Reduction and External Fixation of the Femur

10.5 Knee Surgery

SURGEONS James I. Huddleston III, Stuart B. Goodman

ANESTHESIOLOGISTS Frederick G. Mihm, Christopher J. Painter, Pedro P. Tanaka

Arthroplasty of the Knee
Arthrodesis of the Knee
Open Reduction and Internal Fixation (ORIF) of Patellar Fractures
Repair or Reconstruction of Knee Ligaments
Patellar Realignment
Arthroscopy of the Knee
Knee Arthrotomy
Repair of Tendons—Knee and Leg

10.6 Lower Leg, Ankle, Foot, and Other Lower-Extremity Procedures

SURGEONS Ariel Palanca, David Oji

ANESTHESIOLOGISTS Frederick G. Mihm, Christopher J. Painter, Pedro P. Tanaka

Open Reduction and Internal Fixation (ORIF) of the Tibial Plateau Fracture
Intramedullary Nailing, Tibia
External Fixation, Tibia

Open Reduction and Internal Fixation (ORIF) of Distal Tibia,
Ankle, and Foot Fractures
Repair Nonunion/Malunion, Tibia
Arthroscopy of the Ankle
Ankle Arthrotomy
Ankle Arthrodesis
Total Ankle Replacement
Repair/Reconstruction of Ankle Ligaments
Amputation Through Ankle (SYME)
Amputation, Transmetatarsal
Lengthening or Transfer of Tendons, Ankle, and Foot
Amputation Above the Knee
Amputation Below the Knee
Fasciotomy of the Thigh
Fasciotomy of the Leg
Biopsy, Leg and Foot
Biopsy or Drainage of Abscess/Excision of Tumor

11.0 PLASTIC AND RECONSTRUCTIVE SURGERY

11.1 Facial Cosmetic Surgery

SURGEONS Jeff Chang, David M. Kahn

ANESTHESIOLOGISTS Richard A. Jaffe, Vanila M. Singh

Introduction to Cosmetic Facial Surgery
Facelift and Neck Lift
Brow Lift and Blepharoplasty
Rhinoplasty
Facial Laser Resurfacing

11.2 Nonfacial Aesthetic Surgery

SURGEON David M. Kahn

ANESTHESIOLOGISTS Richard A. Jaffe, Vanila M. Singh

Augmentation Mammoplasty
Reduction Mammoplasty
Mastopexy/Breast Lift
Brachioplasty
Abdominoplasty
Body Lifts

Liposuction

11.3 Craniofacial Surgery

SURGEON Stephen A. Schendel

ANESTHESIOLOGIST Richard A. Jaffe

Repair of Facial Fractures

Le Fort Osteotomies

Mandibular Osteotomies/Genioplasty

11.4 Functional Restoration

SURGEONS James Chang, David M. Kahn, Jacinta Leyden

ANESTHESIOLOGISTS Richard A. Jaffe, Vanila M. Singh

Microsurgery Free-Flap Reconstruction

Microsurgery Replantation

Breast Surgery—Introduction

Breast Reconstruction

Chest Wall Reconstruction

Pressure Sore Reconstruction

11.5 Burn Surgery

ANESTHESIOLOGISTS Richard A. Jaffe, Melissa T. Berhow

Free Skin Graft for Burn Wound (with Tangential Excision,
Excision to Fascia, or Debridement)

12.0 PEDIATRIC SURGERY

12.1 Pediatric Neurosurgery

SURGEONS Gerry A. Grant, Michael S. B. Edwards, Samuel H. Cheshier

ANESTHESIOLOGISTS Michael I. Chen, Lisa Wise-Faberowski

Craniectomy for Craniosynostosis (Craniofacial Surgery)

Repair of Myelomeningocele

Laminectomy, with Release of Tethered Spinal Cord (Spina
Bifida Occulta)

Intracranial Neuroendoscopy (Endoscopic Third Ventriculostomy,
Coagulation of Choroid Plexus, Fenestration of Cyst)

Surgery for Intracranial Arteriovenous Malformation

Craniotomy for Vein of Galen Malformation

12.2 Pediatric Ophthalmic Surgery

SURGEONS Natalia F. Callaway, Darius M. Moshfeghi

ANESTHESIOLOGIST Olga Wolke

Strabismus Surgery

Pediatric Vitreoretinal Surgery

12.3 Pediatric Otolaryngology

SURGEONS Mai Thy Truong, Anna H. Messner

ANESTHESIOLOGISTS Samuel A. Mireles, Ellen Ya-Ping Wang,
Gregory B. Hammer

Myringotomy and Tympanostomy Tube Placement

Tonsillectomy and Adenoidectomy

Bronchoscopy/Esophagoscopy/Removal of Foreign Body

Laryngoscopy, Supraglottoplasty, and Excision of Laryngeal
Lesions

Removal of Branchial Cleft Cyst or Thyroglossal Duct Cyst

Incision/Drainage of Deep Neck Abscess

Laryngotracheal Reconstruction, Cricotracheal Resection, and
Laryngotracheoplasty

Choanal Atresia Repair

Pediatric Tracheostomy

12.4 Pediatric Cardiovascular Surgery

SURGEONS Katsuhide Maeda, Frank L. Hanley

ANESTHESIOLOGISTS Claudia Benkwitz, Lisa Wise-Faberowski,
Komal Kamra, M. Gail Boltz, Chandra Ramamoorthy

Surgery for Atrial Septal Defect (Ostium Secundum)

Surgery for Atrioventricular Canal Defect

Surgery for Ventricular Septal Defect

Surgery for Patent Ductus Arteriosus

Surgery for Coarctation of the Aorta

Surgery for Tetralogy of Fallot

Surgery for Total Anomalous Pulmonary Venous Connection

Surgery for Complete Transposition of the Great Arteries

Surgery for Truncus Arteriosus

Surgery for Tricuspid Atresia

Surgery for Double-Outlet Right Ventricle

Surgery for Hypoplastic Left Heart Syndrome

12.5 Pediatric General Surgery

SURGEONS Jordan S. Taylor, James Wall

ANESTHESIOLOGISTS Andrew J. Giustini, Rebecca E. Claire,
Brenda Golianu

Resection of Cystic Hygroma, Branchial Cleft Cyst, Thyroglossal
Duct Cyst, or Other Cervical Mass

Esophagus—Foreign Body Removal and Dilation

Achalasia, Heller Myotomy, Peroral Endoscopic Myotomy
(POEM), and Advanced Endoscopic Surgeries

Repair of Tracheoesophageal Fistula and Esophageal Atresia

Mediastinal Mass—Biopsy or Resection

Neonatal Lung Resection

Drainage of Empyema

Repair of Pectus Excavatum/Carinatum

Esophageal Replacement, Colon Interposition

Repair of Congenital Diaphragmatic Hernia

Pyloromyotomy for Pyloric Stenosis

Abdominal Tumor: Resection of Neuroblastoma, Wilms Tumor,
and Hepatic Tumors

Laparotomy for Intestinal Perforation and Necrotizing
Enterocolitis

Repair of Biliary Atresia and Choledochal Cysts

Repair of Abdominal Wall Defects: Omphalocele/Gastroschisis

Pull-Through for Hirschsprung Disease

Pull-Through for Imperforate Anus and Cloaca

Repair of Inguinal and Umbilical Hernias and Hydrocele

Surgery for the Undescended Testicle

Resection of Sacrococcygeal Teratoma

Anesthesia for Minimally Invasive Surgery in Pediatric Patients

Ex Utero Intrapartum Treatment (EXIT) Procedure

Pediatric Bariatric Surgery

12.6 Pediatric Urology

SURGEONS Andrew J. Sun, Hsi-Yang Wu

ANESTHESIOLOGIST Radhamangalam J. Ramamurthi

Kidney and Upper Urinary Tract Operations

Percutaneous Stone Procedures
Transurethral Procedures
Simple Open Bladder Operations
Complex Open Bladder Operations
Penile Surgery
Genital Procedures (Clitoroplasty, Vaginoplasty, Urethroplasty)
Inguinoscrotal Procedures
Laparoscopic and Robotic Procedures

12.7 Pediatric Orthopedic Surgery

SURGEONS Amy L. Ladd, Matthew B. Burn, James G. Gamble

ANESTHESIOLOGISTS Genevieve D'Souza, Brenda Golianu,
Radhamangalam J. Ramamurthi

Percutaneous Pinning of Displaced Supracondylar Humerus
Fracture

Closed or Open Reduction of Displaced Lateral Condyle
Humerus Fracture

Aspiration and Injection of Unicameral Bone Cyst

Release for Torticollis

Pollicization of a Finger

Syndactyly Repair

Posterior Spinal Instrumentation and Fusion

Anterior Spinal Fusion for Scoliosis

Pelvic Osteotomy

Acetabular Augmentation (Shelf) and Chiari Osteotomy

Ober Fasciotomy and Yount-Ober Release

Hip, Open Reduction, and Femoral Shortening

Adductor Release or Transfer and Psoas Release

Pinning of Slipped Capital Femoral Epiphysis

Flexible Intramedullary Nailing of Long-Bone Fractures

Proximal Femoral Osteotomy

Epiphysiodesis

Sofield Procedure

Limb Lengthening

Patellar Realignment

Tendon Transfer, Lengthening (Posterior Tibial)

Triple Arthrodesis and Grice Procedure (Extra-Articular Subtalar
Arthrodesis)

Surgical Correction of Clubfoot
Surgery for Epidermolysis Bullosa

12.8 Surgery for Craniofacial Malformations

SURGEON Stephen A. Schendel

ANESTHESIOLOGISTS Nai Yi Sun, Rita Agarwal, Louise K. Furukawa

Surgical Correction of Craniosynostosis
Major Secondary Craniofacial Surgical Procedures
Cleft Lip Repair—Unilateral/Bilateral
Palatoplasty
Pharyngoplasty
Alveolar Cleft Repair with Bone Graft
Secondary Cleft Lip/Nasal Surgery
Otoplasty

12.9 Pediatric Transplantation

SURGEON Waldo Concepcion

ANESTHESIOLOGISTS Julianne Mendoza, Louise K. Furukawa, Gregory B. Hammer

Pediatric Renal Transplantation
Pediatric Liver Transplantation

13.0 NON-OPERATING ROOM PROCEDURES

13.1 Non-Operating Room Procedures: Adult

PROCEDURAL SPECIALISTS Charles DeBattista, Liong-Bing Liem, Michael P. Marks, Daniel Y. Sze, David S. Wang, Andrew Kesselman, Ibrahim A. Idakoji

ANESTHESIOLOGISTS Richard A. Jaffe, John G. Brock-Utne, Timothy P. Angelotti

Anesthesia for Non-Operating Room Procedures
Electroconvulsive Therapy
Interventional Neuroradiology
Direct Current Cardioversion
Implantation of Cardioverter-Defibrillator (ICD) or Cardiac Resynchronization Therapy Defibrillator (CRT-D)
Extraction of Pacemaker and Implantable Cardioverter-

Defibrillator (ICD) Leads
Catheter Ablation of Atrial Fibrillation
Catheter Ablation of Ventricular Tachycardia
Transjugular Intrahepatic Portosystemic Shunt (TIPS)
Endoscopic Retrograde Cholangiopancreatography
Imaging and Image-Guided Procedures
Tracheobronchial Stenting
Thermal Ablation of Tumors and Other Targets
Irreversible Electroporation (IRE Ablation)

13.2 Non-Operating Room Procedures—Pediatric

PROCEDURAL SPECIALISTS Claudia Mueller, William E. Berquist, Sarah S. Donaldson, Anne M. Dubin, Gary E. Hartman, Susan M. Hiniker, Katsuhide Maeda, Lynn Peng, Ann Ming Yeh

ANESTHESIOLOGISTS Claudia Benkwitz, Erin Bushell, Rebecca E. Claire, Louise K. Furukawa, Brenda Golianu, Kristin Sun, Lisa Wise-Faberowski, Komal Kamra

Pediatric Radiation Therapy
Pediatric Cardiac Catheterization and Electrophysiology
Pediatric Oncologic Procedures
Upper/Lower GI Endoscopy
Anesthesia for the Patient with Epidermolysis Bullosa (EB)
Cross-Sectional Imaging (CT, MRI)
Surgical Considerations for ECMO
Anesthetic Management for Surgical Procedures Under ECMO

14.0 OFFICE-BASED ANESTHESIA

PROCEDURAL SPECIALISTS David A. Berman, Vernon J. Adams Jr., Azeem K. Lakha

ANESTHESIOLOGISTS Richard A. Jaffe, Terri D. Homer

Introduction—Anesthesiologist's Perspective
Facial Rejuvenation: Lasers and RF Tissue Tightening
Office Dental Rehabilitation Under Deep IV Sedation
Dental Implants and Bone Grafting

15.0 EMERGENCY PROCEDURES FOR THE ANESTHESIOLOGIST

ANESTHESIOLOGIST Frederick G. Mihm

Emergency Cricothyrotomy
Emergency Pericardiocentesis
Emergency Ultrasound-Guided Pericardiocentesis
Emergent Needle/Catheter Thoracostomy
Emergent Ultrasound-Guided Needle/Catheter Thoracostomy
Emergency Intraosseous Access

16.0 REGIONAL BLOCKS—A BRIEF OVERVIEW

ANESTHESIOLOGISTS Jean-Louis Horn, Mastoora Nasiri

Upper Extremity Blocks
Lower Extremity Blocks
Thoracoabdominal Blocks

APPENDICES

Appendix A: Preoperative Considerations

Clifford A. Schmiesing

Appendix B: Standard Adult Anesthetic Protocols

Richard A. Jaffe, Clifford A. Schmiesing, Steven L. Shafer

Appendix C: Multimodal Perioperative Pain Management

Roya Saffary, Einar Ottestad

Appendix D: Standard Pediatric Anesthetic Management

Birgit Maass, Brenda Golianu

Appendix E: Multimodal Pediatric Postoperative Pain
Management

Genevieve D'Souza, Julie Good, Brenda Golianu

Appendix F: Tables of Drug Interactions and Recommendations
for Preop Management

Carolyn Nguyen

Appendix G: Special Considerations for Latex Allergy

Andrew J. Giustini, Rebecca E. Claire, Anita Honkanen, Brenda Golianu

Appendix H: Perioperative Acupuncture

Brenda Golianu, Jeannie Seybold

Appendix I: Management of the Difficult Airway

C. Philip Larson Jr.

Appendix J: Anticoagulation Guidelines for Neuraxial Procedures

Einar Ottestad, Alex J. Butwick, Brendan Carvalho, Jamie Kuo,
Sharon Jung

[Appendix K: Enhanced Recovery After Surgery](#)

Richard A. Jaffe, Andrew J. Giustini, Brenda Golianu

[Appendix L: High-Flow Nasal Cannula \(HFNC\) and Transnasal
Humidified Rapid-Insufflation Ventilatory Exchange \(THRIVE\)](#)

Erin Bushell

[Appendix M: Abbreviations](#)

SUBJECT INDEX

SECTION **1.0**

NEUROSURGERY

CHAPTER 1.1

Intracranial Neurosurgery

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CRANIOTOMY FOR INTRACRANIAL ANEURYSMS

SURGICAL CONSIDERATIONS

Gary K. Steinberg and Robert L. Dodd

Description

Intracranial aneurysms are focal protrusions arising from weakened arterial walls usually at major bifurcations of the arteries at the base of the brain (some frequent sites of aneurysms are shown in [Fig. 1.1-1](#)) and are most commonly treated by endovascular coiling (>50%) or microsurgical clip ligation. The high rate of mortality (~45% within 30 d) and morbidity from aneurysmal rupture necessitates treatment for symptomatic lesions. Treatment for asymptomatic lesions generally is recommended when the lifetime risk of rupture exceeds the risk of treatment. The most important surgical considerations include clinical presentation, aneurysm size and location, patient age, neurologic status, and medical comorbidities. Aneurysm rupture into the subarachnoid space is the most common clinical presentation; however, symptoms from the mass

effect of enlarging aneurysms or ischemic symptoms from emboli also may occur. Aneurysm morphology, size, and location are important in determining the surgical approach, and these aneurysm characteristics, as well as patient age, condition, and comorbidities, affect the overall outcome. The **Hess and Hunt clinical grading system** (Table 1.1-1) has been proven useful in describing patients with ruptured intracranial aneurysms because it has been shown to have prognostic value in terms of ultimate clinical outcome. Grading is based on the neurologic examination and ranges from grade I (minimal headache, no neurologic deficit) to grade V (moribund) (see Table 1.1-1).

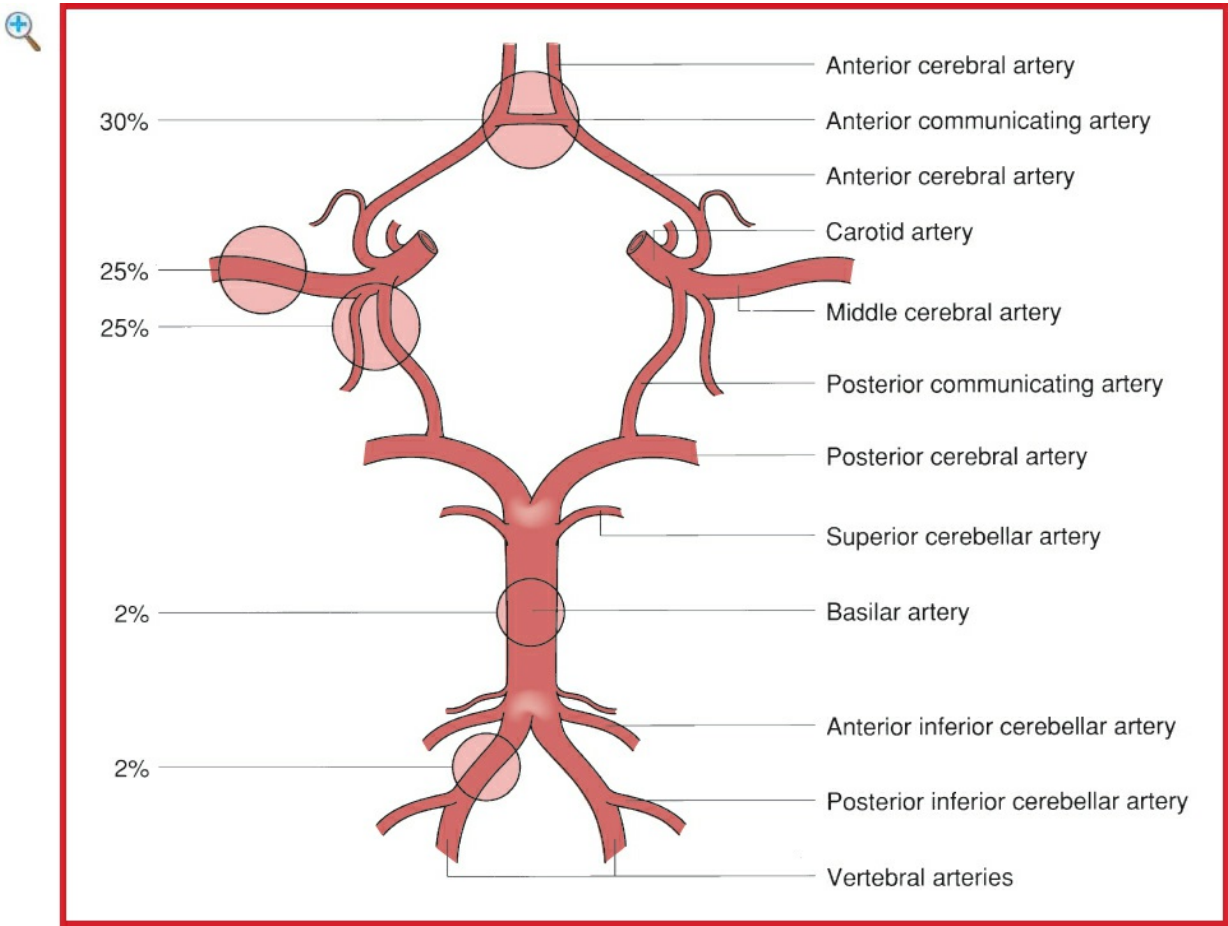


Figure 1.1-1 Locations of aneurysms of the circle of Willis and their relative occurrence. (Reproduced with permission from Greenfield LJ, Mulholland MW, Lillemoe KD, et alSurgery: Scientific Principles and Practice, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Table 1.1-1 Hunt-Hess Grading System for Aneurysmal SAH*	
Grade	Description
I	Asymptomatic or minimal headache and slight neck stiffness (mortality ~2%)
II	Moderate-to-severe headache, neck stiffness, no neurologic deficit

	(except cranial nerve palsy)—(mortality ~5%)
III	Drowsiness, confusion, or mild focal deficit (mortality 15–20%)
IV	Stupor, moderate-to-severe hemiparesis, possible early decerebrate rigidity, and vegetative disturbances (mortality 30–40%)
V	Deep coma, decerebrate rigidity, moribund (mortality 50–80%)

*The presence of serious systemic disease—such as hypertension, diabetes, severe arteriosclerosis, chronic pulmonary disease, and angiographic vasospasm—results in placement in the next less-favorable category.

Patients with ruptured aneurysms require careful preop management to control MAF (CPP), ICP, glucose, and sodium. MAP < 70 or >130, HCQ < 20, glucose >180, and hypoxia have been independently associated with ↑ mortality and morbidity. EVD (ventriculostomy or external ventricular drain) or spinal drain may be placed to manage ICP and/or drain CSF. Unless otherwise required, these drains should be closed during patient transport. On arrival in the OR, level drain at tragus, set height relative to tragus as requested by the surgical team and then open drain. Keep surgeons informed of drainage volume at regular intervals throughout the case.

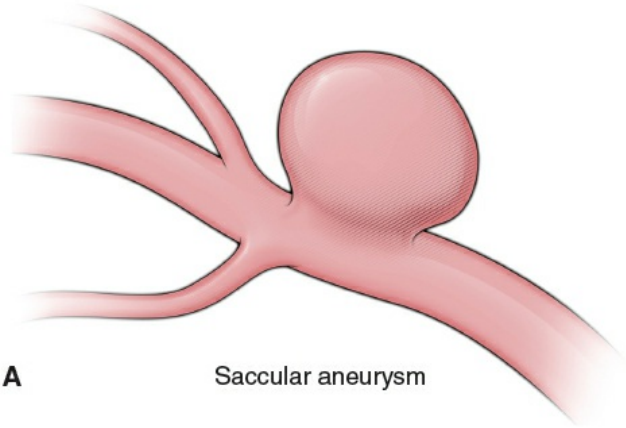
Through a **craniotomy** or **craniectomy**, using microscopic techniques, the parent vessel giving rise to the aneurysm is identified. The aneurysm is isolated, and small, nonferromagnetic alloy spring clips are placed at the origin of the aneurysm, excluding it from the circulation. A **frontotemporal (pterional) craniotomy** normally is used to approach anterior circulation aneurysms. This requires extensive drilling of the medial sphenoid wing (pterion) and allows access to most aneurysms on the anterior and lateral circle of Willis vessels: internal carotid-paraclinoid/superior hypophyseal artery, internal carotid-ophthalmic artery, posterior communicating artery, anterior choroidal artery, internal carotid artery bifurcation, middle cerebral artery, and anterior communicating artery. Posterior circulation aneurysms are approached via a pterional or **subtemporal exposure** (upper basilar artery, posterior cerebral artery, superior cerebellar artery), a **suboccipital exposure** (vertebral artery, posterior inferior cerebellar artery), or a combined subtemporal and suboccipital exposure (basilar trunk, vertebrobasilar junction). Increasingly, aneurysms requiring craniotomy for treatment are complex (see [Fig. 1.1-2](#)) and therefore less suitable for endovascular treatment. Clipping these aneurysms can be time consuming and must be accomplished in a way that preserves flow through all the affected vessels and their branches. This may not be possible without a bypass graft to restore flow distal to the aneurysm ([Fig. 1.1-2C](#)). The use of temporary clips to eliminate flow to the aneurysm can greatly facilitate the placement of permanent clips and will be required to complete a bypass graft.

Temporary clipping may result in focal ischemic injury, which can be minimized by inducing EEG burst suppression in combination with mild hypothermia to decrease metabolic demand. Increasing MAP is often requested during temporary clipping to improve distal supply through collateral flow. Circulatory arrest under CPB with deep hypothermia (16–20°C) may be used for repairing some giant (>2.5 cm) aneurysms.

Usual preop diagnosis

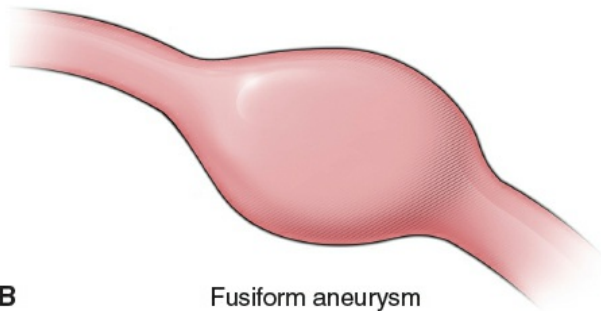
Cerebral aneurysm; subarachnoid hemorrhage (SAH); intracerebral hemorrhage
progressive neurologic deficits (mass effect on cranial nerves or CNS structures); TIAs
cerebral infarct

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and AED medications, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possibility of urinary drain postop. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).



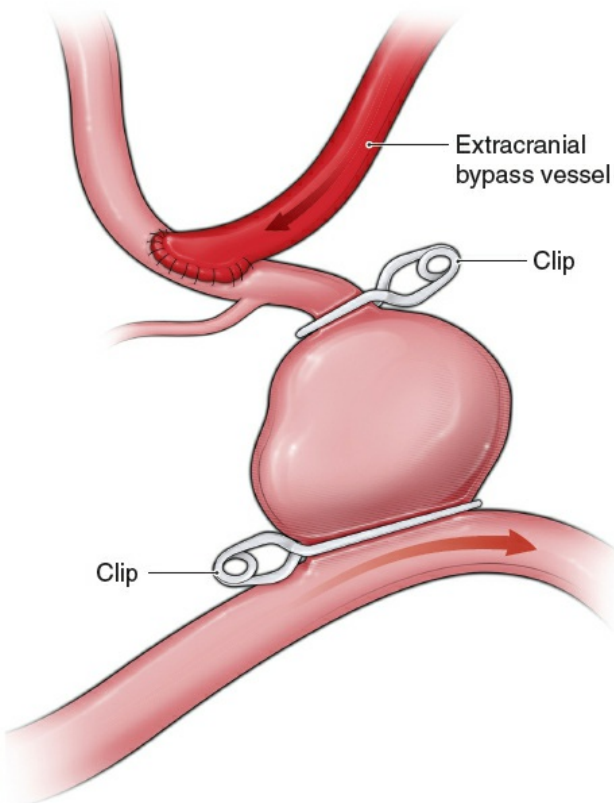
A

Saccular aneurysm



B

Fusiform aneurysm



C

Complex aneurysm

Figure 1.1-2 Aneurysm types. **A:** Saccular; **B:** fusiform; **C:** complex (showing bypass and clip placement).

■ SUMMARY OF PROCEDURES			
	Anterior Circulation Aneurysms	Posterior Circulation Aneurysms	Circulatory Arrest (CPB) W/Deep Hypothermia
Position	Supine, head in Mayfield headrest, turned 30–45° to side away from aneurysm, vertex dropped (Fig. 1.1-6)	⇐ or lateral decubitus, head lateral in Mayfield headrest	⇐ + Both groins must be accessible for arterial + venous cannulation; access to chest for defibrillation
Incision	Frontotemporal	⇐ or temporal, temporosuboccipital, or suboccipital	⇐
Special instrumentation	Operating microscope; aneurysm clips; radiolucent table, and headrest if intraop angiography is planned	⇐ + Aperture clips to accommodate cranial nerves and critical vessels	⇐ + CPB pump; femoral cannulae; defibrillator; CUSA for partially thrombosed aneurysms
Unique considerations	Temporary arterial clipping; mild hypothermia (33°C); intraop angiography with access to femoral artery; electrophysiologic	⇐	⇐ + No manipulation of brain retractors after systemic heparinization; meticulous

	monitoring (SEPs, BAERs, MEPs); brain relaxation; $\pm$ lumbar subarachnoid CSF drainage; dexamethasone 8–12 mg iv; indocyanine green for video angiography. LAs with EPI may $\rightarrow$ $\uparrow$ BP and risk of rupture		attention to hemostasis
Antibiotics	Cefazolin 2–3 g iv q6–8h	$\Leftarrow$	$\Leftarrow$
Surgical time	3–5 h	3–6 h	6–8 h
Closing considerations	Rewarming to $>35^{\circ}\text{C}$ may be necessary	$\Leftarrow$	$\Leftarrow$
EBL	250–1000 mL	$\Leftarrow$	2000–4000 mL
Postop care	ICU 1–2 d; postop $\pm$ CBF monitoring, transcranial Doppler (TDC) monitor for vasospasm	$\Leftarrow$	$\Leftarrow$ + ICP monitoring
Mortality	Unruptured: 0.5%	1.5%	5–10% (giant aneurysms)
	Ruptured:		
	Hunt and Hess grades I–III: $<10\%$ (see Table 1.1-1)	15%	5–15%
	Hunt and Hess grades IV–V: 20–	20–50%	15–50%

40% (see [Table 1.1-1](#))

Morbidity	Neurologic: 5–20%	10–30%	10–50%
	Cranial nerve injury	⇐	⇐
	Stroke: 5.2% (38% if ruptured)	⇐	⇐
	Hydrocephalus: 15–25%	⇐	⇐
	Hyponatremia (SIADH vs. CSW)	⇐	⇐
	Seizures: 2–20%	⇐	⇐
	Respiratory failure:	⇐	⇐
	Rare	⇐	⇐
	Thromboembolism:		
	Rare		
	CSF leak: Rare		
	Infection: Rare		
	Massive blood loss: Rare		
Pain score	3–4	3–4	4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	30–70 yr (Peak incidence in 5th and 6th decades)
Male:Female	44:56
Incidence	8–16/100,000/yr for ruptured aneurysms with SAH (highest in Japan and Finland). Approximately 85% survive initial bleed. Overall incidence of rebleeding is ~11% (4% in the first 24 h).
Etiology	Idiopathic (probably acquired and related to hemodynamic stress at arterial branch points, although may have congenital predisposition to loss of internal elastic lamina); traumatic; infectious; familial
Associated conditions	Polycystic kidney disease; coarctation of the aorta; Marfan syndrome; Ehlers-Danlos syndrome; intracranial AVMs; aortic aneurysm; fibromuscular dysplasia; pseudoxanthoma elastica; Rendu-Osler-Weber syndrome; cocaine/amphetamine use

ANESTHETIC CONSIDERATIONS

(Procedures covered: craniotomy for intracranial aneurysms; craniotomy for cerebral embolectomy)

PREOPERATIVE

Aneurysms may occur in any age group, although they generally become symptomatic and are diagnosed in young or middle-aged adults who are usually in otherwise good health. Most patients have warning Sx before the first major bleed, but these tend to be mild and nonspecific (e.g., headache, dizziness, orbital pain, slight motor or sensory disturbances). The symptoms are generally disregarded by both patients and physicians. Most patients with SAH will be receiving oral nimodipine or iv nicardipine; SBI should be maintained between 120 and 140 mm Hg using vasoactive drug infusions if necessary. Patients with symptomatic vasospasm may benefit from single-H therapy (see below).

System	Considerations
Respiratory	Respiratory complications (e.g., neurogenic pulmonary edema [up to 23%], pneumonia, ARDS, PE) are the most common nonneurologic causes of death following SAH. Pulmonary aspiration may have occurred as the result of a neurologic deficit from an intracranial hemorrhage. Tests: As indicated from H&P. Elevated cTnI may predict pulmonary complications.
Cardiovascular	Generally, these patients do not have other cardiovascular diseases, although intracerebral aneurysms occur more commonly in patients with certain congenital disorders, such as polycystic disease of the kidneys, coarctation of the aorta, fibromuscular hyperplasia, and Marfan or Ehlers-Danlos syndrome. Patients who have had a recent intracranial hemorrhage (ICH) are prone to develop systemic HTN, hypovolemia, ↓ Mg ⁺⁺ , and ECG abnormalities. The HTN is thought to be due to autonomic hyperactivity and is generally treated with antihypertensive medication, which should be

continued up to the time of anesthesia and surgery. ECG abnormalities occur in many patients following ICH and may represent subendocardial injury 2° catecholamine release from intracardiac nerve terminals (contraction band injury). Dysrhythmias (most commonly PVCs) occur in 30–80% of patients, and ischemic changes (typically T-wave inversion and ST-segment depression) are seen in >50%. Appropriate preop preparation includes ECG characterization of the abnormality. If patient has Hx of ischemic heart disease, ECHO and cardiac enzyme studies may be helpful in determining whether the myocardial injury and consequent ECG changes are likely to be of clinical significance.

Tests: ECG; cTnI; others (e.g., ECHO) as indicated from H&P

Neurologic

Occasionally unruptured aneurysms produce neurologic Sx by compressing adjacent neural tissue or cause ↑ ICP. If ICH occurs, the neurologic dysfunction will vary, depending on the site and extent of the hemorrhage. Typically, SAH → ↑↑ ICP → sudden severe HA ± confusion and disorientation, ± motor deficit, ± coma. Admission GCS < 12 predicts ↑ mortality. Significant cerebral vasospasm (~30% incidence) and delayed cerebral ischemia (DCI) may cause worsening of the neurologic deficits. Clinically significant vasospasm commonly begins on day 3 reaching a maximum at 7–8 d and usually is resolved within 2–4 wk. The precipitating event is believed to be clot lysis → release of vasospastic substances. Rx vasospasm with support of BP (occasionally, induced HTN), euvolemia, and systemic Ca⁺⁺ antagonists (e.g., nimodipine [60 mg PO q4h] or iv [3–15 mg/h iv]). Triple-H therapy (↑ volume, ↑ BP, ↓ Hct) should be avoided in favor of euvolemia and a trial of ↑ BP for DCI. Occasionally, intra-arterial angioplasty may be necessary to Rx refractory vasospasm. Control of BP is important: any

	substantial $\uparrow$ BP may $\rightarrow$ serious rebleed, permanent neurologic deficits or death; any substantial $\downarrow$ BP may $\rightarrow$ cerebral ischemia and infarction. Arterial catheterization and continuous beat-to-beat monitoring of BP prior to induction of anesthesia is essential in these patients. Post-SAH Sz is not unusual and may require Rx with AEDs. Tests: CT; MRI; CTA, which the anesthesiologist should examine preop to identify the nature and site of the aneurysm. DSA is the gold standard for detection of vasospasm.
Hydrocephalus	Acute hydrocephalus occurs in 15–20% of patients within 24 h of SAH. External ventricular drainage may be necessary.
Hematologic	Thrombocytopenia may develop following SAH. DVT prophylaxis (e.g., SCD) is beneficial. Tests: Hct; PT; PTT. Target Hb $\sim$ 10 g/dL.
Radiographic findings	CT: Blood in basal cisterns ($\sim$95% of patients) or ventricles; CTA (replacing conventional angiography): anatomical characteristics of the aneurysm.
Laboratory	Cardiac enzymes if ECG changes observed ($\uparrow$ cTnI predicts $\downarrow$ neurologic outcome and $\uparrow$ mortality); serum Mg^{++} and glucose; others as indicated from H&P. Hyponatremia is common and may be 2° SIADH or CSW. In contrast to SIADH, patients with CSW are hypovolemic ($\downarrow$ plasma volume, $\downarrow$ CVP, $\downarrow$ PCWP), $\pm$ $\uparrow$ K^{+}.
Hepatic	Hepatic dysfunction following SAH is not uncommon (24% in one series). Tests: LFTs, as indicated from H&P
Premedication	Small doses of midazolam 1–3 mg iv are preferable to opiates. Detailed discussion with the patient about the anesthetic plan, with appropriate reassurance, is essential. Should an intracranial aneurysm leak or rupture in the immediate preop period, its signs may

	be difficult to distinguish from those associated with excessive response to premedication.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss possibility of scalp block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water especially for posterior fossa craniotomies with their high incidence of PONV.

◆ INTRAOPERATIVE

Anesthetic technique

GETA: The goals of anesthesia for this surgery are to (a) maintain optimum CPF (cerebral MAP minus cerebral venous pressure or ICP, whichever is greater); (b) decrease intracranial volume (blood and tissue) to optimize working space for surgeons, thereby reducing the need for surgical retraction; and (c) decrease metabolic rate and CMRO₂ with the expectation that the brain will tolerate hypotension and ischemia if sudden decreases in CPP become necessary.

Induction	Smooth induction is essential. Fentanyl 7–10 mcg/kg iv to blunt response to intubation and provide analgesia for the first hours of surgery. Propofol 1–2 mg/kg iv to provide amnesia and ↓ cerebral blood volume by inducing cerebral vasoconstriction. Rocuronium 0.6–1.2 mg/kg or vecuronium 0.15 mg/kg to provide muscle relaxation for tracheal intubation and positioning. Succinylcholine acceptable if RSI required. Patients on nimodipine may require pressors (e.g., phenylephrine) during and after induction. Poor H-H grade patients (IV–V) may not tolerate ↓ MAP during induction. Patients may benefit from moderate hyperventilation during induction.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Avoid hypothermia unless indicated for brain protection. Consider preincision placement of scalp blocks. Dexamethasone (4–8 mg iv)

	may ↓ PONV. Avoid long-acting opioids or scopolamine patch, which may confound postop neuro examination. Sequential pneumatic compression devices should be used to ↓ VTE risk.
Maintenance	Isoflurane or sevoflurane (1/2 MAC if EP monitoring is used), inspired with O₂. Avoid N₂O > 50% and entirely in patients with ↑↑ ICP. Propofol (50–100 mcg/kg/min) may be used to replace N₂O and further ↓ cerebral blood volume, ↓ cerebral metabolism, and ↓ CMRO₂. If movement is of concern, rocuronium 4–8 mcg/kg/min will provide adequate neuromuscular blockade. A remifentanil infusion (0.05–0.2 mcg/kg/min) can be used to supplement the anesthetic without interfering with EP monitoring. Contrary to popular belief, TIVA is not necessary for monitoring SSEPs or MEPs.
Emergence	H-H grade IV–V patients should be kept intubated and sedated on ventilator support postop. For grade I–III patients, with the start of dural closure, ↓ inhaled anesthetic concentration, stop any propofol infusion, continue 50% N₂O (if used), supplemented with a low-dose remifentanil infusion (e.g., 0.05 mcg/kg/min). As recovery from anesthesia occurs, BP will increase. Titration of β-adrenergic blocking drugs (e.g., labetalol and esmolol) with vasodilators (e.g., clevidipine or SNP) may be needed; if so, the dose should be stabilized before transport to ICU (see Control of BP, below). Inhalation agents can be D/C'd at the time of dressing application. Most patients will breathe spontaneously and can be extubated uneventfully while on the remifentanil infusion. If the brain has not been injured by the surgical procedure, the patient should awaken within 10 min after cessation of remifentanil administration. As the patient is awakening, it is important to ensure full reversal from neuromuscular blockade (consider sugammadex 2–4 mg/kg) and close regulation of BP. If the patient begins to cough on ETT, either it should be removed or cough reflex suppressed with iv lidocaine

	(0.5–1 mg/kg). Patient is placed in bed in a 30° head-up position and transported to ICU for monitoring overnight. Supplemental O ₂ should be administered and close regulation of BP maintained. Prophylactic antiemetics (e.g., ondansetron 4–8 mg) should be given 30 min before extubation. Consider iv acetaminophen (1 g) if po dose given >4–6 h ago and scalp block placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago.	
Blood and fluid requirements	IV: 14–18 ga × 2 BSS @ <3 mL/kg/h + UO > 1 mL/kg/h Expand blood volume with albumin 5% if Hct > 30% Albumin + PRBC if Hct < 30% Maintain colloid oncotic pressure Hetastarch may → coagulopathy and renal dysfunction	Maintain euvolemia (see zero balance fluid management, Appendix K, p. K-2). If blood volume is normal, crystalloid fluid should not exceed 3 mL/kg/h beyond that required to replace UO. Rapid, massive blood loss is possible. If blood volume is low (↑ SVV or PPV), albumin 5% is given; combinations of albumin and blood, if Hct is <30%. Albumin is the preferred colloid.
Brain relaxation	Hyperventilate to PaCO ₂ = 35 mm Hg (PetCO ₂ = 30 mm Hg). PaO ₂ > 100 mm Hg	↓ PaCO ₂ → ↓ cerebral vascular volume (better surgical access) + ↑ CBF to ischemic areas ("Robin Hood" effect) + ↓ anesthetic requirements + ↑ lactic acid buffering.
Brain relaxation (cont'd)	Consider propofol infusion (50 mg/kg/h iv) to replace N ₂ O ↓ isoflurane/sevoflurane	Propofol → cerebrovasoconstriction → ↓ ICP. Remifentanyl infusion (0.05–

	to <0.5 MAC Mannitol 0.5–1 g/kg ± Furosemide 0.3 mg/kg Maximum osmolarity = 320 mOsm ± Steroids ± Lumbar CSF drain Head up to provide venous drainage Minimize neck flexion/rotation	0.2 mcg/kg/min) may then be required to provide adequate analgesia. Mannitol/furosemide → ↓ K⁺; monitor level and replace as necessary. If mannitol is administered too rapidly, ↓ BP may occur, 2° peripheral vasodilation. For example, 8 mg dexamethasone at start of surgery. CSF drain may be placed after induction of anesthesia, avoid rapid drainage → aneurysm rupture or remote hemorrhage. Otherwise → ↑ venous pressure → ↑ ICP and ↓ CBF and ↑ venous bleeding.
Monitoring	Standard monitors Arterial line Bladder/esophageal temperature Blood glucose (keep <180 mg/dL) UO (Foley catheter) CVP line (subclavian preferred) ± Evoked potentials/EEG	Direct monitoring of arterial BP is essential because marked BP fluctuations may occur, necessitating rapid drug therapy. Transducers should always be leveled at the head. Monitoring CVP is desirable in virtually all patients to assess adequacy of fluid therapy, for infusion of vasoactive drugs both intraop and postop, and for aspiration of VAE. Localization of the catheter can be determined by CXR, ECG tracing (noting P-wave

		changes), or pressure wave morphology as the catheter is withdrawn from the right atrium.
Hypothermia	Water-circulating pads Cold operating room	Mild hypothermia (33–34°C) is used in some centers to ↓ CMRO ₂ and to ↓ susceptibility to ischemic injury during temporary clip application. CMRO ₂ decreases ~30% @ 33°C. This level of hypothermia has minimal effect on coagulation or the incidence of cardiac dysrhythmias.
	Bladder irrigation	Rewarming using surface means can be quite slow. The use of bladder irrigation (40–42°C saline) or a heat exchange catheter is useful.
	Thermogard XP®, InnerCool® or equivalent	A heat exchange catheter may be placed in the inferior vena cava (via femoral vein) or superior vena cava via the subclavian vein to facilitate patient cooling and rewarming.
	IHASt study ⁸	Although it implied that mild hypothermia was not useful in aneurysm surgery, the IHASt study was not designed to assess the effectiveness of mild hypothermia in the patient group most likely to benefit: otherwise intact patients requiring long temporary clip times. Generalizing the IHASt

	Delayed response to peripherally administered drugs	findings to all aneurysm patients is an unfortunate disservice to many of them. Administer drugs through CVP line in hypothermic patients to ensure prompt effect.
Single H therapy (for vasospasm)	Hypertension	Goal: SBP 120–150 mm Hg (preclipping); 160–200 mm Hg (postclipping). Hypervolemia: No benefit. ↑ cardiopulmonary complications, ↑ infection. Euvolemia is recommended. Hemodilution: Not recommended. Optimal Hct is difficult to predict but is probably 30–33% for most patients.
Control of BP	During application of head fixation device (Mayfield): remifentanyl: 100–250 mcg iv bolus 1–2 min in advance During aneurysm exposure: ↓ MAP to ~80% of baseline During temporary clipping: ↑ MAP to ~120% of baseline	Control of BP is critical. ↑↑ BP → ↑↑ transmural pressure across the aneurysmal wall → rupture of the aneurysm. Many neurosurgeons apply a temporary clip on the major feeding vessel(s) in advance of clipping the aneurysm. This technique depressurizes the aneurysm and makes clipping easier and less likely to cause inadvertent rupture. During temporary clipping, keep MAP ~20% above baseline to maximize collateral flow. Phenylephrine is preferred because it has minimal dysrhythmogenic potential. EEG/EP monitoring can be useful to guide BP

	Postclipping: MAP typically 70–90 mm Hg	management. If it becomes necessary to ↓ BP, use esmolol 50–200 mcg/kg/min to ↓ HR to 50–60, supplemented as necessary with clevidipine 1–21 mg/h or SNP 0.1–4 mcg/kg/min to desired effect. Labetalol (5–100 mg total dose) is a useful adjunct for postop BP control.
Video angiography	Indocyanine green (ICG) NB: Usually contains iodine	ICG is an iv fluorescent dye for video angiography. Usual dose is 2.5–7.5 mg through a central line depending on tissue thickness. Rarely anaphylactic or other allergic reactions may occur.
Aneurysmal rupture	↓ MAP to 40–50 mm Hg Consider carotid compression Adenosine via CVP line Hypothermic dose: 12 mg iv Normothermic dose: 0.5–1 mg/kg	Bolus SNP: MAP will ↓ ~20 mm Hg/15 mcg SNP. Ipsilateral or bilateral carotid occlusion is often effective in controlling hemorrhage while a temporary clip is applied. Adenosine will produce asystole, allowing time for the application of a temporary clip. Escalate dose until asystole has been achieved ($t_{1/2} < 10$ sec). An external pacemaker should be available for hypothermic patients.
Positioning	For most aneurysms: Supine, head turned, Mayfield fixation (NB: ↑ BP with use of pins)	Anesthetic circuit and all monitoring and vascular catheter lines are directed to patient's side or feet, where

	Use shoulder roll ✓ and pad pressure points ✓ eyes	the anesthesiologist is positioned during surgery. SCDs used to minimize DVT. Shoulder roll to ↓ brachial plexus stretch. Remifentanyl (100–250 mcg bolus) to minimize ↑ BP during skull pinning.
Complications	Aneurysm rupture (intraop) Hypothermia (mild) DVT	6–18% incidence; up to 2% rupture during induction. Most patients can be extubated safely at core T ≥ 35°C with active rewarming in progress. SCDs should be used for DVT prophylaxis.

POSTOPERATIVE

Complications	Intracranial hemorrhage Stroke Cerebral vasospasm SIADH/CSW (less common)	New deficits or delayed emergence may necessitate urgent transport to CT scanner and/or return to the OR. Avoid prophylactic triple-H therapy, usually not evident intraop. Rx:NS.
Multimodal pain management	Acetaminophen 1 g iv ± Meperidine (10–20 mg iv prn) ± Fentanyl 25–50 mcg prn	If not given preop or if 4–6 h have elapsed since previous dose. Emphasize multimodal approach (see p. K-5). Meperidine will ↓ postop shivering in hypothermic patients. Minimize opioid use to ↓ PONV and to avoid unnecessary sedation.

	Place/Reinforce scalp blocks	0.5% bupivacaine with epi is as effective as liposomal bupivacaine.
ERAS	Ondansetron 4 mg iv ± Metoclopramide 10–20 mg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	CT scan, if any change in neurologic status	Be prepared to resecure airway.

Suggested Readings

1. Ayling OG, Ibrahim GM, Drake B, et al: Operative complications and differences in outcome after clipping and coiling of ruptured intracranial aneurysms. *J Neurosurg* 2015; 123(3):621–8.
2. Chalouhi N, Hoh BL, Hasan D: Review of cerebral aneurysm formation, growth, and rupture. *Stroke* 2013; 44(12):3613–22.
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ANESTHETIC CONSIDERATIONS FOR CRANIOTOMY FOR GIANT INTRACRANIAL ANEURYSMS (REQUIRING DEEP HYPOTHERMIC CIRCULATORY ARREST)

PREOPERATIVE

Aneurysms are classified as “giant” when they are >2.5 cm in diameter. These represent ~5% of all aneurysms. They occur twice as often in women, usually become symptomatic in the 4th or 5th decade of life, and present particularly difficult surgical challenges: (a) their large size makes direct visualization of the vascular anatomy difficult; (b) vascular branches essential to maintaining flow to normal brain may arise from the aneurysmal/SAC and cannot be included in the clipping without causing permanent neurologic injury; (c) standard aneurysm clips may not occlude a large, turgid aneurysm, or may slip or move, once applied; and (d) giant aneurysms may easily rupture during dissection or clip application. Many of these aneurysms are amenable to coiling or other interventional radiologic techniques. For those requiring craniotomy, temporary clips applied during mild hypothermia often provide sufficient opportunity to decompress and occlude the aneurysm. Rarely, deep hypothermia to 18°C, achieved with fem-fem CPB/ECMO and temporary circulatory arrest, may be required to safely decompress the aneurysm and protect the brain during circulatory arrest. The duration of asystole may be as long as 45 min.

Respiratory	None unless patient has Hx of smoking or has sustained pulmonary aspiration as a result of a neurologic deficit from an intracranial hemorrhage. Tests: As indicated from H&P; elevated cTnI may predict pulmonary complications.
Cardiovascular	Generally, these patients do not have other cardiovascular diseases (see Anesthetic Considerations for Intracranial Aneurysms). Tests: ECG; cTnI; others (e.g., ECHO) as indicated from H&P
Neurologic	These patients usually present with complaints of intermittent or persistent headaches or visual disturbances that are probably due to aneurysmal compression of adjacent neural tissue or ↑ ICP. If an intracranial hemorrhage occurs, neurologic dysfunction varies, depending on site and extent of the hemorrhage. Cerebral vasospasm is a major complication of intracranial hemorrhage (see discussion in Anesthetic Considerations for Intracranial Aneurysms). Tests: CT; MRI; angiogram. The anesthesiologist should examine the cerebral angiogram preop to

	visualize the size and site of aneurysm.
Hematologic	T&C for 6 U PRBCs. Tests: Hct; PT; PTT; CBC; others as indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	Premedication is usually desirable, with small doses of midazolam (e.g., 1–2 mg) preferable to opiates. Detailed discussion with patient about the anesthetic plan, with appropriate reassurance, is effective in reducing premedication requirements. Should an intracranial aneurysm leak or rupture in the immediate preop period, it may be difficult to distinguish this event from changes associated with excessive responses to premedication.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss possibility of scalp block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water especially for posterior fossa craniotomies with their high incidence of PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: The goals of anesthesia for this procedure are to (a) provide adequate surgical anesthesia; (b) ↓ intracranial volume (blood and tissue) to optimize working space within the cranial compartment, thereby minimizing the need for surgical retraction of brain tissue; and (c) ↑ tolerance of the brain to ischemia by decreasing CMRO₂, which occurs with the use of deep hypothermia, barbiturate or propofol therapy, ± isovolemic hemodilution.

Induction	Propofol 2–3 mg/kg iv to provide anesthesia and ↓ CBV by inducing cerebral vasoconstriction. Fentanyl 7–10 mcg/kg iv to blunt the response to laryngoscopy/intubation and provide analgesia for the first hours. Vecuronium 0.15 mg/kg or rocuronium 0.6–
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	1.2 mg/kg to provide relaxation for intubation and positioning.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Consider preincision placement of scalp blocks. Dexamethasone (4–8 mg iv) may ↓ PONV. Avoid long-acting opioids or scopolamine patch, which may confound postop neuro examination. Sequential pneumatic compression devices should be used to ↓ VTE risk.	
Maintenance	Propofol 100–200 mcg/kg/min administered by constant infusion pump. These doses provide additional amnesia and ↓ CBV and CMRO ₂ . Isoflurane ≤1%. N ₂ O not used. An additional dose of NMB is administered just prior to the start of CPB.	
Emergence	Because of the length and nature of the operation and the potential for temporary neurologic injury, it is advisable to leave the ETT in place immediately postop and send patient to the ICU on controlled ventilation. If patient begins to cough, the reflex should be suppressed with opiates, NMBs, and/or LTA sprayed down the ETT. The patient is placed in bed in a 30° head-up position and transported to ICU for overnight monitoring. Supplemental O ₂ should be administered and close regulation of BP maintained. Prophylactic antiemetic (e.g., ondansetron 4 mg) should be given 30 min before extubation. Consider iv acetaminophen 1 g iv and scalp block placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago.	
Blood and fluid requirements	IV: 14–16 ga × 2 BSS @ 1–2 mL/kg/h PRBC 4–6 U	Maintain euvoolemia. Cold BSS up to 10 mL/kg is administered during surgery. Zero balance or goal-directed fluid management is recommended.
Isovolemic	5% albumin: 1000–2000	Albumin is placed in a

hemodilution	mL CPD bags (for blood collection)	refrigerator at 4°C the night before surgery to be used as necessary for cooling during isovolemic hemodilution. After induction of anesthesia, a 2nd arterial or large-vein cannula is placed for removal of blood into CPD bags. Generally, about 1000 mL of blood is removed and replaced with 1 L of cold albumin 5%. This usually → ↓ Hct to 22–26%. Frequent intraop Hct checks are appropriate. The withdrawn blood is held for reinfusion at the conclusion of operation. In addition, the perfusate from the CPB unit is spun down, and packed cells are returned to patient.
Control of ICP (brain relaxation)	Hyperventilate to $\text{PetCO}_2 = \sim 30$ mm Hg Limit crystalloid <10 mL/kg + UO Limit isoflurane $\leq 1\%$ ± Mannitol 0.5–1 g/kg ± Furosemide 0.3 mg/kg Dexamethasone 8–12 mg ± Lumbar CSF drainage Position head to promote venous drainage	Ventilation is controlled and TV and RR adjusted such that $\text{PaCO}_2 = \sim 35$ mm Hg. There are some advantages to mild hypocarbia, including ↓ CBV, and better buffering of brain lactic acid that may form as a result of ischemia. Elevate head of bed, and minimize neck flexion/rotation if possible.
Monitoring	Standard monitors Temperature: esophageal, bladder and brain surface Arterial line	Frequent checks are made of Hct, electrolytes, and ACT values, before, during, and after CPB.

	CVP (triple-lumen) line	CVP line is used for infusions of esmolol, SNP, and phenylephrine. Subclavian preferred.
	Blood glucose	Keep glucose between 80 and 180 mg/dL.
	UO	Keep UO > 0.5 mL/kg/h.
Control of BP	Maintain BP normal to 20% below normal Esmolol infusion SNP infusion	↑ BP during induction or prior to CPB will ↑ likelihood of rupture. Prior to CPB, BP is controlled using anesthetic agents ± an esmolol infusion to ↓ HR to a range of 50–60 bpm, ± SNP infusion. SNP facilitates both cooling and rewarming because of its peripheral vasodilatory effect.
	Phenylephrine	If a vasoconstrictor is needed, particularly during CPB while patient is still cold, phenylephrine is preferred because of its minimal dysrhythmogenic potential.
Positioning	Shoulder roll 180° table rotation ✓ and pad pressure points ✓ eyes Mayfield skull fixation (NB: ↑BP)	Anesthetic circuit and all monitoring and vascular catheter lines are directed to patient's feet. Make sure that all will reach the foot of operating table.
		Remifentanyl 100–250 mg bolus will blunt BP response to pinning.
	SCD	SCDs used to minimize DVT.
Deep hypothermia	Initial cooling: Circulating water	Surface cooling and/or central cooling is begun as soon as

and CPB

blankets
Ice packs
Thermogard®-type
device

induction is complete, using either a central venous heat exchange device (e.g., Thermogard XP®) or thermal blankets above and below patient, ice packs, and infusion of cold fluids during establishment of isovolemic hemodilution.

SNP infusion

An SNP infusion (if tolerated) will promote surface cooling.

Heparinization
ACT monitoring
Goal: 3–4× baseline

Once the aneurysm is exposed and the decision is made that CPB is required, systemic **heparinization** (load: 300 U/kg; maintenance: 100 U/kg/h) is established, and patient is put on CPB using fem-fem bypass and cooled to ~18°C. During CPB cooling, the heart will usually fibrillate between 22°C and 26°C. After 18°C is reached, the CPB unit is shut off to deflate the aneurysm; it may be activated and shut off several times during clipping to evaluate adequacy of the surgical occlusion of the aneurysm and to apply additional clips. **Total circulatory arrest time should not exceed 45 min.**

Rewarming

When clipping is complete, CPB is resumed and warming instituted. Partial CPB is continued until normal

		cardiac rhythm is established and body temperature reaches $\sim 36^{\circ}\text{C}$. Once partial CPB is D/C, patient will tend to cool unless vigorous efforts at warming are continued. Warming the OR and iv fluids, use of a Bair Hugger, or a Thermogard®-type device will facilitate the warming process. ACT analysis is performed to establish that heparin reversal is complete 5–10 min after protamine (1 mg/100 U heparin activity). Blood is sent for clotting studies, and Plts, FFP, and calcium gluconate are administered as needed.
Complications	↓↓ BP 2° failure to maintain circulating volume Dysrhythmias 2° ↓ K⁺ from diuresis and cold	Restore circulating volume. ✓ K⁺ level and replace as indicated.

POSTOPERATIVE

Complications	HTN Vasospasm Intracranial hemorrhage, stroke Hypothermia Hypervolemia Coagulopathy DVT Seizures PE	HTN Rx: Esmolol + SNP titrated to effect. Vasospasm Rx: Euvolemia, trial of ↑ MAP; consider nimodipine. Patient should be rewarmed to $36\text{--}37^{\circ}\text{C}$ before terminating CPB. ✓ coag status. SCDs may be useful. Sz Rx: Phenytoin (1 g iv
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		slowly to avoid ↓BP) or levetiracetam (Keppra®) 1 g iv. NB: Phenytoin is incompatible with dextrose containing solutions.
Multimodal pain management	Scalp block Acetaminophen 1 g iv ± Fentanyl 25–50 mcg prn Reinforce scalp blocks	PT usually remains intubated and sedated (e.g., propofol infusion) in immediate postop period. If not given preop or if 4–6 h have elapsed since previous dose. 0.5% bupivacaine with epi is as effective as liposomal bupivacaine.
Tests	CT scan Coag panel	If any question about neurologic status arises, a CT scan is performed postop. Coagulation studies are needed early postop to assure normal coagulation.
ERAS	Ondansetron 4 mg iv ± Metoclopramide 10–20 mg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

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CRANIOTOMY FOR CEREBRAL EMBOLECTOMY

SURGICAL CONSIDERATIONS

Gary K. Steinberg and Robert L. Dodd

Description

Although endovascular thrombectomy (first choice—see [p. 493](#)) and intravenous or endovascular intra-arterial thrombolysis are the current standard therapies for intracranial intravascular clots, embolic occlusion of a major intracranial vessel occasionally requires microsurgical embolectomy. In particular, when the embolus is a large atherosclerotic plaque or foreign body (such as a balloon or microcoil from endovascular treatment), surgery may be the treatment of choice. Because cerebral ischemia often proceeds to irreversible infarction before the surgeon can restore blood flow, early diagnosis is of the utmost importance, and several studies have demonstrated that the best results from embolectomy occur when the procedure is performed within 6–24 h following the onset of a neurologic deficit.

A standard craniotomy is fashioned as previously described for other lesions involving the vasculature at the skull base, and the occluded intracranial artery is exposed using microsurgical techniques. The involved arterial segment is isolated and temporarily occluded with miniature clips, and an arteriotomy is performed to remove the thrombus or embolus ([Fig. 1.1-3](#)). The arteriotomy is then closed and blood flow reestablished.

Usual preop diagnosis

Stroke; TIA; intracranial arterial occlusion; catheter embolization to intracranial artery

ERAS: These procedures are usually urgent/emergent and present little opportunity for ERAS considerations.

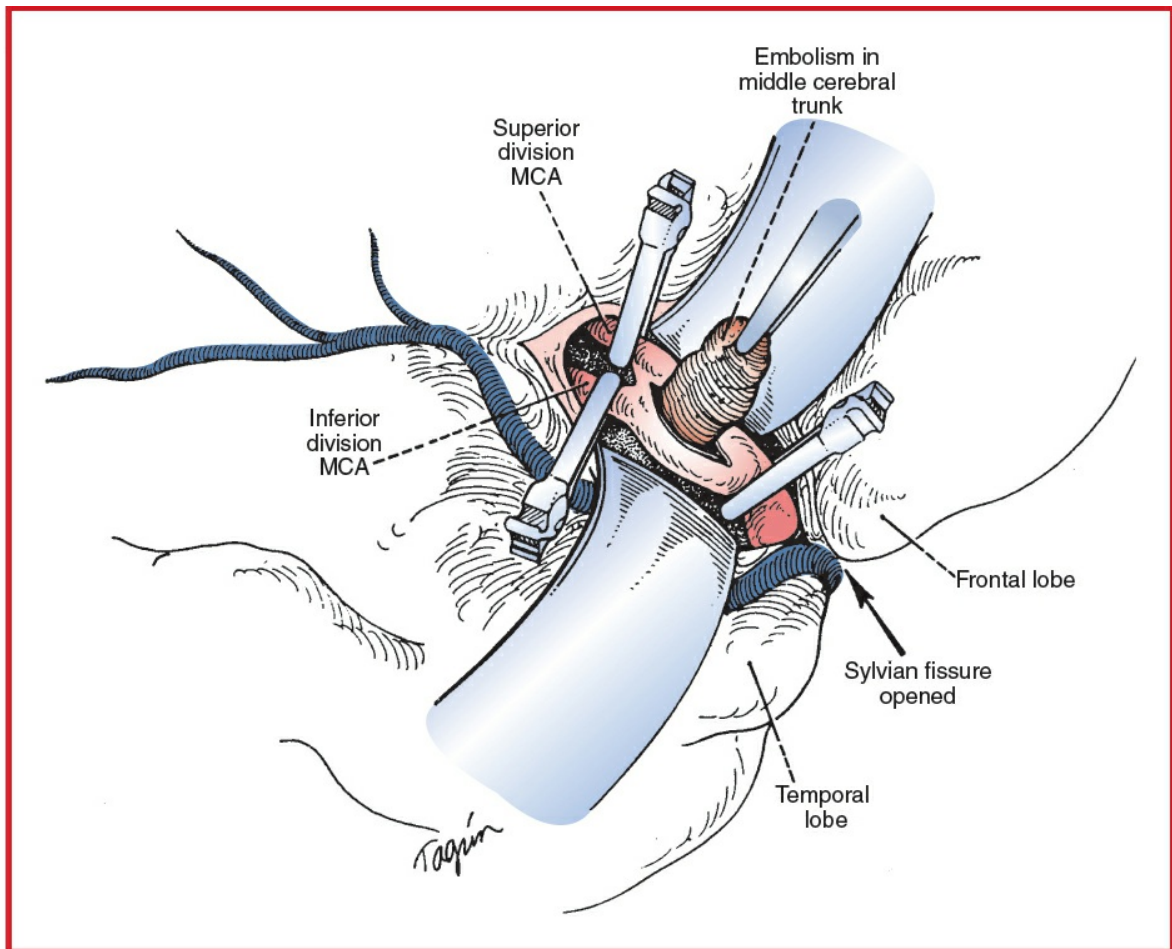


Figure 1.1-3 Middle cerebral artery (MCA) embolectomy. Exposure of right MCA in the sylvian fissure and removal of an embolus from the MI segment. (Reproduced with permission from Ojemann RG, Ogilvy CS, Crowell RM, et al. *Surgical Management of Neurovascular Disease*, 3rd edition. Williams & Wilkins, Baltimore: 1995.)

■ SUMMARY OF PROCEDURE

Position	Supine or lateral decubitus
Incision	Frontal, temporal, or occipital
Special instrumentation	Microscopic instruments (fine forceps, miniature vascular clips, operating microscope)
Unique considerations	Neuroprotective agents during arterial segment occlusion (barbiturates, mannitol), mild hypothermia (33°C)
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	3–4 h
Closing considerations	Avoid ↓ BP (MAP 80–100 mm Hg). Induced mild HTN (MAP 90–110) during temporary arterial occlusion

EBL	100–250 mL
Postop care	Control BP (MAP 80–100 mm Hg); start aspirin postop day 1; ICU: 1–2 d
Mortality	5–10%
Morbidity	Intracerebral hemorrhage Stroke MI: Rare Thromboembolism: Rare Respiratory failure: Rare Infection: Rare
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	50–80 yr
Male:Female	1:1
Incidence	Rare
Etiology	Atherosclerosis; carotid artery disease; atrial fibrillation; iatrogenic endovascular catheter complication
Associated conditions	HTN; CAD; PVD; carotid artery disease; hyperlipidemia; smoking; alcohol abuse; obesity; atrial fibrillation

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Craniotomy for Intracranial Aneurysms.

Suggested Readings

1. Albers GW, Marks MP, Kemp S, et al: Thrombectomy for stroke at 6 to 16 hours with selection by perfusion imaging. *N Engl J Med* 2018; 378(8):708–18.
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CRANIOTOMY FOR INTRACRANIAL VASCULAR MALFORMATIONS

SURGICAL CONSIDERATIONS

Gary K. Steinberg

Description

Intracranial vascular malformations are congenital abnormalities that cause intracranial hemorrhage, seizures, headaches, progressive neurologic deficits, or audible bruits. Intracranial vascular malformations comprise high-flow arteriovenous malformations (AVMs); low-flow angiographically occult vascular malformations (AOVMs) including cavernous malformations (a collection of enlarged capillaries with thin inelastic walls prone to leaking), “cryptic” AVMs, capillary telangiectasias, and transitional malformations; and low-flow venous angiomas (developmental venous anomalies). **Microsurgical resection** is the optimal treatment for these lesions, although preop endovascular embolization and preop or postop focused **stereotactic radiosurgery** (heavy particle or photon) may be useful adjuncts.

Most moderate-sized and large **AVMs** (>3 cm diameter) will have been partially embolized in the cath lab prior to surgery. Resection is accomplished using a standard **scalp flap** and with **craniotomy** centered over the area of the AVM using image guidance techniques. The patient is positioned appropriately to place the craniotomy site uppermost in the field and parallel to the floor. For example, a patient with a left frontal AVM would be positioned supine, head turned to the right, a left frontal or bicranial scalp flap raised, and a left frontal craniotomy bone flap removed. A patient with a right medial occipital AVM would be positioned in the left lateral decubitus position with head turned semiprone and a right occipital scalp flap and craniotomy performed. Smaller AVMs (<3 cm diameter), many low-flow AOVMs, and many deep-seated vascular malformations (AVMs and AOVMs) require only a small **stereotactic craniotomy**. This is performed by attaching fiducial markers or (less frequently) a stereotactic base frame to the patient’s skull (using local anesthetic and sedation). Next, a CT or MRI scan is obtained. The location of the AVM in relation to the markers (or frame) is calculated, using a computer and stereotactic geometric principles. The patient is taken to the OR, intubated (fiberoptically if a frame is used), and positioned for surgery. The surgical navigation system reference is attached to the headrest and

microscope and calibrated. A small scalp flap and a small craniotomy (a few cm in diameter) can be fashioned precisely for microscopic exposure of the malformation. Microsurgical resection of brain stem and thalamic vascular malformations often necessitate special positioning. In frame-based surgery, a three-dimensional arc frame is fixed to the base frame, and coordinates are set to localize the vascular malformation within the brain.

Usual preop diagnosis

Cerebral AVM; dural AVM (an A-V shunt within the dura mater); cavernous malformation (low-flow hemangioma); angiographically occult vascular malformation; intracerebral hemorrhage; subarachnoid hemorrhage; seizures; epilepsy; progressive neurologic deficit; migraine or vascular headaches

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and AED medications, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURES			
	Standard Craniotomy (High-Flow AVM)	Stereotactic Craniotomy (Low-Flow AVM)	Brain Stem/Thalamic Vascular Malformations
Position	Supine, lateral, Concorde (modified prone; Fig. 1.1-4)	⇐	Lateral, Concorde, or semisitting (Fig. 1.1-5)
Incision	Frontal, temporal, parietal, occipital, suboccipital, or combination	⇐	Suboccipital (midline), paramedian, or occipital
Special instrumentation	Operating microscope; radiolucent table and headrest. Access to femoral artery for intraop angiography.	⇐	⇐

	Surgical navigation system.		
Unique considerations	Induced hypotension (MAP 60–65 mm Hg) during resection, use of neuroprotective agents. Relaxed brain. Mild hypothermia (33°C); ± lumbar CSF drain.	Use of neuroprotective agents and relaxed brain	⇐
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)	⇐	⇐
Surgical time	4–10 h	2–5 h	3–6 h
Closing considerations	Maintain MAP 65–75 mm Hg to ↓ postresection hyperemia. Avoid ↑ venous pressure. Consider loading dose of phenytoin (1 g iv for adults) if not previously on anticonvulsants.	Keep MAP 70–90 mm Hg ⇐ ⇐	⇐ No anticonvulsants necessary
EBL	500–3000 mL	<250 mL	⇐
Postop care	ICU × 1–2 d. Maintain MAP 65–75 mm Hg for 1 d. ICP monitoring, ventricular drain; normovolemic in ICU.	ICU × 1 d; normovolemic in ICU	⇐
Mortality	1–10%, depending on AVM size, location, and venous drainage pattern	<0.5%	<2%
Morbidity	Overall: 5–30% Breakthrough bleeding: <5% Intracranial hemorrhage	<5% ⇐ ⇐ ⇐	10–50% (transient) <5% ⇐

	Cerebral edema	⇐	⇐
	Stroke <15%	⇐	⇐
	Hydrocephalus	⇐	⇐
	Massive blood loss:	⇐	⇐
	Occasional		⇐
	Thromboembolism:		⇐
	Rare		
	Infection: Rare		
Pain score	3–4	3–4	3–4

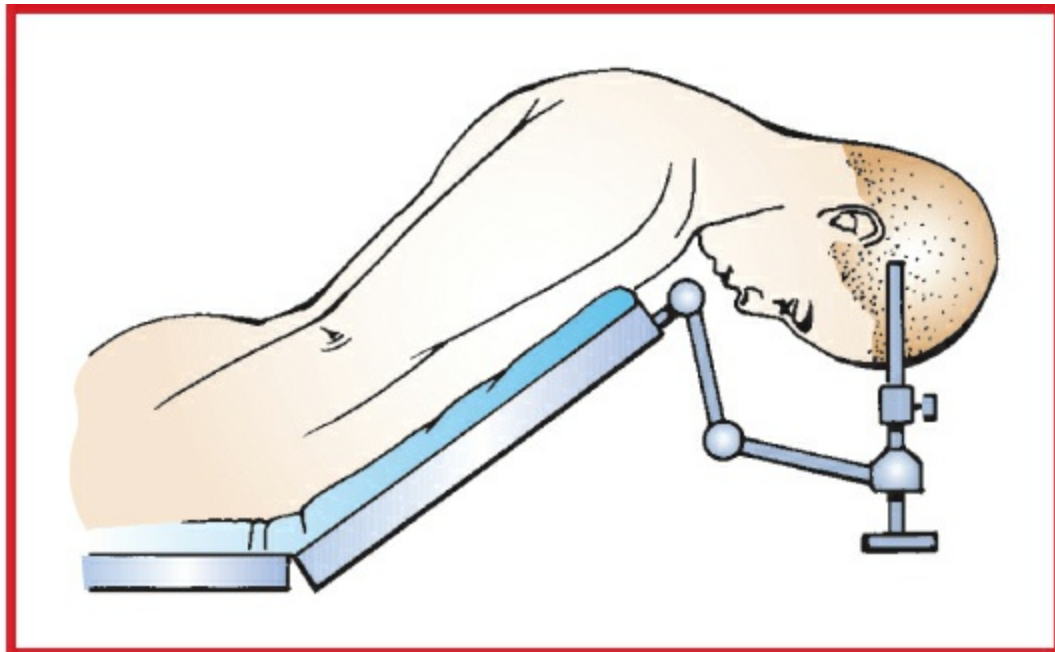


Figure 1.1-4 Concorde (modified prone) position for resection of posterior fossa vascular malformations.

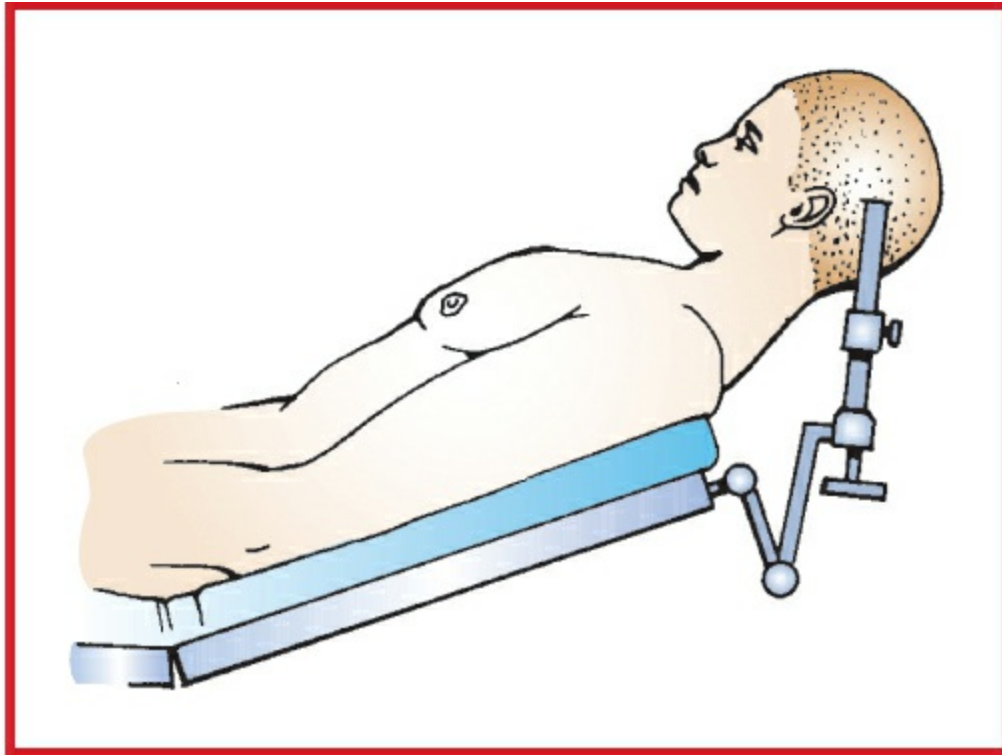


Figure 1.1-5 Semisitting position for resection of deep posterior corpus callosum or thalamic vascular malformations.

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–40 yr (most common), 41–60 yr (less frequent). Average age: 33 yr
Male:Female	1:1 (slight male predominance)
Incidence	0.2% of the U.S. population (AVM); 0.1% (cavernous malformation)
Etiology	Congenital; traumatic for dural AVM
Associated conditions	Intracranial aneurysm (~10% of AVM patients [up to 58% in some studies]); von Hippel-Lindau disease; Rendu-Osler-Weber syndrome; familial cavernous malformation syndrome

■ ANESTHETIC CONSIDERATIONS

■ PREOPERATIVE

AVMs are typically direct arterial-to-venous communications without intervening normal capillary circulation. On histologic examination, the vessel walls are thin and lack a muscular layer; consequently, the vessels exhibit loss of normal vasomotor

control or responsiveness to changes in PaCO₂. **Stereotactic localization** is essential for safe excision of deep-seated AVMs (e.g., those located in the corona radiata, basal ganglia, visual center, cerebellar white matter, or corpus callosum). Untreated, the overall risk of rupture is ~2–4% per year (varies by size). AVMs > 6 cm in maximum diameter and located near eloquent areas of the cortex or with deep venous drainage (grades IV and V) have a high complication rate and may not be suitable for surgical resection without prior partial embolization. Review imaging studies and discuss surgical approach with surgeon.

Respiratory	Not usually significant unless patient has Hx of smoking or has aspiration pneumonitis as a result of a neurologic deficit from an intracranial hemorrhagic or ischemic event. Tests: As indicated from H&P
Cardiovascular	Generally, these patients do not have other cardiovascular diseases. Occasionally, ECG changes are noted following intracranial hemorrhage and may represent subendocardial injury 2° catecholamine release. Tests: ECG; others as indicated from H&P
Neurologic	Presenting Sx depend on location and size of AVM and whether it is a low- or high-flow lesion. Hemorrhage with resultant headache and/or neurologic deficit is the most common Sx, although patients also may present with intractable seizure disorder, recurrent HAs, or Sx of cerebral ischemia 2° high-flow AV shunts, causing an intracerebral steal. Surgical treatment is essential to prevent future hemorrhage (incidence of 2–4%/yr) with substantial mortality (6–30%) or severe morbidity (15–80%). Unlike hemorrhages from an intracerebral aneurysm (generally subarachnoid), hemorrhages from an AVM are usually ventricular or intraparenchymal; hence, they are seldom associated with cerebral vasospasm. Tests: CTA; MRI; cerebral angiogram. Preop cerebral angiogram indicates size and location of the AVM and whether it is likely to be a low- or high-flow lesion.

Hematologic	After surgery for AVM, it is fairly common for surrounding brain tissue to swell and resection sites to bleed. The cause of this is unknown, but it may be related to diversion of former AVM blood flow into the surrounding vasculature, producing a fragile hyperemic state. Thus, it is advisable to attempt embolization of the lesion before surgery; obtain coag studies preop, and consider staging resection over more than one sitting when the residual AVM is large.
Laboratory	Tests: Hct; PT; PTT; Plt count CBC; other tests as indicated from H&P
Premedication	If medication is desirable, small doses of midazolam (e.g., 1–3 mg iv) are useful. Detailed discussion with patient about the anesthetic plan, with appropriate reassurance, is essential.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss possibility of scalp block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water especially for posterior fossa craniotomies with their high incidence of PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: The goals of anesthesia for this operation are to (a) ↓ CMRO₂ to lessen the dependence of normal brain on vessels feeding the AVM; (b) decrease intracranial volume (blood and tissue) to optimize surgical working space within the cranial compartment and minimize the need for surgical retraction; and (c) reduce CPP (10–20% below normal) to lessen blood loss during excision of the AVM (postcraniotomy CPP = cerebral arterial pressure minus cerebral venous pressure). For stereotactic image-guided surgery, scalp localizing markers are attached. The patient is then sent to MR or CT, where the exact coordinates defining the AVM and critical adjacent structures are established. The patient is brought to OR, and anesthesia is induced.

Induction	Propofol 1–2 mg/kg iv provides amnesia and ↓ CBV by
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inducing cerebral vasoconstriction. Fentanyl 7–10 mcg/kg iv blunts the response to laryngoscopy and provides analgesia for the first hours. High-dose opiates used as a primary anesthetic technique do not alter CBF or CMRO₂ enough to provide any special benefits. Rocuronium (0.6–1 mg/kg) or vecuronium (0.15 mg/kg) provides muscle relaxation for intubation and patient positioning. Occasionally, the patient may be in a stereotactic frame, and ET intubation must be accomplished before anesthesia is induced because the frame may partially occlude the mouth, making conventional laryngoscopy impossible. Awake oral fiberoptic intubation of the trachea is the easiest method for accomplishing this.

ERAS

Verify preincision antibiotics. Assess volume status (see [Appendix K](#)). Maintain normovolemia. Avoid hypothermia unless indicated for brain protection. Consider preincision placement of scalp blocks. Dexamethasone (4–8 mg iv) may ↓ PONV. Avoid long-acting opioids or scopolamine patch, which may confound postop neuro examination. Sequential pneumatic compression devices should be used to ↓ VTE risk.

Maintenance

Isoflurane ≤1% or sevoflurane ≤2% (limit to 1/2 MAC maximum if EP monitoring is used) with 1:1 O₂/N₂O. With EP monitoring, a remifentanyl infusion (0.05–0.2 mcg/kg/min) may be necessary to supplement the anesthetic. Propofol (50–150 mcg/kg/min) by continuous infusion may be administered to provide ↓ CBV and ↓ CMRO₂ and allow for reduction in inhalation agent concentration or elimination of N₂O. Mild hypothermia (33°C) provides additional cerebral protection (see below). Additional neuromuscular blocking drugs are usually not necessary but can be administered if patient movement is of concern. Induced mild hypotension is often useful during AVM resection. Following resection, induced brief hypertension (e.g., MAP = 90 mm Hg) may be requested to assess hemostasis.

Emergence

With the start of dural closure, consider changing the anesthetic to low-dose sevoflurane (e.g., 0.5%) in 50% N₂O, supplemented with a low-dose remifentanyl infusion (e.g., 0.05 mcg/kg/min). A propofol infusion (if used) should be discontinued at the start of the scalp closure. The patient's BP generally will increase, and titration of β -adrenergic blocking drugs (e.g., labetalol or esmolol) and/or vasodilators (e.g., clevidipine or SNP) may be needed (see Control of BP, below). The inhalation agents can be D/C'd at the time of dressing application. Most patients will breathe spontaneously and can be extubated uneventfully while on a 0.05 mcg/kg/min remifentanyl infusion. Reverse any residual neuromuscular blockade. If the brain has not been injured by the surgical procedure, the patient should awaken within 10 min after cessation of remifentanyl administration. Close regulation of BP is essential. If the patient begins to cough on ETT, either it should be removed or cough reflex suppressed with iv lidocaine (1 mg/kg). Patient is placed in bed in a 20–30° head-up position and transported to ICU for monitoring overnight. Supplemental O₂ should be administered and close regulation of BP maintained (typically at ~10% below baseline values). Prophylactic antiemetics (e.g., metoclopramide 10–20 mg and ondansetron 4 mg) should be given 30 min before extubation. If not given preincision, consider iv acetaminophen (1 g) and/or local anesthetic scalp blocks for intra- and postop analgesia. Consider scalp block reinforcement if initially placed >4 h ago. Consider additional acetaminophen iv if po dose given >4–6 h ago.

Blood and fluid requirements

IV: 14–16 ga $\times$ 2
BSS @ <3 mL/kg/h + UO
Massive blood loss possible
Maintain euvolemia
Maintain colloid oncotic pressure

If blood volume is normal, BSS—not to exceed 3 mL/kg/h beyond that required to replace UO—is given. If hypovolemic, albumin 5% is given, if Hct > 30%; combination of albumin and

Hypothermia	Water-circulating blankets Cold OR ± Thermogard®-type device Bladder irrigation	blood, if Hct < 30% Mild hypothermia (33–34°C) is used in some centers for cerebral protection. This level of hypothermia does not interfere appreciably with coagulation, nor is it generally associated with ↑ SSI or cardiac dysrhythmias. Warming is begun as soon as possible after lesion resection and is greatly facilitated by the use of a central warming device (e.g., Thermogard® or equivalent) or bladder irrigation.
Brain relaxation (control of ICP)	Hyperventilate to PaCO ₂ = ~35 mm Hg or PetCO ₂ = 30 mm Hg Limit isoflurane/sevoflurane ≤ 0.5 MAC Maintain euvolemia Consider propofol infusion (50 mg/kg/min) to replace N ₂ O Mannitol 0.5–1 g/kg Furosemide 10–20 mg q6h Lumbar CSF drainage	↓ PaCO ₂ → ↓ cerebral vascular volume → ↑ working space and lessens need for vigorous retraction of brain tissue. ↓ PaCO ₂ may also improve the regional distribution of CBF by preferentially diverting blood to potentially ischemic areas of the brain. If AVM is superficial, decreasing brain volume is less important, and the first five techniques listed (at left) are usually sufficient. If AVM is deep, the additional listed therapies may be needed.
Monitoring	Standard monitors Core temp: deep esophageal best Arterial line CVP line, subclavian	Direct monitoring is essential for rapid control of BP. Level transducer at the head. Monitoring CVP via a near right atrial catheter is

	preferred UO Blood glucose (100–180 mg%)	desirable in virtually all patients to assess adequacy of fluid therapy, for infusion of vasoactive drugs and aspiration of VAE. Localization of the catheter can be determined by CXR, ECG tracing, noting P-wave changes, or pressure wave morphology. NB: If patient has a high-flow AVM causing a large AV shunt, venous blood may appear arterialized (bright red) during central venous catheterization.
Control of BP	Isoflurane/sevoflurane Esmolol infusion SNP/clevidipine infusion Maintain normovolemia Labetalol (emergence)	Close regulation of BP during induction and prior to excision of AVMs is important, both to prevent bleeding ($\uparrow$ BP) and to avoid ischemia 2° steal ($\downarrow$ BP). After surgical excision is underway, however, modest decreases in MAP ($\leq 20\%$ below normal) using esmolol and/or SNP/clevidipine should be achieved to prevent excessive bleeding.
Positioning	Shoulder roll Three-point fixation (Mayfield) ✓ and pad pressure points ✓ eyes 180° rotation SCDs	For most AVMs, patient is supine, head turned laterally in three-point fixation, a roll under the shoulder on the side of operation (Fig. 1.1-6). Anesthetic hoses and all monitoring/vascular catheter lines are directed toward patient's feet or side. Make sure that all will reach the foot of operating table. SCDs

are used to minimize DVT. Remifentanyl (2–4 mcg/kg iv bolus) should be used to minimize $\uparrow$ BP during skull pinning.

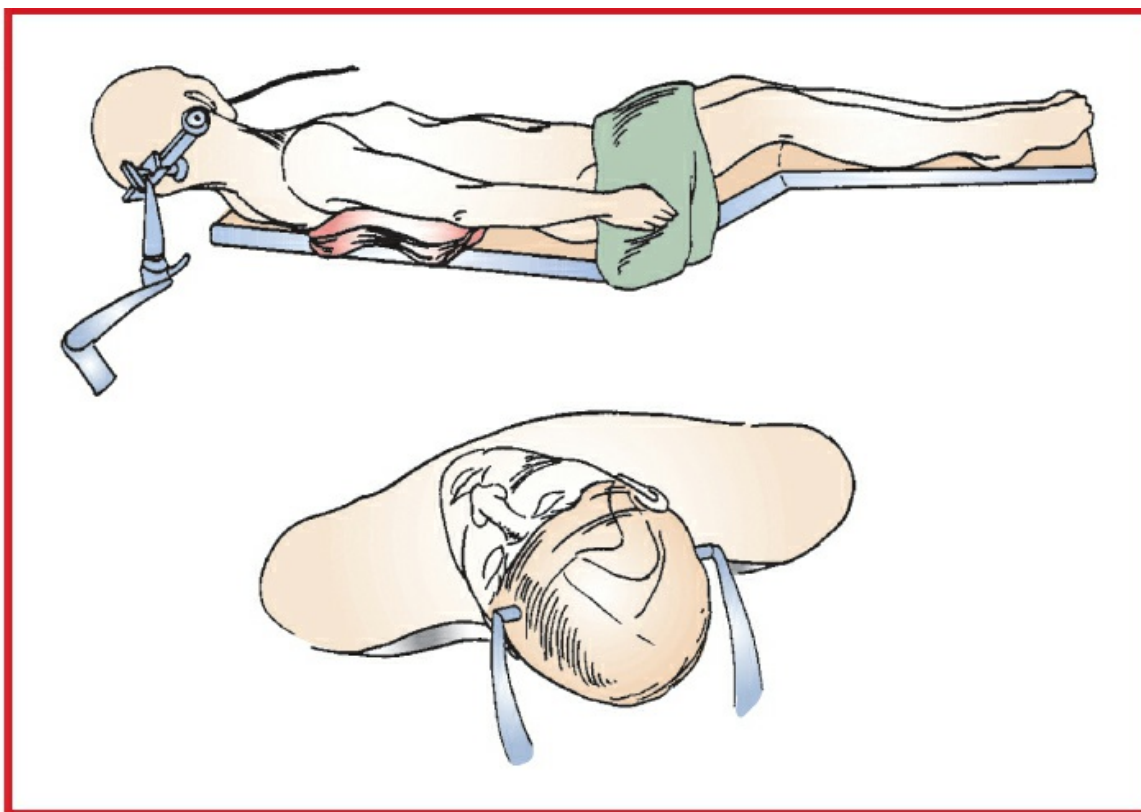


Figure 1.1-6 Supine position, head elevated above heart, turned 30–45° to side, vertex dropped for approach to anterior circulation aneurysms and frontal vascular malformations. (Reproduced with permission from Long DM: Atlas of Operative Neurosurgical Technique, Vol. 1. Williams & Wilkins, Baltimore: 1989.)

▼ POSTOPERATIVE

Complications

Neurologic deficits

If these complications occur, the patient likely will have to be reintubated and transported to the CT scanner for further neurologic evaluation or possible reoperation.

Cerebral edema and $\uparrow$ ICP Careful regulation of BP is

	Intracerebral hemorrhage	essential to avoid postop hemorrhage.
	Seizures	*Sz Rx: Levetiracetam (1 g) or phenytoin (1-g loading dose). NB: Incompatible with dextrose-containing solutions.
Multimodal pain management	Acetaminophen 1 g iv	If not given preop or if 4–6 h have elapsed since previous dose.
	± Meperidine (10–20 mg iv prn) ± Fentanyl 25–50 mcg prn	Emphasize multimodal approach (see p. K-5). Meperidine will ↓ postop shivering in hypothermic patients. Minimize opioid use to ↓ PONV and to avoid unnecessary sedation.
	Place/Reinforce scalp blocks	0.5% bupivacaine with epi is as effective as liposomal bupivacaine.
Tests	CT scan	If neurologic status changes.
ERAS	PONV protocol	Ondansetron 4 mg iv ± metoclopramide 10–20 mg iv. Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Chang SD, Lopez JR, Steinberg GK: The usefulness of electrophysiologic monitoring during resection of central nervous system vascular malformations. *J Stroke Cerebrovasc Dis* 1999; 8:412–22.
2. Derdeyn CP, Zipfel GJ, Albuquerque FC, et al: Management of brain arteriovenous malformations: a scientific statement for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke* 2017; 48(8):e200–24.
3. Gross BA, Du R: Natural history of cerebral arteriovenous malformations: a meta-analysis. *J Neurosurg* 2013; 118(2):437–43.
4. Miller C, Mirski M: Anesthesia considerations and intraoperative monitoring during surgery for arteriovenous malformations and dural arteriovenous fistulas. *Neurosurg*

Clin N Am 2012; 23(1):153–64.

5. Morgan MK, Davidson AS, Assaad NNA, et al: Critical review of brain AVM surgery, surgical results and natural history in 2017. *Acta Neurochir (Wien)* 2017; 159(8):1457–78.
6. Rangel-Castilla L, Spetzler RF, Nakaji P: Normal perfusion pressure breakthrough theory: a reappraisal after 35 years. *Neurosurg Rev* 2015; 38(3):399–404; discussion 404–5.
7. Solomon RA, Connolly ES Jr: Arteriovenous malformations of the brain. *N Engl J Med* 2017; 376(19):1859–66.
3. Steinberg GK, Stoodley MA: Surgical management of intracranial arteriovenous malformations. In: Schmidek HH, ed. *Operative Neurosurgical Techniques*, 4th edition. WB Saunders, Philadelphia: 2000, 1363–91.

CRANIOTOMY FOR EXTRACRANIAL-INTRACRANIAL REVASCULARIZATION (EC-IC BYPASS)

SURGICAL CONSIDERATIONS

Gary K. Steinberg and Robert L. Dodd

Description

Extracranial-intracranial (EC-IC) revascularization procedures are performed when (a) deliberate occlusion of a major cervical artery (carotid or vertebral) is necessary and inadequate collateral CBF is available or (b) stenosis or occlusion of major cervical or intracranial arteries causes TIA or stroke, despite the use of maximum medical therapy (e.g., antiplatelet drugs, heparin, or Coumadin). The chief causes of stenosis or occlusion are atherosclerotic disease, radiation injury, and moyamoya disease. The subset of patients who benefit from revascularization are those whose radiographic and metabolism studies demonstrate that they have ↓ CBF and poor or absent vascular reserve. The most important surgical considerations include site of stenosis, adequacy of donor graft, and patient age.

The donor vessel (usually a scalp artery) is identified using ultrasound mapping and microscopically dissected from the surrounding tissue taking care to maintain flow. This portion of the procedure may require 1–2 h. A standard craniotomy is then fashioned as previously described for other lesions involving the vasculature around the skull base, and the intracranial site of anastomosis is exposed using microscopic techniques.

Typically, a donor superficial temporal artery is anastomosed to an intracranial artery (e.g., middle cerebral artery) distal to the site of stenosis. When scalp vessels are inadequate, an interposition vein segment can be sutured to a cervical artery and then anastomosed to the designated intracranial artery. The most common EC-IC procedure is a **superficial temporal artery (STA)-to-middle cerebral artery (MCA) branch anastomosis** (see Figs. 1.1-7 to 1.1-11). Other grafts include STA-to-posterior cerebral artery, STA-to-superior cerebellar artery, occipital artery-to-posterior inferior cerebral artery or interposition saphenous vein segment graft from the cervical external carotid artery to the middle cerebral artery, posterior cerebral artery, or superior cerebellar artery. During anastomosis, the recipient artery must be temporarily clamped for 10–20 min. The potential for ischemic injury during this interval can be minimized by inducing burst suppression on EEG combined with ongoing mild hypothermia (33–34°C).

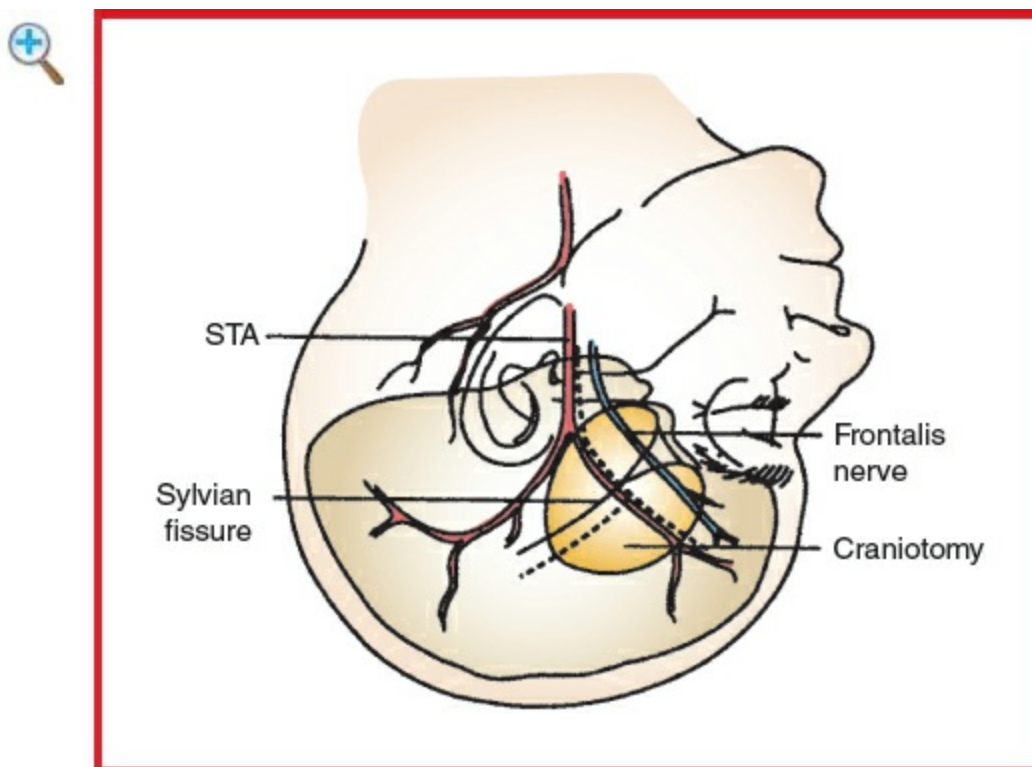


Figure 1.1-7 Typical skin incision for EC-IC bypass. The main incision is planned over the superficial temporal artery (STA) with a T extension to allow exposure of the bone. (Reproduced from Chang SD, Steinberg GK: Superficial temporal artery to middle cerebral artery anastomosis. *Tech Neurosurg* 2000; 6(2):86–100.)

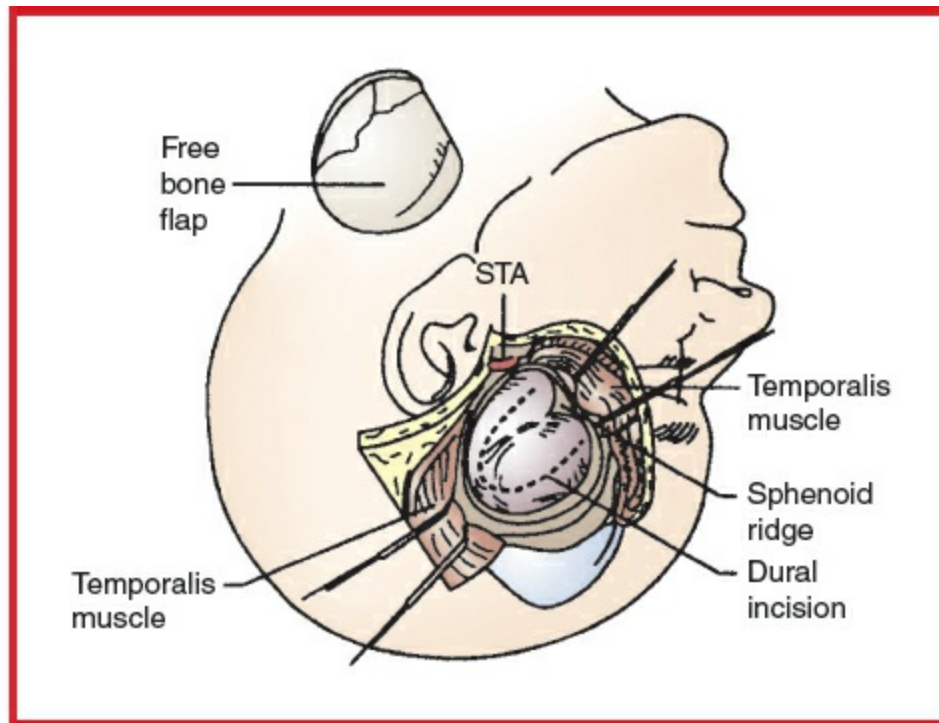


Figure 1.1-8 After the STA is dissected out, the temporalis muscle is divided and the bone flap is made to allow exposure of the brain over the anterior sylvian fissure. (Reproduced from Chang SD, Steinberg GK: Superficial temporal artery to middle cerebral artery anastomosis. Tech Neurosurg 2000; 6(2):86–100.)

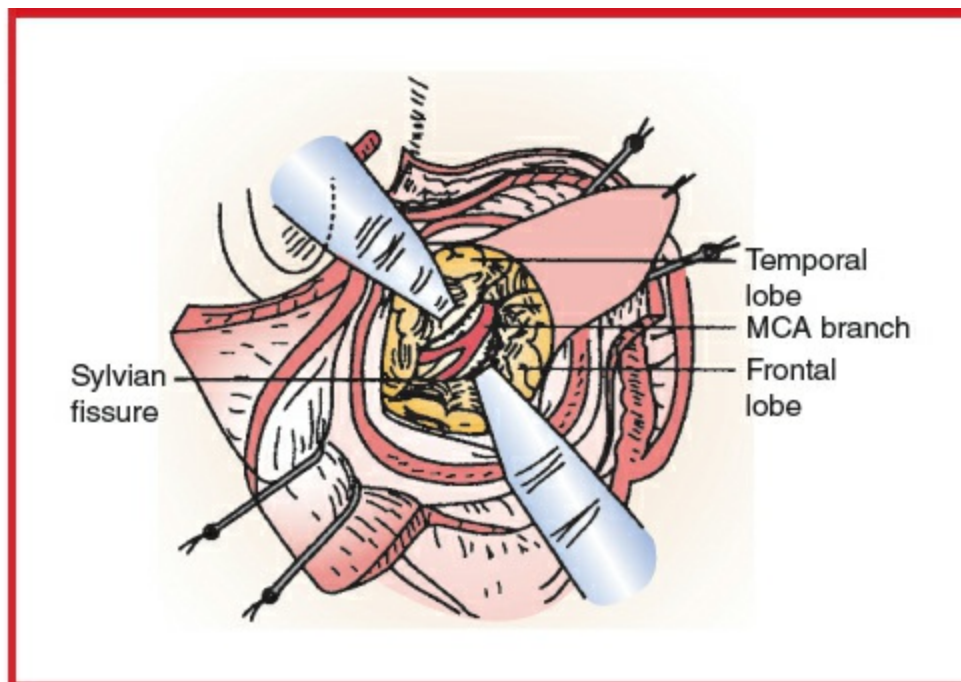


Figure 1.1-9 A middle cerebral artery (MCA) recipient vessel is identified. The sylvian fissure can be split to allow identification of a larger, more proximal branch of the MCA, preferably an M3 branch. (Reproduced from Chang SD, Steinberg GK: Superficial temporal artery to middle cerebral artery anastomosis. Tech Neurosurg 2000; 6(2):86–100.)

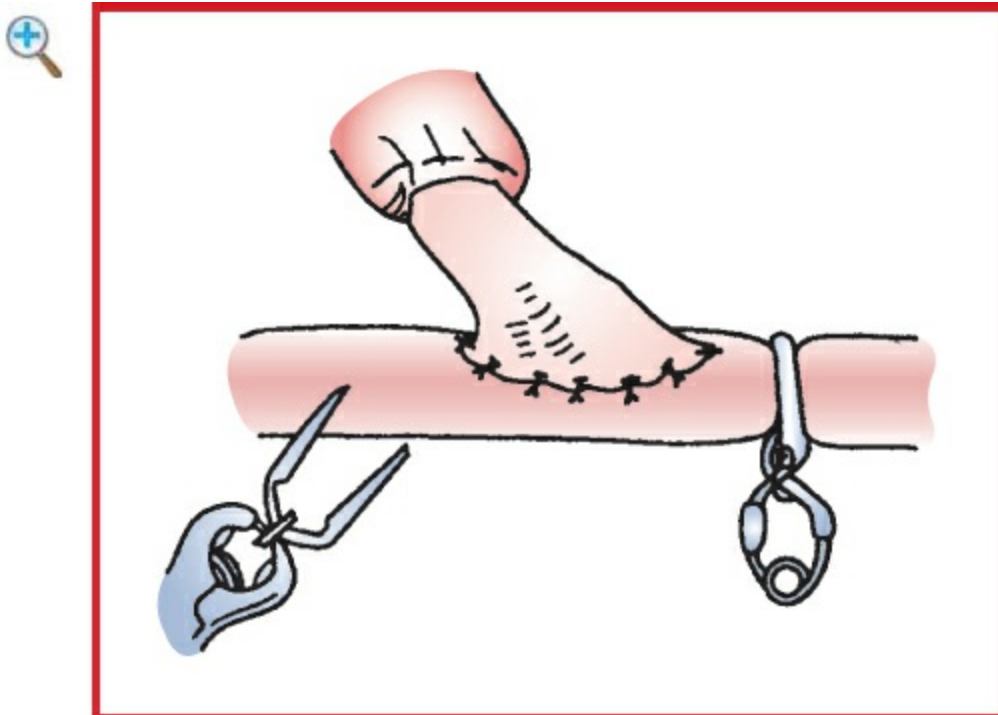


Figure 1.1-10 When the anastomosis is complete, the vascular clips are removed. (Reproduced from Chang SD, Steinberg GK: Superficial temporal artery to middle cerebral artery anastomosis. *Tech Neurosurg* 2000; 6(2):86–100.)

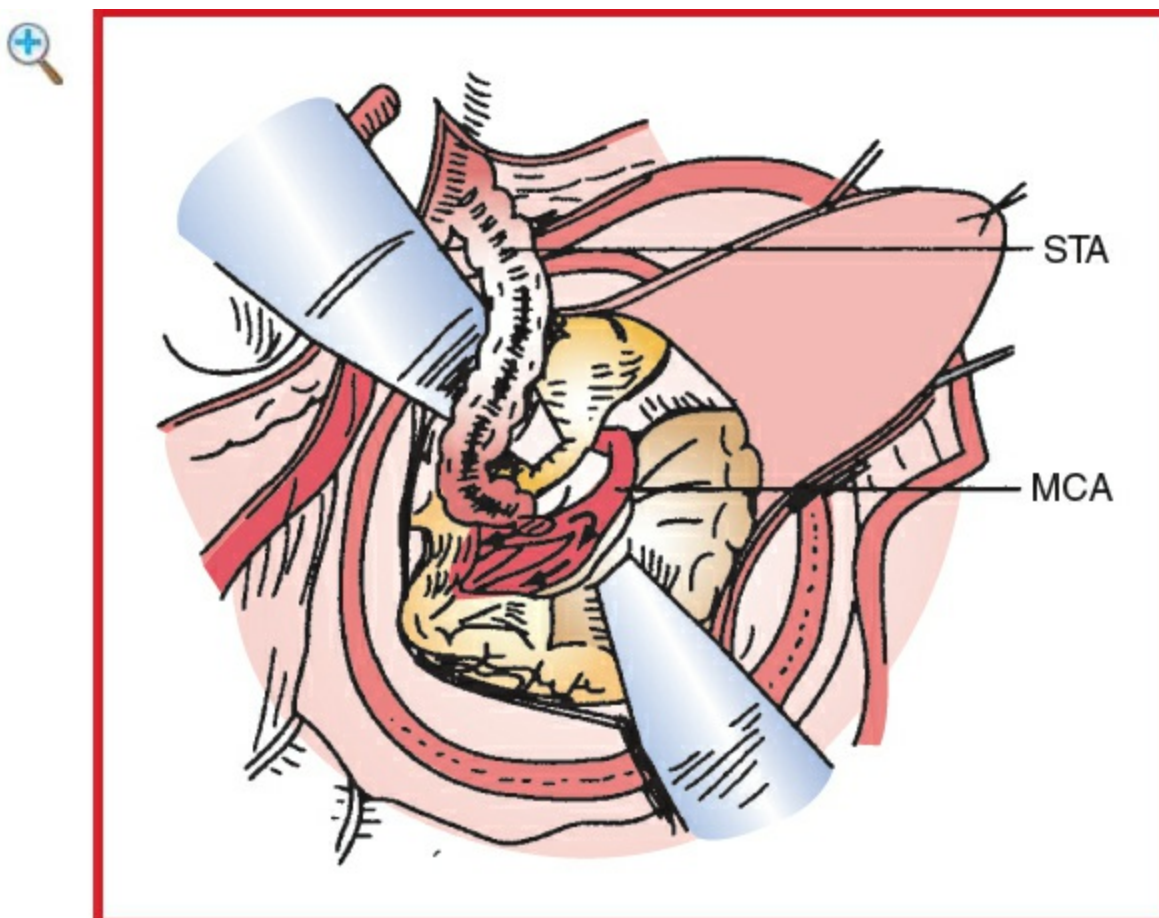


Figure 1.1-11 The completed anastomosis shows the STA positioned such that flow is directed toward

the proximal portions of the MCA. (Reproduced from Chang SD, Steinberg GK: Superficial temporal artery to middle cerebral artery anastomosis. Tech Neurosurg 2000; 6(2):86–100.)

Variant procedure or approaches

Indirect revascularization is commonly employed in patients in whom the graft vessels are too small for direct anastomosis. Vascular source tissues include a temporalis muscle flap and a flap of dura folded under so that its outer vascular surface is apposed to the cortical surface. **Encephalo-duro-arterio-synangiosis (EDAS)** is an indirect variant procedure wherein the STA is dissected circumferentially with its adventitia in the scalp, left in continuity, and laid on the surface of the brain after opening the dura. **Omentum-to-brain transposition** is becoming a more frequently used variant, wherein a portion of the omentum, with its luxuriant blood supply, is isolated using endoscopic technique, left attached to the right gastroepiploic artery, tunneled subcutaneously in the chest and neck, and laid over an area of poorly vascularized cerebral cortex after opening the dura. Revascularization is induced by angiogenesis factors and growth substances secreted by the brain and omentum.

Usual preop diagnosis

Moyamoya disease (cerebral ischemia due to occlusion of vessels at base of the brain); stroke; TIA; carotid artery stenosis (inaccessible to carotid endarterectomy); carotid artery occlusion; middle cerebral artery stenosis or occlusion; vertebral artery stenosis or occlusion; basilar artery stenosis or occlusion

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and AED medications, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	EC-IC Bypass	EC-IC Bypass with Vein Graft	Indirect Procedures
Position	Supine or lateral decubitus	⇐	⇐
Incision	Frontal, parietal, temporal, occipital, or a combination of	⇐ + Medial aspect of the leg and thigh for	⇐ + possible abdominal incisions for

	these, depending on area to be vascularized.	harvesting greater saphenous vein	endoscopic harvesting omentum and chest/neck incision for tunneling
Special instrumentation	Microscopic instruments; microvascular Doppler to identify scalp donor artery course and confirm graft patency. Videoangiography with indocyanine green	⇐ + Tunneling instruments	⇐
Unique considerations	Neuroprotective agents (propofol) and induced mild HTN (MAP 90–110 mm Hg) during cross-clamp of recipient intracranial artery. Avoid hypocapnia. Mild hypothermia (33°C). Dexamethasone 8–12 mg iv	⇐ + Attention to proper alignment of vein when tunneled to avoid kinking; heparinization if major cervical artery (carotid, vertebral) is temporarily occluded.	⇐ + Avoid devascularizing source tissue during dissection or compromise to blood supply during closure.
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)	⇐	⇐
Surgical time	4–6 h	⇐	⇐
Closing considerations	Careful attention to hemostasis. Avoid compromise of graft with dural closure, bone replacement, or scalp closure	⇐	⇐
EBL	<100 mL	⇐	100–200 mL

Postop care	Start aspirin on POD 1; monitor for subdural hygroma (CSF fluid collection in subdural space); ICU × 1 d	⇐	⇐
Mortality	<0.5%	⇐	⇐
Morbidity	Transient neurologic deficit: 20–40%	⇐	
	Subdural hygroma:	⇐	⇐
	Rare	⇐	⇐
	Wound infection: Rare	⇐	Abdominal
	Stroke: Rare		hernia: Rare
Pain score	3	3–5	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	40–80 yr; 2–20 yr for moyamoya disease
Male:Female	1:1 for atherosclerotic disease; 1:1.4 for moyamoya disease
Incidence	Thromboembolic stroke common, but indications for EC-IC bypass rare; 1/million/yr for moyamoya disease
Etiology	Atherosclerosis; embolism from heart or carotid artery
Associated conditions	HTN; CAD; PVD; hyperlipidemia; smoking; alcohol abuse; obesity; moyamoya disease

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients range in age from pediatric (classic moyamoya disease) to adult. These patients may have significant preexisting neurologic deficits. HTN in these patients is often an adaptation to cerebral vascular insufficiency → hypoperfusion. Inappropriate treatment of this compensatory HTN → cerebral ischemia and stroke. Patients with symptomatic moyamoya disease or bilateral carotid stenosis or occlusion may be good candidates for EC-IC bypass, especially since there are no alternative forms of therapy that have proven to be effective. Most patients present with h/o TIAs or stroke, and most are receiving antiplatelet medication.

Respiratory	None unless patient has Hx of smoking or has sustained pulmonary aspiration 2° neurologic deficit. Tests: As indicated from H&P
Cardiovascular	HTN is a common adaptive mechanism to maintain cerebral perfusion; therefore, “normalization” of BP preop may be undesirable. Adult patients may have generalized vascular disease, including CAD, so a careful cardiac Hx, physical examination, and ECG analysis should be done. If findings are positive, consider a more complete evaluation, including ECHO and coronary angiography. Be aware that cardiac insufficiency is the cause of about half of the deaths in patients with cerebrovascular disease. Tests: Consider ECG; others as indicated from H&P
Neurologic	Patients present with Sx of focal ischemic lesions. Cerebral angiography is necessary to r/o other causes of TIAs and characterize collateral circulation. Regional CBF studies are generally not helpful because measurement does not distinguish between low flow due to cerebrovascular obstruction and that due to low metabolic demands 2° prolonged cerebral ischemia. Preexisting deficits should be well characterized. Tests: CTA; MRI; angiogram; Diamox® scan
Hematologic	For most patients, a platelet-suppressive dose of aspirin should be continued through the day of surgery; drugs affecting PT/PTT should be D/C'd at least 1 wk before surgery to avoid excessive bleeding; clopidogrel (Plavix) should be D/C'd 5 d before surgery. Tests: Hct; PT; PTT; C8C
Laboratory	Tests as indicated from H&P
Premedication	If medication is desirable, small doses of midazolam (e.g., 1–3 mg iv) are useful. Detailed discussion with the patient about the anesthetic plan with appropriate reassurance is essential.
ERAS	Check compliance with presurgical ERAS

recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

■ INTRAOPERATIVE

Anesthetic technique

GETA: The goals of anesthesia for this procedure are to (a) maximize flow to the ischemic area through collateral channels by maintaining BP at normal or somewhat elevated values; (b) ↑ tolerance of the brain to ischemia by decreasing CMRO₂ with the use of mild hypothermia (33–34°C) and anesthesia; (c) ↓ intracranial volume (blood and tissue) to optimize working space within the cranial compartment, thereby minimizing the need for surgical retraction; and (d) provide adequate surgical anesthesia.

Induction	Propofol 1–2 mg/kg iv to provide amnesia and ↓ cerebral blood volume by inducing cerebral vasoconstriction. Fentanyl 7–10 mcg/kg iv to blunt response to intubation and provide analgesia for the first hours of surgery. Rocuronium 0.6–1.0 mg/kg or vecuronium 0.15 mg/kg to provide muscle relaxation for tracheal intubation. Bolus ephedrine and/or phenylephrine are often necessary to maintain CPP during induction.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Avoid hypothermia unless indicated for brain protection. Dexamethasone (4–8 mg iv) may ↓PONV. Avoid long-acting opioids or scopolamine patch, which may confound postop neuro examination. Sequential pneumatic compression devices should be used to ↓VTE risk.
Maintenance	Isoflurane ≤0.6% or sevoflurane ≤1% with 1:1 O ₂ /N ₂ O and a remifentanyl infusion (0.05–0.2 mg/kg/min). A phenylephrine infusion is usually necessary to maintain CPP during the procedure. Continued neuromuscular blockade is usually not necessary. Burst suppression is induced with propofol (~1 mg/kg) just before cross-clamping of the cerebral artery in preparation for

	anastomosis. A bolus dose of phenylephrine and/or ephedrine is usually given concomitantly to offset the hypotensive effects of propofol. If mild hypothermia was employed, rewarming should start as soon as revascularization is complete.	
Emergence	With the start of dural closure, reduce remifentanyl infusion to 0.05 mcg/kg/min. A propofol infusion (if used) should be discontinued at this time. The patient's BP generally will increase and titration of β-adrenergic blocking drugs (e.g., labetalol or esmolol) and/or vasodilators (e.g., clevidipine or SNP) may be needed (see Control of BP, below). The inhalation agents can be D/C'd at the time of dressing application. Reverse any residual neuro muscular blockade. Most patients will breathe spontaneously and can be extubated uneventfully while on the remifentanyl infusion. If the brain has not been injured by the surgical procedure, the patient should awaken promptly. Close regulation of BP is essential. If the patient begins to cough on ETT, either it should be removed or cough reflex suppressed with iv lidocaine (1 mg/kg). Patient is placed in bed in a 20–30° head-up position and transported to ICU for monitoring overnight. Supplemental O₂ should be administered and close regulation of BP maintained (typically at baseline values). Prophylactic antiemetics (e.g., ondansetron 4 mg $\pm$ metoclopramide 10–20 mg iv) should be given 30 min before extubation. Consider iv acetaminophen for postop analgesia if po dose given >4–6 h ago.	
Blood and fluid requirements	IV: 18 ga $\times$ 2 BSS @ <1–3 mL/kg/h + UO Zero balance fluid management (see Appendix K, p. K-2)	In hypothermic patients, iv drugs should be administered through the central line to ensure timely entry into the central circulation.
	CVP (triple lumen)	Use low-volume (1 mL) extension for drug infusions. Reserve distal lumen for

		pressure monitoring and air aspiration.
Control of brain volume (ICP)	Avoid hypovolemia ↓ Isoflurane $\leq 0.6\%$ or sevoflurane $< 1\%$ $N_2O \leq 50\%$ $\pm$ Furosemide 10–20 mg iv $\pm$ Mannitol 0.5–1.0 g/kg	Maintain normovolemia. Generally, vigorous control of brain volume is not necessary because surgeon is working with cerebral vessels on the surface of the brain. Keep $PaCO_2 = \sim 40$ mm Hg. Hypocarbica may cause unwanted cerebral vasoconstriction in these patients.
Monitoring	Standard monitors EEG Core temperature Arterial line CVP (triple-lumen catheter) UO	Establish burst suppression before X-clamp. Core temperature is best approximated by a deep esophageal temperature probe. Bladder temp tends to lag behind core temp. The distal CVP lumen should be used for pressure monitoring.
Control of BP	Maintain normal BP (MAP) Phenylephrine infusion Normovolemia	Maintenance of MAP is important because cerebral perfusion is often impaired in these patients. If a vasoconstrictor is needed, phenylephrine is preferred because it has minimal dysrhythmogenic potential. Responses to vasoactive drugs are much easier to regulate if a normal blood volume has been maintained throughout the anesthetic period.

Positioning	✓ and pad pressure points ✓ eyes - Shoulder roll - SCDs - Mayfield 3-point skull fixation	Anesthetic hoses, monitoring, and vascular lines directed to patient's feet where anesthesiologist is usually positioned during surgery. Used to minimize DVT. Remifentanyl (2–4 mcg/kg) should be used to minimize ↑BP during pinning.
Hypothermia	- Cold-water circulating blankets - Core temp goal: 33–34°C - Thermogard®-type device - Bladder irrigation	Surface cooling is quite effective in patients with high surface-to-volume ratios and should be started as soon as induction of anesthesia is complete, using a cold-water circulating blanket underneath and above the patient. When anastomosis is nearly complete, vigorous efforts at warming, including bladder irrigation, are initiated. Cooling and especially rewarming can be greatly facilitated by the use of a central heat exchange device (e.g., Thermogard®).
Complications	- Seizures - Stroke - Hemorrhage at anastomosis	*S_z Rx: Levetiracetam (1 g) or phenytoin (1-g loading dose). NB: Incompatible with dextrose-containing solutions.
	Brain swelling	May be 2° hyperemia in revascularized territory.

POSTOPERATIVE

Complications	Localized scalp necrosis	Major complications uncommon; localized scalp
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	Brain swelling	necrosis unique to this procedure. May be 2° to delayed hyperemia. Careful control of BP is important.
Multimodal pain management	Avoid scalp blocks Meperidine (10–20 mg iv prn)	May interfere with donor vessel blood flow. May be useful to suppress shivering. Postop pain is usually not severe.
	Acetaminophen 1 g iv ± Fentanyl 25–50 mcg prn	If not given preop or if 4–6 h have elapsed since previous dose.
Tests	Cerebral angiogram	Cerebral angiography documents patency of graft and collateral flow.
	Regional blood flow studies	Regional blood flow studies are often useful.
	CT scan	If any question about neurologic status, a CT scan is performed.
ERAS	PONV protocol	Ondansetron 4 mg iv ± Metoclopramide 10–20 mg iv Encourage early enteral nutrition, Foley catheter removal, and mobilization.

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CRANIOTOMY FOR TUMOR

SURGICAL CONSIDERATIONS

Lawrence M. Shuer, Steven D. Chang, and Navjot Chaudhary

Description

Many classification systems exist for brain tumors. Generally, brain tumors can be classified by their location, and a differential diagnosis can be generated based on tumor location and patient age. Brain tumors are either supratentorial or infratentorial ([Table 1.1-2](#)) or intraaxial or extraaxial ([Table 1.1-2](#)). The surgical approach depends on the location of the lesion, the need for brain relaxation, and whether exposure will require brain resection. Patient positioning generally depends on the location and surgical approach to the tumor ([Table 1.1-3](#)). Once positioned, the patient's head typically is placed in a Mayfield pin fixation system to prevent head movement during surgery and to allow for intraoperative image-guided stereotactic navigation.

Several types of incisions are used for these procedures. **Linear incisions** can be used to resect small tumors over the convexity or when using a midline approach to the posterior fossa and often have the advantage of a more rapid wound closure. **Curvilinear** or **horseshoe-shaped incisions** are commonly used for larger tumors. After the skull is exposed, burr holes are made with a perforating drill, and the bone flap is

cut with the craniotome. Most surgeons routinely use a **free bone flap**, in which the bone is completely removed and stored for the duration of the case. Other surgeons turn an **osteoplastic flap**, where the bone is left attached to muscle and/or pericranium to keep it partially vascularized. When performing certain posterior fossa resections (e.g., a retromastoid craniotomy or low suboccipital craniotomy), the surgeon may choose to remove the bone without replacement, performing a **craniectomy** instead of a craniotomy. After the bone is removed, a few small holes are drilled near the edge of the craniotomy. These holes are used to suspend the dura using sutures. This helps prevent blood from accumulating in the epidural space during the rest of the case. The dura is then opened either in a stellate or curvilinear fashion. The method of dural opening generally is based on the size of the bone opening and its proximity to venous sinuses. The surgeon then proceeds with tumor removal if it is on the surface or with brain retraction/resection if the tumor is deep to the surface. At this point, the surgeon may request anesthetic interventions for brain relaxation (osmotic diuresis, hyperventilation) and specific BP control. After the tumor is removed, hemostasis is achieved and the dura is closed. The bone flap is replaced and fixated, the galea is closed with sutures, and the skin incision is closed with staples. During closure, the surgeon may ask for specific BP parameters to ensure that there is hemostasis in the resection bed and surgical field. Patients typically are extubated after cranial surgery because it is paramount to obtain a neurologic examination as soon after surgery as possible. Patients undergoing a craniotomy almost always require a postoperative ICU course.

Table 1.1-2 Common Tumor Location

Supratentorial	Infratentorial	Intraaxial Tumors	Extraaxial Tumors
Metastatic tumors	Metastatic tumors	Astrocytoma/glioblastoma	Meningeal tumors
Astrocytoma/glioblastoma	Acoustic neuromas	Metastatic tumors	Acoustic neuromas
Oligodendroglioma	Meningiomas	Oligodendroglioma	Choroid plexus papilloma
Ependymoma	Hemangioblastoma	Ependymoma	Craniovertebral junction tumors
Meningioma	Medulloblastoma	Hemangioblastoma	Hemangioblastoma
Choroid plexus papilloma		Medulloblastoma	
Craniopharyngioma		PNET	
Primitive neuroectodermal			

tumor (PNET)			
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Table 1.1-3 Common Patient Positions Based on Tumor Location	
Supine:	Most supratentorial tumors in the frontal, temporal, or anterior parietal lobe. Tumors of the lateral ventricles and 3rd ventricle. Tumors of the anterior two-thirds of the interhemispheric fissure
Lateral:	Tumors of the posterior or lateral parietal lobe, posterior temporal lobe, cerebellopontine angle, or lateral cerebellum. Tumors of the posterior ventricular horn
Prone:	Tumors of the occipital lobe, most midline cerebellar tumors, and tumors of the 4th ventricle. Tumors of the posterior one-third of the interhemispheric fissure and the tentorium
Sitting:	Tumors of the pineal region. Tumors of the 4th ventricle or midline cerebellum

Electrophysiologic monitoring detects changes in SSEP, brain stem auditory evoked potentials (BAEP), and motor evoked potentials. Such monitoring is commonly used for tumors of the brain stem and skull base or when resecting tumors in critical locations. Early changes in electrophysiologic monitoring potentials may alert the anesthesiologist to manipulate BP and may help the surgeon to further define the extent of safe retraction and tumor resection to minimize the likelihood of a postop deficit.

Image-guided navigation involves the use of a computerized workstation that can track the position of specific instruments before and during the operation. This allows the surgeon to (a) plan appropriate skin and bone openings, (b) choose an optimal trajectory to the tumor, and (c) achieve a volumetric resection of the tumor. Image-guided navigation involves obtaining a CT or MRI scan prior to the start of the surgical procedure; the time required to obtain this study is usually offset by a shorter operative time. Data from the operative microscope can also be incorporated into the navigation system to aid in tumor dissection.

Mild hypothermia is often utilized when resecting tumors in eloquent areas of the brain because hypothermia has been shown to provide neuroprotection.

CSF drainage usually is accomplished by use of a lumbar drain. Lumbar drainage typically is used for tumors in the cerebellopontine angle, in the pineal region, or tumors associated with significant edema. CSF drainage can also be accomplished by placing a

ventriculostomy or by draining various cisterns during the operation.

Awake craniotomies typically are reserved for tumors adjacent to or within fine motor or speech areas of the language dominant cortex. Patients are sedated during cranial opening and are awakened once the dura is open. Direct cortical stimulation is performed to determine the relationship between the tumor and the speech or motor centers. The surgeon then maps out eloquent and noneloquent areas of the brain prior to beginning the tumor resection. The patient continues to converse during the tumor resection to ensure that language function is not altered.

Interstitial chemotherapy or brachytherapy can be performed after the tumor resection by implanting chemotherapy impregnated wafers or radioactive seeds in the resection cavity. In addition, a catheter can be placed in the resection bed and injected with radioactive material at a later date.

Intraoperative fluorescent dyes can assist in achieving gross or near-total resection. Maximal safe resection is thought to have an important role in improving overall survival and progression-free survival, but identifying tumor borders can be challenging in some glioma cases. For this reason, imaging adjuncts, such as 5-aminolevulinic acid (5-ALA) or fluorescein sodium, have been proposed as strategies for better delineating the limits of surgical resection.

Variant procedure or approaches

Patient position varies depending on tumor location and surgeon preference ([Table 1.1-3](#)). Tumors on the convexity, or surface, of the brain may require minimal brain relaxation or exposure. Deep tumors, or tumors around the brain stem or skull base, may require substantial brain relaxation and retraction for optimal exposure. For deep tumors or tumors within or adjacent to critical structures, the operating microscope is commonly used. Large vascular tumors may undergo preop embolization of the tumor blood supply to minimize intraop blood loss.

Minimally invasive techniques exist for tumor resection. Access points include supraorbital (“eyebrow craniotomy”), transnasal, minipterional, retromastoid, and transventricular routes. In addition, MRI-guided laser ablation techniques have provided a method for stereotactic access to thermocoagulate brain tumors through a small twist drill hole placed with the patient in the MRI scanner. Advantages of these minimally invasive techniques include smaller incision, less postop pain, and faster recovery times. Disadvantages include limited access and visibility.

Intraop MRI (iMRI) is an advance, which has allowed surgeons to better define brain

tumor margins in the operating room in conjunction with surgical navigation systems (e.g., StealthStation™). This allows for more accurate confirmation of complete surgical resection of brain tumors. iMRI systems may include those that are moved into the OR or those that require transporting the patient while anesthetized to a nearby location for scanning.

Usual preop diagnosis

Glioma; glioblastoma multiforme; astrocytoma; oligodendroglioma; ependymoma; PNET; meningioma; craniopharyngioma; choroid plexus papilloma; hemangioblastomas; medulloblastoma; acoustic neuroma; brain metastasis; hemangiopericytoma

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and AED medications, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine, lateral, prone, or sitting, based on location of tumor
Incision	Linear or curvilinear, based on location of tumor
Special instrumentation	Operating microscope, laser, CUSA, electrophysiologic monitoring, image-guided navigation, lumbar drain, ventriculostomy
Unique considerations	ETT must be taped securely in a location satisfactory to the surgeon; anode or RAE tube is helpful in certain situations. Decadron is usually given preop and dosed every 6 h to minimize brain edema. Brain relaxation techniques may be required. The patient with ↑ ICP may require special consideration for induction of anesthesia. Awake craniotomy may be required for tumors involving eloquent brain areas.
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	3–5 h typically
Closing	Possible requirement for dural graft. Drain often left in the

considerations	epidural or subgaleal space. Good BP control during closing and extubation to prevent hemorrhage into tumor resection bed.
EBL	50–500 mL (Meningiomas and renal cell mets are highly vascular tumors, → blood loss may be substantial.)
Postoperative care	ICU or close observation unit × 1–3 d. Fluid and electrolytes require frequent monitoring. BP may need to be controlled with antihypertensives.
Mortality	0–5% (mortality higher for tumors in critical locations)
Morbidity	Infection: 1% Neurologic: neurologic disability, nerve injury: 0–10% CSF leak: 1–3% Venous sinus injury, air embolus Endocrine disorder Massive blood loss
Pain score	2–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Infant to 85 yr (usually 20–60 yr)
Male:Female	~1:1
Incidence	Common neurosurgical procedure (~40,000 dx/yr in the United States)
Etiology	Neoplastic

ANESTHETIC CONSIDERATIONS

(Procedures covered: craniotomy for tumor; craniotomy for skull tumor)

PREOPERATIVE

Typically, this is a healthy patient population, apart from Sx attributable to intracranial pathology (↑ ICP, Sz, HA, N/V, visual disturbances). Review imaging studies and discuss surgical approach, positioning, tumor type, and potential for blood loss with surgeon. These patients are often receiving steroids to ↓ ICP. The patient may have received po 5-aminolevulinic acid 3 h before anesthetic induction. Side effects can

include phototoxic skin reactions (reduce exposure to room light or sunlight for 24 h postop), severe hypersensitivity reactions, lactic acidosis, and N/V.

Respiratory	Occasionally, neurogenic pulmonary edema may occur. H/o chemotherapy or radiation therapy may affect pulmonary function and airway management. Otherwise no special considerations, unless indicated from H&P.
Cardiovascular	Benign or malignant brain tumors cause edema formation in adjacent normal brain tissue, which may → ↑ ICP. If ICP increases sufficiently to cause herniation of the brain stem, patients develop the “Cushing triad” of HTN, bradycardia, and respiratory irregularity. These changes will resolve when ICP is reduced, so vigorous attempts to regulate BP and HR prior to craniotomy are not warranted. After the diagnosis of brain tumor is made, most patients are placed on high-dose steroid therapy to lessen edema in surrounding normal brain. Steroids are extremely effective in this setting, and Sx of ↑ ICP will often abate. Tests: Consider ECG; others as indicated from H&P
Neurologic	Patients may present with complaints related to ↑ ICP (e.g., HA, N/V, visual changes, recent onset of Sz), neurologic deficits from compression of motor area, or as a result of hemorrhage from the tumor or edema in surrounding normal brain. Document preop physical findings. Tests: A CT scan or MRI will delineate the site and size of the tumor, especially if iv contrast material, such as gadolinium, is administered to enhance the margins of the tumor. Look for intracranial mass effects: midline shift, ↓ ventricular volume, ↓ peri-brainstem (basal cisterns) CSF space, obstructive hydrocephalus.
Laboratory	Coags should be normal. Hct, electrolytes, and other tests as indicated from H&P.

Premedication	Standard premedication (except for patients with the possibility of ↑ ICP: no sedation).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water for patients undergoing posterior fossa craniotomy or with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Small tumors, particularly those located in deeper brain structures, may be localized and resected using stereotactic or image guidance techniques. Generally, scalp markers (fiducials) or a stereotactic frame are placed on the patient's head; then, the patient is taken to CT/MR for determination of the exact tumor site. GETA is almost invariably used for tumor removal, although MAC (see Awake Craniotomy) is used on rare occasions when the surgeon needs to assess motor or sensory function during the resection of the tumor adjacent to critical motor or speech areas. Surgeons may visualize certain tumors by utilizing iv fluorescent dyes (e.g., fluorescein—a marker for areas with compromised BBB, or 5-aminolevulinic acid—a Hb precursor that causes the accumulation of fluorescent porphyrins in malignant glioma cells). When used in conjunction with a specialized operating microscope, these compounds allow greater definition of the tumor and its boundaries with normal brain tissue. Anaphylactic reactions can occur with the use of these agents.

Induction	If the patient is in a stereotactic frame, or a difficult intubation is anticipated, orotracheal intubation will need to be accomplished before induction of GA. Awake fiberoptic intubation is the best choice. Ensure adequate airway anesthesia to minimize sudden ↑ICP and ↑MAP. After the airway is secured, anesthesia usually is induced with propofol (1–3 mg/kg) and fentanyl (5 mcg/kg) in combination with an NMR (e.g., rocuronium 0.6 mg/kg or vecuronium 0.1 mg/kg). To minimize ↑↑ BP and ↑ ICP with ET intubation, it is important that the patient be well anesthetized (and paralyzed) before undertaking laryngoscopy. Induction doses of propofol or fentanyl
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	may not be sufficient to abolish increases in ICP that may occur with laryngoscopy and tracheal intubation. Consider hyperventilating the patient and using remifentanyl (2–3 mcg/kg) as part of the induction technique to further blunt the response to laryngoscopy in patients with ↑ ICP.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Consider pre-incision placement of scalp blocks. Dexamethasone (4–8 mg iv) may ↓PONV. Avoid long-acting opioids or scopolamine patch, which may interfere with postop neuro examination. Sequential pneumatic compression devices should be used to ↓VTE risk.
Maintenance	Isoflurane or sevoflurane ≤ 1 MAC ($\leq 1/2$ MAC if EP monitoring is used) inspired with 50:50 O₂/N₂O. Propofol (50–100 mcg/kg/min) can replace N₂O and may be used to further ↓ cerebral blood volume, ↓ cerebral metabolism, and ↓ CMRO₂. A remifentanyl infusion (0.05–0.2 mcg/kg/min) can be used to supplement the anesthetic without interfering with EP monitoring. Excessive EMG activity can be Rx with 5–10 mg rocuronium bolus.
Emergence	Surgeons should request transient ↑ MAP to 90–100 mm Hg to test hemostasis after the tumor has been resected. With the start of dural closure, D/C propofol infusion if in use, continue a low-dose remifentanyl infusion (e.g., 0.05 mcg/kg/min). The patient's BP generally will increase, and titration of β-adrenergic blocking drugs (e.g., labetalol or esmolol) and/or vasodilators (e.g., clevidipine or SNP) may be needed (see Control of BP, below). Reverse NMB as necessary. The anesthetic agents can be D/C'd at the time of dressing application while continuing the remifentanyl infusion. Most patients will breathe spontaneously and can be extubated uneventfully while on 0.05 mcg/kg/min remifentanyl. If the brain has not been injured by the surgical procedure, the patient should awaken promptly.

	Keep close regulation of BP. Patient is placed in bed in a 20–30° head-up position and transported to ICU for monitoring overnight. Supplemental O₂ should be administered and BP control maintained. Prophylactic antiemetics (e.g., metoclopramide 10–20 mg and ondansetron 4 mg) should be given iv ~30 min before extubation. Consider scalp block placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po or iv dose given >6 h ago.	
Blood and fluid requirements	IV: 16–18 ga × 2 Type and screen (or cross) BSS @ 2–3 mL/kg/h	Brain tumors (e.g., meningioma) can be highly vascular. To minimize postop cerebral edema, limit BSS to ≤3 mL/kg/h + replacement of UO. If volume is needed, administer albumin 5% as required. Zero balance or goal-directed fluid management is recommended.
Monitoring	Standard monitors Arterial line CVP line ± Precordial Doppler ± BAER, SSEP, MEP UO	If the head of the OR table is significantly elevated (≥30°) or if the patient is in the seated position, a precordial Doppler monitor is necessary to monitor for VAE. Evoked potentials can be monitored easily in the presence of inhaled anesthetics (e.g., ≤0.6% isoflurane with 50% N₂O), contrary to popular belief.
Positioning	Supratentorial: supine/lateral Posterior fossa: prone/sitting	For brain tumors in the frontal, parietal, or temporal lobes, patient will be supine with head in Mayfield pin

✓ and pad pressure points
✓ eyes

fixation, turned to the side with a roll under the shoulder on the operative side (Fig. 1.1-6). For occipital or posterior fossa tumors, patient may be prone or, occasionally, sitting (see Anesthetic Considerations for Cervical Neurosurgical Procedures, Positioning). Acoustic neuromas are generally most easily removed with patient in the lateral ("park-bench") position with a roll under the axilla. The patient generally lies on a "beanbag," which holds her/him firmly in the lateral position.

Control of ICP

Maintain normovolemia
Control BP in normal range for patient
 $\text{PaO}_2 > 100$ mm Hg
Limit isoflurane or sevoflurane to 1/2 MAC for EP monitoring
Steroids: 8–12 mg Decadron iv

$\text{PaCO}_2 = 25\text{--}30$ mm Hg

Consider replacing N_2O

Patients with intracranial tumors may be on the steep portion of the intracranial compliance curve such that any increase in intracranial volume may $\rightarrow \uparrow\uparrow$ ICP. Transient increases in ICP—even up to 50–60 mm Hg—are tolerated, provided they are promptly terminated. Sustained increases in ICP $> 25\text{--}30$ mm Hg are associated with severe neurologic injury and poor outcome.
 $\downarrow \text{PaCO}_2 \rightarrow \downarrow \text{CBV}$ (providing better surgical access) + $\uparrow$ CBF to ischemic areas ("Robin Hood" effect).
If further reduction in ICP is

	with propofol infusion Mannitol 0.5–1.0 g/kg ± Furosemide 10–20 mg ± Lumbar CSF drain Head up 20–30°	needed, replace N₂O with a propofol infusion. Mannitol/furosemide → ↓ K⁺; monitor level and replace as necessary. If mannitol is administered too rapidly, profound ↓ BP will occur, probably from peripheral vasodilation. CSF drain may be placed after induction of anesthesia and may be opened as required to ↓ CSF volume and pressure. Keep surgeon informed of volume drained.
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POSTOPERATIVE

Complications	Seizures Neurologic deficits Tension pneumocephalus Hemorrhage requiring reexploration Edema and ↑ ICP	*S_z Rx: Levetiracetam (1 g) or phenytoin (1-g loading dose). NB: Incompatible with dextrose-containing solutions. In seated position, additional rare, but possible complications include quadriplegia from excessive flexion of the head or tension pneumocephalus from intracranial air. Severe tension pneumocephalus may delay emergence from anesthesia or cause postop neurologic deficits. CT scan may be necessary for Dx.
Multimodal pain management	Acetaminophen 1 g iv	If not given preop or if 4–6 h have elapsed since previous dose.

	Scalp blocks (see Fig. 1.1-12) ± Meperidine (10–20 mg iv prn) ± Fentanyl 25–50 mcg bolus prn	0.5% bupivacaine with epi is as effective as liposomal bupivacaine. Meperidine reduces postop shivering. Avoid excessive sedation.
Tests	CT scan	If patient exhibits any delay in emergence from anesthesia and surgery, or any new neurologic deficits emerge postop, a CT scan is invariably obtained.
ERAS	PONV protocol	Ondansetron 4 mg iv, ± metoclopramide. Encourage early enteral nutrition, Foley catheter removal, and mobilization.

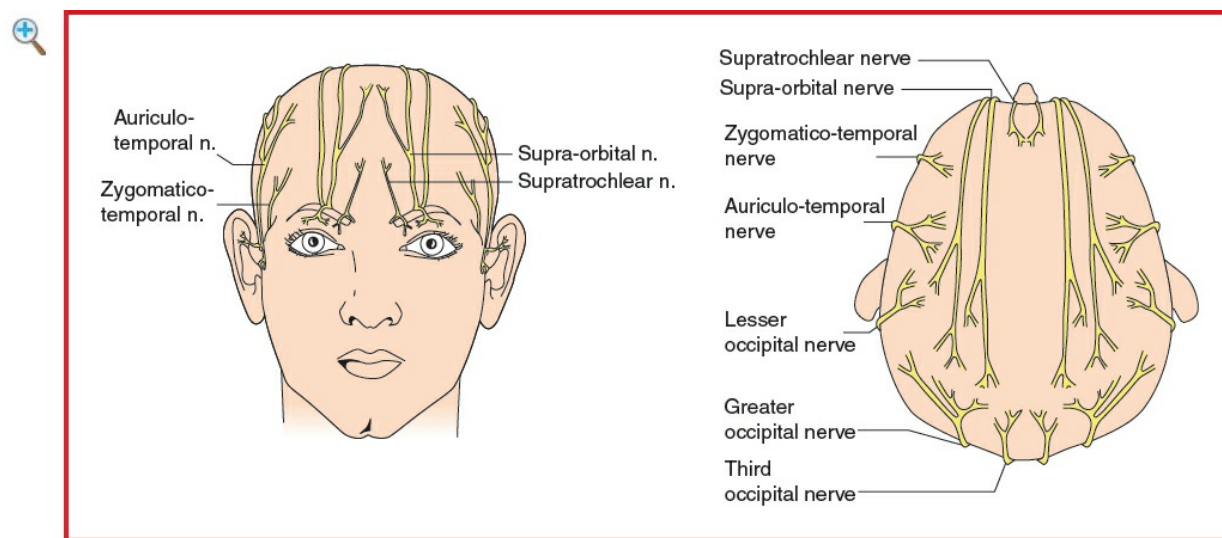


Figure 1.1-12 Scalp innervation. Typically, the supraorbital, auriculotemporal, lesser occipital, and greater occipital nerves are blocked near their scalp origins.

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CONSIDERATIONS FOR AWAKE CRANIOTOMY

PREOPERATIVE

These patients typically are otherwise healthy except for a tumor (or AVM) involving eloquent (speech or motor) areas of the brain. In some centers, awake craniotomies are still used for epilepsy surgery. Awake craniotomies for functional procedures (e.g., deep brain electrode placement) usually involve 1 or 2 burr holes with small dural punctures and are discussed on [p. 79](#).

In addition to the preop anesthetic considerations for brain tumors or vascular malformations, it is essential that these patients (and family) meet with their anesthesiologist and monitoring neurologist to discuss in detail both the anesthetic and surgical aspects of the procedures, as well as the specific speech or motor mapping techniques that will be used. The patient should be told that a Foley catheter will be placed while they are sedated; however, after the sedation is stopped, the catheter will become annoying, and the patient may be convinced that he/she has a full bladder. Intraop cooperation is critical, and there is no pharmacologic substitute for a well-

informed patient.

ERAS: Preop counseling by the surgeon/anesthesiologist should include a detailed description of the awake craniotomy procedure and what to expect. Patients should also be advised to continue chronic pain and AED medications, limit NPO duration, and advised regarding alcohol and smoking cessation, carb loading, postop pain management options, urinary drain postop, description of antimicrobial prophylaxis, and skin preparation. See [Appendix K, p. K-1](#).

Premed	Midazolam 1–3 mg iv is usually tolerated. Beware of respiratory depression → ↑ ICP. Antiemetic: for example, ondansetron 4 mg iv slowly and q4h during the procedure. Patients who are caffeine dependant will benefit from replacement caffeine (100- to 200-mg tablet with sip of water).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Although some centers use an asleep-awake-asleep technique with airway control (ETI or LMA) during the sleep phases, we prefer the following technique that minimizes airway manipulation and its inherent risk of coughing and bucking.

Premapping	Most patients with significant space-occupying lesions will benefit from the early administration of mannitol 0.5 g/kg by slow iv infusion (check with surgeon for possible contraindications). With the patient in a lounge-chair position and following the application of standard monitors and nasal prong O ₂ , initial sedation is provided by incremental midazolam titrated to effect (1–10 mg iv). A loading dose of dexmedetomidine (1 mcg/kg iv over 10–15 min as tolerated) is given while the sites for arterial line and
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central line (subclavian or antecubital PICC line) placement are infiltrated with local anesthetic. Following the loading dose, a maintenance infusion of dexmedetomidine (0.2–0.7 mcg/kg/h) typically 0.4 mcg/kg/h is started. The dexmedetomidine produces cooperative sedation and analgesia without respiratory depression. A ↓ BP and ↓ HR are common and occasionally will require treatment with vagolytics and/or pressors. Sedation is usually deepened with a propofol infusion (25–75 mcg/kg/min) to cover the more invasive aspects of the procedure. A nasal airway or THRIVE device (e.g., Optiflow®) may be useful in some patients with mild airway obstruction. Scalp blocks (supraorbital, temporal, and occipital nerves) are placed using 0.5% bupivacaine with epinephrine (see [Fig. 1.1-12](#)). Arterial line, central line, and Foley catheter are placed. The arterial line and iv are ideally placed on the ipsilateral upper extremity to minimize interference with mapping. It is important that the nurse administer intraurethral lidocaine gel, holding it in place for 2–3 min before inserting the Foley catheter. Pin sites for the Mayfield skull fixation are infiltrated with local anesthetic, and the clamp is applied. The incision is infiltrated with local anesthetic, and after patient comfort has been ensured, the craniotomy is allowed to proceed. Drapes should be arranged so that the anesthesiologist and neuromonitoring team have an unobstructed view of, and access to, the patient's face and airway.

Mapping

The dexmedetomidine (and propofol) infusions are stopped at the end of the craniotomy. Initial mapping (e.g., phase reversal localization) can be accomplished while the patient is recovering from the sedation. Some direct cortical stimulation can produce seizures; it is important that sterile, iced saline be immediately available for cortical irrigation

	to suppress the seizure activity. If seizure activity spreads, it will be necessary to administer a bolus dose of propofol (20–30 mg iv) through the central line. Airway support (e.g., LMA) may be necessary in the postictal phase. The patient should be awake and cooperative within 20 min after the dexmedetomidine and propofol infusions have stopped. Occasionally, flumazenil may be necessary to reverse residual midazolam effects. Detailed speech or motor mapping can then be readily accomplished. We use a miniature wireless microphone taped near the patient's mouth to amplify the patient's voice during testing and tumor resection.
Resection	If the lesion is located safely away from the mapped speech or motor areas, the resection can be accomplished after reinstituting the dexmedetomidine and propofol infusions. Otherwise, the patient must be kept awake and talking throughout the resection. Often the patient will benefit from a low-dose remifentanyl infusion (e.g., 0.05 mcg/kg/min) to improve ability to tolerate positional discomfort.
Closing	Closing is best accomplished by resuming the previously effective infusions of dexmedetomidine and propofol. The infusions should be stopped during the final phase of wound closure. Careful control of BP may require the use of labetalol, esmolol, and SNP, which should be titrated during emergence. Typically, the patient will be awake and alert for transport to the ICU.
ERAS	Reinforce scalp blocks as necessary. Encourage early ambulation, alimentation, and Foley catheter removal.

Suggested Reading

1. Brown T, Shah AH, Bregy A, et al: Awake craniotomy for brain tumor resection: the

- rule rather than the exception? J Neurosurg Anesthesiol 2013; 25(3):240–7.
2. Hernández G, Roca O, Colinas L: High-flow nasal cannula support therapy: new insights and improving performance. Crit Care 2017; 21(1):62.
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CRANIOTOMY FOR SKULL TUMOR

SURGICAL CONSIDERATIONS

Lawrence M. Shuer, Steven D. Chang, and Navjot Chaudhary

Description

Tumors of the skull fall into the classification of other bony tumors. Examples of types of skull tumors often requiring surgery include eosinophilic granuloma, histiocytosis, hemangioma, osteoma, epidermoid, dermoid, metastases, osteosarcoma, fibrous dysplasia, chordoma, chondrosarcoma, and meningioma. These tumors may occur anywhere on the skull, except chordoma and chondrosarcoma, which usually arise from the skull base. The exact positioning of the patient depends on the location of tumor. For example, the sitting or prone position often is used for tumors in the occipital or suboccipital regions, and the supine position is used for frontal, temporal, or parietal tumors. Some surgeons prefer the lateral position for temporal or parietal bone lesions and the prone position for occipital and some suboccipital bone lesions. Complete immobilization of the head is often not necessary, so the patient's head can be placed in a horseshoe or a Shea headrest. If complete immobilization is required, Mayfield pin fixation is used. The tumor is removed by placing burr holes near or around the tumor. Then, the affected skull is removed by cutting a bone flap around the tumor with the craniotome. In a suboccipital or posterior fossa craniectomy, the bone often is removed piecemeal with either a drill or a series of rongeurs. The dura usually is not opened unless it is involved with the tumor. The surgeon may elect to perform a **cranioplasty** to cover the defect, depending on the size and location. The defect can be repaired using a variety of methods, which include the use of a polymer, such as methyl methacrylate, or a bone cement, such as hydroxyapatite. In addition, autologous bone may be harvested from another location on the skull or another site (hip or rib) for reconstruction. Last, the surgeon may elect to repair the cranial defect at a later date, using a custom implant made from titanium, methyl methacrylate, or polyethylethylketone (PEEK). After the tumor is removed and the reconstruction is completed, the skin incision is closed. Occasionally, the skull tumor will be approached intracranially, if its location favors

that approach. Examples include tumors of the petrous portion of the temporal bone, fibrous dysplasia involving the optic canal, or skull base tumors, such as chordomas and chondrosarcomas.

Usual preop diagnosis

Eosinophilic granuloma; histiocytosis; hemangioma; osteoma; epidermoid; dermoid tumor; metastatic tumors; osteosarcoma; fibrous dysplasia; chordoma; chondrosarcoma; meningioma

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine, lateral, prone, or sitting, based on the location of the tumor
Incision	Dependent on location of tumor
Special instrumentation	Neuro drill, craniotome, image-guided navigation (if intracranial)
Unique considerations	ETT must be taped securely in a location satisfactory to the surgeon. Reinforced or RAE tube may be helpful in certain situations.
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	1–4 h
Closing considerations	Drain often left in epidural space. Surgeon often requests control of BP to avoid hemorrhage into the bed of tumor.
EBL	25–500 mL
Postop care	ICU or close observation unit
Mortality	0–2% (higher for tumors in critical locations)
Morbidity	Usually <5% Infection Neurologic disability CSF leak

Massive blood loss 2° venous sinus injury

Pain score 2–5

■ PATIENT POPULATION CHARACTERISTICS

Age range Infant to 85 yr (usually 20–60 yr)

Male:Female ~1:1

Incidence Unknown

Etiology Neoplastic

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Craniotomy for Tumor. Note, however, that patients with skull tumors rarely have problems with ICP.

Suggested Readings

1. Cheng CH, Chuang HY, Lin HL, et al: Surgical results of cranioplasty using three dimensional printing technology. Clin Neurol Neurosurg 2018; 9(168):118–23.
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3. Verret DJ, Ducic Y, Oxford L, et al: Hydroxyapatite cement in craniofacial reconstruction. Otolaryngol Head Neck Surg 2005; 133(6):897–9.

CRANIOTOMY FOR TRAUMA

■ SURGICAL CONSIDERATIONS

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Description

Head injuries occasionally require emergent surgical procedures to evacuate mass lesions or débride contused or contaminated brain. The majority of these injuries are supratentorial. The surgical procedure depends on the exact type and location of the injury (e.g., epidural, subdural, intraparenchymal hematomas, or depressed skull fracture); however, most traumatic injuries can be addressed through a wide frontotemporoparietal craniotomy. Often the entire head is shaved and placed in a headrest (pins, suction cups, or horseshoe). If the C-spine has not been cleared, the

patient may be placed in a lateral position to minimize neck movement. Timing is often important for these procedures because the patient usually has a component of $\uparrow$ ICP, either from a hematoma or from cerebral edema due to the trauma. Hyperventilation, mannitol, hypertonic saline, and diuresis may be necessary to control the patient's ICP until the patient can be taken into the OR and undergo a wide decompressive hemicraniectomy (frontotemporoparietal craniectomy). For a standard trauma craniotomy, the scalp incision starts anterior to the tragus and continues superiorly in a question-mark type path, ending in the frontal area (Fig. 1.1-13). This incision can be modified according to the location and extent of the injury. The skin is reflected, and the skull is perforated with a cranial drill. The temporal burr hole should be placed first because if the patient's condition deteriorates rapidly, the temporal burr hole can be enlarged quickly to a craniectomy for decompression before continuing with the craniotomy. A chronic subdural hematoma may be drained through burr holes. However, if there is an acute subdural hematoma, epidural hematoma, intraparenchymal hemorrhage, depressed fracture, or penetrating wound, a craniotomy bone flap is elevated. Wide exposure is used to visualize and control sources of bleeding. Gentle warm irrigation may be used to aid in removal of an acute subdural hematoma. Mass lesions and associated contused brain parenchyma are identified and removed (Fig. 1.1-14). For penetrating head injuries or depressed skull fractures, the wound is débrided, foreign bodies are removed, bleeding is controlled, and the dura is repaired if lacerated. It is important to débride the wound as much as possible, but it is often not possible to safely remove all fragments from the wound. Depending on the injury and the presence of brain swelling, it may be necessary to close the dura with autologous pericranium or fascia lata. If these are not available, a synthetic dural substitute or bovine pericardium can be used. Alternatively, the dura may be left open or multiple relaxing incisions may be made in the dura to allow for brain swelling. The surgeon can often repair and reconstruct a depressed skull fracture after removing a bone flap that surrounds the fracture. In addition, the bone flap may be left out to compensate for brain swelling and replaced later. A subgaleal drain may be inserted and brought out through a separate incision. An ICP monitor or a ventriculostomy may be placed at the end of the procedure.

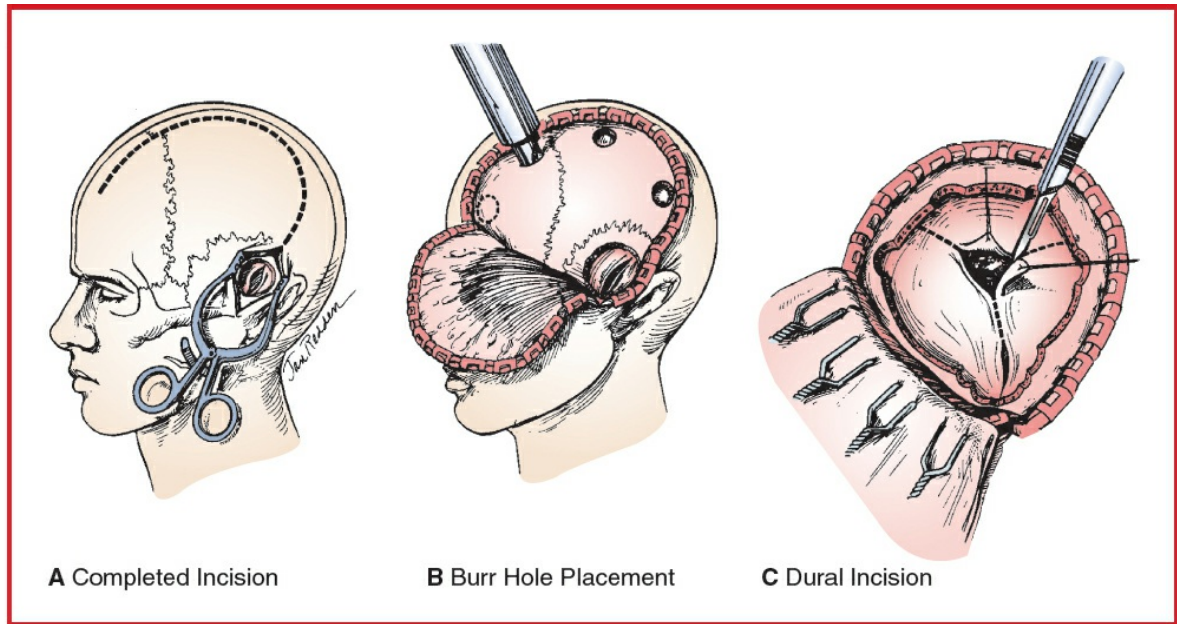


Figure 1.1-13 **A:** In an acute subdural hematoma: burr hole is made, followed by craniotomy, and the incision is extended upward to form a large question mark, the medial extent of which follows the midline. **B:** Additional burr holes are made, with the medial ones 1.5 cm off the midline to avoid injury to the major venous structures and granulations. The anterior burr hole is placed above the frontal sinus (the size of which can be estimated from preop radiographs). **C:** The dura can be opened with a Y- or X-shaped incision, with a flap being based on the superior sagittal sinus. (Reproduced with permission from Grossman RG, Loftus CM Principles of Neurosurgery, 2nd edition. Lippincott-Raven, Philadelphia: 1999.)

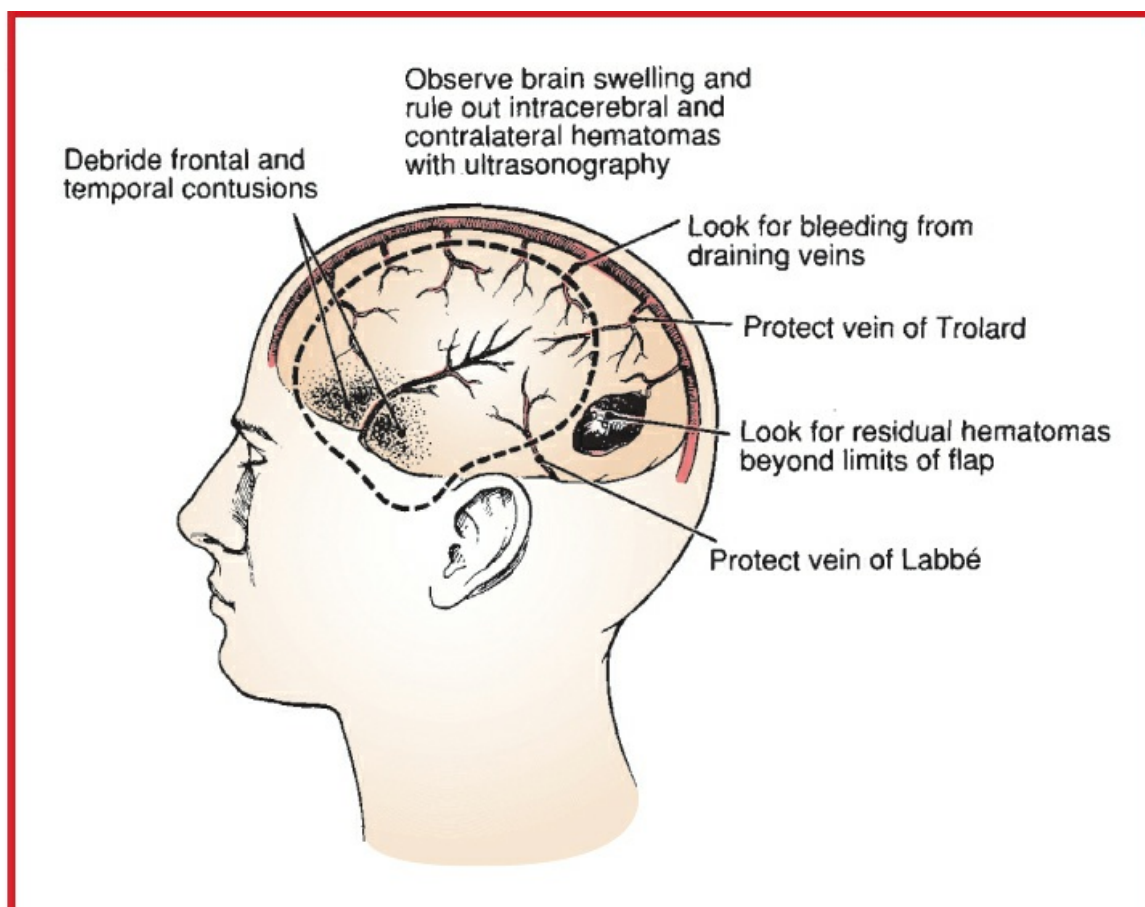


Figure 1.1-14 Precautions to be taken during a craniotomy for subdural hematoma. (Reproduced with permission from Grossman RG, Loftus CM Principles of Neurosurgery, 2nd edition. Lippincott-Raven, Philadelphia: 1999.)

Frontal sinus fracture

If a skull fracture involves the frontal sinus and/or there is a CSF leak, the sinus may need to be obliterated or cranialized. Obliteration involves removal of the mucosa of the sinus and filling the sinus with antibiotic-soaked Gelfoam® sponges. Cranialization of the frontal sinus involves removing the posterior bony wall of the sinus and removing the mucosa. If there is a CSF leak, the dura will need to be repaired.

Variant indication

Some patients, with a large ischemic stroke, will develop malignant cerebral edema, necessitating a decompressive hemicraniectomy (frontotemporoparietal craniectomy) to control rising ICPs and prevent herniation.

Usual preop diagnosis

Epidural hematoma; acute subdural hematoma; intracerebral hematoma; depressed skull fracture; cerebral contusion; gunshot wound of the brain; malignant cerebral edema

ERAS: These procedures are usually urgent/emergent and present little opportunity for ERAS considerations.

■ SUMMARY OF PROCEDURE

Position	Supine, lateral, prone, or sitting, depending on site of injury
Incision	Varies with location of injury (typical incision shown in Fig. 1.1-13)
Unique considerations	The patient may have ↑↑ ICP. Because incipient herniation and/or associated injuries may be a concern, timing is critical. C-spine precautions may be necessary. Patients may also be on anticoagulation medications, necessitating transfusions of blood products or recombinant activated factor VII.
Antibiotics	Cefazolin 2 g iv (within 30–60 min before incision)
Surgical time	1.5–4 h
Closing considerations	Application of head dressing may jostle ETT at end of case → ↑ BP. Patient may stay intubated postop. ICP monitor or external ventricular drain may be placed. Antiepileptic medications may be given for prophylaxis, especially if the patient had a seizure.
EBL	25–500 mL
Postop care	ICU or close observation unit until stable. Fluid and electrolytes require frequent monitoring, as the patient may develop SIADH. BP may need to be controlled with vasodilator and β-blocker infusions.
Mortality	10–50%, depending on lesion; higher for acute subdural hematomas, lower for epidural hematomas.
Morbidity	Morbidity depends on lesion; morbidity from craniotomy itself is low. Infection Neurologic disability Nerve injury CSF leak Endocrine disorders: SIADH, CSW, panhypopituitarism, DI

	Massive blood loss: venous sinus injury
Pain score	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Infant to 85 yr (usually 15–40 yr); peaks at 0–4 yr and 15–19 yr
Male:Female	2:1
Incidence	Relatively common: >1.7 million TBI/yr in the United States (mostly concussion)
Etiology	Trauma
Associated conditions	Abdominal injuries; C-spine fractures

■ ANESTHETIC CONSIDERATIONS

■ PREOPERATIVE

Traumatic brain injury (TBI) is the leading cause of death of persons <24 yr old. A penetrating injury of the skull usually will cause major damage to the brain as a result of both focal and diffuse neuronal injury with hemorrhage into brain tissue. Surgery is necessary to evacuate hematomas, control intracranial bleeding, débride the wound, and remove bone fragments, foreign material, and damaged brain so that the cranial vault can better accommodate the brain swelling that inevitably occurs. Epidural hematomas form between the skull and dura and are usually due to bleeding from an artery (e.g., middle meningeal). Hence, time is of the essence, and rapid evacuation and control of the bleeding is essential if permanent neurologic injury is to be avoided. Subdural bleeding occurs between the dura and the leptomeninges lining the brain surface. This bleeding is usually venous in origin and usually occurs more gradually. Focal intracranial hemorrhages may be either arterial or venous, and, as with subdural hematomas, must be evacuated if they are enlarging.

Respiratory	Ensure a secure airway and adequate ventilation. Localized injuries to the frontal or parietal lobes may not cause any respiratory changes. If ↑ ICP, respirations may become phasic (Cheyne-Stokes) or slow (<10 min) and deep resulting in substantial
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hypocapnia. Many patients with head injuries demonstrate partial airway obstruction from the tongue falling back into the posterior pharyngeal space. If this occurs, or if the patient is comatose and unable to protect the airway and prevent aspiration of gastric contents, immediate tracheal intubation should be performed. Head injuries in the region of the occipital lobes may → hypoventilation or apnea. Neurogenic pulmonary edema may occur in up to 20% of TBI patients.

Tests: As indicated from H&P and as time allows

Cardiovascular

Most patients with head injuries evidence ↑ BP and ↑ HR. Even transient hypotension (SBP < 90 mm Hg) → ↑ morbidity and mortality. If ICP increases sufficiently to cause herniation of brain stem, patients develop the “Cushing triad” of ↑ BP, ↓ HR, ± irregular respiration. These changes resolve when ↑ ICP is relieved, so vigorous attempts to regulate BP prior to craniotomy are not warranted. However, the patient should be taken to OR as quickly as possible.

Tests: As indicated from H&P and as time allows

Neurologic

Neurologic evaluation of the head-injured patient is based on the Glasgow coma scale ([Table 1.1-4](#)). The scale involves evaluation of three functions: eye opening, verbal response, and motor response. Using this scoring system, the severity of brain injury may be classified as mild (13–15 points), moderate (9–12 points), or severe (8 points or less). By definition, any patient having 8 points or less is in a coma. Additional useful neurologic examinations include assessment of pupillary size and reactivity to the light, reflex responses, and evidence of asymmetry or flaccidity of the extremities or decerebrate (e.g., rigid arm extension) or decorticate (e.g., rigid arm flexion) posturing. Head-injured patients whose neurologic function is deteriorating rapidly, and in whom an epidural or subdural hemorrhage is suspected, should be taken to OR immediately.

	Hyperventilation is appropriate if herniation is eminent. Tests: CT scan
Hematologic	Severe head injury may be associated with a progressively worsening coagulopathy, resulting in a clinical picture similar to that of DIC. The reason for this is not known, but the brain is rich in thromboplastin and other coagulation factors. Tests: Hct; PT; PTT; others as indicated from H&P
Associated injuries	Multiple organ system injury must be considered. ATLS assessment if time permits. Evidence of intrathoracic or intraperitoneal hemorrhage should be sought and repaired first if the patient is hemodynamically unstable. Tests: May include MRI, CT, ultrasound, and others as indicated.
Laboratory	Other tests as indicated from H&P, and as time permits
Premedication	Usually none. Steroids are not recommended to ↓ICP in TBI.

Table 1.1-4 Glasgow Coma Scale (GCS)	
Category	Score
I. Eyes open:	
Never	1
To pain	2
To verbal stimuli	3
Spontaneously	4
II. Best verbal response:	
None	1
Incomprehensible sounds	2
Inappropriate words	3

Patient disoriented and converses	4
Patient oriented and converses	5
III. Best motor response:	
None	1
Extension (decerebrate rigidity)	2
Flexion abnormal (decorticate rigidity)	3
Flexion withdrawal	4
Patient localizes pain	5
Patient obeys	6
	I + II + III Total = 3–15

INTRAOPERATIVE

Anesthetic technique: GETA

Induction

↑ ICP is likely in most patients with head injury requiring operation, and induction of anesthesia is best accomplished with drugs that ↓ ICP. A mannitol infusion (0.25–1 mg/kg iv over 10–20 min) may be necessary to control ICP. If the patient is hemodynamically stable and not hypovolemic, titrated induction with propofol (1–3 mg/kg) + fentanyl (2–5 mcg/kg) is satisfactory. If hemodynamically unstable, etomidate (0.1–0.4 mg/kg) is suitable for induction. Ketamine is not used because of its ability to ↑ ICP. If the patient is comatose, anesthetic requirement is less, needing only O₂ and muscle relaxant, or low-dose isoflurane (0.5%) or sevoflurane (1%). A nondepolarizing muscle relaxant (vecuronium [0.1 mg/kg] or rocuronium [1 mg/kg]) is administered for ET intubation. Succinylcholine may ↑ ICP transiently but is still appropriate for emergency airway management. A defasciculating dose of a nondepolarizing NMB may attenuate the ICP ↑. Nasotracheal intubation is not recommended for patients

with the possibility of maxillary and/or basilar skull fractures because of the potential for inserting the tube through the fracture site into the brain. To minimize $\uparrow\uparrow$ BP and $\uparrow\uparrow$ ICP with ET intubation, consider bolus remifentanyl (2–5 mcg/kg iv) 1–2 min before laryngoscopy. In hypovolemic patients, hydration with a mixture of crystalloid (e.g., NS) and colloid (e.g., 5% albumin) should be initiated prior to induction.

Maintenance

The ideal drug for maintenance of anesthesia decreases ICP and CMRO_2 , maintains cerebral autoregulation, redistributes flow to potentially ischemic areas, and provides protection for the brain from focal ischemia. Unfortunately, there is no ideal anesthetic.

Both isoflurane and sevoflurane are good choices for patients undergoing neurosurgical procedures. Although these agents cause dose-dependent increases in CBF and volume, and hence $\uparrow$ ICP, these effects tend to be mitigated by the prior administration of propofol, by hyperventilation, and by limiting the inspired concentrations to <1 MAC. At these concentrations, CBF responses to changes in PaCO_2 are maintained, and cerebral autoregulation remains intact.

N_2O is best avoided in patients with significantly elevated ICP and in patients with potential for pneumothorax or pneumocephalus. A propofol infusion is a reasonable alternative, although its use has been associated with a $\downarrow$ CBF $>$ $\downarrow$ $\text{CMRO}_2 \rightarrow \uparrow$ potential for ischemia. Jugular bulb O_2 saturation may be reduced during propofol anesthesia, consistent with an unfavorable supply-demand balance.

BP goals: Keep SBP $>$ 100 mm Hg (50–69 yr old), ≥ 110 mm Hg (15–49 and >70 yr old), and CPP ~ 70 mm Hg. Avoid prophylactic hyperventilation. Mild hypothermia may be beneficial if maintained >48 h. DVT prophylaxis is recommended. High-dose steroid administration is associated with worsened outcomes. Muscle relaxation can be maintained as necessary.

Emergence	Because recovery from head injury is so unpredictable, it is generally advisable to leave the ETT in place and maintain controlled ventilation until there is sufficient clinical evidence that normal neurologic recovery is occurring. Postop sedation can be accomplished using dexmedetomidine (0.2–0.7 mcg/kg/h) or propofol (20–75 mcg/kg/min) not to exceed 5 mg/kg/h to minimize the occurrence of propofol-related infusion syndrome (including metabolic acidosis, $\uparrow$ K⁺, rhabdomyolysis).	
Blood and fluid requirements	Possible severe blood loss IV: 14–16 ga $\times$ 1–2 BSS @ 1–3 mL/kg/h $\pm$ Albumin 5% $\pm$ RBCs for Hct < 27 Coagulopathy is common	Blood transfusion to keep Hct $\sim$30. Minimize postop cerebral edema by limiting crystalloid volume to <3 mL/kg/h + replacement of UO. Glucose-containing solutions should be avoided; blood glucose levels should be maintained between 80 and 180 mg/dL. If volume is needed, albumin 5% should be administered. Consider Rx with tranexamic acid (load = 1 g over 10 min followed by 1 g over 8 h infusion) for persistent bleeding. Initiate within 3 h of injury.
Control of blood pressure	Keep CPP $\sim$70 mm Hg Keep SBP > 100–110 mm Hg Maintain normovolemia Vasopressors	Controlled hypotension generally is not used unless bleeding becomes profuse and difficult to control. $\downarrow$ BP is better treated initially with volume replacement than vasopressors. Target SBP goals are age dependent. Phenylephrine or norepinephrine infusions can be used to maintain CPP.

Monitoring	Standard monitors (p. B-1). Avoid hypothermia Arterial line ± CVP line UO	If the injury is extensive or unknown, or if the patient is unstable, invasive monitoring is mandatory. Monitor for VAE if the head is elevated above the heart. Level the arterial line at the head. EEG, EP, and SjO ₂ monitoring may be desirable.
Positioning	✓ and pad pressure points ✓ eyes	For occipital or posterior fossa lesions, patient may be prone or sitting (see Anesthetic Considerations for Cervical Neurosurgical Procedures, Positioning). Otherwise, patient will be supine with head in Mayfield skeletal fixation and turned to the side, with a roll placed under the shoulder on the operative side.
Control of ICP	Adequate anesthesia Head up 20–30° Keep SBP ≥ 90 mm Hg Propofol infusion Hyperventilation to	Transient increases in ICP—even up to 50–60 mm Hg—are generally tolerated, provided they are promptly terminated. Sustained increases in ICP > 25–30 mm Hg are associated with severe neurologic injury and poor outcome. Patients with skull fractures or intracranial bleeding may be on the steep portion of intracranial compliance curve such that any increase in intracranial volume may cause ↑↑ ICP. ↓ CO ₂ → cerebral

$\text{PaCO}_2 = 25\text{--}30 \text{ mm Hg}$
 $\text{PaO}_2 > 100 \text{ mm Hg}$

vasoconstriction $\rightarrow \downarrow \text{CBV}$, $\downarrow \text{CBF}$, and $\downarrow \text{ICP}$. Patients with diffuse brain injury may have lost cerebrovascular sensitivity to PaCO_2 . Despite maintaining adequate ventilation, oxygenation, and BP, patients with diffuse head injury often exhibit arterial and CSF lactic acidosis, a further indication of the metabolic derangement that exists in the brain from the injury.

Keep CPP $\sim 70 \text{ mm Hg}$

Control MAP and cerebral venous pressure so that CPP = $\sim 70 \text{ mm Hg}$. In severe diffuse head injury with loss of autoregulation, some parts of the brain may exhibit "luxury perfusion," while other areas exhibit severe ischemia.

Mannitol $0.25\text{--}1.0 \text{ g/kg}$

Vigorous diuresis will commence in 20–30 min (if blood volume and renal perfusion are adequate), and brain shrinkage will follow. Max serum osmolarity = 320 mOsm/L . If mannitol is administered too rapidly, profound refractory hypotension will occur.

Furosemide $10\text{--}20 \text{ mg}$

Simultaneous administration of furosemide ($10\text{--}20 \text{ mg}$) is recommended to avoid the transient $\uparrow \text{CBV}$ and $\uparrow \text{ICP}$

Hypertonic saline

following mannitol.
Hypertonic saline (e.g., 3%
@: 2–3 mL/kg) may provide
better control of ICP than an
equiosmolar dose of
mannitol.

POSTOPERATIVE

Complications

Seizures

Sz Rx: Levetiracetam (1 g) or
phenytoin (1-g loading dose).
NB: Incompatible with
dextrose-containing solutions.

Neurologic deficits
Hemorrhage

Some patients with severe
head injury remain
unconscious for weeks or
months, without evidencing
any substantial neurologic
recovery. A late complication
of head injury is
hydrocephalus requiring a
shunt procedure.

SIADH/CSW
DI

These causes of ↓ Na⁺ must
be differentiated for proper
Rx.

↑ UO + ↓ Urine Osm + ↑
serum Na

NPE
↑ ICP

Neurogenic pulmonary edema
may result from massive
sympathetic discharge 2° ↑↑
ICP. Rx: ↓ ICP, alpha-blockers
to control BP, respiratory
support.

DIC

TBI is a risk factor for DIC.

Multimodal pain management

Propofol or
dexmedetomidine
infusion

Most patients will remain
intubated and sedated
postop.

Tests	CT scan ICP monitor	Unless neurologic recovery is rapid, periodic CT scans are obtained postop to follow the intracranial changes. In addition, in many institutions, a device for monitoring ICP postop is placed at the time of operation.
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Suggested Readings

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MICROVASCULAR DECOMPRESSION OF CRANIAL NERVE

SURGICAL CONSIDERATIONS

Lawrence M. Shuer, Steven D. Chang, and Navjot Chaudhary

Description

Microvascular decompression (MVD) is used to treat various disorders of the cranial nerves, including trigeminal neuralgia (TN), hemifacial spasm and, more rarely, glossopharyngeal neuralgia (GN). Trigeminal neuralgia is characterized by brief episodes of intense, stabbing facial pain along the distribution of the trigeminal nerve. This pain usually can be elicited by gentle stimulation of the affected area. Trigeminal neuralgia is further subdivided into type 1 (episodic pain) and type 2 (continuous pain) variants. Continuous pain is a poor prognostic factor when treating trigeminal neuralgia with any modality including surgery. Hemifacial spasm is characterized by paroxysmal repetitive twitching of the facial muscles. The twitching usually starts with the muscles around the eye and can progress to involve the rest of the facial muscles. Glossopharyngeal neuralgia is characterized by paroxysmal pain that involves the ear and throat. Typically, the pain is described as “stabbing” and radiates from one site to the other. Swallowing, drinking cold beverages, talking, or coughing can elicit the pain. A Chiari malformation may mimic the symptoms of glossopharyngeal neuralgia. Combinations of vascular compression syndromes involving multiple nerves and multiple offending vessels have also been reported.

All of these conditions are usually unilateral and are often caused by compression of a cranial nerve by a vascular structure. In trigeminal neuralgia, the superior cerebellar artery is the usual culprit. In hemifacial spasm, the anterior inferior cerebellar or vertebral artery is usually compressing the facial nerve. In glossopharyngeal neuralgia, the posterior inferior cerebellar or vertebral artery is usually the offending artery. Large veins can also compress the cranial nerves. The goal of an MVD is to remove the pressure on the cranial nerve. To perform an MVD, a linear incision is made behind the ear on the affected side ([Fig. 1.1-15A](#)). Dissection is taken down to the skull, and several burr holes are made. The burr holes can be enlarged into a **craniectomy** ([Fig. 1.1-15B](#)) or used for elevation of a bone flap. The craniectomy is placed below the transverse sinus and medial to the sigmoid sinus to allow access to the cerebellopontine angle. The mastoid air cells are sealed with bone wax to prevent CSF leak. Any venous sinus bleeding is controlled, and the dura is opened ([Fig. 1.1-15C](#)). With brain relaxation, the cerebellum is retracted. The operating microscope allows the surgeon to explore the involved cranial nerve. If an offending vessel is identified, it is carefully dissected away from the nerve and shredded Teflon felt or a small plastic sponge is placed to keep the vessel away from the cranial nerve. In the case of trigeminal or

glossopharyngeal neuralgia, if no offending vessel is identified, a **partial section of the nerve** may be performed. Partial section of the 9th or 10th cranial nerve may cause some vasomotor instability. BAERs, facial nerve monitoring, and EMGs can be monitored intraop to protect the cranial nerves. Once the cranial nerve has been decompressed, the dura is closed, and the wound is closed in layers. The main variations in this procedure are in patient positioning and surgeon's preference of monitoring modalities. Patient positioning may be lateral, prone, supine, or rarely sitting. Intraop, mannitol (0.5–1 g/kg) and/or a lumbar drain (for CSF removal) may be needed for brain relaxation. Success rate is typically 75–95%, with a recurrence rate as high as 3.5% per year.

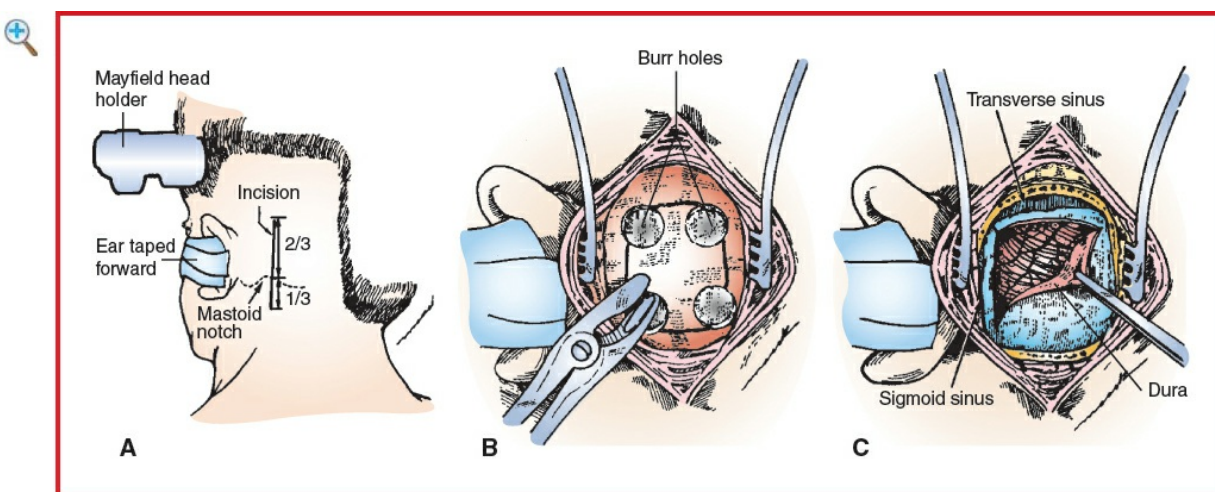


Figure 1.1-15 Microvascular decompression. **A:** Location of incision. **B:** Enlargement of burr holes into craniectomy with a rongeur. **C:** Dural opening on the left side. Note position of venous sinuses. (Reproduced with permission from Wilson CB: *Neurosurgical Procedures: Personal Approaches to Classic Operations*. Williams & Wilkins, Baltimore: 1992.)

Alternative procedure

Endoscopic-assisted or fully endoscopic approaches to MVD offer the advantage of a panoramic view of the surgical field using a small craniotomy and with less brain retraction. Disadvantages include the need for greater care during exposure and during insertion or withdrawal of instruments to avoid collateral damage. Stereotactic radiosurgery is less effective than microvascular decompression but may be appropriate for patients unwilling or unable to undergo surgery. Other, less frequently used options include radiofrequency rhizotomy, glycerol injection, and balloon compression.

Usual preop diagnosis

Trigeminal neuralgia; tic douloureux; hemifacial spasm; tinnitus; glossopharyngeal neuralgia

ERAS: Preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and other meds, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Normally, lateral rotation with shoulder roll, head elevated; upper shoulder retracted caudally
Incision	Retroauricular (mastoid) (see Fig. 1.1-15A)
Special instrumentation	Operating microscope; cranial perforator; ± facial nerve monitoring; ± EMG; ± BAER
Unique considerations	Risk of air embolus in head-elevated positions. Precordial Doppler or TEE required for head elevation $\geq 30^\circ$
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	2–3 h
EBL	25–250 mL
Postop care	ICU or close observation unit. Observe for change in neurologic status (e.g., level of alertness, response to commands), usually for 12–24 h
Mortality	~1%
Morbidity	Facial sensory deficit: 25% (often transient) Aseptic meningitis: $\leq 20\%$, usually occurring 3–7 d postop Hearing loss: 3% Diplopia (usually transient) Facial weakness: 2% Deafness: 1% Infection CSF leak Massive blood loss due to vertebral artery injury or venous sinus laceration
Pain score	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	40–85 yr (usually 60–70 yr)
Male:Female	1:2
Incidence	TN: 4.3/100,000/yr; GN: 25.6/100,000/yr at age >70 yr; prevalence: 16/100,000
Etiology	Vascular compression of cranial nerve demyelination; multiple sclerosis plaque; cerebellopontine angle tumor
Associated conditions	HTN; multiple sclerosis; rheumatoid arthritis; Charcot-Marie-Tooth disorder; Sjögren syndrome

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Microvascular decompression involves a full craniotomy for decompression of a cranial nerve that is causing facial pain and/or spasm of facial muscles. Generally, these patients have trigeminal neuralgia or tic douloureux that has not been responsive to medical management (e.g., carbamazepine [Tegretol] therapy), stereotactic radiosurgery (Gamma Knife), and percutaneous rhizotomy or glycerol injection has failed.

Respiratory	None unless the patient has a long-standing Hx of smoking and has COPD.
Cardiovascular	Many patients will have Hx of idiopathic HTN requiring antihypertensive medications. Good preop control of BP is important because it will make intraop and postop management of BP easier. Anticipate exaggerated hemodynamic response to anesthetic medications.
Neurologic	The presenting symptom is usually pain ± muscle spasm in the maxillary and/or mandibular division of the trigeminal nerve, unaccompanied by any motor or sensory deficits. Identify trigger points so that they may be avoided during face mask placement and induction.
Laboratory	None, except for routine preop studies.
Premedication	Generally, patients for these procedures are elderly and do not require any special premedication.

	Midazolam 1–3 mg iv will provide amnesia for the preop events, if that is desired by patient or surgeons.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water for these patients.

INTRAOPERATIVE

Anesthetic technique

GA is necessary because a full craniotomy is performed. ICP is not increased in these patients. Brain volume reduction (typically using mannitol 0.25–0.5 g/kg), however, is important to provide the surgeon with sufficient space to identify and relieve the pressure on the offending nerve without requiring excessive brain retraction in the process.

Induction	Induction is best accomplished with propofol (1–2 mg/kg), followed by neuromuscular blockade and ET intubation. Facemask placement may trigger intense pain in susceptible patients. Consider O ₂ insufflation for preoxygenation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Dexamethasone (4–8 mg iv) may ↓PONV. Avoid long-acting opioids or scopolamine patch, which may interfere with postop neuro examination. Sequential pneumatic compression devices should be used to ↓VTE risk.
Maintenance	Maintain anesthesia with 0.5 MAC sevoflurane or isoflurane with 50 mcg/kg/min propofol. BP is maintained in the normal range during the operation. Hyperventilation to achieve a PaCO ₂ of 25–30 mm Hg is helpful to ↓ brain volume and ↑ workspace. Once the nerve has been isolated, stop hyperventilation. A spinal drain may be used to remove CSF during the operation and improve exposure. The drain usually is opened at

	the time of dural opening and closed as soon as surgery on the nerve is complete.	
Emergence	Reverse any residual neuromuscular blockade. The ETT is removed following return of protective airway reflexes. Postop HTN may need to be controlled with esmolol and/or clevidipine by continuous pump infusion. Prophylactic antiemetic (e.g., metoclopramide 20 mg and ondansetron 4–8 mg) should be given 30 min before extubation. The incidence of PONV is high following this procedure. Consider acetaminophen (1 g iv) and/or local anesthetic scalp infiltration for postop analgesia. Consider iv acetaminophen if po dose given >4–6 h ago.	
Blood and fluid requirements	Large blood loss is rare IV: 18 ga × 1 BSS @ 1–3 mL/kg/h	Mannitol 0.5–1 mg/kg is sometimes necessary to provide sufficient brain volume reduction to permit adequate surgical exposure. Zero balance or goal-directed fluid management is recommended (see Appendix K, p. K-2).
Monitoring	Standard monitors Arterial line ± CVP line ± EMG and/or SSEP	Invasive monitoring will facilitate intraop and postop BP management. Sometimes EMG and/or SSEP monitoring of the facial nerve is performed.
Positioning	Lateral, rotation with shoulder roll ± Pillow between legs ✓ and pad pressure points ✓ eyes	Patients usually will be positioned with their head in a “far lateral” position. Padding of the shoulder and placing a pillow between the legs may be necessary. A beanbag can be used to hold patient stable in the lateral (“park bench”) position.

POSTOPERATIVE

Complications	PONV	PONV is common and usually requires multimodal treatment including aprepitant.
	Bleeding Brain edema	Postop recovery is usually uneventful. On rare occasion, significant brain edema or bleeding may be experienced.
Multimodal pain management	Acetaminophen 1 g iv Fentanyl 25–50 mcg iv prn Reinforce scalp blocks	If not given preop or if 4–6 h have elapsed since previous dose. 0.5% bupivacaine with epi is as effective as liposomal bupivacaine.
Tests	CT scan	If neurologic recovery is delayed.
ERAS	Ondansetron 4 mg iv ± Metoclopramide 10–20 mg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Ashkan K, Marsh H: Microvascular decompression for trigeminal neuralgia in the elderly: a review of the safety and efficacy. *Neurosurgery* 2004; 55(4):840–8; discussion 848–50.
2. Fariselli L, Ghielmetti F, Bosetti D, et al: The role of radiosurgery in trigeminal neuralgia. *Neurol Sci* 2017; 38(Suppl 1):63–5.
3. Meng L, Quinlan JJ: Assessing risk factors for postoperative nausea and vomiting: a retrospective study in patients undergoing retromastoid craniectomy with microvascular decompression of cranial nerves. *J Neurosurg Anesthesiol* 2006; 18(4):235–9.
4. Minahan RE, Mandir AS: Neurophysiologic intraoperative monitoring of trigeminal and facial nerves. *J Clin Neurophysiol* 2011; 28(6):551–65. Review.
5. Oesman C, Mooji JJ: Long-term follow-up of microvascular decompression for trigeminal neuralgia. *Skull Base* 2011; 21(5):313–22.
6. Piazza M, Lee JY: Endoscopic and microscopic microvascular decompression

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8. Sheehan J, Pan HC, Stroila M, et al: Gamma Knife surgery for trigeminal neuralgia: outcomes and prognostic factors. J Neurosurg 2005; 102(3):434–41.

BIFRONTAL CRANIOTOMY FOR CSF LEAK

SURGICAL CONSIDERATIONS

Lawrence M. Shuer, Steven D. Chang, and Navjot Chaudhary

Description

Trauma (44% accidental; 29% surgical) is the leading cause of CSF leaks. CSF leaks can also be caused by tumors (22%), congenital malformations, or as a result of transsphenoidal surgery. Most CSF leaks involve the floor of the anterior cranial fossa (e.g., cribriform plate: 35%), with drainage through the auditory canal (otorrhea), nasopharynx (rhinorrhea), sinus air cells, or other less common routes. CSF leaks can occasionally stop on their own or by management with a lumbar drain. However, to lessen the risk of meningitis, surgical repair is the usual treatment.

The surgical repair of CSF leaks has changed with the advent of the endoscope. The success rates of endoscopic closure are similar to open surgical approaches while avoiding the morbidity associated with craniotomies. Although it is common to attempt closure of a CSF leak endoscopically before proceeding on to an open repair, selected patients may benefit by proceeding directly to an open surgical approach.

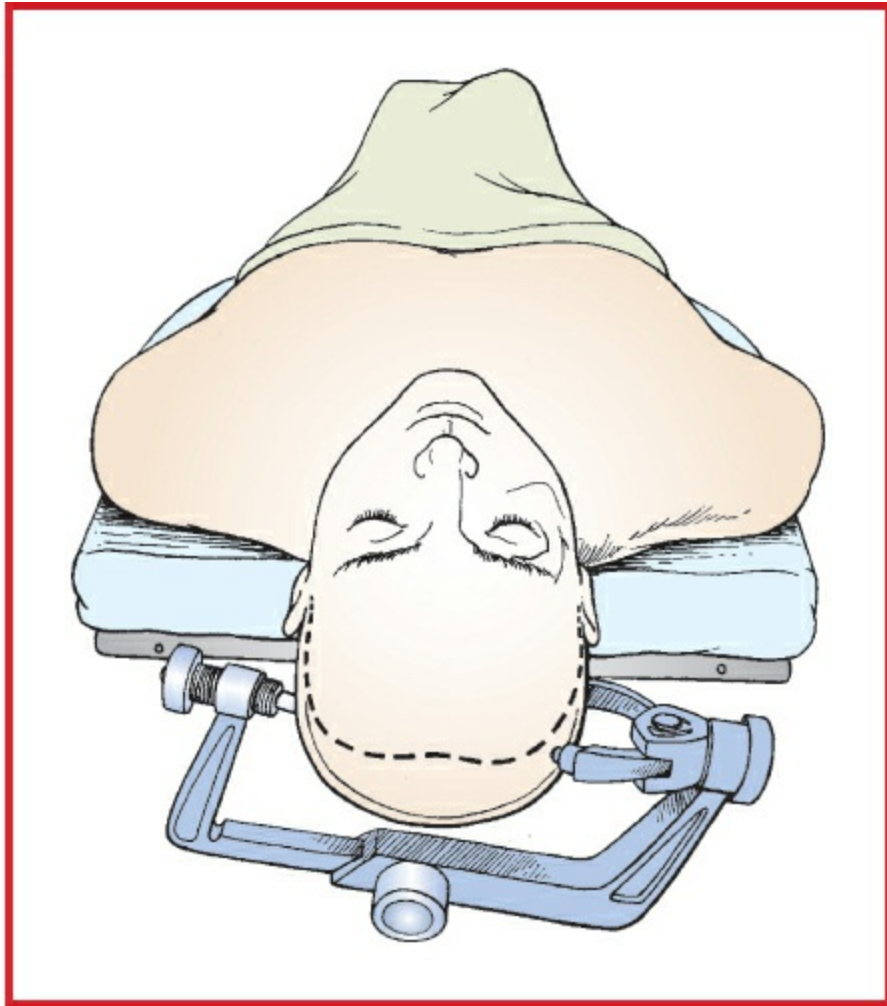


Figure 1.1-16 Patient positioning for bifrontal craniotomy with bicoronal skin incision outlined. This position allows the frontal lobes to separate from the anterior cranial fossa with minimal retraction on the frontal lobes. (Reproduced with permission from Tindall GT, Cooper PR, Barrow DL *The Practice of Neurosurgery*. Williams & Wilkins, Baltimore: 1996.)

An open surgical repair is typically done through a bifrontal craniotomy. The patient is placed in a supine position, and the head is stabilized with a Mayfield clamp. A bicoronal skin incision ([Fig. 1.1-16](#)) is made, and the pericranium is often turned as a separate flap and kept moist during the case. Next, burr holes are drilled and a bifrontal free bone flap is elevated. Care is taken to avoid injury to the underlying superior sagittal sinus. If the frontal sinuses are entered, the mucosa is stripped and the sinus is packed with antibiotic-soaked Gelfoam. The dura is opened over the inferomedial frontal lobes, and the sagittal sinus is ligated. An intradural exploration is undertaken to determine the site of the leak. Identification of the leak site is essential for successful outcome. The source of the leak may be identified using radioactive tracer or CT metrizamide studies. Brain relaxation is usually necessary to reduce the need for mechanical retraction and may require mannitol, hyperventilation, and/or the placement of a lumbar drain. The frontal lobes are elevated, and the olfactory tracts may be

sacrificed. If the site of the leak can be determined, the dural repair can be performed using fascia lata, pericranium, bovine pericardium, or a synthetic dural substitute. It usually is necessary to strip the dura off the anterior cranial fossa to complete the repair. Defects in the bone can be plugged with a variety of materials (e.g., pericranial flap, fat, temporalis muscle flap, bone, or wax). After the repair is complete, the dura can be closed and the bone flap replaced, often with a drain left in the epidural space. In addition, tissue sealants such as fibrin glue can be sprayed over the dural suture line to help create a watertight seal. At this point, brain relaxation is no longer required. The wound is closed with stitches, and skin is closed with staples.

A combined approach may be used for CSF leaks 2° to a tumor that invades the dura and the cribriform plate. In these cases, the procedure often is performed with an otorhinolaryngologist. The intracranial part of this procedure is identical to the bifrontal craniotomy, except that bone and tumor are removed at the floor of the anterior cranial fossa. The extracranial part of this procedure is performed by the otorhinolaryngologist, who approaches the sinonasal region endoscopically or through a lateral rhinotomy or degloving gingival incision. The extracranial approach is chosen to complement the cranial exposure, and a common space is created between the two operative fields. Closure involves isolating the two operative fields once again. The dural repair is as above. The mucosa of the nasal cavity is replaced with the use of a skin graft.

Alternate procedure

Transnasal endoscopic repair of selected CSF leaks has been shown to be effective with significantly ↓ morbidity compared with craniotomy.

Usual preop diagnosis

CSF leak or rhinorrhea; fracture of the anterior cranial fossa; intracranial encephalocele; cribriform plate tumor; olfactory neuroblastoma (tumor of the olfactory epithelium). β -2 transferrin is a relatively sensitive and specific marker for CSF and can be used to establish the diagnosis

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and other meds, limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, postop pain management options, possibility of urinary drain postop, and description of antimicrobial prophylaxis and skin preparation. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position

Supine

Incision	Bicoronal (Fig. 1.1-16); transnasal for endoscopic
Special instrumentation	Operating microscope; endoscope; image guidance system
Unique considerations	Brain relaxation desired; lumbar subarachnoid catheter; fascia lata graft
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	2–3.5 h, depending on extent of leak or lesion
Closing considerations	Drain often left in epidural space. Anticipate that the application of a full head dressing will jostle patient and ETT → ↑ BP + coughing/bucking.
EBL	75–500 mL
Postop care	ICU or close observation unit; Valsalva maneuver, coughing and vomiting can cause ↑↑↑ ICP → recurring CSF leak.
Mortality	≤5%
Morbidity	Usually <5% for all complications Infection (ascending meningitis) Recurrent CSF leak Neurologic disability Nerve injury Massive blood loss: venous sinus injury
Pain score	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–65 yr
Male:Female	~3:2
Incidence	Relatively uncommon neurosurgical procedure
Etiology	Traumatic (73%); congenital; neoplastic

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Transsphenoidal Resection of Pituitary

Tumor.

NB

Positive pressure mask ventilation in patient with an anterior CSF leak may produce severe pneumocephalus.

Suggested Readings

1. Abuabara A: Cerebrospinal fluid rhinorrhea: diagnosis and management. *Med Oral Patol Oral Cir Bucal* 2007; 12(5):E397–400.
2. Martin TJ, Loehrl TA: Endoscopic CSF leak repair. *Curr Opin Otolaryngol Head Neck Surg* 2007; 15(1):35–9.
3. Psaltis AJ, Schlosser RJ, Banks CA, et al: A systematic review of the endoscopic repair of cerebrospinal fluid leaks. *Otolaryngol Head Neck Surg* 2012; 147(2):196–203.
4. Sharma SD, Kumar G, Bal J, et al: Endoscopic repair of cerebrospinal fluid rhinorrhoea. *Eur Ann Otorhinolaryngol Head Neck Dis* 2016; 133(3):187–90.

TRANSORAL OR TRANSNASAL APPROACH TO THE CERVICOMEDULLARY JUNCTION AND ODONTOID

SURGICAL CONSIDERATIONS

Lawrence M. Shuer, Steven D. Chang, and Navjot Chaudhary

Description

The transoral or transnasal approaches provide excellent access to the odontoid process of the C2 vertebral body, as well as the skull base just anterior to the brain stem ([Fig. 1.1-17](#)). This is important for conditions where there is pressure on the ventral brain stem or spinal cord → Sx: HA, impaired ambulation, hyperreflexia, paresthesias, neurogenic bladder, etc. Common indications for this approach include basilar impression (upward translocation of the odontoid process; [Fig. 1.1-18A](#)), degeneration of the odontoid due to rheumatoid disease ([Fig. 1.1-19](#)), fracture of the odontoid, and resection of extradural tumors, such as chordomas and metastases. In these cases, the operation is performed through the oral cavity with an incision in the posterior wall of the pharynx. Special retractors hold the mouth open and keep the tongue out of the way. Fluoroscopic or image guidance helps the surgeon maintain optimal trajectory. Intraop monitoring of evoked potentials (i.e., SSEP or BAER) and use of the operative microscope helps avoid injury to the brain stem and spinal cord. The dissection is

carried down to expose the anterior arch of C1 and body of C2. To decompress the region, the anterior arch of C1 and the odontoid process are removed. For basilar impression, a portion of the clivus may need to be removed to gain access to the dens (Fig. 1.1-18B). For rheumatoid arthritis, the pannus (thickened fibrous tissue) surrounding the dens is also removed (Fig. 1.1-18B). If an extradural tumor is found, it is resected in a piecemeal fashion. If a CSF leak occurs, the dura can be closed with fascia, muscle, fibrin glue, or thrombin-soaked Gelfoam. Upon completion of the decompression, the mucosa is closed. It may be necessary to fuse the occiput to the upper C-spine to correct any cervical instability. This may take place at the same time or at a later date. C-spine precautions normally are used during and after the case. The patient may be in traction with tongs or a halter.

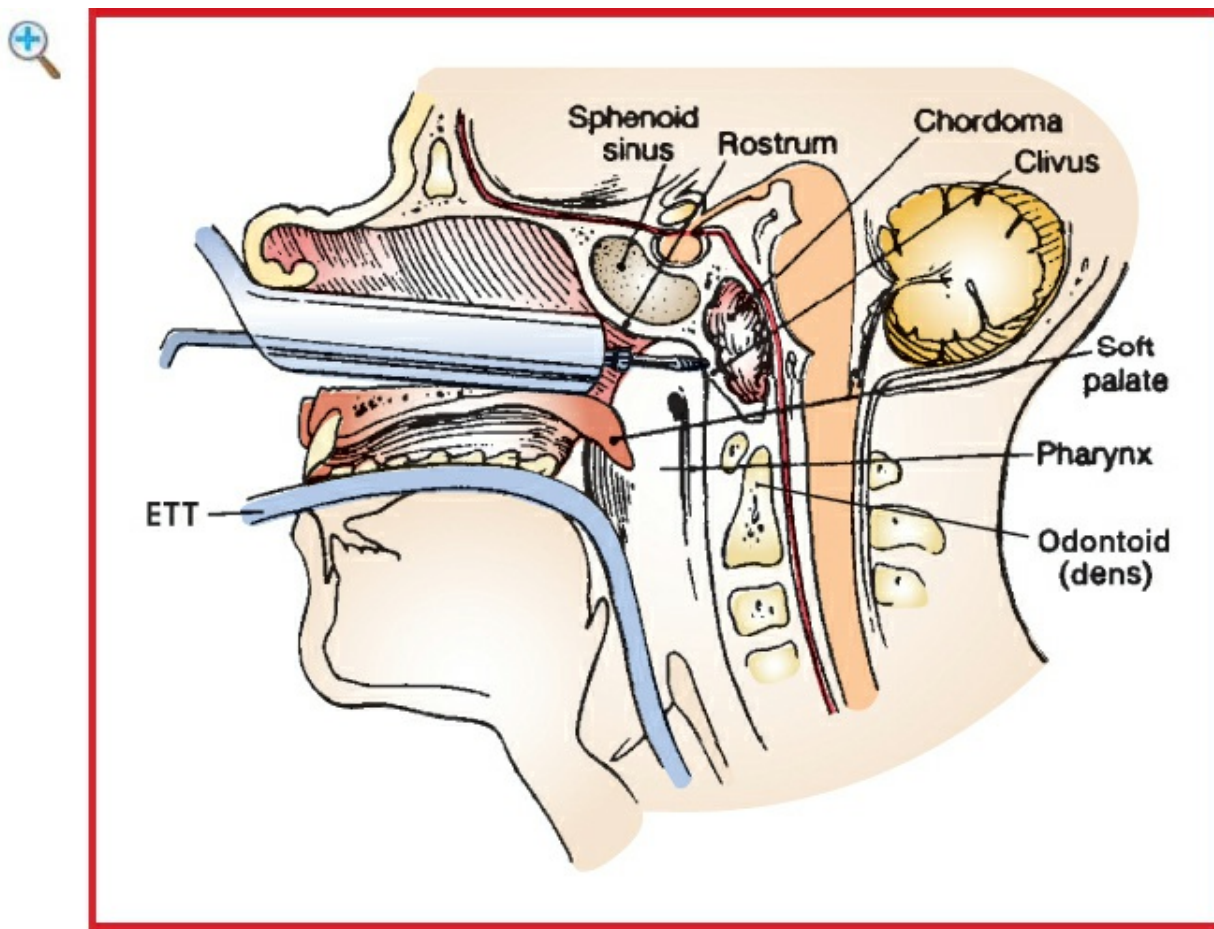


Figure 1.1-17 Transnasal inferior exposure of the clivus and chordoma. This procedure may be performed with endoscopic assistance, eliminating the need for a nasal speculum. (Reproduced with permission from Grossman RG, Loftus CM Principles of Neurosurgery, 2nd edition. Lippincott-Raven, Philadelphia: 1999.)

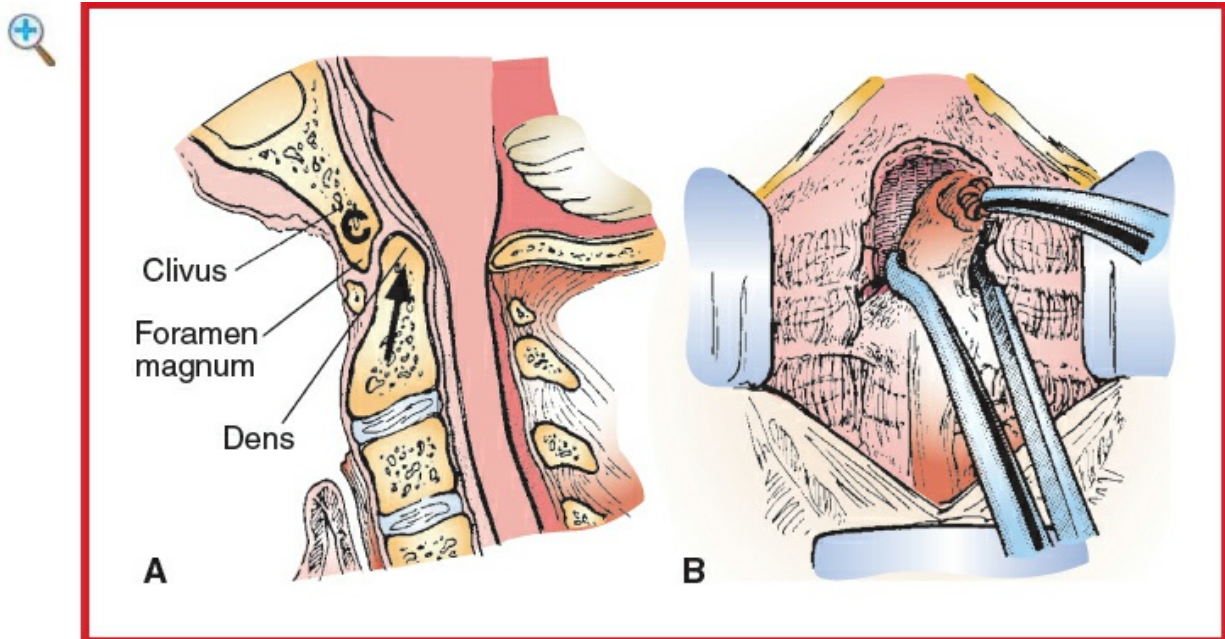


Figure 1.1-18 **A:** Basilar impression. The dens has moved superiorly into the foramen magnum (arrow). **B:** Removal of the dens with a drill. The arch of C1 and a portion of the clivus have been removed. (Reproduced with permission from Donald PJ: *Surgery of the Skull Base* Lippincott-Raven, Philadelphia: 1998.)

Variations in procedure

A fiberoptic nasal intubation or tracheostomy may be necessary to gain adequate exposure and maintain ventilation. It also may be necessary to split the soft palate, hard palate, tongue, and/or mandible to obtain adequate exposure. Because of the morbidity associated with a transoral decompression of the craniocervical junction, some surgeons recommend a posterior decompression with stabilization first, followed by an anterior decompression if necessary. An anterior resection of the odontoid creates an unstable craniocervical junction that requires at least a halo stabilization and more commonly an occipital-cervical fusion. This procedure may also be performed endoscopically.

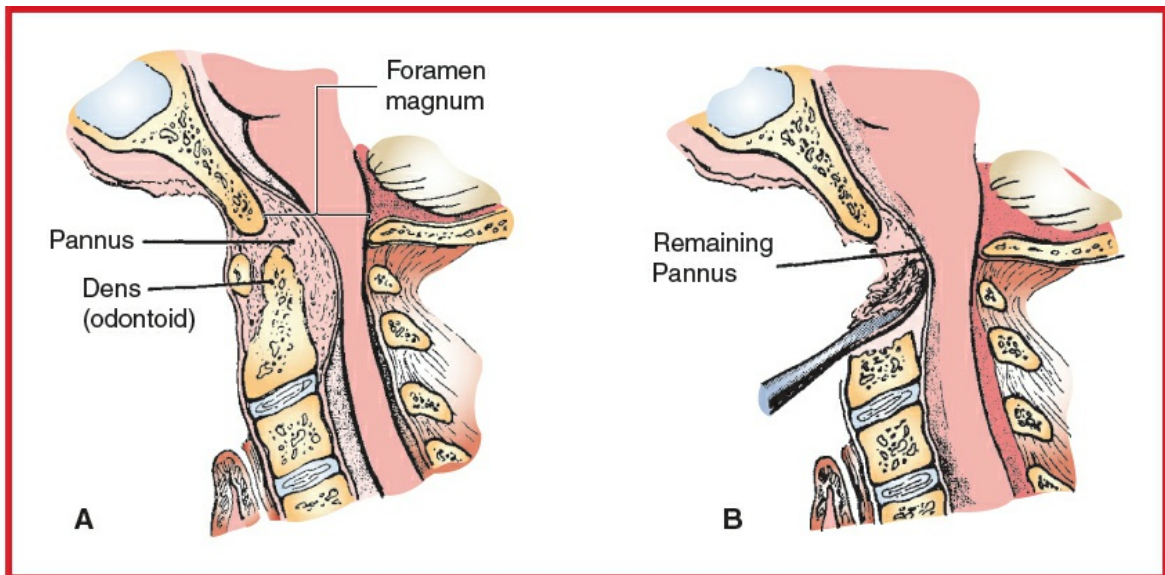


Figure 1.1-19 **A:** Pannus formation in a patient with rheumatoid arthritis. **B:** Removal of pannus. (Reproduced with permission from Donald P.J. Surgery of the Skull Base. Lippincott-Raven, Philadelphia: 1998.)

Usual preop diagnosis

Basilar impression (platybasia); odontoid fracture; rheumatoid arthritis with atlantoaxial instability and anterior impingement of the cord; chordoma; metastatic tumor

ERAS: Preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and other meds, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K](#), p. K-1.

■ SUMMARY OF PROCEDURE

Position	Supine; head in traction or pins
Incision	Back of oropharynx or nasopharynx
Special instrumentation	Operating microscope; I.I.; intraop monitoring = SSEP or BAER; micro drill; intraoral/intranasal retractors for exposure; image-guided navigation system (optional)
Unique considerations	FOL may be necessary. ETT must be taped securely in a location satisfactory to surgeon. Anode or RAE tube may be helpful. Occasionally, a tracheostomy is performed in advance. Odontoidectomy creates an unstable craniocervical junction, which may need to be temporarily

	stabilized with a halo or immediate occipital cervical fusion.
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	2.5–3 h
Closing considerations	Patient may remain intubated postop. C-spine precautions may be necessary.
EBL	25–250 mL
Postop care	ICU or close observation unit; monitor airway for swelling (✓ for stridor).
Mortality	0–3%
Morbidity	All <5%: Infection CSF leak Neurologic Massive blood loss
Pain score	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–85 yr (usually 20–60 yr)
Male:Female	~1:2
Incidence	Rare
Etiology	Neoplastic; traumatic; congenital; degenerative
Associated conditions	Rheumatoid arthritis (atlantoaxial subluxation in 25%; basilar compression in 8%); traumatic injury

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Transsphenoidal Resection of Pituitary Tumor.

Suggested Readings

1. Dlouhy BJ, Dahdaleh NS, Menezes AH: Evolution of transoral approaches, endoscopic endonasal approaches, and reduction strategies for treatment of craniovertebral junction pathology: a treatment algorithm update. *Neurosurg Focus* 2015; 38(4):E8.
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3. Shriver MF, Kshetry VR, Sindwani R, et al: Transoral and transnasal odontoidectomy complications: a systematic review and meta-analysis. *Clin Neurol Neurosurg* 2016; 148:121–9.

TRANSSPHENOIDAL RESECTION OF PITUITARY TUMOR

SURGICAL CONSIDERATIONS

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Description

The transsphenoidal approach to the sella turcica is a direct procedure used to gain access to the pituitary gland and sella region and is associated with relatively fewer complications than a craniotomy. The procedure usually is performed transnasally; although a sublabial transseptal approach is out of favor, it may be used in patients with large tumors. An otorhinolaryngologist often will participate in obtaining the exposure, transnasally with an endoscope. Once the sphenoid sinus is reached, it is entered by removing the face and rostrum. The thin bone of the lateral wall of the sphenoid sinus protects portions of the optic nerve and carotid artery. The sella is entered by removing a portion of the sellar floor ([Fig. 1.1-20](#)). Under image guidance and with the aid of the operating microscope and/or endoscope, the surgeon can operate safely on the pituitary gland, respecting the midline and avoiding injury to the carotid arteries in the cavernous sinus on either side of the exposure. After the sellar floor is removed, the dura is opened and the tumor is removed with a series of microdissectors and curettes. Following tumor removal, if there is a CSF leak, the surgeon may harvest fat, muscle, or fascia from the thigh or abdomen to pack into the sella to reduce free intrasellar space and serve as a graft to prevent a leak. Resorbable materials, such as Gelfoam and fibrin glue, also may be used to obliterate dead space within the sella. The floor of the sella can be reconstructed with bone salvaged from the exposure or artificial material. During the closure, the sphenoid sinus also may be packed and sealed with the same materials used to close the sella. The nostrils may or may not be packed, depending on the details of the technique used.

Alternative procedure

Endoscopic transsphenoidal approach has several advantages (improved surgical view and access, fewer complications, and more rapid recovery) and is associated with the same success rate as the microscopic approach.

Usual preop diagnosis

Pituitary tumor; prolactin-secreting tumor; growth hormone–secreting tumor (acromegaly); ACTH-secreting tumor (Cushing disease); visual compromise, intrasellar tumor; craniopharyngioma; meningioma; Forbes-Albright syndrome (galactorrhea-

amenorrhea syndrome)

ERAS: Preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and other meds, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

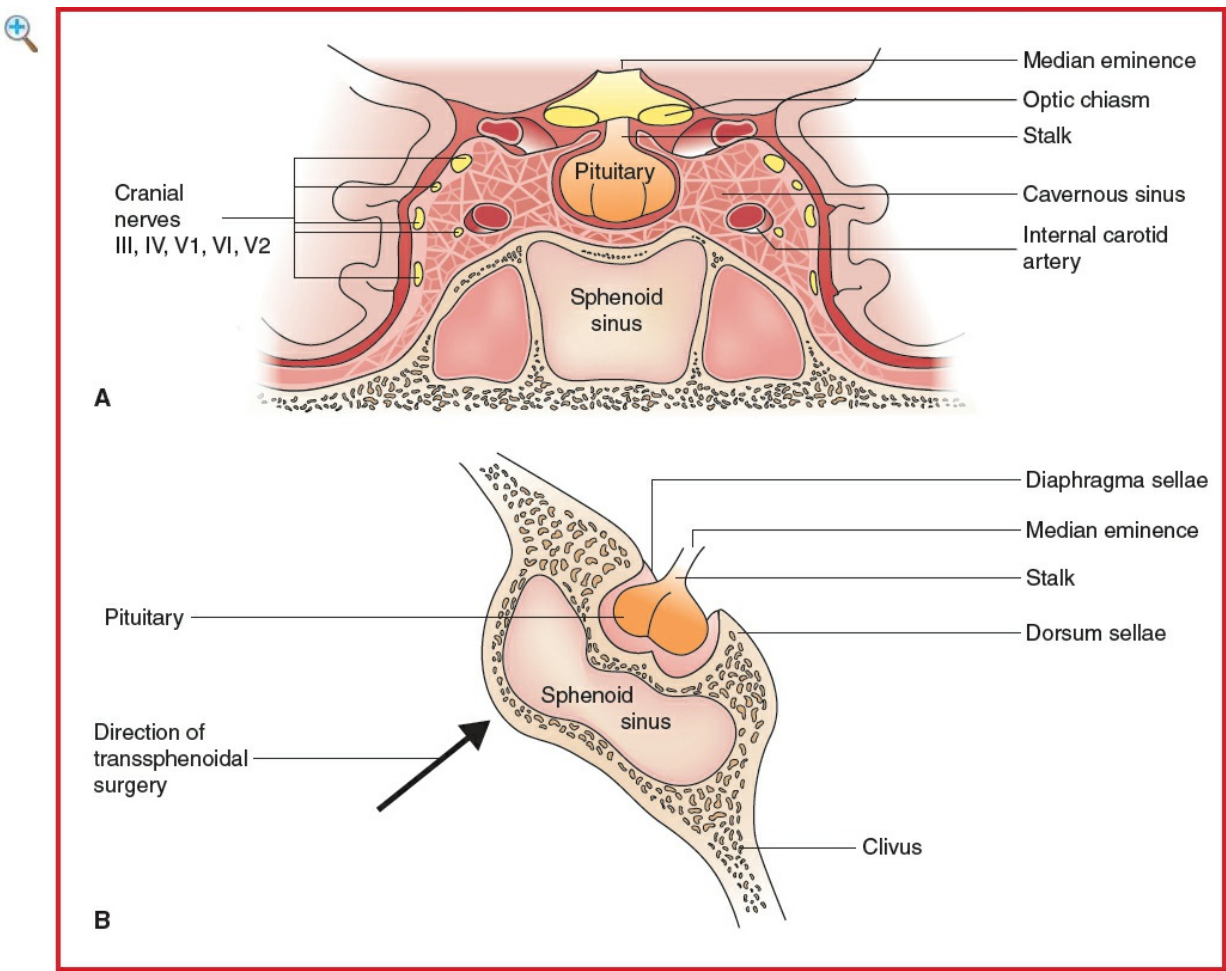


Figure 1.1-20 **A:** Midpituitary coronal view of parasellar region. The sphenoid sinus is below, and the cavernous sinuses are lateral. **B:** Midsagittal view of the pituitary and surrounding bony structures. Note the approach for transsphenoidal surgery. Anterior is to the left.

■ SUMMARY OF PROCEDURE	
Position	Supine, head elevated 20–30°
Incision	Endonasal; +abdomen (or thigh) for fat graft; occasionally sublabial, maxillary gingiva
Special instrumentation	Operating microscope; operating endoscope; micro drill; neuronavigation (image guidance)

	Throat packs may be used but are usually unnecessary.
Unique considerations	ETT must be taped securely in a location away from the surgeon (usually left side of the mouth). Anode or RAE tube may be helpful but is not essential. The generous use of topical vasoconstrictors and the injection of epinephrine-containing solutions may → ↑ BP and dysrhythmias. Dissection in the nasal cavity can be painful → ↑ BP and ↑ ICP. There is a risk of VAE following inadvertent entry into the cavernous sinus. Valsalva maneuver, jugular compression, or injection of saline through a lumbar intrathecal catheter may be requested to facilitate tumor exposure. Injuries to sphenopalatine or carotid arteries are rare but may be dramatic when they occur.
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	2.5–3 h
Closing considerations	Possible abdominal or thigh fat graft. Closure quite fast. Prevent straining and bucking if there has been an intraoperative CSF leak. Remove any throat pack before extubation.
EBL	25–250 mL
Postop care	Fluid and electrolytes require frequent monitoring as the patient may develop diabetes insipidus (DI), usually transient, following this procedure.
Mortality	1%
Morbidity	Anterior pituitary insufficiency: 19.4% DI: 0.5–31%; SIADH: 9–25% Sinusitis: 8.5% Septum perforation: 3% CSF leak: 3.5% Epistaxis: 3.0% Loss of vision: 1.8% Meningitis: 1.5% Carotid artery injury: 1.1% stroke (rare)

Pain score 2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–85 yr (usually 30–50 yr)
Male:Female	2:3 (microadenomas)
Incidence	16.7% (autopsy studies); 1.5/100,000/yr diagnosed
Etiology	Neoplastic; traumatic
Associated conditions	Cushing disease (ACTH); acromegaly (GH); amenorrhea/galactorrhea (prolactin); hyperthyroidism (TSH); hypergonadotropism (FSH; LH); multiple endocrine neoplasia I

■ ANESTHETIC CONSIDERATIONS

(Procedures covered: bifrontal craniotomy for CSF leak; transoral approach to cervicomedullary junction, odontoid; transsphenoidal resection of pituitary tumor)

■ PREOPERATIVE

Patients presenting with pituitary tumors can be divided into two categories, those with functional (hormone secreting) tumors and those with nonfunctional (nonsecreting) tumors. Nonfunctional tumors are usually large at the time of diagnosis (headache, visual changes, panhypopituitarism), producing Sx as a result of compression. Functional tumors can be subcategorized:

Prolactin secreting: Few anesthetic implications beyond possible mass effect and ↑ ICP.

ACTH secreting: HTN is very common and is often associated with diastolic dysfunction. OSA is common and may have implications for postop airway management. Hyperglycemia is common and should be treated if blood glucose >180 mg/dL. ↑ cortisol → fragile skin and veins → difficult iv access.

Growth hormone secreting: Hypertrophy of the mandible, tongue, and other facial tissues is common and complicates airway management. Hoarseness suggests laryngeal stenosis or recurrent laryngeal nerve dysfunction. OSA is very common with its attendant risk of postop airway compromise. HTN and diastolic dysfunction (with a need for ↑ filling pressure) are very common. The incidence of intraop ventricular ectopy and SVT is higher in these patients. ECG changes (e.g., ST/T

wave changes, conduction abnormality) are common.

Pituitary tumors may present with the sudden onset of Sx (including severe headache, ophthalmoplegia, coma) as the result of hemorrhagic infarction (pituitary apoplexy). This is a life-threatening emergency often requiring urgent transsphenoidal decompression following stabilization with volume and high-dose corticosteroids.

Endocrine	Rarely, the mass effect may cause the clinical picture of panhypopituitarism requiring preop treatment with thyroxine, glucocorticoid, and vasopressin. Functional tumors commonly secrete prolactin (→ lactation), ACTH (→ adrenal hyperplasia), or growth hormone (→ acromegaly). Hyperprolactinemia may occur with any tumor 2° loss of inhibition of prolactin secretion. Tests: Preop endocrine studies, including serum and urinary levels of pituitary, thyroid, and adrenal hormones; appropriate replacement therapy established before proceeding with surgery. "Stress-dose" hydrocortisone (e.g., 100 mg iv q8h) is often unnecessary; treatment should be guided by symptoms and periop cortisol levels. Dexamethasone will not interfere with cortisol assays, but some surgeons continue to avoid this steroid.
Respiratory	No special requirements, unless patient has acromegaly, in which case large facial features, long neck, large tongue, and redundant soft tissue in the oropharynx may make mask fit and ET intubation difficult. If these patients evidence hoarseness or inspiratory stridor, they should have a full clinical and radiologic evaluation of the upper airway. Tests: As indicated from H&P
Cardiovascular	No special requirements unless patient has acromegaly, in which case they may have refractory HTN, ischemic heart disease, cardiomyopathy, dysrhythmias, valvular disease, and diabetes. Tests: As indicated from H&P
Neurologic	Secretory tumors of the pituitary are usually small,

	confined to the sella, rarely cause ↑ ICP (2° direct tumor effects or by obstruction of 3rd ventricle), and commonly produce Sx of endocrine dysfunction early in their growth. In contrast, nonfunctional pituitary tumors may not produce Sx until they extend beyond the boundaries of the sella, causing HAs (the most common presenting symptom) or compression of the optic chiasm, producing visual field defects (temporal or bitemporal hemianopsia). Tests: A CT or MRI will delineate the site and size of the tumor, especially if iv contrast material, such as gadolinium, is administered to enhance the margins of the tumor.
Musculoskeletal	If growth hormone is the primary secretant, patient will exhibit Sx of acromegaly, including large hands, feet, head, and tongue. Cushing myopathy (UE proximal muscle weakness) may affect apparent recovery from muscle relaxant. Tests: As indicated from H&P
Laboratory	Metabolic panel, Hct, and others as indicated from H&P. ↓ Na⁺ may indicate DI; ↑↑ Ca⁺⁺ may indicate MEN I.
Premedication	If ↑ ICP is not a concern, patients may benefit from midazolam 1–2 mg iv. In patients with ↑ ICP consider pretreatment with mannitol (e.g., 0.25 g/kg 20 min before induction). Acromegalic patients may develop airway obstruction following even small doses of sedatives or narcotics.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA is required for these operations because the surgical approach is commonly through the nose or rarely through the mouth above the maxillary gum line and behind the nose. A lumbar drain may be placed to permit intraop management of ICP and improve access to the tumor.

Induction	If a difficult intubation is anticipated (e.g., severe acromegalic patients), orotracheal intubation will need to be accomplished before induction of GA. LMA placement may be difficult, and awake FOL is often the best choice. Because pituitary tumors generally are confined to the sella turcica and, hence, ICP usually is not increased, a standard induction technique with standard ETT is appropriate. An oral RAE tube may be preferred by some surgeons. If increased ICP is of concern, induction should be similar to that used for patients with other kinds of brain tumors (see Anesthetic Considerations for Craniotomy for Tumor). Because the surgeon will be working from the patient's right side, the ETT must be positioned at the far left side of the mouth. The surgeon may infiltrate the nasal mucosa with lidocaine and epinephrine → ↑ BP + dysrhythmias → possible myocardial ischemia.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Consider dexamethasone (4–8 mg iv) may ↓ PONV. Sequential pneumatic compression devices should be used to ↓ VTE risk.
Maintenance	Standard maintenance with sevoflurane (or propofol infusion) supplemented with remifentanyl. Generally, no further NMBs are administered beyond ET intubation. With adequate anesthesia, and the head in a Mayfield clamp, patient movement of any consequence is highly unlikely. Ventilation is controlled with PaCO ₂ maintained in the normal range. Hyperventilation is not desirable because it makes it more difficult to access the tumor in the sella and establish that it has been removed in its entirety. Following the resection, the surgeon may request a Valsalva maneuver to check for CSF leak.

Emergence	If the patient evidences normal emergence from anesthesia, the ETT should be removed before vigorous coughing ensues. Before removing the ETT, however, make certain that all blood accumulated in the back of the throat is suctioned out and that any oropharyngeal packs that may have been placed in the back of the throat have been removed. An OG tube should be used to suction out the stomach. The surgeon may have packed the nose at the end of operation, making the patient an obligatory mouth breather. If there is any question about airway patency because of a large tongue, small mouth, or soft tissue redundancy in the oropharynx, the ETT should be left in place until patient is fully awake from anesthesia. Prophylactic antiemetics (e.g., ondansetron 4–8 mg and metoclopramide 20 mg) should be given 30 min before extubation. Consider iv acetaminophen if po dose given >4–6 h ago.	
Blood and fluid requirements	Minimal blood loss usual Potential large blood loss IV: 16–18 ga × 1–2 BSS @ 1–3 mL/kg/h	Blood loss is minimal, unless the surgeon inadvertently enters the internal carotid artery or cavernous sinus. Additional iv access is prudent. Zero balance or goal-directed fluid management is recommended (see Appendix K, p. K-2).
Monitoring	Standard monitors ± Arterial line ± UO ± CVP line ± Precordial Doppler	Monitor for VAE in semisitting position if >30° head up. Ulnar artery flow may be compromised in acromegaly—document Alan test before placing radial artery catheter. CVP line should be placed in patients at significant risk for VAE or as medically indicated.
Positioning	✓ and pad pressure	The surgeon will use an

	points ✓ eyes Shoulder roll Table turned 90–180°	operating microscope or endoscope, which means that the anesthesiologist will be positioned at the side or foot. Anesthetic hoses and intravascular lines must be long enough to be accessible.
Complications	Carotid injury VAE	Potential for large-volume hemorrhage and intraop stroke. Deliberate hypotension may facilitate repair. Carotid compression may → ↓ CBF → ischemia.

POSTOPERATIVE

Complications	Hypopituitarism Diabetes insipidus (24–48 h postop) CSF leak SIADH Cranial nerve injury	Replacement therapy with steroids is necessary in hypopituitary patients until normal pituitary function returns. Occasionally, patients will develop DI postop → polyuria and ↓ urine specific gravity <1.005. Rarely, this may occur in the PACU necessitating fluid replacement and/or DDAVP.
Multimodal pain management	Acetaminophen Fentanyl 25–50 mcg prn	Frontal HA is the most common complaint. Consider acetaminophen (1 g iv) for postop analgesia if not given preop or if 4–6 h have elapsed since previous dose.
Tests	CT scan Serum cortisol Visual acuity/fields	If a patient exhibits any delay in emergence from anesthesia and surgery, or any new neurologic deficits

		emerge postop, a CT scan is ordinarily obtained.
ERAS	Ondansetron 4 mg iv ± Metoclopramide 10–20 mg iv	Metoclopramide will promote gastric emptying of swallowed blood. Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Alsumali A, Cote DJ, Regestein QR, et al: The impact of transsphenoidal surgery on neurocognitive function: a systematic review. *J Clin Neurosci* 2017; 42:1–6.
2. Alzhrani G, Sivakumar W, Park MS, et al: Delayed complications after transsphenoidal surgery for pituitary adenomas. *World Neurosurg* 2018; 109:233–41.
3. Dunn LK, Nemergut EC: Anesthesia for transsphenoidal pituitary surgery. *Curr Opin Anaesthesiol* 2013; 26(5):549–54.
4. Lim M, Williams D, Maartens N: Anaesthesia for pituitary surgery. *J Clin Neurosci* 2006; 13(4):413–8.
5. Murad-Kejbou S, Eggenberger E: Pituitary apoplexy: evaluation, management, and prognosis. *Curr Opin Ophthalmol* 2009; 20(6):456–61. Review.

VENTRICULAR SHUNT PROCEDURES

SURGICAL CONSIDERATIONS

Lawrence M. Shuer, Steven D. Chang, and Navjot Chaudhary

Description

Many conditions exist where it is necessary to shunt CSF from the ventricles to another body cavity where it can be absorbed readily. The most common condition is hydrocephalus, where there is dilation of the ventricular system due to an obstruction in the flow of CSF (Fig. 1.1-21) or decreased absorption of CSF by the arachnoid villi. Hydrocephalus is commonly treated by diverting CSF to the peritoneal cavity via a ventriculoperitoneal shunt (**VP shunt**). Most shunt systems have a one-way, pressure-dependent valve to regulate the flow of CSF. These valves usually have a preset pressure setting (e.g., low, medium, high) at which they open and allow CSF to flow. Newer valves have externally adjustable opening pressures, flow-regulating devices, and/or antisiphon systems to prevent overshunting. In addition, antibiotic impregnated

catheters can be used to reduce the risk of infection.

To insert a VP shunt, the patient is positioned so that the cranial incision and abdominal incision are aligned in the same plane. The scalp is shaved (usually over the frontal or parietal region), and a continuous surgical field is created from head to abdomen. The cranial incision is made over the intended region of cannulation of the ventricle, and a burr hole is made in the cranium. A subgaleal pocket is created for the valve, usually behind the ear. A separate incision is made in the abdomen, and the dissection is carried down to the level of the peritoneum. A catheter is then passed subcutaneously from the abdominal incision to the cranial incision with a special tunneling instrument ([Fig. 1.1-22A](#)). Alternatively, the catheter can be threaded from the cranial incision to the abdominal incision. It may be necessary to use one or more incisions between the head and the abdomen to thread the catheter. The valve is connected to the peritoneal catheter and placed in the subgaleal pocket. A ventricular catheter is then inserted into the ventricle ([Fig. 1.1-22B](#)), and a small amount of CSF is drained to check placement and patency of the catheter. The ventricular catheter is then connected to the valve, and CSF flow through the entire shunt system is checked by draining some CSF from the distal end of the peritoneal catheter. The distal end is then placed into the peritoneal cavity and all wounds are closed.

Any component of the shunt may malfunction; thus, it may be necessary to test each component at the time of revision to identify the problem. To test the patency of the ventricular catheter, it is disconnected from the valve, and CSF should flow freely through it. A saline-filled manometer can be attached to the valve to measure the opening pressure of the valve. The manometer should drain spontaneously until the opening pressure of the valve is reached. An elevated opening pressure may indicate a valve malfunction or a distal occlusion. If a distal occlusion is suspected, gentle irrigation may clear it, but externalization may be necessary.

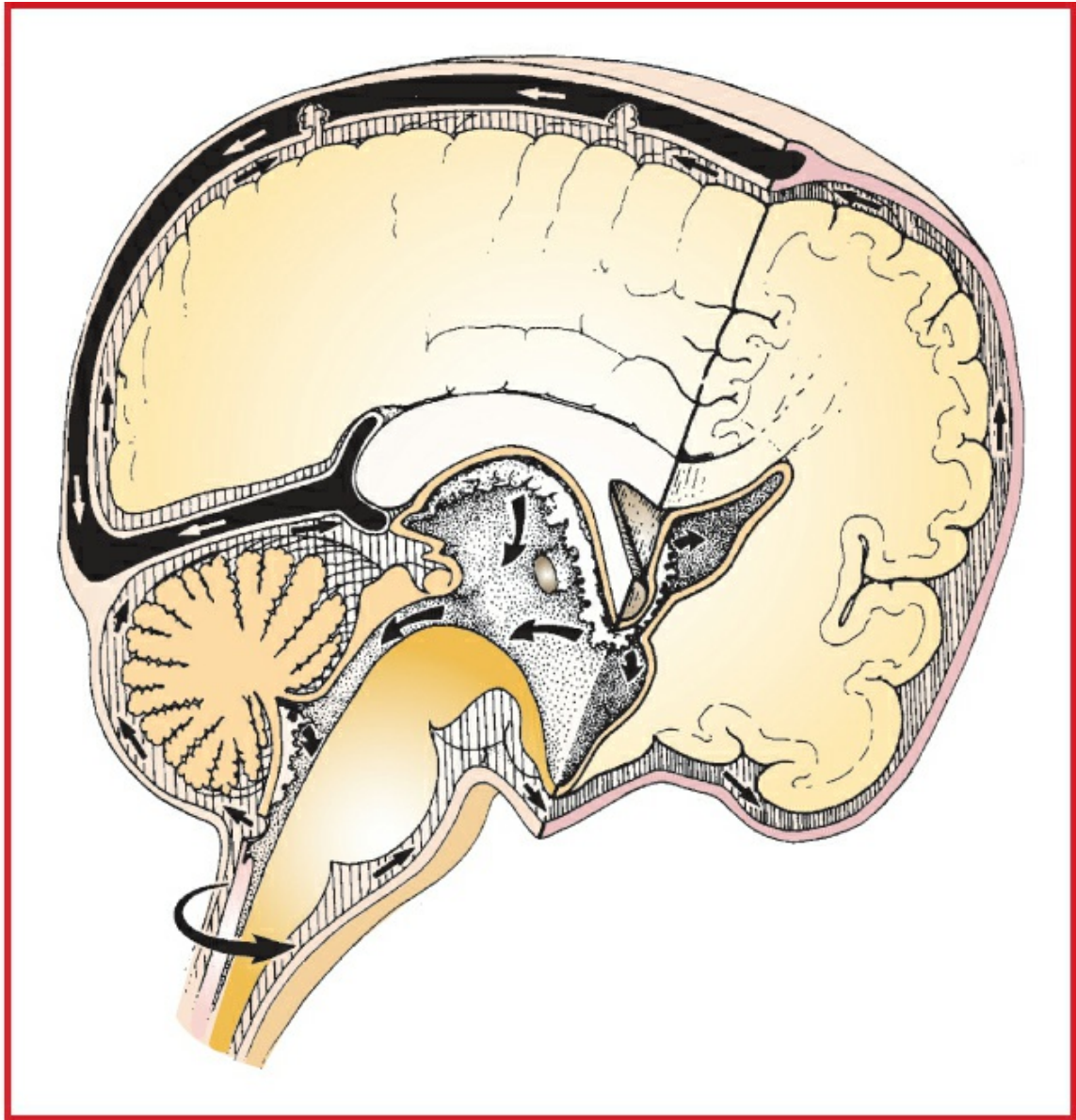


Figure 1.1-21 Pathways of CSF flow and potential sites of obstruction. (Reproduced with permission from Tindall GT, Cooper PR, Barrow DL. *The Practice of Neurosurgery*, Vol III. Williams & Wilkins, Baltimore: 1996. Originally from Scott RM: *Hydrocephalus*. Concepts in Neurosurgery, Vol III. Williams & Wilkins, Baltimore: 1990.)

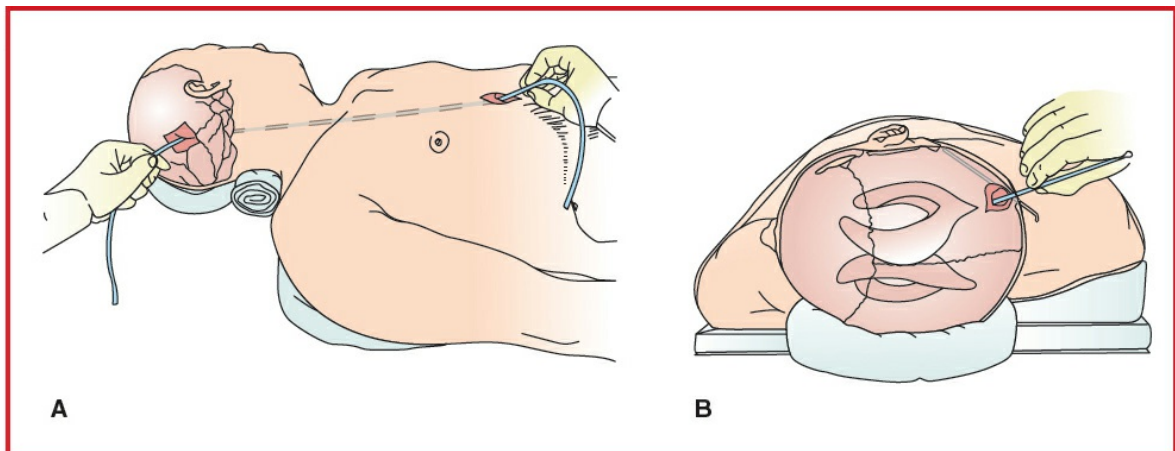


Figure 1.1-22 A: The catheter is threaded from the cranial incision to the abdominal incision. **B:** Insertion of the ventricular catheter. (Reprinted from Meyer FB:Atlas of Neurosurgery: Basic Approaches to Cranial and Vascular Procedures, 1st edition. Churchill Livingstone, Philadelphia: 1999. Copyright © 1999 Elsevier. With permission.)

Variant procedure or approaches

Endoscopy, fluoroscopy, ultrasound, image-guided navigation, or intraoperative MRI may be used to help place the catheter into the ventricle. The ventricular catheter can be placed in either lateral ventricle; occasionally, both lateral ventricles are cannulated. This procedure also is used to shunt the 4th ventricle and, sometimes, subarachnoid cysts and subdural hygromas. The distal end alternatively may be placed in the pleural cavity or the right atrium (ventriculoatrial or **VA shunt**). If the distal end of the catheter is placed into the pleural cavity, a Valsalva maneuver is performed upon closure to reinflate the lung. To place the distal end into the atrium, a vein is cannulated in the neck (usually IJ, EJ, or facial vein), and the catheter is fed into the atrium under fluoroscopic guidance. It may be necessary to inject radiopaque contrast to verify proper placement. Rarely, the distal end of the catheter can be placed in an intracranial venous sinus, the gallbladder, or the ureter.

Endoscopic third ventriculostomy (ETV)

Endoscopic third ventriculostomy is being used with increasing frequency to treat both obstructive and communicating forms of hydrocephalus. Head fixation is generally not needed, and the procedure involves a small skin incision over the frontoparietal lobe to gain access to the skull. A small burr hole is then made. The dura is opened over the burr hole, permitting the insertion of the endoscope into the lateral ventricle. Under endoscopic guidance, the tuber cinereum is then fenestrated with a balloon tip catheter. CSF is diverted through the membranous floor of the 3rd ventricle into the basal cisterns. The endoscope is then removed, and the wound is closed.

Usual preop diagnosis

Hydrocephalus; obstructive or communicating hydrocephalus; aqueductal stenosis; Dandy-Walker malformation (cystic dilation of the 4th ventricle and incomplete formation of the cerebellar vermis); occult hydrocephalus; normal pressure hydrocephalus; subarachnoid cyst; subdural hygroma; pseudotumor cerebri

ERAS: Preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and other meds, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	VP Shunt	VA Shunt	ETV
Position	Supine, with head turned to contralateral side	⇐	⇐
Incision	Scalp, either coronal and retroauricular or parietal; + neck and abdomen	Scalp, either coronal and retroauricular or parietal; + neck	Scalp over frontoparietal lobe
Special instrumentation	Ventricular endoscope (optional)	⇐ + I.I.	Ventricular endoscope
Unique considerations	Patient to be treated as if there is ↑ ICP	⇐	⇐
Antibiotics	Cefazolin 2 g iv	⇐	⇐
Surgical time	1 h	⇐	⇐
EBL	5–25 mL	⇐	⇐
Postop care	PACU → room; usually kept flat for 24 h	⇐	⇐
Mortality	<1%	⇐	⇐
Morbidity	Infection: <15% Neurologic: Intracranial bleed: <1% Subdural hematoma: <1% Hardware failure: <1%	⇐	Basilar artery injury ↑↑ ICP → ↓↓ HR or asystole
Pain score	4–6	2–4	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn-elderly
Male:Female	1:1

Incidence Etiology	Common Congenital (3–4/1000 live births); acquired; neoplastic; infectious; posthemorrhagic
Associated conditions	Subarachnoid hemorrhage; myelodysplasia; spina bifida (95% incidence); intraventricular hemorrhage; intraventricular tumor

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Ventricular shunts are inserted to ameliorate hydrocephalus or cyst formation, which may be either congenital or acquired. EVD (if present) may have been clamped by surgical team to enlarge ventricles to facilitate insertion of VP shunt. Resulting ↑ICP may exacerbate N/V and/or drowsiness.

Cardiovascular	↑ ICP → ↑ BP + ↓↓ HR (Cushing response) Tests: As indicated from H&P
Neurologic	The most common presenting Sx is HA. If hydrocephalus is severe, Sx of ↑ ICP (>15 mm Hg); for example, N/V, drowsiness, papilledema, Sz, and focal neurologic defects) develop.
Laboratory	Tests as indicated by H&P
Premedication	Usually not required; should be avoided in patients with ↑ ICP.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique: GETA

Induction	If ↑ ICP, iv induction with propofol (1–2 mg/kg) is preferred because of its ability to ↓ CBV → ↓ ICP. ET intubation is accomplished with the use of an NMB (e.g.,
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ERAS	rocuronium [0.6–1 mg/kg]). Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Dexamethasone (4–8 mg iv) may ↓PONV. Avoid long-acting opioids or scopolamine patch, which may confound postop neuro examination. Sequential pneumatic compression devices should be used to ↓VTE risk.	
Maintenance	Isoflurane ≤1% or sevoflurane <2%, inspired with 50:50 N₂O/O₂ mixture. Depending on duration of operation, additional doses of rocuronium (0.2 mg/kg) may be needed. Maintain normal temperature in children by keeping OR warm (78°F) and using a forced-air warming blanket. Ventilation should be controlled. TV and frequency should be adjusted such that the PetCO₂ = 35–40 mm Hg. Hyperventilation and hypocarbia are undesirable because they make cannulation of the ventricle(s) more difficult. Maintain normotension. During endoscopic third ventriculostomy (ETV), puncture of the floor of the 3rd ventricle may be associated with significant ↓↓ HR or asystole. The surgeon should be immediately informed if bradycardia occurs. Catheter tunneling will be stimulating → ↑HR and ↑BP, which can be prevented with bolus propofol or remifentanyl.	
Emergence	Ensure reversal of any residual neuromuscular blockade. ETT is removed at the conclusion of the anesthetic. Prophylactic antiemetic (e.g., metoclopramide 10–20 mg and ondansetron 4 mg) should be given 30 min before extubation.	
Blood and fluid requirements	IV: 18–20 ga × 1 BSS @ 1–3 mL/kg/h	Zero balance fluid management (see Appendix K).
Monitoring	Standard monitors	Invasive monitors as indicated.
Positioning	Table turned 180° ✓ and pad pressure points ✓ eyes	Supine with a bolster under the shoulder on the operative side. The head, chest, and abdomen are prepped, so all

		anesthesia equipment and lines must be at the sides of the patient.
Complications	Valve malfunction ↓↓ HR or asystole Basilar artery injury	Major intraop complications from this operation are rare. ETV may be associated with severe bradycardia and injury to the basilar artery or its branches.

▼ POSTOPERATIVE

Multimodal pain management	Children <2 yr: acetaminophen suppositories (10–15 mg/kg q4h) Adults: acetaminophen 1 g iv q6h ± Fentanyl 25–50 mcg prn	If not given preop or if 4–6 h have elapsed since previous dose.
ERAS	Ondansetron 4 mg iv ± Metoclopramide 10–20 mg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. He M, Ouyang L, Wang S, et al: Laparoscopy versus mini-laparotomy peritoneal catheter insertion of ventriculoperitoneal shunts: a systematic review and meta-analysis. *Neurosurg Focus* 2016; 41(3):E7.
2. Spennato P, Ruggiero C, Aliberti F, et al: Third ventriculostomy in shunt malfunction. *World Neurosurg* 2013; 79(2 Suppl):S22.e21–6.
3. Woodworth GF, McGirt MJ, Elfert P, et al: Frameless stereotactic ventricular shunt placement for idiopathic intracranial hypertension. *Stereotact Funct Neurosurg* 2005; 83(1):12–6.
4. Wu Y, Green NL, Wrensch MR, et al: Ventriculoperitoneal shunt complications in California: 1990–2000. *Neurosurgery* 2007; 61(3):557–62.

CRANIOCERVICAL DECOMPRESSION (CHIARI MALFORMATION)

SURGICAL CONSIDERATIONS

Lawrence M. Shuer, Steven D. Chang, and Navjot Chaudhary

Description

Historically, congenital hindbrain abnormalities characterized by cerebellar descent were known collectively as **Arnold-Chiari malformations**; now, they are known as **Chiari malformations**. In these abnormalities, portions of the cerebellum protrude through the foramen magnum and may compress the brain stem and upper cervical spinal cord. Frequently, the malformation is accompanied by **syringomyelia**, a condition in which CSF is located abnormally within the spinal cord. In **type I Chiari** malformation, the cerebellar tonsils are herniated through the foramen magnum. In **type II Chiari** malformation, the tonsils, vermis, 4th ventricle, pons, and medulla are displaced caudally through the foramen magnum. Type II Chiari malformations are usually found in infants and are associated with myelomeningoceles. In **type III Chiari** malformations there is a defect in the back of the head or neck through which the cerebellum and/or brain stem is herniated. Patients with **type IV Chiari** malformations usually do not survive past infancy.

To make room for the brain stem and caudally displaced cerebellum, decompression of the craniocervical junction is necessary. This procedure may be performed in the prone, three-fourths prone, or seated position. The head is placed in Mayfield pin fixation. A midline incision is made, and the dissection is carried down to the skull and the posterior arch of C1. Paramedian burr holes are placed to aid in the suboccipital craniectomy. The craniotome is then used to remove an $\sim 4 \times 3$ -cm bone flap that includes the foramen magnum. Care is taken to avoid injury to the underlying venous sinuses when performing the craniectomy.

The posterior arch of C1 is removed, and as many upper cervical lamina as are needed to fully decompress the Chiari malformation are removed. Next, any constricting fibrous bands are cut, and the dura is opened. Any subarachnoid adhesions should be lysed to allow the free flow of CSF. The tonsils are dissected apart under microscopic guidance to gain an opening to the 4th ventricle ([Fig. 1.1-23](#)). The cerebellar tonsils can be reduced with electrocautery or by resection to facilitate the outflow of CSF from the 4th ventricle. A stent or a shunt may be placed in the 4th ventricle and brought out to the subarachnoid space to ensure adequate drainage of the 4th ventricle. The dura is closed

loosely with autologous pericranium, fascia lata, or bovine pericardium ([Fig. 1.1-23B](#)). This creates a widely patent cisterna magna and prevents constriction at the cervicomedullary junction.

Usual preop diagnosis

Chiari malformation I or II or Arnold-Chiari malformation; syringomyelia

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of lines and urinary catheter postop. See [Appendix K](#).

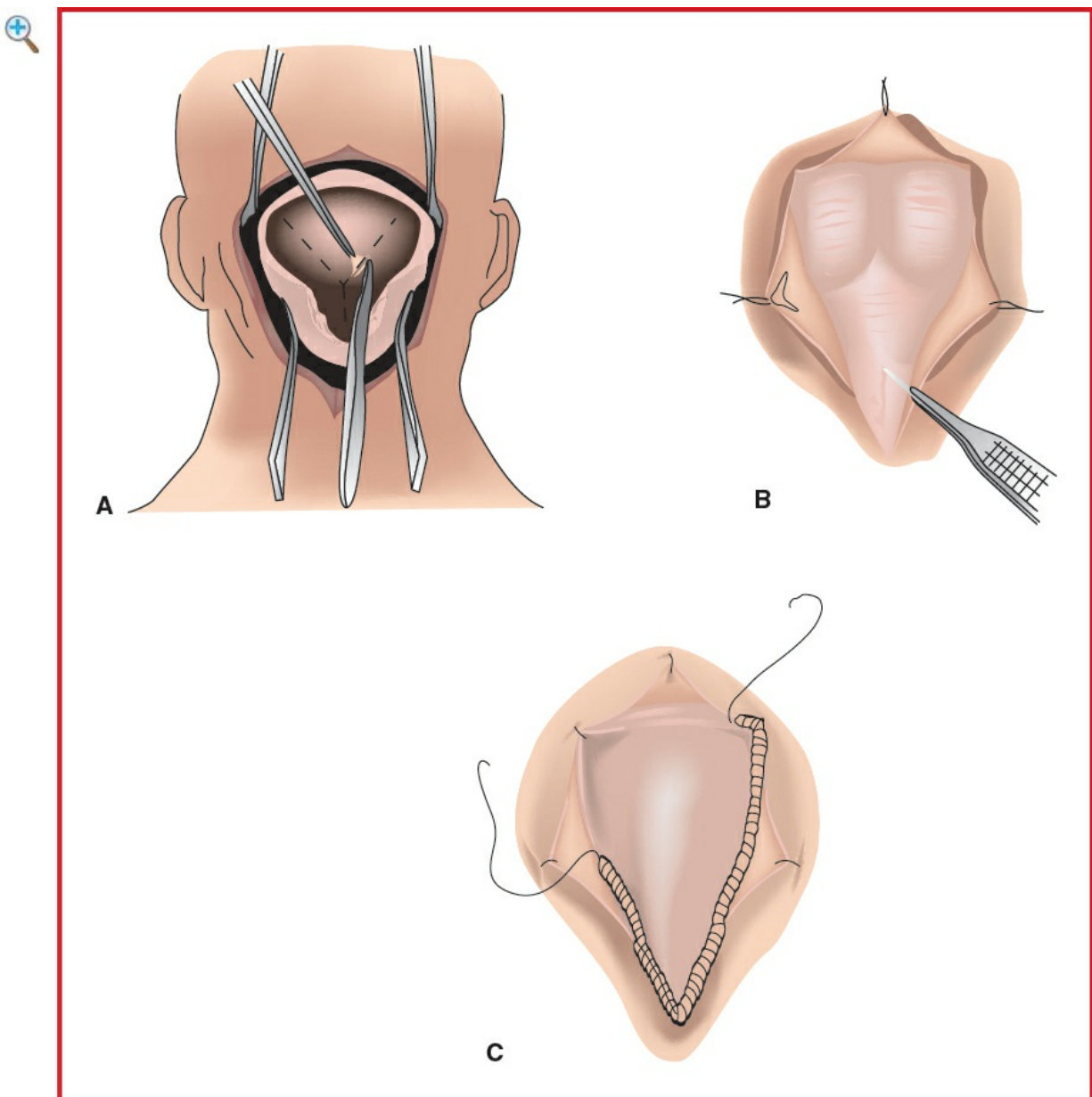


Figure 1.1-23 A and B: Surgical anatomy in Chiari I decompression. The use of the no. 12 blade is demonstrated. Because the cutting surface is drawn up and away from the central nervous system

tissue, the risk of injury is minimized with patch graft. (Adapted with permission from Barrow Neurological Institute.) **C:** Surgical anatomy in Chiari II malformation. Demonstration of the surgical anatomy of the Chiari II malformation showing a low-lying torcular and a medullary kink that is more caudally displaced than the cerebellar vermis.

■ SUMMARY OF PROCEDURE

Position	Prone or sitting
Incision	Midline posterior, posterolateral thigh for fascia lata graft (optional)
Special instrumentation	Operating microscope
Unique considerations	Risk of air embolus; brain stem manipulation can cause BP and pulse instability; dural patch used to expand dura at foramen magnum
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	2.5–3.5 h
Closing considerations	BP control to ensure adequate hemostasis
EBL	25–250 mL
Postop care	ICU or constant observation unit; neurologic function monitored
Mortality	0–3% (usually respiratory arrest)
Morbidity	Respiratory depression: 14% up to 5 d postop All others <5%: Infection Neurologic: dysphagia and ↓ gag reflex Aseptic meningitis CSF leak Massive blood loss; vertebral artery or transverse sinus injury
Pain score	5–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Infant to 70 yr; Chiari I: average age = 41 yr
Male:Female	~1:1
Incidence	Relatively rare neurosurgical procedure
Etiology	Congenital; acquired, S/P lumboperitoneal shunting
Associated conditions	Hydrocephalus; syringomyelia; scoliosis; myelodysplasia

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Cervical Neurosurgical Procedures.

Suggested Readings

1. Cesmebasi A, Loukas M, Hogan E, et al: The Chiari malformations: a review with emphasis on anatomical traits. Clin Anat 2015; 28(2):184–94.
2. Hankinson TC, Klimo P Jr, Feldstein NA, et al: Chiari malformations syringohydromyelia and scoliosis. Neurosurg Clin N Am 2007; 18(3):549–68.
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STEREOTACTIC NEUROSURGERY

SURGICAL CONSIDERATIONS

Steven D. Chang and Navjot Chaudhary

Description

Stereotaxis applies simple rules of geometry to radiographic images to allow precise localization within the brain. Such techniques can be applied to a variety of neurosurgical procedures, providing up to 1 mm accuracy. This precision makes it possible to perform certain intracranial procedures less invasively. Methods include both frame-based and frameless navigation systems, as well as specific stereotactic applications for functional neurosurgery and radiosurgery.

Frame-based stereotaxy

Stereotactic localization was developed using frames; however, with the widespread use of image-guided, “frameless” stereotactic systems, frame-based stereotaxy is rarely

used. These procedures begin with the attachment of a frame to the patient's head using four screws that anchor it to the skull. This is typically done outside the OR, using local anesthetic and, sometimes, light sedation. In the cooperative adult, frame application takes only 5–10 min; however, GA typically is used for children. Once the frame is in place, access to the patient's airway is restricted. A key for emergency removal of the frame must be kept with the patient at all times. With the stereotactic frame in place, CT, MRI, and/or cerebral angiography are used for target identification and localization. During imaging, a set of fiducials (radiographically visible markers) is attached to the frame. These fiducials provide the geometric reference points needed for localization. Data from imaging studies are used to calculate the spatial coordinates of the target(s). In nearly all cooperative patients, there is no need for sedation or analgesia during this stage of the procedure.

Frame-based stereotaxy is used most frequently for brain biopsies; however, drainage of cystic lesions or placement of a catheter or depth electrodes also can be performed. These common procedures involve relatively minor surgery, usually requiring only light sedation and local anesthesia. After prepping and draping, specialized operating tools are attached to the stereotactic frame. These instruments provide spatial guidance throughout the surgical procedure. A burr or twist drill hole is the usual form of intracranial access. Maximal sedation is needed during drilling and dural opening. Anesthetic considerations include ensuring adequate oxygenation and ventilation, which may be compromised by the sterile drapes. In addition, sedation should be light enough at times to allow the neurosurgeon to test gross neurologic function, such as extremity movement, both during and immediately after biopsy. At the completion of the case, the frame is removed and the patient is brought to the recovery room.

Image-guided “frameless” stereotaxy

Over the last several years, techniques have been developed to provide neurosurgeons with a real-time computer display of position and trajectory. Spatial information is displayed on high-speed computer workstations as two-dimensional images and three-dimensional renderings obtained from CT and/or MRI. These devices have the added advantage of being frameless; instead of a frame providing a system of reference, small markers (fiducials) are affixed to the scalp and forehead with adhesive. Advances in image-guided navigation also allow the incorporation of ultrasound and endoscopy to provide intraoperative updating of the preoperative radiographic images.

Image-guided surgery begins with an imaging study performed with fiducials in place. The remainder of the case is performed in the OR, usually under GA. After the patient's head is appropriately fixed and positioned, the locations of the fiducials are entered into

a computer, using any one of several different commercially available digitizing techniques. The most common system uses triangulation of infrared light from LEDs to determine the position of a pointer in space. The computer calculates the position of the pointer with respect to the patient and displays on the monitor the images from the scan with a representation of the pointer superimposed. Because the process of localization assumes a constant frame of reference, it is important that fiducials do not move between the time of imaging and registration in the OR. A further refinement, the dynamic reference frame, may be affixed to the craniotomy headrest to allow intraoperative repositioning of the patient. The OR must be set up to accommodate the computer and monitor as well as to allow a direct line-of-sight between the operative site and an array of infrared sensors (see [Fig. 1.1-24](#)). Care must be taken to avoid patient movement during the initial registration of the fiducials because this introduces a component of error with respect to the accuracy of the navigation system. The camera system can be repositioned during the case and still maintain accuracy.

Robotic stereotaxy: clinical application of robotic systems include stereotactic placement of intracranial electrodes, biopsy needles, laser probes, and intraventricular catheters. ROSA™ (Zimmer Biomet) and Renaissance® (Mazor Robotics), two of the more widely employed robotic systems, have been used for both frame-based (ROSA™) and frameless stereotactic procedures. Targeting errors are reported to be <2 mm. Complication rates are similar to traditional surgery, but robot assistance may offer the benefit of significantly reduced operating time in experienced hands.

The spatial information provided to neurosurgeons by image guidance can be used to access deep or critically located lesions through the most direct trajectory and a smaller craniotomy. These techniques have been used primarily during craniotomy for tumor or vascular malformation; however, image guidance also can be used for brain biopsy, shunt placement, cyst drainage, or other smaller procedures. As with most craniotomies, GA is used. Maneuvers that alter the spatial relationship of the intracranial contents, such as hyperventilation, diuresis, or CSF drainage, will result in brain shift relative to the presurgical images and should be avoided if possible, particularly during the early stages of surgery.

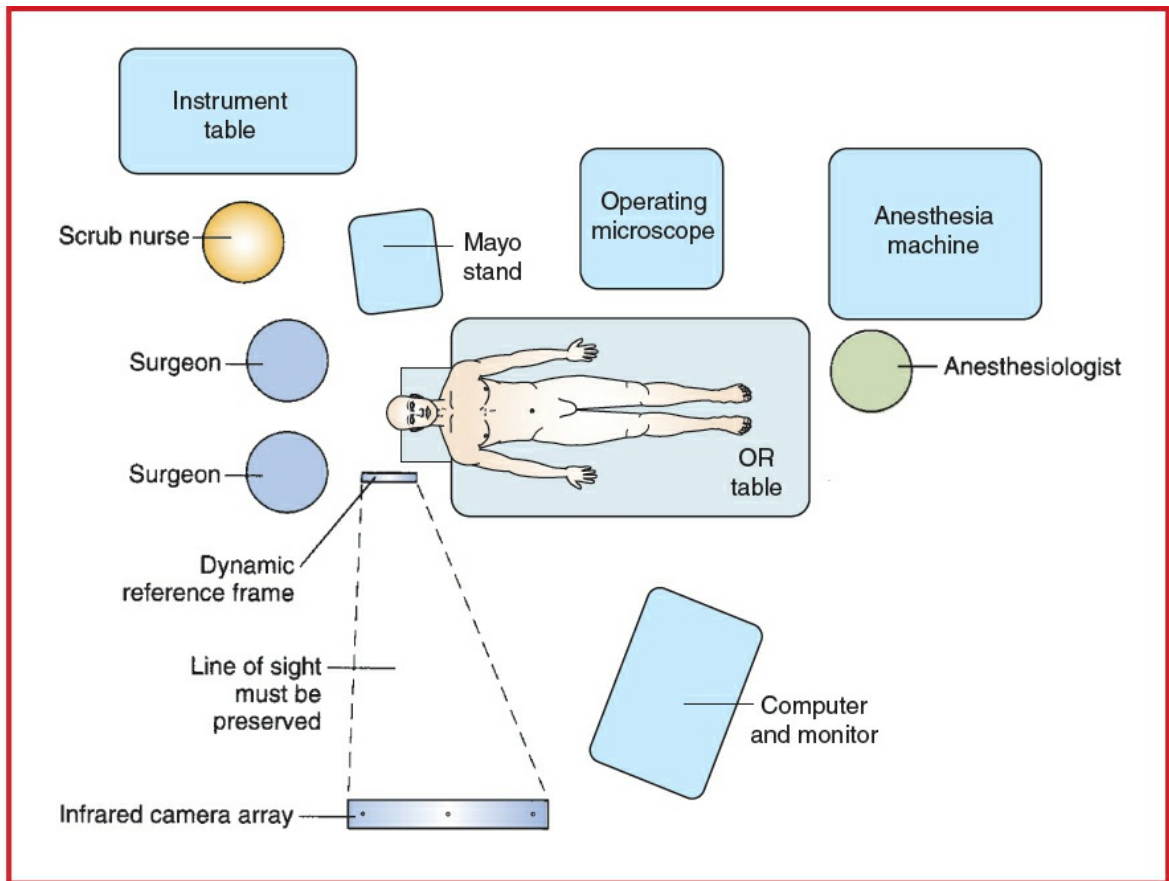


Figure 1.1-24 OR layout for frameless stereotactic surgery.

Radiosurgery

This technique uses focused radiation to ablate small or residual tumors and vascular malformations. It can also be used to treat some pain, psychiatric, and movement disorders. In contrast to other stereotactic operations, radiosurgery is performed outside the OR, using a linear accelerator or other specialized irradiation device. Cooperative adults need little or no sedation during the procedure. Children, however, almost always require GA from the beginning of frame placement through the actual treatment, which may last from 5 to 8 h. Consequently, anesthesia must be administered at several locations in the hospital, as well as during transport between these sites. Ideally, GA is initiated in the imaging suite just prior to frame placement. After transport to the recovery unit, the child is maintained under anesthesia while radiosurgical treatment is being planned, which takes from 1 to 3 h. The patient is then transported to the radiosurgery suite, still anesthetized, and treatment is performed. Depending on complexity, radiosurgery itself may last from 1 to 2 h. At the completion of treatment, the stereotaxic frame is removed and the child is awakened.

Newer, frameless radiosurgical systems can obviate the need for GA, and most children can be set up and treated with iv sedation alone. However, the setup and treatment are

usually performed on different days, and the child can go home while the treatment plan is created. These systems also allow for fractionation of the total dose, which may spare damage to surrounding critical structures.

Functional neurosurgery

Surgery using precisely placed lesions (e.g., **pallidotomy**, **thalamotomy**) or deep brain stimulators for the treatment of movement disorders, pain, and certain psychiatric syndromes are encompassed by the term “**functional stereotaxy**.” These techniques depend on absolutely precise localization of specialized instruments. These surgeries traditionally have required the use of a stereotactic frame. However, advances in imaging and image-guided navigation have obviated the need for a stereotactic frame. Concurrent neuroanatomic localization using electrophysiologic techniques is routinely used, regardless of the stereotactic system employed. To preserve neural potentials and allow the patient to cooperate with neurologic testing, GA is not used. Frameless procedures begin with placement of the fiducials (small screws) and imaging. Typically, the fiducials are implanted in the patient’s skull using local anesthetic and a special screwdriver. Then, the patient undergoes a CT scan, which is uploaded into the image-guided navigation system. The surgeon can then plan the trajectory in the computer. In the OR, the patient’s head is placed in Mayfield pins, and the image-guided navigation system is set up and the fiducials are registered. A small skin incision is made, and a burr hole is placed with the patient under iv sedation. The patient is then awakened and must remain awake and cooperative throughout a several-hour procedure, which may require patient participation in various neurologic tests. The recording electrodes are positioned using the image-guided navigation system and the electrophysiologic monitoring. After the proper electrophysiologic position is obtained, the recording electrode is removed and the deep brain stimulator is placed in the exact same position and depth. After the deep brain stimulator is placed, a subgaleal pocket may be created to house the leads until they can be connected to the pulse generator. Frame-based procedures begin with the attachment of a rigid frame to the patient’s skull with local anesthesia. The stereotactic frame provides an immobile reference that aids in targeting. The remainder of the procedure is similar to the frameless procedure previously described.

Usual preop diagnosis

Brain disease requiring biopsy; brain tumor; vascular malformation; Parkinson disease; hydrocephalus; pseudotumor cerebri; pain syndromes; psychiatric syndromes; cyst drainage

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should

include continuing chronic pain and AED medications, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K](#).

■ SUMMARY OF PROCEDURES				
	Frame-Based Stereotaxy	Frameless Stereotaxy	Radiosurgery	Functional Neurosurgery
Position	Supine or prone	Sitting, supine, prone, or lateral	Supine or prone	Supine; frameless or head fixed in frame
Incision	0.5–2 cm, scalp	1–12 cm, scalp	None	1–3 cm, scalp
Special instrumentation	Stereotactic treatment arc ± Robotic assistance	Image guidance system ± Robotic assistance	Linear accelerator (LINAC) CyberKnife	Electrophysiologic monitoring equipment, image-guided navigation, C-arm ± Robotic assistance
Unique considerations	Key for emergency removal of frame	No patient movement during registration. Clear line-of-sight between camera and dynamic reference frame	Extended anesthesia in several different locations for pediatric patients	Patient is usually awake, able to cooperate with testing. Key for emergency removal of frame (if used)
Antibiotics	Cefazolin 2 g iv	⇐	None	Cefazolin 2 g iv
Surgical time	0.5–1.5 h	2–6 h	5–8 h	2–6 h
EBL	<10 mL	20–1000 mL	None	<100 mL
Postop care	PACU	ICU or PACU	PACU	PACU
Mortality	<0.5%	1%	None	<1%
Morbidity	Overall: <1% Symptomatic intracerebral hemorrhage	Infection Air emboli Hemorrhage Stroke Sz Worsening edema	N/V: 2–3%	N/V: 1–10% Intracerebral hemorrhage New neurologic deficits
Pain score	2	4	2	2

PATIENT POPULATION CHARACTERISTICS				
Age range	All ages	⇐	0–12 yr	18–90 yr
Male:Female	1:1	⇐	⇐	⇐
Incidence	Uncommon	Unusual	⇐	Rare
Etiology	Tumor; infection; AIDS; demyelinating syndromes	Tumor; vascular malformations; epilepsy	⇐	Parkinson disease; pain syndromes; psychiatric syndromes

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for specific neurosurgical procedures, e.g., tumor resection, functional neurosurgery, etc.

A variety of neurosurgical procedures are performed using stereotactic guidance especially when the lesion is small and/or is located deep within brain tissue or as a means of obtaining a biopsy of a lesion for diagnosis. For example, focal, deep-seated AVMs may be resected under stereotactic, or image-guided, control. These patients are often otherwise healthy. In adults, fiducial markers or a stereotactic frame are placed, and the patient is taken to the radiologic suite for CT/MRI scan to determine stereotactic coordinates before surgery. The patient is then brought to OR with the frame or fiducial markers in place. The tool for removing the stereotactic frame must be readily available in the event of an airway emergency. Biopsies may be accomplished under local anesthesia with MAC or GETA. If a complete resection is planned (e.g., AVM resection), GETA is used.

Suggested Readings

1. Fomenko A, Serletis D: Robotic stereotaxy in cranial neurosurgery: a qualitative systematic review. *Neurosurgery* 2018; 83(4):642–50.
2. Holloway KL, Gaede SE, Starr PA, et al: Frameless stereotaxy using bone fiducial markers for deep brain stimulation. *J Neurosurg* 2005; 103(3):404–13.
3. Kemmerer E, Shah S: An update on radiation therapy of brain metastases. *Surg Oncol Clin N Am* 2017; 26(3):347–55.
4. Risholm P, Golby AJ, Wells W III: Multimodal image registration for preoperative planning and image-guided neurosurgical procedures. *Neurosurg Clin N Am* 2011; 22(2):197–206, viii.
5. Venkatraghavan L, Manninen P, Mak P, et al: Anesthesia for functional neurosurgery: review of complications. *J Neurosurg Anesthesiol* 2006; 18(1):64–7.
6. Woodworth GF, McGirt MJ, Samdani A, et al: Frameless image guided stereotactic brain biopsy procedure: diagnostic yield, surgical morbidity, and comparison with

the frame based technique. J Neurosurg 2006; 104(2):233–7.

Suggested Viewing

Links are available online to the following videos:

Endoscopic Pituitary Surgery: https://www.youtube.com/watch?v=D_UQS345C8w

Adenosine for Temporary Flow Arrest During Intracranial Aneurysm Surgery
<https://www.youtube.com/watch?v=2KkgbtocnNY>

Cavernous Malformations of the Brainstem: A Supracerebellar Infratentorial Approach
<https://www.youtube.com/watch?v=1jl62S2xH6c>

Chronic Subdural Hematoma: https://www.youtube.com/watch?v=_ct7S72NzPQ

Neurosurgery: Craniopharyngioma Supra-Orbital Removal: <https://www.youtube.com/watch?v=gyVALITTFQQ>

Superficial Temporal Artery Dissection: https://www.youtube.com/watch?v=djL_utmksEI

Surgery for Giant PCOM Aneurysms Video 1: <https://www.youtube.com/watch?v=QhJ8NElcLIw>

The Middle Fossa Approach and Extended Middle Fossa Approach Video 1
<https://www.youtube.com/watch?v=OLgkeJ1u3kw>

Intra-operative Real-time Querying of White Matter Tracts: Frameless Stereotactic Neuronavigation
<https://www.youtube.com/watch?v=KrGkGPR5hs0>

Bypass for Complex Basilar Aneurysms: <https://www.youtube.com/watch?v=yQu0w4xr1aU>

Neuroendoport (SM) Technique—Minimally Invasive Brain Surgery: <https://www.youtube.com/watch?v=z6gidGVkzno>

Aneurysm Clip Surgery—Middle Cerebral Artery: <https://www.youtube.com/watch?v=8vCBMiEgXQg>

Fluorescein for Glioma Surgery: <https://www.youtube.com/watch?v=uljTeU6Y6e4>

CHAPTER 1.2

Functional Neurosurgery

SURGEONS

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FUNCTIONAL NEUROSURGERY—INTRODUCTION: THE SURGICAL TREATMENT OF PAIN, MOVEMENT DISORDERS, AND EPILEPSY

INTRODUCTION

Jaimie M. Henderson, MD

Functional neurosurgery differs from most surgical subspecialties in that it aims to alter the function of the nervous system, rather than addressing anatomical abnormalities. The most common treatment areas include chronic pain, epilepsy, and movement disorders. Each of these disease states has a distinctive anesthetic requirement dictated by the specific pathophysiology and subsequent intraoperative objectives. Procedures may be performed under local anesthesia, conscious sedation, or GA and can be divided into ablative (destructive) or augmentative (nondestructive) techniques. Ablative techniques are less frequently performed because of their unpredictable duration of effect and irreversible nature of the lesion. Neuroaugmentation (electrical stimulation or continuous local drug administration) tends to produce more lasting benefits and, more importantly, can be changed to meet fluctuations in the patient's symptoms.

One of the keys to successful functional neurosurgical therapies is target

identification (Fig. 1.2-1). Many targets can be identified anatomically on appropriate radiographic studies, whereas others may be difficult to visualize even with special techniques. However, even areas that can be targeted anatomically may vary in terms of their physiologic function. Thus, it is important to incorporate physiologic testing in many functional neurosurgical procedures. Most physiologic confirmation techniques require that the patient be awake and cooperative during testing. Thus, the need for physiologic identification often dictates a specific anesthetic technique.

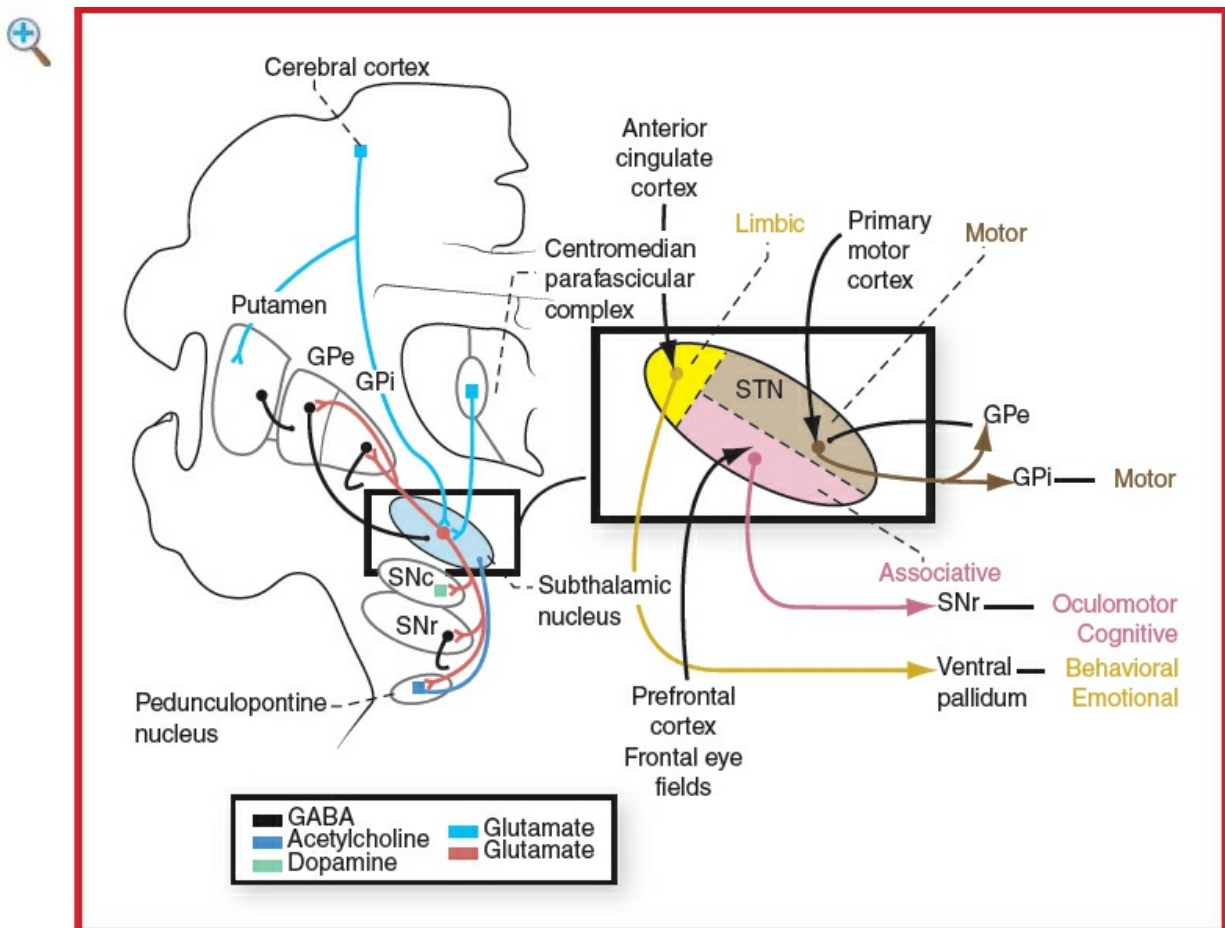


Figure 1.2-1 Topographical anatomy of the ventrolateral thalamus. GPe, external segment of the globus pallidus; GPi, internal segment of the globus pallidus; SNc, substantia nigra pars compacta; SNr, substantia nigra pars reticulata; STN, subthalamic nucleus.

STEREOTACTIC PROCEDURES: DEEP BRAIN STIMULATION, LASER INTERSTITIAL THERMAL THERAPY, AND MR-GUIDED FOCUSED ULTRASOUND

SURGICAL CONSIDERATIONS

Description

Many procedures are currently performed with **frame-based stereotactic techniques**, with intracranial access achieved via a burr hole or twist drill. Recently, the development of **MR-guided focused ultrasound** has allowed precise lesioning of deep brain structures without a cranial opening (see below). There is a growing trend to use “frameless” stereotactic approaches, with two major systems currently in use. Real-time frameless localization can be achieved by replacing the stereotactic frame with rigidly attached skull markers (“fiducials”) that provide a fixed reference. Another innovative alternative is to produce a custom-made fixed trajectory guide using rapid prototyping techniques. Both of these approaches have been used successfully and appear to be equivalent in accuracy to a stereotactic frame. Deep brain stimulation (DBS) electrodes and lesioning probes such as lasers can also be placed with real-time guidance in the MRI suite. These procedures are done under GA, with anatomic rather than physiologic target confirmation.

Two approaches to identification of the functional target are used following initial stereotactic CT or MRI radiographic localization; both require patient cooperation. In one approach, the target is confirmed by assessing symptomatic resolution during high-frequency macrostimulation and by identifying surrounding structures using their characteristic stimulation-evoked responses. In the other approach, before stimulation testing, single-neuron recordings are performed to localize the appropriate target through somatotopic kinesthetic and/or somatosensory responses. In an awake patient, appropriate stimulation is delivered to the skin or by passive and active movement of the joints. This technique utilizes specialized high-impedance microelectrodes and amplifiers susceptible to interference from monitoring equipment, which may necessitate manual measurements of BP and clinical assessment of oxygenation. Anesthesia or sedation can modify neuronal activity significantly and, thus, interfere with functional mapping. For example, propofol has been shown to inhibit globus pallidus and subthalamic neurons for several minutes beyond its behavioral effects. In addition, many of these agents have been shown to change evoked potential (EP) responses, raising the possibility that they could alter stimulation thresholds for either the internal capsule or optic tract.

A variety of movement disorders are amenable to surgical treatment. They can be divided into akinetic (e.g., Parkinson’s) and hyperkinetic (e.g., dystonias, tremor, spasticity) syndromes. Idiopathic Parkinson disease (PD) is best treated with deep

brain stimulation (DBS) targeted to a functional subcomponent of either the subthalamic nucleus or globus pallidus interna (Fig. 1.2-2). Tremor syndromes are treated by targeting the ventral intermedius (Vim) thalamic nucleus (MR-guided focused ultrasound or stimulation), whereas dystonias are treated by stimulation of the globus pallidus interna. Spasticity is not treated stereotactically but through the administration of intrathecal baclofen (GABA-like synaptic inhibition) in the lumbar cistern via an internalized pump system. Abrupt loss of intrathecal infusion of baclofen, however, can quickly lead to a serious withdrawal syndrome with dysautonomia, circulatory collapse, and death within hours. Spasticity refractory to intrathecal baclofen may be amenable to selective dorsal root rhizotomy performed through an open laminectomy.

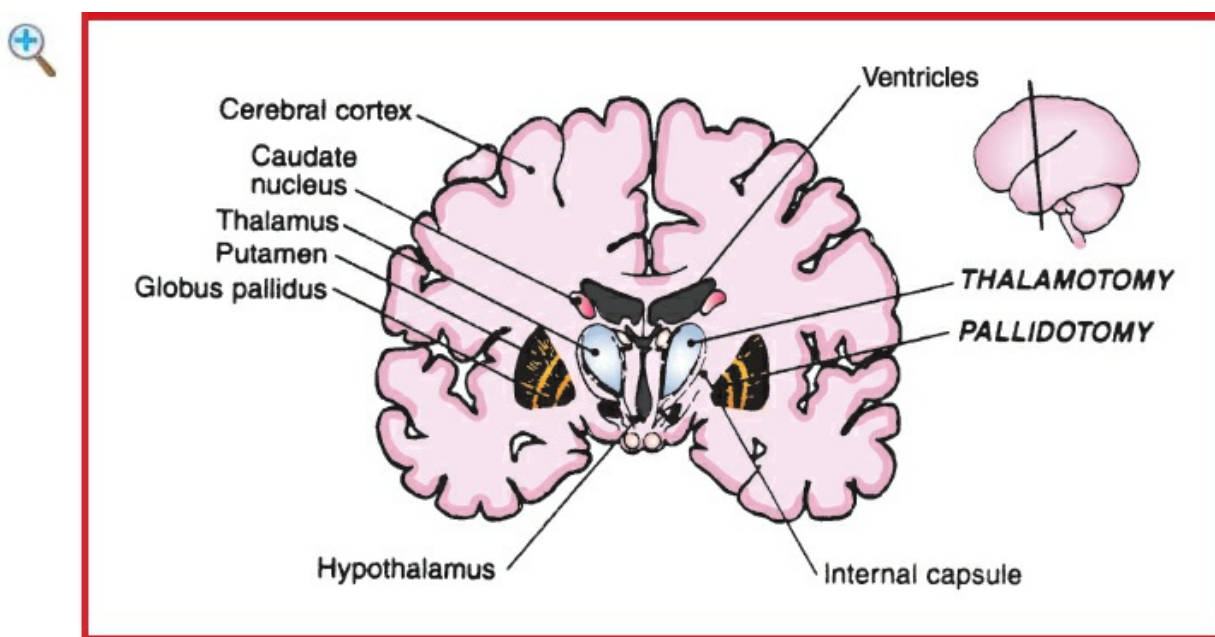


Figure 1.2-2 Anatomical locations for therapeutic lesions (thalamotomy and pallidotomy) for the surgical treatment of Parkinson disease. Inset shows the plane of the coronal section through the diencephalon, identifying the lesions. (Reproduced with permission from Mason LJ, Cojocaru TT, Cole DJ: Surgical intervention and anesthetic management of the patient with Parkinson's disease. In: Jaffe RA, Giffard RG, eds. *International Anesthesiology Clinics: Topics in Neuroanesthesia*. Little, Brown, Boston: 1996; 4(34):141.)

For DBS, mapping and implantation of the intracranial stimulating electrode is performed under local anesthesia to permit monitoring of behavioral and physiologic responses. A 2- to 3-cm linear incision is placed near the coronal suture and 20–50 mm from the midline. This can be accomplished with local anesthesia under propofol sedation for patient comfort. Subsequently, minimal or no sedation is used, as patient cooperation is necessary during the functional mapping component of the case. If single-neuron recordings are used, propofol should be

discontinued at least 20 min in advance of mapping because it can produce prolonged suppression of target neuronal activity. If analgesia is needed, use a low-dose (0.01–0.05 mcg/kg/min) remifentanyl infusion. Local anesthetics should not contain epinephrine. Avoid central-acting β -blockers (e.g., propranolol), which suppress tremor activity in target neurons. Sedation with meperidine in patients taking selegiline is contraindicated. No dopaminergic antagonist (e.g., metoclopramide, droperidol) should be given to parkinsonian patients or patients with dopamine-responsive dystonia. To enhance single-cell responses, withhold medications prescribed for target symptoms for 8–24 h. In some cases (e.g., Parkinson disease), this may result in a rebound HTN requiring active treatment; however, patients will resume their medications postop (often in the postanesthesia recovery area), and the HTN will resolve. The implantable pulse generator (IPG) may be placed in an infraclavicular subcutaneous pocket during a separate surgical procedure or at the time of the intracranial electrode implantation. Implantation of the IPG and subcutaneous tunneling of the electrode lead can be done under MAC but is best tolerated under GETA. To facilitate intubation, the calvarial wound is closed temporarily, and the stereotactic localizing apparatus is removed. Closure consists of a single, interrupted suture for stab wounds or two-layer suture/staple closure for a burr hole. IPG pockets are closed in two layers. The subcutaneous layer is closed with absorbable sutures, and the skin is closed with staples or sutures.

MR-guided focused ultrasound is performed in an interventional MRI scanner equipped with an array of several hundred high-intensity ultrasound transducers in a helmet that surrounds the patient's head. A stereotactic frame is placed under local anesthetic, following which the patient is placed into the transducer helmet on the scanner table. An elastic membrane allows cooled water to circulate around the head in order to provide a conductive medium for the ultrasound energy. The target in the Vim thalamus is placed at the focal point of the ultrasound array, and a lesion is made by gradually increasing the ultrasound energy over a number of 10- to 20-second sonications. Temperature is monitored in real time using MR thermography, and the patient is frequently evaluated during the procedure for side effects and reduction of tremor. Patients frequently experience near-complete tremor relief immediately following the procedure, with an average reduction in tremor over the long term of ~50%.

Usual preop diagnosis

Medically intractable idiopathic PD with disabling L-dopa-induced dyskinesia, bradykinesia, or rigidity; severe fluctuations in medication responses; dystonia

musculorum deformans; post-CVA dystonia; occasionally, torticollis; tremor; movement disorders

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and other medications (Parkinson medications typically held), limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Linear (burr holes); stab wound (twist drill)
Unique considerations	Minimal/no sedation. BP ↑ in patients who have their antihypertensive medications withheld. Manual vital sign monitoring if electronic monitors interfere with single-cell recordings. Be aware that BP cuff or iv's may produce sensations that distract the patient during awake stimulation testing for sensory side effects.
Antibiotics + other meds	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	3–5 h
EBL	25–150 mL
Postop care	Continue antibiotics at surgeon's discretion, Parkinson medications. BP may ↓. Maintain MAP < 90 mm Hg to prevent postop bleeding. PACU
Mortality	<0.5%
Morbidity	Overall: 8–27% Infection: <10% Hardware failure: Rare Intracranial hemorrhage: 3.3–6% Cognitive disturbances
Pain score	2–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	12–80 yr; PD: 45–60 yr at onset
Male:Female	PD: M > F
Incidence	PD: 1% > 60 yr; 4% > 80 yr
Etiology	Idiopathic; genetic (10–15% will have a 1st- or 2nd-degree relative with PD)
Associated conditions	Dystonia; PD: cognitive impairment, depression, dysphagia, orthostatic hypotension

■ ANESTHETIC CONSIDERATIONS

Parkinson disease (PD) is the most common movement disorder, affecting ~1% of the population >60 yr. It is the result of the loss of dopaminergic neurons in the basal ganglia → ↓ dopamine (dopamine/acetylcholine imbalance) in basal ganglia → movement disorder. Medical treatment is directed primarily to restoring dopamine levels by increasing the availability of the dopamine precursor (L-dopa), by inhibiting liver dopa decarboxylase (carbidopa, usually given in combination with L-dopa as Sinemet), by releasing endogenous dopamine (amantadine [Symmetrel]), and by blocking MAO-B (selegiline [Eldepryl], rasagiline [Azilect]). Medical treatment also may include dopamine agonists (pergolide [Permax], bromocriptine [Parlodel]) and acetylcholine antagonists (amantadine [Symmetrel], benztropine [Cogentin]) to correct the dopamine/acetylcholine imbalance. Patients presenting for surgery usually have failed medical therapy. They will have been taken off their antiparkinsonian medications 8–24 h before surgery. This will maximize their symptoms to help assess treatment effects intraop; thus, preop assessment on the day of surgery may be difficult.

■ PREOPERATIVE

Respiratory	Autonomic dysfunction → esophageal dysfunction → ↑ risk of aspiration. Patients typically have ↓ vital capacity 2° rigidity, dyskinesia → ↓ respiratory function, and ↓ cough. Laryngospasm and respiratory failure may occur following abrupt withdrawal of antiparkinsonian medications. Tests: CXR; consider ABG; PFTs may be difficult to obtain.
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Cardiovascular	Autonomic dysfunction → orthostatic hypotension. Dopamine replacement therapy → cardiac dysrhythmias, ↓ BP, and hypovolemia. In patients taking selegiline (MAOI): avoid meperidine → ↑↑ BP, rigidity, agitation; sympathomimetics → exaggerated ↑ BP.
Neurologic	The primary Sx of Parkinson disease includes rigidity, tremor, bradykinesia, and muscle weakness. Secondary symptoms include dementia, depression, and speech difficulty. Patients may alternate between a state of immobility and one of exaggerated tremor (which may interfere with intraop monitoring). PD medications are sedating—modafinil may be used to improve daytime function.
Renal	Urinary retention is common.
Gastrointestinal	Autonomic dysfunction → gastroparesis, ↑ incidence of reflux. Poor nutrition. Pharyngeal muscle dysfunction → dysphagia.
Laboratory	As indicated from H&P.
Premedication	None. These patients may have received small doses of fentanyl and/or midazolam to facilitate placement of the stereotactic frame. For frameless procedures, fiducial markers or mounting screws will have been placed before surgery using local anesthesia.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water if tolerated by the patient.

INTRAOPERATIVE

Anesthetic technique

MAC, with minimal sedation limited to surgical opening and closing, as patient cooperation and typical neuronal activity are essential for intraop mapping and neurologic examination. The surgeon should have generously infiltrated the incision sites with local anesthetic.

Induction	Low-dose propofol (25–75 mcg/kg/min), or dexmedetomidine (0.1–0.4 mcg/kg/h), may be used for sedation during burr-hole placement. If propofol or dex is used, it should be discontinued 20 min before electrophysiologic recording or mapping. In the event of an airway emergency in framed procedures, a means of releasing the patient from the stereotactic frame must be readily available.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Avoid hypothermia. Surgeon will place scalp field blocks.	
Maintenance	BP control is essential to ↓ the risk of intracranial hemorrhage during passage of the recording and stimulating electrodes. If not otherwise contraindicated, MAP should be kept ~20% below normal for that patient. This usually can be accomplished without an arterial line by using an infusion of NTG and/or esmolol titrated to effect or by using small doses of atenolol (0.5–1 mg increments iv). Esmolol and atenolol are preferred over other parenteral β-blockers because CNS penetration is limited → minimal effect on central tremor.	
Emergence	Postop BP control can be continued by using atenolol or labetalol titrated to effect at the end of the procedure. The patient is usually transported to the PACU and then to the neurology ward for postop monitoring. Antiparkinsonian medication should be given in the PACU and may result in ↓ BP. Consider iv acetaminophen if po dose not given or given >4–6 h ago.	
Blood and fluid requirements	Minimal blood loss IV: 18–20 ga × 1 BSS @ 1–3 mL/kg/h	For unilateral procedures, iv should be placed in the ipsilateral arm (relative to side of surgery). Zero balance or goal-directed fluid management is

		recommended.
Monitoring	Standard monitors VAE monitoring if head is significantly elevated ($\geq 30^\circ$)	Exaggerated tremors and/or neurologic testing in these patients may interfere with monitoring. It may be helpful to place the BP cuff on a leg (less tremor). For unilateral procedures, monitors should be placed on the ipsilateral arm and leg.
Positioning	✓ and pad pressure points ✓ stereotactic frame clearance	Patient comfort may be improved by placing pillows under the knees to relieve lower back strain. A massage therapist can greatly improve patient comfort during the procedure.
Complications	Intracerebral hemorrhage Loss of airway in stereotactic frame VAE	Dx: ↓ mental status and hemiparesis. CT scan usually required to confirm Dx. Emergency craniotomy may be necessary. Rx: may require rapid removal of the stereotactic frame. Sx: cough, pleuritic chest pain, ↓BP, ↑HR.

POSTOPERATIVE

Complications	Intracranial hemorrhage Motor deficit Visual field deficit	Intracranial hemorrhage may require emergency craniotomy. The surgeon
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	Aphasia	should be notified immediately if any new deficits are noted.
Tests	None	Unless otherwise indicated.
Multimodal pain management	Usually not necessary Acetaminophen 1 g iv ± Fentanyl 25–50 mcg prn	Acetaminophen po if not given preop or if 4–6 h has elapsed since previous dose.
ERAS	Ondansetron 4 mg iv (no metoclopramide)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

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4. Elias WJ, Lipsman N, Ondo WG, et al: A randomized trial of focused ultrasound thalamotomy for essential tremor. *N Engl J Med* 2016; 375(8):730–9.
5. Fasano A, Daniele A, Albanese A: Treatment of motor and non-motor features of Parkinson's disease with deep brain stimulation. *Lancet Neurol* 2012; 11(5):429–42.
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SURGICAL ANALGESICS: SPINAL CORD STIMULATION, INTRATHECAL PUMPS, AND CORTICAL STIMULATION

SURGICAL CONSIDERATIONS

Description

Chronic pain arises from a variety of etiologies. It can involve both neuropathic and nociceptive processes and occur in a variety of anatomical distributions (e.g., radicular versus a hemibody pattern). Therapeutic interventions are dictated by the pathophysiology of the pain, its qualitative nature, etiology, and the patient's prognosis. Many common procedures for chronic pain are directed at the spinal cord and may consist of epidural or intrathecal medications or electrical stimulation.

For chronic spinal cord stimulation (e.g., postlaminectomy syndrome), an epidural electrode is implanted either percutaneously or via an open laminectomy. Percutaneous electrodes are easier to implant and have less associated surgical pain. The surgical electrode (implanted via laminectomy) confers greater mechanical stability in the epidural space. In addition, given its larger contact size, it can generate higher current densities with less drain on the implanted system. For percutaneous electrodes, a small skin incision is made two to three vertebral levels caudal to the target region of the spinal cord. For “surgical paddle” electrodes, the skin incision is made one to two levels caudal to the target zone of the spinal cord, and a laminectomy is performed to provide access to the epidural space. Incisions are closed in the standard fashion.

Most percutaneous electrode placements and some paddle electrode placements are done awake so that intraoperative test stimulation can be performed. The procedures are done in the prone or lateral position with consequent implications for airway management in the sedated patient. Localization of the electrodes is

accomplished initially based on radiographic criteria; however, these localizations are only approximate, and it is recommended that the electrode placement be confirmed by intraop stimulation. The patient needs to be sufficiently alert to communicate the quality, distribution, and intensity of the stimulation-induced paresthesias. Surgical paddle placement requires a variable extent of muscle dissection and laminectomy, which may be done under general anesthesia or with epidural anesthesia for patient comfort. If done under general anesthesia, response to paddle test stimulation may be measured by manual palpation of musculature stimulated at higher test voltage and/or using intraoperative electrophysiologic monitoring. To assess efficacy, the electrode may be externalized and percutaneous stimulation used for assessment. If there is $\geq 50\%$ reduction in pain, the patient may return to the OR for internalization of the implantable pulse generator (IPG). Postop, these patients can have an exacerbation of pain particularly if neuropathic in nature. They may require iv lidocaine or ketamine infusions to return them to their preop baseline, even with a functioning and appropriately located stimulating electrode.

For pain of central origin (e.g., post stroke, multiple sclerosis, and trigeminal nerve pathology), surgical procedures are often directed at the thalamus and cortex. Thalamic interventions include stereotactic insertion of stimulating electrodes into sensory thalamus. Thalamic DBS requires awake mapping of the somatotopic representation of the affected body part through either microelectrode mapping or stimulation-induced paresthesias. For medically intractable neuropathic pain syndromes, epidural motor cortex stimulation has shown mixed results. Functional mapping of the motor cortex for epidural stimulation can be done with anatomical and physiologic mapping under GA via a craniotomy, with the anesthetic tailored to minimize interference with EP recording. The incision consists of a 5- to 10-cm linear or 5×10 -cm trapezoidal incision placed over and paralleling the motor cortex.

Identification of the appropriate location on the motor strip can be done initially with epidural mapping using SSEPs elicited from the part of the body where the pain originates. In cases of denervation syndromes, SSEP may not be present, and the minimum intensity point for epidural stimulation-induced movements is used. In rare cases, the surgeon may elect to perform the surgery awake to facilitate mapping. To assess efficacy, the electrode may be externalized and percutaneous stimulation assessed for overall therapeutic efficacy. If there is $\geq 50\%$ reduction in pain, the patient will return to the OR for internalization of the IPG. Closure consists of calvarial reconstruction with externalization of leads. The craniotomy

incision is closed in the standard fashion. If a percutaneous trial is done, the IPG implant requires a second GETA with reopening of wounds.

The mainstay of surgical treatment of trigeminal neuralgia is microvascular decompression of the trigeminal nerve in the prepontine cistern. In patients who are poor surgical candidates, treatment is accomplished with ablative procedures aimed at the gasserian ganglion, using chemolytic, mechanical, or RF techniques. Patients who fail gasserian ganglion interventions or have atypical facial pain may be candidates for thalamic DBS or motor cortex stimulation. Stereotactic radiosurgical techniques are increasingly utilized as an ablative treatment for trigeminal neuralgia, particularly in the elderly (>65 yr old) population or in other surgically averse candidates. Radiosurgery employs highly focused beams of radiation to partially ablate the intracisternal portion of the trigeminal nerve while sparing surrounding structures (brain stem and other cranial nerves). The downside to radiosurgery is that the effects and consequent pain relief may take weeks to months to manifest, as opposed to other surgical and ablative treatments whose effects are immediate.

For pain unresponsive to spinal cord stimulation or because of unacceptable side effects of parental medications, continuous intrathecal administration of analgesics can be accomplished with an implantable medication delivery system. An incision is made over the L3-L4, L4-L5, or L5-S1 spinous process and a second incision is placed in the RLQ of the abdomen. An intrathecal catheter is then placed and anchored to the lumbodorsal fascia. A tunneling tool is used to bring the catheter from the lumbar spinal region to the abdomen. This tunneling procedure is rarely tolerated without GETA or deep sedation. A reservoir or continuous-delivery pump is then placed in the abdomen and attached to the catheter. Both incisions are closed in two layers.

Intrathecal narcotics (typically morphine) can produce satisfactory results because therapeutic drug concentrations can be achieved at the level of spinal opiate receptors without influencing higher CNS opiate systems. This therapeutic intervention can be assessed through a percutaneous catheter trial. Patients who experience $\geq 50\%$ reduction in pain are candidates for a totally implanted pump system. These are low-morbidity procedures (5–10%) with infections and hardware failures constituting the greatest problems. Abrupt cessation of medications—through either patient noncompliance with refill schedules (10–12 wk) or hardware failure—however, can lead to a serious withdrawal syndrome.

Usual preop diagnosis

Chronic pain

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and AED meds, limiting NPO duration alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Prone/lateral
Incision	2–3 levels above target area (spinal cord stimulation) + laminectomy; L3, L4, L5 (intrathecal pump) + 2nd incision RLQ of the abdomen
Unique considerations	Patient communication important; therefore, avoid over sedation. Copious use of local anesthetics (avoid overdosing)
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	1.5–3 h
EBL	25 mL
Postop care	May have ↑ pain during recovery period
Mortality	<0.5%
Morbidity	Overall: 5–10% Infection Hardware failure Withdrawal Sx
Pain score	3–8

■ PATIENT POPULATION CHARACTERISTICS	
Age range	20–75 yr
Male:Female	1:1
Incidence	Common
Etiology	Iatrogenic; neuropathic; traumatic

Associated conditions

Opiate dependency

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients with chronic pain often present unique challenges to the anesthesiologist and surgeon. They may be intolerant of any additional pain (e.g., iv placement, positioning on the OR table, postop pain) and seemingly immune to typical doses of systemic analgesics while remaining vulnerable to respiratory depression. In most cases, patients should continue their usual pain medications and co-analgesics until ~2 h before surgery. Typically any transdermal fentanyl patch should be left in place unless its location will be subject to intraop heating, compression, or other manipulation. A detailed h/o of patient pain, effective (and ineffective) pain management techniques and drugs, and the current medication regime (including the use of both over-the-counter and under-the-counter drugs and herbals) is essential.

Respiratory	Chronic pain → ↓ physical activity → ↓ respiratory reserve.
Cardiovascular	Ventricular dysrhythmias have been associated with methadone at doses >200 mg/d. Chronic pain → ↓ physical activity → ↓ cardiovascular reserve.
Neurologic	A careful neurologic assessment is essential to detect preexisting deficits.
Psychological	These patients often benefit from a detailed explanation of the surgical and anesthetic procedures including expectations and contingency plans for intraop pain management if an awake procedure is planned. These patients may have a higher incidence of coexisting psychiatric disease and often have high scores on standardized measures of hysteria and hypochondria. Patients on serotonergic drugs or antipsychotic medication may develop serotonin syndrome or neuroleptic malignant syndrome (hyperthermia, muscle rigidity, encephalopathy, autonomic dysfunction).

Gastrointestinal	Chronic opioid use may →↓ gastric emptying and ↓ GI motility with chronic constipation. Inactivity 2° chronic pain → obesity.
Laboratory	As indicated from H&P. For patients on anticonvulsants, drug levels are useful to exclude toxic effects.
Premedication	Chronic opioid use may make the patients at risk for gastric aspiration 2° ↓ bowel motility. Consider metoclopramide 10–20 mg iv 30-min preinduction with Na citrate 30 mL po on transport to the OR. Midazolam iv titrated to effect can be beneficial.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water.

INTRAOPERATIVE

Anesthetic technique

The placement of epidural electrodes, intrathecal cannulae, and cortical stimulators is often best accomplished under GA using anesthetic techniques appropriate for spinal or intracranial procedures. Accurate electrode placement may require an awake patient during at least a portion of the procedure. For intracranial electrode placement, see the anesthetic technique for awake craniotomy. These same general principles can also be used for spinal cord electrode placement, although regional techniques are usually simpler and just as effective. The following description pertains to the placement of electrodes for spinal cord stimulation.

Thoracic epidural	Following the application of standard monitors and with an iv catheter in place, the patient should be seated carefully on the edge of the operating table. Provide support for feet and arms as necessary. The epidural injection should be at the level of the surgery (usually 2–3 levels below the electrode target site). A single bolus injection of 8–10 mL of 0.75% bupivacaine usually provides excellent surgical anesthesia without compromising the ability
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	to map the more rostral portion of the spinal cord. It is important to avoid over sedation and to minimize the use of systemic analgesics in order to ensure patient cooperation with accurate spinal cord mapping.	
Blood and fluid requirements	IV: 18 ga × 1 BSS @ 1–3 mL/kg/h	Zero balance or goal-directed fluid management is recommended. Recommend zero balance fluid management (see Appendix K, p. K-2).
Monitoring	Standard monitors	
Positioning	Prone ✓ and pad pressure points ✓ eyes	The patient should self-position on bolsters (hips and upper chest) or Wilson frame.
Complications	Spinal cord injury Serotonin syndrome (SS) vs. neuroleptic malignant syndrome (NMS)	May be precipitated by meperidine, droperidol, metoclopramide, oxycodone among others. Ddx can be difficult. Dantrolene may be effective for NMS but not for SS.

POSTOPERATIVE

Complications	Spinal cord injury Epidural hematoma Respiratory distress Drug withdrawal symptoms	Prompt postop neurologic assessment is essential. May occur in response to the patient's usual opiate dose if the spinal stimulation has produced
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		significant pain relief.
Tests	None unless otherwise indicated	Surgeons may request specific tests.
Multimodal pain management	Typically complex and multimodal	Patient often intolerant of postop pain while remaining susceptible to narcotic overdose.
	Acetaminophen 1 g iv	Acute pain service consult recommended.
	± Fentanyl 25–50 mcg prn	If not given preop or if 4–6 h has elapsed since previous dose.
	Reinforce scalp blocks	0.5% bupivacaine with epi is as effective as liposomal bupivacaine.
ERAS	Ondansetron 4 mg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Aaronson ST, Sears P, Ruvuna F, et al: A 5-year observational study of patients with treatment-resistant depression treated with vagus nerve stimulation or treatment as usual: comparison of response, remission, and suicidality. *Am J Psychiatry* 2017; 174(7):640–8.
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VAGUS NERVE STIMULATION

SURGICAL CONSIDERATIONS

Description

The surgical treatment of medically intractable epilepsy consists of either **surgical resection** of the epileptic site or **vagus nerve stimulation (VNS)**. For patients who are not candidates for resective surgery, VNS can be a viable surgical strategy. Although it is a low-morbidity procedure, its efficacy also is low. VNS tends to produce a 50% or greater reduction of Sz in ~48% of patients after 18 mo of stimulation. By contrast, **temporal lobectomy** (often with amygdalohippocampectomy) for patients with medial temporal lobe epilepsy has an 80–85% chance of seizure freedom in carefully selected patients whose seizures are refractory to medications. Surgery for **extra temporal epilepsy** originating in neocortex has a 50% chance of making the patient seizure free, in the absence of a structural lesion. (If an anatomical abnormality is present, then that figure increases to ~75%.) In general, the complication rate is ~7%, dependent on the exact location of the Sz focus and its relationship to eloquent cortex ([Fig. 1.2-3](#)).

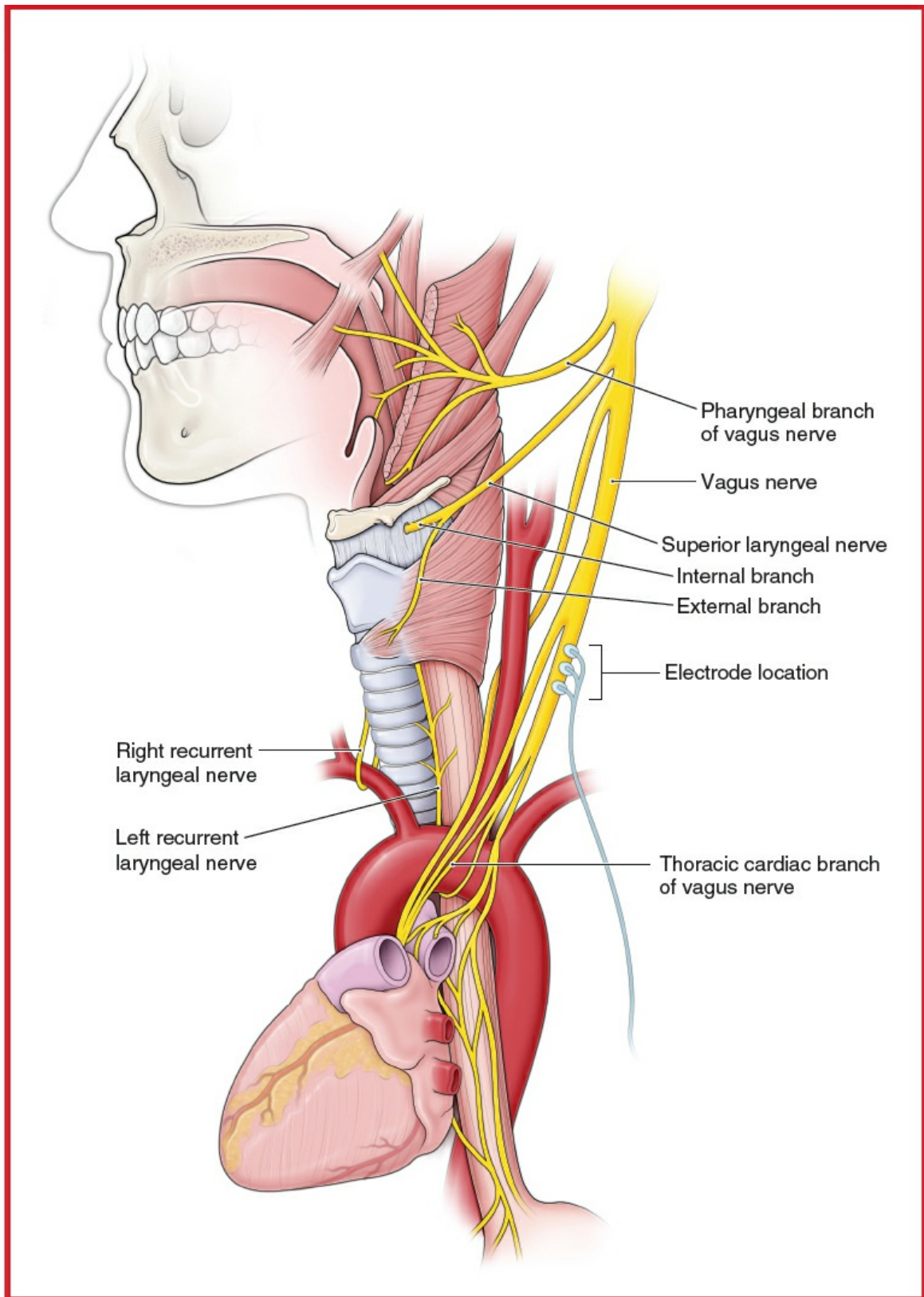


Figure 1.2-3 Surgical anatomy for placement of vagal nerve stimulator electrodes.

Left vagus nerve stimulation is also used to treat some types of medically refractory depression. As with its use in epilepsy, the exact mechanism of action

remains unclear.

Two incisions are used for VNS: one left anterior cervical, placed approximately at the C6-C7 level, with a second incision in the left infraclavicular region for placement of the implantable pulse generator (IPG). Most surgeons perform the procedure through a carotid-type incision and place the IPG caudally through blunt dissection. The left vagus nerve is isolated, and the electrode assembly is wrapped around the nerve below the origin of the superior and inferior cervical cardiac branches. During surgery, the interface between the vagus nerve and the electrode is tested with electrical stimulation. Fortunately, this is associated with a very low incidence of bradycardia and extremely rare reports of asystole. All resolve with cessation of stimulation and administration of atropine. The surgeon should inform the anesthesiologist when vagus stimulation is about to begin. The incisions are closed using the standard technique; however, because many patients are developmentally delayed, the final skin layer is often closed with a subcutaneous technique that does not require subsequent suture removal.

Usual preop diagnosis

Partial-onset seizures refractory to medication; chronic or recurrent depression refractory to other medical treatment

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and AED meds, limiting NPO duration alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Anterior cervical (C6-C7) + infraclavicular pocket (for IPG)
Unique considerations	Possible bradycardia → asystole during stimulation testing
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	1.5 h
EBL	<50 mL
Postop care	Observe for ↑ Sz activity postop × 8 h

Mortality	<0.5%
Morbidity	Hoarseness: ~50%, may be 2° vagal nerve injury or fatigue, device malfunction Cough: ~40% in first 3 mo after surgery Infection: 2% Dysphagia during stimulation → ↑ aspiration risk Dyspnea during stimulation; patients with OSA may have ↑ apneic events Nerve/vascular injury (similar to carotid surgery)
Pain score	1–2

■ PATIENT POPULATION CHARACTERISTICS

Age range	12–80 yr (≥18 yr for treatment of depression)
Male:Female	1:1
Incidence	20–40% of epilepsy patients are medically refractory
Etiology	Idiopathic, typically
Associated conditions	Developmental delay; severe depression

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Antiepileptic medications can control seizure disorders in most (~70%) patients, but some develop intolerable side effects; others are refractory to medical therapy. Surgical resection of the Sz focus may be the only effective therapy for some patients. For patients who are not candidates for surgical resections, the placement of a vagus nerve stimulator may be a viable alternative. These patients typically take multiple antiepileptic medications that should be continued throughout the morning of surgery. These same general considerations are also applicable to patients with refractory depression.

Neurologic	The only common neurologic finding in epilepsy patients is a Hx of uncontrollable Sz, either focal or generalized. Obtain a description of Sz and any
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	prodromal Sx.
Psychiatric	Medications used to treat depression (e.g., MAOIs, SSRIs) may have significant interactions with other drugs (e.g., meperidine → serotonin syndrome). Antiepileptic medications may have additive effects with anesthetics and significantly alter the pharmacodynamics of other drugs (e.g., ↓ duration of NMB).
Gastrointestinal	Abnormal liver function may be associated with use of valproate and carbamazepine. Tests: LFT and others as indicated from H&P
Hematologic	Phenytoin/phenobarbital → ↓ Hct; carbamazepine/valproate/ethosuximide/primidone → ↓ Plt; carbamazepine/primidone → ↓ WBC. Tests: CBC and others as indicated from H&P
Laboratory	Tests as indicated from H&P
Premedication	Standard premedication is usually appropriate.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water.

INTRAOPERATIVE

Anesthetic technique

GETA: Chronic use of antiepileptic medications may cause hepatic enzyme induction → accelerated drug metabolism; resistance to muscle relaxants including succinylcholine.

Induction	Standard induction. Once anesthesia is induced, a nondepolarizing NMR is administered.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Avoid hypothermia. Consider preincision placement of scalp blocks for open procedures.
Maintenance	Standard maintenance. Generally, no further NMBs

	are administered beyond those used for tracheal intubation. Hyperventilation may promote Sz activity. Sevoflurane and enflurane can induce epileptiform EEG activity.	
Emergence	No special considerations. Prophylactic antiemetic (e.g., ondansetron 4 mg) should be given 30–60 min before extubation. For open procedures, consider scalp block placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga BSS @ 4–6 mL/kg/h	Possibility of injury to carotid artery or jugular vein may result in significant blood loss. Zero balance or goal-directed fluid management is recommended.
Monitoring	Standard monitors EEG monitoring not necessary	O ₂ sat and ETCO ₂ monitoring will indicate the adequacy of ventilation and oxygenation.
Positioning	✓ and pad pressure points ✓ eyes	Head turned to the right with neck extension.
Complications	Asystole	Rarely (<1%), the SA node is innervated by the left vagus nerve. In those patients, vagal stimulation may result in severe bradycardia or asystole. Rx: stop stimulation; atropine 0.5 mg iv ± CPR.
	Bronchospasm	Bronchospasm may occur

Sz	as a result at vagal nerve stimulation.
Vascular injury	

POSTOPERATIVE

Complications	Seizure Vocal cord paralysis Hoarseness	Sz precautions may be necessary. Monitor carefully for altered mental status.
	Peritracheal hematoma	Airway obstruction may occur.
Multimodal pain management	Usually mild discomfort Acetaminophen 1 g iv ± Fentanyl 25–50 mcg prn Reinforce scalp blocks	Rx: If not given preop or if 4–6 h has elapsed since previous dose. 0.5% bupivacaine with epi is as effective as liposomal bupivacaine.
Tests	No routine tests	
ERAS	Ondansetron 4 mg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Aaronson ST, Sears P, Ruvuna F, et al: A 5-year observational study of patients with treatment-resistant depression treated with vagus nerve stimulation or treatment as usual: comparison of response, remission, and suicidality. *Am J Psychiatry* 2017; 174(7):640–8.
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EPILEPSY SURGERY

SURGICAL CONSIDERATIONS

Lawrence M. Shuer

Description

In the United States, the prevalence of epilepsy is ~5–20/1,000 (0.5–2%), meaning that at least 1.5 million people have epilepsy. In childhood, the incidence and prevalence are higher, with 90% of all new cases occurring before the age of 20. Intractable epilepsy is defined as persistent seizure activity of such frequency or severity that it prevents normal function and/or development. This diagnosis is made only after an adequate trial of anticonvulsant medication(s), with therapeutic levels, has been documented. Of all those with epilepsy, 10–20% prove to be intractable; it is estimated that ~20–30% of patients with intractable epilepsy may benefit from a surgical procedure.

The causes of epilepsy are varied, ranging from idiopathic to neoplastic. Epilepsy surgery is most beneficial in patients with partial epilepsy 2° a structural lesion. Most commonly, this lesion is located in the temporal lobe, and the most common operation is a temporal lobectomy, in both children and adults. Cerebral dominance and, hence, the location of speech may be determined using a preop Wada test (intracarotid amobarbital or other anesthetic injection to localize language function). Functional MRI mapping of speech centers is replacing this invasive test at many centers. Studies to define the epileptogenic focus include simultaneous recordings of video/EEG and high-resolution MRI and PET scans. Temporal lobe surgery may involve removal of only the structural lesion and associated epileptogenic cortex, cortical resection alone, excision of the amygdala and hippocampus, or removal of the entire anterior temporal lobe, with the extent of posterior resection dependent on dominance. Depending on the local protocol, intraop electrocorticography may be used, requiring modification of the anesthetic

technique. In addition, the speech center may need to be identified intraop, necessitating an awake procedure. These differing options will significantly alter the choice of anesthesia and must be established before surgery.

For a standard temporal lobectomy, the patient is placed supine on the operating table with the head turned 90° and held with pin fixation. A “question mark” temporal incision is often used, and hemostasis is achieved with skin clips. A flap—either a free temporal bone flap or an osteoplastic flap, based on the temporalis muscle—is elevated with a high-speed craniotome. A subtemporal craniectomy allows visualization of the entire anterior temporal lobe. The dura is opened, widely exposing the anterior 6–6.5 cm of the temporal lobe. Labbé vein must be preserved. At this point, surface and/or depth electrocorticography may be employed, and inhalation anesthetics must not be used. After mapping the lesion, amygdala and hippocampus or anterior temporal lobe is removed. Temporal lobectomy involves resection of both the lateral and medial temporal structures and is commonly performed in two steps. Often an operating microscope will be used to completely resect medial structures, including the uncus and hippocampal formation. Injury to the brain stem, 3rd and 4th cranial nerves, and either the middle cerebral or posterior cerebral arteries can occur; these are known complications of this surgery. Closure of the dura, bone flap, and scalp concludes the surgery.

Variant procedure or approaches

There are several variant procedures. The first is sectioning of the corpus callosum, known as a **corpus callosotomy**. This is commonly used for patients with atonic seizures or partial seizures with secondary generalization. Either the anterior two-thirds or the entire corpus callosum is divided in the midline. The approach is the same as any transcallosal, intraventricular procedure and uses a bifrontal, paramedian scalp incision and elevation of free-bone flap adjacent to the midline in the region of the coronal suture. Injury to the sagittal sinus is possible and may result in massive VAE or hemorrhage. In addition, numerous bridging veins across the interhemispheric fissure must be preserved to avoid venous congestion and possible infarction. The right cerebral hemisphere is gently retracted from the falx, exposing the paired anterior cerebral arteries and underlying corpus callosum. If an anterior two-thirds transection is performed, an intraop x-ray is required to determine the posterior border.

The second variant involves a frontal, temporal, or occipital craniotomy for resection of a structural epileptogenic focus, such as a tumor or AVM. This procedure may use stereotaxic localization, and the resultant craniotomy may be

performed with image guidance or in a stereotaxic head frame, which affects the method of intubation. The subsequent craniotomy is similar to the excision of any structural lesion, with the exception of intraoperative electrocorticography of surrounding cortex, which if used will affect the choice of anesthetic.

The third variant is most common and consists of a diagnostic procedure involving placement of surface and/or depth electrodes. This may be performed with stereotaxic localization and robotic assistance. Often only burr holes, outlining the future craniotomy flap, are used. After placement, the electrodes are externalized, and postop, the patient's naturally occurring seizures are recorded, in conjunction with video monitoring, to register the clinical presentation with the onset of ictal activity. This recording/observation period may last for several days. The patient is then returned to the OR and the epileptogenic focus resected as described earlier.

The fourth alternative is selective amygdalohippocampectomy, a variation of the standard anterior temporal lobectomy. In this procedure, the surgeon makes a cortical incision in the anterior temporal lobe and exposes and resects the amygdala and hippocampus, sparing the remaining portions of the temporal lobe. This procedure is sometimes used on the dominant side of the brain in an effort to lower the risk of postop speech and language dysfunction.

Stereotactic laser ablation (laser interstitial thermotherapy [LiTT]) is a recent alternative to amygdalohippocampectomy. In this procedure, a 1.65-mm laser catheter/applicator (Visualase®, Medtronic) is introduced stereotactically through a small (3.2-mm) burr hole under general anesthesia in the OR or MRI suite. The patient is then repositioned and moved into the MRI scanner where the laser applicator is advanced to target. Thermal ablation is then performed under direct MR thermography visualization. The laser tip may be repeatedly repositioned to treat larger tissue volumes. This technique has the advantages of being minimally invasive and allowing the lateral temporal structures to remain unharmed, thus further decreasing the risk of speech and language dysfunction and other lateral temporal lobe functions. Once ablation is complete, the laser catheter/applicator is removed, the incision is closed (usually with a single suture), and the patient is moved from the scanner, awakened and extubated.

Another option is responsive neurostimulation (RNS): In this procedure, electrodes are stereotactically placed in the seizure-onset zone(s) in the hippocampus or cortex and connected to a battery operated module implanted in the skull. When the neurostimulator module senses pre-seizure activity, it delivers an electrical

stimulus intended to disrupt the development of that seizure. This technique is particularly useful for bilateral seizure foci or for foci located in eloquent brain areas. Long-term efficacy appears to be comparable to other modalities and with minimal side effects.

Usual preop diagnosis

Temporal lobe epilepsy; partial epilepsy; intractable epilepsy

ERAS: Preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and AED meds, limiting NPO duration, alcohol and smoking cessation review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine, occasionally prone for occipital lesions; table turned 180°
Incision	Temporal question mark, reverse question mark, paramedian, frontal, or occipital
Special instrumentation	Operating microscope; Cavitron; bipolar cautery; surface electrode grids and strips; depth electrode
Unique considerations	Electrocorticography requiring awake craniotomy; awake procedures for mapping of temporal and/or frontal speech areas; stereotaxic craniotomy for lesion resection
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)
Surgical time	3 h
EBL	Minimal for diagnostic procedures; 250–500 mL with craniotomy (adults)
Postop care	After craniotomy, ICU for 12–24 h
Mortality	<1%
Morbidity	Hemiplegia Dysphasia Ophthalmoplegia Brain stem injury

Pain score	2–4
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■ PATIENT POPULATION CHARACTERISTICS

Age range	New-onset Sz: pediatric (<2 yr) and geriatric (>65 yr) populations most commonly
Male:Female	1:1
Incidence	200,000/yr (new cases of epilepsy): ~10% eventually present for surgery per year
Etiology	Idiopathic (mesial temporal sclerosis); infectious (brain abscess, encephalitis); traumatic (glial scar); vascular (AVM, infarct); neoplastic (glioma, hamartoma, ganglioglioma); congenital (cortical dysplasia)
Associated conditions	Tuberous sclerosis; Sturge-Weber syndrome; infantile hemiplegia; encephalitis; hemimegalencephaly; infections; tumors; trauma

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Epilepsy is a common neurologic disorder among young adults. Antiepileptic medications will abolish seizure disorders in many patients, but some develop intolerable side effects to such medications; others are refractory to any medical therapy. Surgical ablation of the seizure focus may be the only effective therapy for these patients. Several operations may be done, the most common being placement of surface or depth electrodes to determine the focus of the Sz, with subsequent temporal lobectomy or amygdalohippocampectomy for removal of the focus. In some cases, the lesion may be very focal and amenable to stereotactic localization and removal. At some centers, both seizure mapping and resection are carried out during the course of an awake craniotomy. If wake-up testing is planned, the procedure should be explained to the patient in detail, including what the patient can expect to hear and feel during the test. More recently, less invasive procedures (stereotactic laser ablation and responsive neurostimulation) have been shown to be beneficial in select patient populations.

Neurologic

Usually the only neurologic finding is a Hx of uncontrollable seizures, either partial or generalized.

	Obtain a description of Sz and any prodromal Sx. Sz medications should be continued through the morning of surgery. Antiepileptic medications may have additive effects with anesthetics and significantly alter the pharmacodynamics of other drugs (e.g., ↓duration of NMB). Tests: Preop fMRI mapping and/or a Wada test (intracarotid injection of anesthetic) is performed to determine whether the area of proposed surgery has any cerebral dominance or speech function.
Gastrointestinal	Abnormal liver function may be associated with valproate and carbamazepine use. Vomiting is associated with levetiracetam. Tests: LFT and others as indicated from H&P
Hematologic	Phenytoin/phenobarbital →↓ Hct Carbamazepine/valproate/ethosuximide/primidone →↓ Plt Carbamazepine/primidone/levetiracetam →↓ WBC Tests: CBC and others as indicated from H&P
Laboratory	Tests as indicated from H&P
Premedication	Standard premedication is usually appropriate.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water.

INTRAOPERATIVE

Anesthetic technique

GETA (± wake-up testing), conscious sedation or local anesthesia (awake craniotomy). Typically, the placement of surface or depth electrodes is done under GETA, as is a temporal lobectomy in the nondominant hemisphere. If the seizure focus is in the dominant hemisphere and/or if there is any question about possible injury to speech or motor areas, the procedure may be performed under local anesthesia (awake craniotomy), with intraop localization of the seizure focus.

Induction	Standard induction. After anesthesia is induced, a nondepolarizing neuromuscular relaxant is administered. If a difficult intubation is anticipated (e.g., stereotactic frame), orotracheal intubation is best accomplished before induction of GA. An awake fiberoptic intubation is the best technique.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Avoid hypothermia. For open procedures, consider preincision placement of scalp blocks.
Maintenance	Standard maintenance. Generally, no further NMBs are administered beyond that used for tracheal intubation. With adequate anesthesia, and the head fixed in pins (Mayfield-Kees skeletal fixation), patient movement of any consequence is highly unlikely. Furthermore, it may be useful to see movement of an extremity as an indicator of inadequate depth of anesthesia.
Emergence	No special considerations. Prophylactic antiemetic (e.g., ondansetron 4 mg iv, ± metoclopramide 10–20 mg iv) should be given 30 min before extubation. Surgeon should infiltrate wound with 0.5% bupivacaine (max = ~30 mL) or 0.75% ropivacaine (max = ~40 mL) with epinephrine (1:200K) to improve postop analgesia. For open procedures, consider scalp block placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago.
Conscious sedation	Most commonly accomplished using infusions of propofol (25–75 mcg/kg/min) with remifentanyl (0.02–0.05 mcg/kg/min) or dexmedetomidine (1 mcg/kg load over 10 min + 0.2–0.7 mcg/kg/h infusion).

Wake-up testing

At some institutions, an asleep-awake-asleep technique is used in which the patient is

placed under GETA for positioning and craniotomy. After surgical exposure of the seizure area, the patient is allowed to awaken to assess neurologic function while areas of the brain are stimulated. When the seizure focus has been adequately delineated, GA is reinstituted for the remainder of the operation. An excellent anesthetic technique under these circumstances includes the use of sevoflurane ($\pm 50\%$ N₂O) for amnesia and remifentanyl (0.05–2 mcg/kg/min) by continuous infusion for analgesia. The advantage of these drugs is that they are quickly eliminated, allowing for rapid emergence for the awake component of the procedure, followed by rapid reinduction of anesthesia when the testing period is over. To allow the patient to talk during the awake portion of the procedure, the ETT must be removed. This is best accomplished by inserting a small or medium-sized tube changer through the ETT before it is removed. The tube changer can then be used as a guide for reinsertion of the ETT, and it will not prevent normal vocalization by the patient during the awake phase. Alternatively, an LMA can be used to secure the airway initially, removed for the awake testing and reinserted for closure.

Blood and fluid requirements	Moderate blood loss IV: 18 ga $\times$ 2 BSS @ 4–6 mL/kg/h	Two iv cannulae are useful: one for fluid administration and another for infusion of anesthetic and other drugs. Alternatively, one peripheral iv and a CVP catheter are inserted. Blood transfusions are seldom needed. Zero balance or goal-directed fluid management is recommended (see Appendix K, p. K-2).
Monitoring	Standard monitors. Arterial line $\pm$ CVP line UO	O ₂ sat and ETCO ₂ monitoring will indicate the adequacy of ventilation and oxygenation in awake patients.
Positioning	Table rotated 180° ✓ and pad pressure	Semisitting position with head held in Mayfield-

	points ✓ eyes	Keeps 3-pin clamp and rotated laterally with a roll under the shoulder on the operative side. Anesthetic hoses and intravascular lines must be long enough to be accessible.
Complications	Anxiety/agitation Sz	Local anesthetic toxicity may produce agitation and seizures. Iced NS should be immediately available to flood cortical surface for Sz control.

▼ POSTOPERATIVE

Complications	Seizure/cerebral edema Bleeding	Monitor carefully for altered mental status. May require urgent reexploration and removal of electrode grid if present.
Multimodal pain management	Acetaminophen 1 g iv ± Fentanyl 25–50 mcg prn Reinforce scalp blocks PCA (selected patients)	Avoid oversedation. LA infiltration of the incision effectively reduces postop pain. If not given preop or if 4–6 h has elapsed since previous dose. 0.5% bupivacaine with epi is as effective as liposomal bupivacaine. Hydromorphone PCA (e.g., 0.2–0.4 mg, 5- to 10-min lockout) combined with ondansetron 4 mg iv

		q4h.
Tests	CT scan	If a patient exhibits any delay in emergence from anesthesia and surgery, or any new neurologic deficits appear, a CT scan is invariably obtained.
ERAS	Ondansetron 4 mg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Bloor M, Nandi R, Thomas M: Antiepileptic drugs and anesthesia. *Paediatr Anaesth* 2017; 27(3):248–50.
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SURGERY FOR SPASTICITY

SURGICAL CONSIDERATIONS

Description

Neurologic conditions associated with spasticity of the extremities include spinal cord injury, multiple sclerosis, stroke, etc. The spasticity is often managed with oral medications and botulinum toxin treatment. An alternative approach is to infuse baclofen (GABA agonist that decreases frequency and amplitude of tonic impulses to the muscle spindles) into the spinal subarachnoid space via an implanted pump. When these therapeutic modalities fail, it may be necessary to perform surgery. Some of the procedures are destructive in that a lesion is placed in certain nerves or the spinal cord to destroy the reflex arc that is contributing to the spasticity. There are both percutaneous and open techniques for placing the lesion. In the **open procedures**, a **laminectomy** is performed (see Posterior Lumbar Spine Surgery). In a case where a **myelotomy** is to be performed, the lower spinal cord is exposed and incised at the appropriate location to interrupt the reflex arc. In certain cases, selective electrical stimulation can be performed on isolated dorsal rootlets from the involved extremity. Abnormal responses in the extremities are monitored via EMG and direct observation. When abnormal responses are detected, that particular rootlet is divided (**selective dorsal rhizotomy**). This procedure is somewhat tedious and requires that stable anesthetic conditions be maintained so that motor and sensory nerve monitoring can be carried out during the procedure. The surgeon usually must sacrifice 50–75% of the dorsal rootlets in cases of spastic diplegia found in cerebral palsy. Selective dorsal rootlet rhizotomy reduces spasticity by decreasing facilitatory input to spinal motor neurons.

Variant procedure or approaches

Percutaneous **RF rhizotomy** is another procedure used for spasticity. A thermal lesion is placed in the appropriate dorsal roots with an RF generator. Needles are passed into the neural foramen via a posterolateral trajectory for levels L1-L5 and then from a midline approach to the S1 root. The needle position is determined by A-P and lateral fluoroscopy and by stimulus mapping. Stimulating current is delivered via an electrode passed through the needle. A low-level stimulating current causes muscle twitching in the appropriate leg if the needle is in the proper location. Once placed and verified, the

RF generator is used to produce the lesion. At the termination of this procedure, the patient's legs should be flaccid. Patients with spinal cord injury producing anesthesia below T10 may not require additional anesthetic for the procedure.

Usual preop diagnosis

Spasticity; multiple sclerosis; spinal cord injury

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and other meds, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Open Rhizotomy or Myelotomy	RF Rhizotomy
Position	Prone	⇐
Incision	Midline	Needles placed in lumbar region
Special instrumentation	EMG monitor; nerve stimulator	RF generator; fluoroscope
Unique considerations	Anesthetic that allows EMG recordings/no muscle relaxation	May not require anesthesia if anesthetic below waist (2° spinal cord surgery)
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)	None
Surgical time	4 h	2 h
Closing considerations	Surgeon may wish to test dural closure with Valsalva maneuver	
EBL	50–250 mL	Negligible
Postop care	Head flat; PACU room	⇐
Mortality	<1%	⇐

Morbidity	All <5%: CSF leak Infection Hemorrhage Neurologic impairment	←
Pain score	4	2

■ PATIENT POPULATION CHARACTERISTICS

Age range	3–55 yr
Male:Female	~3:2
Incidence	Relatively uncommon neurosurgical procedure
Etiology	Hyperactivity of gamma stretch reflex
Associated conditions	Multiple sclerosis; cerebral palsy; spinal cord injury

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Surgical Correction of Spinal Dysraphism, Pediatric Neurosurgery.

Suggested Readings

1. Aquilina K, Graham D, Wimalasundera N: Selective dorsal rhizotomy: an old treatment re-emerging. *Arch Dis Child* 2015; 100(8):798–802.
2. Sitthinamsuwan B, Nunta-Aree S: Ablative neurosurgery for movement disorders related to cerebral palsy. *J Neurosurg Sci* 2015; 59(4):393–404.
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PERCUTANEOUS PROCEDURES FOR TRIGEMINAL NEURALGIA

SURGICAL CONSIDERATIONS

Lawrence M. Shuer

Description

Three percutaneous procedures are commonly used to treat trigeminal neuralgia (a well-defined pain disorder of the face). Each involves placing a needle percutaneously from the cheek into the foramen ovale at the base of the skull under iv anesthetic ([Fig. 1.2-4](#)). For **glycerol injection** (rarely used), it is necessary for the surgeon to verify that the needle is placed in the cistern of the trigeminal (AKA gasserian) ganglion. This usually is done via I.I. or x-ray films with contrast instilled into the cistern by the surgeon. There should be free flow of CSF through the needle. The patient is then placed in the seated position with head flexed, and sterile glycerol is injected into the cistern (a painful event). The patient is taken to the recovery room sitting up with the head still flexed for 1 h. The glycerol damages neurons in the ganglion, which usually causes mild sensory loss and relieves the tic pain in most cases. **Percutaneous balloon compression** of the ganglion is a procedure done in some centers for this condition. The needle is placed similarly to the previously mentioned procedure, but in this case, a balloon catheter is placed through the needle, and the balloon is inflated after it is successfully located in the gasserian ganglion cistern. The patient may be kept sedated throughout this procedure. Compression of the ganglion will relieve the pain in many patients.

Radiofrequency (RF) rhizotomy is the most common percutaneous procedure used for trigeminal neuralgia. This procedure differs from glycerol injection in that an RF generator is used to place a thermal lesion (sparing the less heat-sensitive sensory neurons) in the appropriate portion of the gasserian ganglion. The needle is actually an insulated electrode with a portion of the tip exposed. The proper needle position is determined by applying stimulating current, with the patient awake, while assessing patient's responses. Multiple brief periods of anesthesia will be required to adjust the needle position or to lesion the nerve. It is important for the patient to awaken quickly and be able to cooperate with the stimulus localization throughout this procedure. This same approach may be used to place a stimulating electrode in the ganglion (ganglionic stimulation) with the leads subsequently tunneled to an implantable pulse generator. Direct stimulation has been shown to provide pain relief in many patients.

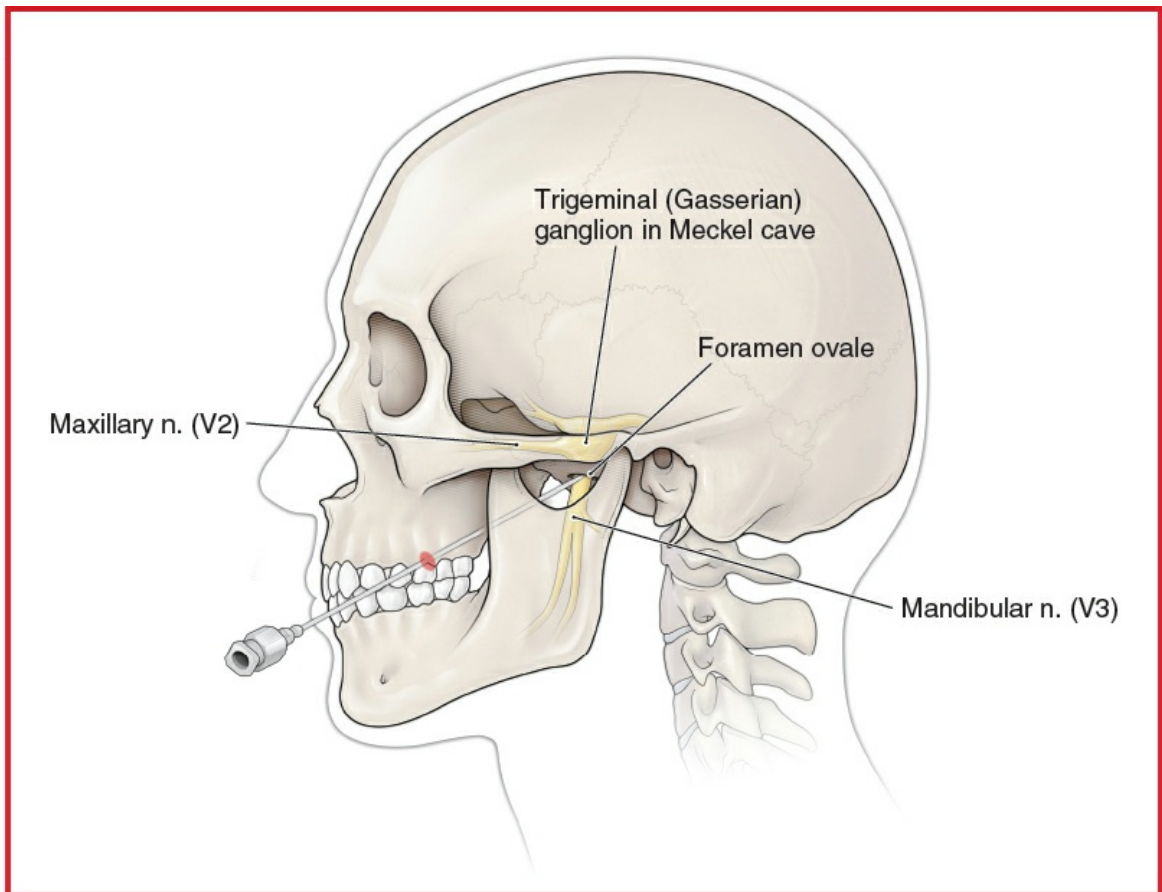


Figure 1.2-4 Percutaneous approach to the foramen ovale and the trigeminal (gasserian) ganglion used in the RF ablation procedure for trigeminal neuralgia.

RF neuromodulation is a nondestructive variant of RF ablation. Both percutaneous RF electrode placement at the trigeminal ganglion and treatment using pulsed RF energy can be accomplished under MAC with fluoroscopic control. This technique has a much lower complication rate compared to RF ablation but is also much less efficacious.

Usual preop diagnosis

Trigeminal neuralgia; tic douloureux; hemifacial spasm

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include continuing chronic pain and other meds, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Balloon
Compression

	Glycerol Injection		RF Rhizotomy
Position	Supine	⇐	⇐
Incision	Needle placed lateral to mouth on cheek	⇐	⇐
Special instrumentation	I.I.	⇐	⇐ + RF generator
Unique considerations	Hypertensive response to needle placement	⇐	⇐ + Requirement for periodic deep sedation with rapid awakening for RF lesioning
Antibiotics	Usually none	⇐	⇐
Surgical time	0.5 h	1 h	1–1.5 h
EBL	None	⇐	⇐
Postop care	Seated position with head flexed for 1 h	PACU	PACU
Mortality	rare	⇐	⇐
Morbidity	Infection: rare	⇐	⇐
	Complete facial numbness or anesthesia dolorosa: ~4%	⇐	⇐
	Extraocular muscle paresis (diplopia): <4%	⇐	⇐
	CSF leak: rare	⇐	⇐
	Masticatory problem: common	⇐	⇐
	Facial hematoma	⇐	⇐
	Sensory loss:	⇐	⇐

	common Recurrence: >50% in 5 yr	~50% in 5 yr	⇐
Pain score	3–7	2–4	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	40–85 yr (usually 60–70 yr)
Male:Female	1:2
Incidence	Trigeminal neuralgia: 4.3/100,000/yr (25.6/100,000/yr > 70 yr)
Etiology	Vascular compression of trigeminal nerve → demyelination; multiple sclerosis plaque; cerebellopontine angle tumor; AVMs
Associated conditions	HTN; multiple sclerosis; rheumatoid arthritis; Charcot-Marie-Tooth disorder; Sjögren syndrome; hemifacial spasm

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Trigeminal neuralgia, or tic douloureux, is a condition that develops in adults usually >60 yr old. It is more common in women in whom an intermittent, severe, lancinating pain arises over the maxillary and/or mandibular divisions of the trigeminal nerve. The ophthalmic division of the trigeminal nerve is rarely involved. The pain is unilateral and often can be precipitated by stimulating a trigger point, such as by contact with the cheek, mastication, or brushing teeth. The cause of this condition is not known. Medical management consists of therapy with carbamazepine (Tegretol) or oxcarbazepine (Trileptal), anticonvulsants, and analgesics specific for this condition. The failure rate for medical treatment is 50–60%. Stereotactic radiosurgery (dose = ~60 Gy) has been shown to have long-term effectiveness in ~50% of patients who failed drug therapy. Surgery is considered when medical or radiotherapy management fails to control pain or complications of drug therapy develop (anemia, bleeding disorders, dizziness, etc.). One of three types of percutaneous procedures is performed: **glycerol injection**, **RF rhizotomy** (or pulsed RF treatment) of the symptomatic branches of the trigeminal ganglion, or direct **ganglionic stimulation**. If these treatments fail, microvascular

decompression of the trigeminal nerve is considered (see [p. 50](#)).

Respiratory	No common associations or special considerations, unless patient has a long-standing Hx of smoking and has COPD. Tests: As indicated from H&P
Cardiovascular	Most patients have Hx of idiopathic HTN and take antihypertensive medications. Good control of BP preop is important because most patients become hypertensive during the operative procedure. Intraop HTN is often unavoidable because of the surgical need to have the patient awake during much of the procedure. Tests: As indicated by H&P
Neurologic	The presenting symptom is pain (shock-like = better outcome) in the maxillary and/or mandibular division of the trigeminal nerve, unaccompanied by motor or sensory deficits. (Dx should be questioned in the presence of sensory deficits). Patients likely will have failed first-line Rx and now be partially treated with second-line drugs including phenytoin, baclofen, gabapentin, and various antidepressants, which may have anesthetic management implications.
Laboratory	As indicated from H&P. CBC/platelets if Rx carbamazepine. Anticonvulsant drug levels
Premedication	Midazolam (1–4 mg iv)
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water.

INTRAOPERATIVE

Anesthetic technique

GA, regardless of which percutaneous technique is to be used. O₂ by nasal cannula should be administered before induction of anesthesia. Patients with OSA may benefit

from the use of high-flow nasal cannulae (THRIVE device, e.g., OptiFlow™). To keep the O₂ cannula out of the surgeon's field, supply tubing must be taped above the eye on the side of operation.

Induction

Because the surgeon will want the patient awake to check pain distribution as soon as the needle or electrode is in place, the induction drug must be potent, but short acting. Methohexital (0.5 mg/kg) is preferred for this purpose, but propofol (1 mg/kg) alone or at a reduced dose in combination with remifentanyl (0.5 mcg/kg) has been used successfully. The drug should be injected by bolus into a rapidly flowing iv to achieve a high concentration in the brain quickly. Continuous infusion is not satisfactory because it fails to achieve a high brain concentration quickly, and its continuous administration prolongs the time before the patient is sufficiently able to communicate with the surgeon. In experienced hands, the needle is placed within a matter of 2–3 min. If difficulty is encountered in placing the needle, additional bolus doses of WW drug ± midazolam may be needed. Patients may develop apnea for 1–2 min, followed by partial or total airway obstruction while the surgeon has his hand in the mouth positioning the needle. Prior to induction of anesthesia, therefore, it is essential that the anesthesiologist optimize ventilation and oxygenation by asking the patient to take a series of deep breaths through the nose with the mouth closed. This will ↑ the O₂ level and ↓ the CO₂ level in the lungs before induction. To maintain adequate spontaneous ventilation and oxygenation, it may be necessary to institute forward displacement of the mandible while the surgeon inserts the needle. Another alternative is to insert a nasal airway. If **glycerol injection** is used, needle position is verified by radiologic imaging, using a radiopaque dye. Patient is awakened and placed in a seated

	position with head flexed. The glycerol is injected, causing transient severe pain. Patient is then moved to recovery room still in seated position with the head forward to keep the glycerol localized to the region of the trigeminal ganglion. Fortunately, this procedure is now rarely used. If RF rhizotomy electrode placement is performed, the patient is awakened and the position of the electrode is verified by stimulating the ganglion with heat or directly and determining the site of pain. The patient is then reanesthetized with a smaller bolus of the same drug to permit electrode adjustment or local heating for ~1 min. This process may be repeated several times. Each time, the patient needs less anesthetic because of both the cumulative effects of the drug and the fact that the ganglion is becoming damaged by the heat and manipulation.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia. Avoid hypothermia.	
Emergence	Following glycerol injection, patients are maintained in a seated position with an ice pack on the cheek at the site of needle insertion to minimize postop bleeding and swelling. These patients often complain of pain in the face, requiring opiate analgesics. Following RF rhizotomy, the affected area of the face is usually numb, which is the end point for concluding the operation.	
Blood and fluid requirements	IV: 18–20 ga × 1 BSS @ 1–3 mL/kg/h	Zero balance fluid management is recommended (see Appendix K, p. K-2).
Monitoring	Standard monitors.	HTN patients may require an arterial line.
Control of BP	Labetalol Esmolol bolus dose	Patients almost invariably become hypertensive

	Clevidipine infusion may be necessary	during this therapy. Attempts to lessen these episodes with a clonidine patch preop or use of intermittent doses of esmolol or labetalol prophylactically or therapeutically are only partially successful.
Positioning	Table turned 180° ✓ and pad pressure points ✓ eyes	Patient in reclining position with head in the midline.
Complications	Failure of needle placement Bleeding Respiratory arrest (glycerol effect) Oculocardiac reflex	Major complications from this operation are uncommon but include (a) failure to identify the foramen ovale, through which the needle must be inserted to reach the trigeminal ganglion. If the needle cannot be placed within 30–40 min, the procedure usually is delayed pending additional imaging studies. (b) Bleeding into cheek from puncture of a branch of the facial artery. (c) Apnea from spillover of the glycerol solution into CSF, presumably affecting the respiratory center in the 4th ventricle.

POSTOPERATIVE

Multimodal pain management	Parenteral opiates Acetaminophen 1 g iv ± Fentanyl 25–50 mcg prn Reinforce scalp blocks	Usually needed for postglycerol pain Rx. If not given preop or if 4–6 h has elapsed since previous dose.
ERAS	Ondansetron 4 mg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

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2. Monstad P: Microvascular decompression as a treatment for cranial nerve hyperactive dysfunction—a critical view. *Acta Neurol Scand Suppl* 2007; 187:30–3.
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Suggested Viewing

Links are available online to the following videos:

Microvascular Decompression for Atypical Hemifacial Spasm:

<https://www.youtube.com/watch?v=4aOuANvc2-g>

Deep Brain Stimulation (DBS) Parkinson's (Australia 60 Minutes segment):

<https://www.youtube.com/watch?v=j3NjNKm0pio>

CHAPTER 1.3

Spinal Neurosurgery

SURGEONS

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ANESTHESIOLOGISTS

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INTRODUCTION: GUIDELINES FOR THE INTRAOPERATIVE MANAGEMENT OF PATIENTS UNDERGOING SPINE SURGERY

Karl Zheng and Martin S. Angst

Anesthesia for spine surgery today is more common and more complicated than ever before. Many different surgical approaches, OR tables, frames, and operative systems are used in spine procedures, each with their own set of anesthetic considerations. Some of these patient positions can decrease venous return, alter pulmonary compliance, drastically restrict iv access, or introduce significant measurement errors in noninvasive blood pressure (NIBP) measurements. Some spine surgeries are performed with navigation technology including CT guidance and robotic systems like Mazor X and Mazor Renaissance, while minimally invasive techniques are replacing larger incisions and more invasive procedures. While these changes may increase fidelity of pedicle screw placement and decrease patient recovery times, they also increase technical complexity, potentially increase surgical times, and require more radiation exposure. There are spine surgeries that are so extensive they require multiple positions or are staged across more than 1 d of surgery. Patients for spine surgery have changed as well. Chronic pain and opioid tolerance is particularly increased in the spine surgery patient population, making management of postop analgesia more

challenging than before. Anesthesiologists today need to be able to navigate this evolving landscape of spine surgery by identifying relevant details and adapting their anesthetic plans. A good rule of thumb is always to communicate with the surgeon ahead of time to avoid any mismatch of expectations.

GENERAL CONSIDERATIONS

Surgical procedures are divided into three classes based on the extent of surgery: minor, major, and complex. Considerations for the intraop management of patients undergoing spine surgery are class specific. Procedures are listed in the table:

Minor (EBL < 200 mL)	Major (EBL 200–1000 mL)	Complex (EBL 1–10 L)
• 1–2 level ACDF • ≤2 level decompression or microdiscectomy without instrumentation	• 3–4 level ACDF/PCDF • 1–3 level ALIF/XLIF • 1–2 level TLIF • 1–2 level anterior/posterior • Degenerative corpectomy	• 6–18 level instrumentation • ≥3 level anterior/posterior fusion • Pedicle subtraction osteotomy (PSO) • Vertebral column resection (VCR) • Tumor corpectomy • Tumor debulking

Access

Consider establishing vascular access as suggested:

- Minor surgery: 18- to 20 ga iv cannula, A-line only if medically indicated
- Major surgery: 16- and 18- to 20 ga iv cannulas, A-line recommended
- Complex surgery: 2× 14–16 ga iv cannulas or central venous line (Cordis), A-line

Fluid management

The overall aim is to keep patients euvolemic, avoiding excessive fluid administration. Fluids should be titrated on an “as-needed basis” to maintain euvolemia. Integration of various clinical indicators status (arterial blood pressure, pulse pressure variation, acid/base balance, clinical assessment of peripheral perfusion, trends in central venous pressure [CVP]) will allow assessment of fluid. Consider using colloids early in major and complex surgery (after administration of 1–2 L crystalloids) to avoid tissue edema.

- Crystalloids: Balanced solutions (e.g., LR) are preferred in combination with normal saline to limit sodium and chloride load.
- Colloids: Colloids preferentially expand intravascular volume and limit volume load of the interstitial space. This is particularly relevant in patients who are in prone position for extended periods of time. These patients have an increased risk of postop

visual loss. Significant tissue edema is also associated with slower bowel recovery and airway edema. The use of albumin 5% is recommended. The use of hetastarch is discouraged in major and complex surgery as it may impair renal function and worsen coagulopathy.

Blood pressure management

The overall aim is to avoid decreases in mean arterial pressure (MAP) $>20\%$ from baseline (typical blood pressure at rest without stress). Elevated MAP ($\geq 80\text{--}90$ mm Hg) may be required in patients with myelopathy or trauma to the spine. Consider taking the following sequential actions if MAP decreases $>20\%$:

- ↓ Anesthetic level (use age-adjusted monitored anesthesia care [MAC] and evaluate individual patients' true anesthetic requirements). The use of 50% nitrous oxide (minimal impact on systemic vascular resistance [SVR]) is suggested to spare volatile anesthetics. A total of 0.7–1.0 MAC is often sufficient to prevent awareness and provide adequate surgical conditions.
- Opioids are mainly used to control hemodynamic variables during GETA. Opioids do not exert preemptive analgesic effects. Toward the end of surgery, opioids should be titrated to provide a component of postop pain control.
- Consider starting a continuous infusion of phenylephrine at a rate ≤ 0.4 mcg/kg/min. The use of higher rates has to be balanced against the risk of concealing intravascular volume depletion. This is particularly important in patients with significant blood loss. Fluids should be used to restore euolemia if a decrease in MAP $>20\%$ is due to intravascular volume depletion (normal SVR).

Transfusion of blood products

Unexpected significant bleeding can occur during any spine surgery from direct vascular injury, for example, of the common iliac vessels during an anterior lumbar approach. During a routine prone decompression surgery, injury to epidural veins can also result in abrupt unexpected blood loss. However, surgeries where potentially life-threatening bleeding is expected typically fall into two major categories. The first is a large fusion surgery that involves many levels (>8) and subsequently a large incision and many hours to complete. Bleeding potential is always higher if the surgery involves revising a previous fusion. The second is a major osteotomy (e.g., corpectomy) of a vertebral body (for tumor metastasis or traumatic fracture), pedicle subtraction osteotomy (full-thickness osteotomy for deformity correction), or vertebral column resection (major deformity correction). Tumor metastases, renal cell carcinoma and breast cancer in particular, can be highly vascular and often benefit from preop embolization if time permits.

While there is some controversy regarding the threshold Hct/Hb that should trigger administration of blood, a threshold of 24% (Hct) or 8 g/dL (Hb) is suggested for major and complex surgeries. Consider administering blood at Hct ~27% or Hb ~9 g/dL if blood loss is rapid and continuous to provide a margin of safety. If blood loss is rapid and large, administering blood products may be dictated by hemodynamic parameters rather than specific Hct or Hb targets.

- Prophylactic use of tranexamic acid (10 mg/kg bolus) should be considered for major surgery (discuss with surgeons) and is strongly recommended for complex surgery (10 mg/kg bolus followed by an infusion of 1 mg/kg/h). Use with caution in patients with a history of thromboembolic events or malignancy.
- A cell saver should be available for major and complex surgery except for cancer-related surgeries.
- Consider the use of fresh frozen plasma (FFP) after administering 4–6 units of packed red blood cells (PRBCs) or as clinically indicated. Administration of FFP and PRBCs in a ratio of 1:1–2 is suggested.
- Consider administration of platelets (one apheresis product) after transfusion >6 PRBCs, a platelet count <100,000/mcL, or as clinically indicated. Consider administration of platelets for each additional 6 PRBCs.
- Consider administration of cryoprecipitate (10 U) after transfusion of 10 PRBCs to maintain a plasma fibrinogen level >100 mg/dL.
- Consider calling for help and initiating massive transfusion protocol in case of massive and/or uncontrollable bleeding.

Prophylaxis for PONV

A special consideration concerns the use of steroids (dexamethasone). Steroids have to be avoided in patients undergoing spinal fusion as they may impact bone healing. Discussing the use of dexamethasone with surgeons is strongly recommended. PONV prophylaxis should follow guidelines using risk stratification (see [Appendix B: Standard Adult Anesthetic Protocols, p. B-1](#)). The risk for developing PONV is increased by the use of postop opioids, being a nonsmoker, being female, and having a history of PONV or motion sickness. Each factor adds 1 point to a maximum score of 4. The number of agents for preventing PONV is equal to the risk score. Agents to consider include the following:

- Ondansetron (4 mg)
- Dexamethasone (4 mg) in nonfusion procedures
- Propofol-based anesthetic
- Scopolamine patch (1.5 mg)
- Haloperidol (1 mg)
- Metoclopramide (10–20 mg)

Pain management

The major aim is to provide adequate pain control (pain at rest $\leq 3/10$, pain during movement $\leq 5/10$) using a multimodal approach. While opioids are an important component of the multimodal approach, opioid-sparing techniques are encouraged. Certain components of a typical multimodal approach require special considerations in patients undergoing spine surgery. In particular, the use of NSAIDs and COX-2 inhibitors (e.g., celecoxib) are usually avoided in patients undergoing spinal fusion.

- Patients should be started on acetaminophen (1 g po) before surgery, then 1 g po q6h postop until they are switched to an oral combination product containing an opioid and acetaminophen. Dose adjustments may be necessary in the very elderly and in patients with impaired liver function.
- Consider the use of celecoxib (200 mg po) before surgery for nonfusion procedures.
- Consider the use of an intraop ketamine infusion (0.5 mg/kg bolus followed by an infusion rate of 0.25 mg/kg/h). Stopping the infusion 30–45 min before the end of surgery is recommended to facilitate emergence from anesthesia. The use of ketamine is encouraged in patients undergoing major and complex surgeries and in patients with history of chronic pain complaints. Consider the use of an intraop lidocaine infusion (100 mg bolus followed by an infusion rate of 1–2 mg/kg/h). A lidocaine infusion is recommended for patients undergoing major or complex surgery.
- The use of gabapentin before surgery is controversial. Gabapentin can cause postop drowsiness and impair early mobilization. Consider preop administration of gabapentin (600 mg po) in patients with a history of chronic pain.

Neuromonitoring

The use of neurophysiologic monitoring during spine surgery is highly dependent on surgeon preference. Patients requiring neuromonitoring including motor evoked potentials (MEPs) should not receive neuromuscular blockers, except for tracheal intubation. EMG is primarily performed to assess electrical stimulation thresholds for pedicle screw placement as an indicator of proximity to the spinal canal or exiting nerve structures. If baseline potentials will be recorded before turning a patient prone use succinylcholine to facilitate intubation. A standard anesthetic plan not interfering with the monitoring of evoked potentials includes the use of 0.5 MAC of a volatile anesthetic, $<50\%$ N₂O, and a remifentanyl infusion or TIVA targeted to a patient's anesthetic needs. Patients requiring EMG monitoring (Nuvasive XLIF, MIS TLIF) should only receive succinylcholine for intubation.

Perioperative visual loss (POVL) is a rare surgical complication (estimated incidence: <1 in 1000) that primarily occurs after prone spine surgery. This is due to ischemic injury to the optic nerve by either one of two mechanisms. The less common

etiology is central retinal artery occlusion (CRAO) whereby direct pressure on the eye leads to retinal ischemia. This is almost always due to poor positioning and can be avoided by ensuring that there is no direct pressure on the eyes. The more common etiology of POVl today is ischemic optic neuropathy (ION). ION is essentially compartment syndrome within the optic nerve canal because of venous congestion that restricts perfusion of the optic nerve leading to ischemic injury. POVl due to ION will most likely affect both eyes, whereas CRAO typically leads to unilateral vision loss. Should any visual loss occur, the change is usually permanent.

A landmark publication by the ASA in 2011 looking at POVl after prone spine surgery identified the following risk factors as statistically significant: male gender, obesity, diabetes, duration of surgery, blood loss during surgery, and use of Wilson frame. Additionally, the use of colloid in combination with crystalloid for resuscitation was found to be protective presumably due to less edema. General guidelines for reducing the risk of POVl would be to minimize periods of hypotension and anemia particularly near the end of a lengthy case when presumably edema would be most severe. The technique of deliberate hypotension during major spine surgery is no longer advised.

ANTERIOR FUSION/FIXATION OF THE UPPER CERVICAL (C1-C2) SPINE

SURGICAL CONSIDERATIONS

Description

Transoral odontoid excision: The **transoral approach** is indicated primarily to relieve ventral irreducible compression of the cervicomedullary junction due to extradural lesions involving the lower part of the clivus, C1 and C2 vertebral bodies. This approach provides direct access to the C1 anterior arch and odontoid process of C2. The anesthetized patient is positioned supine usually with cervical traction. A Dingman or similar retractor is used to facilitate surgical access. The soft palate is retracted upward with stay sutures. Through a posterior midline incision over the pharyngeal wall, the C1 anterior arch and C2 vertebra are exposed ([Fig. 1.3-1](#)). Using fluoroscopic guidance, bony decompression of the clivus, C1 anterior arch, odontoid process, and C2 vertebral body is performed. Instrumentation of C1-C2 may be performed with plate and screws. The wound is closed in two layers after securing hemostasis. As the procedure often results in significant instability at the craniovertebral junction, posterior occipitocervical fusion often is required.

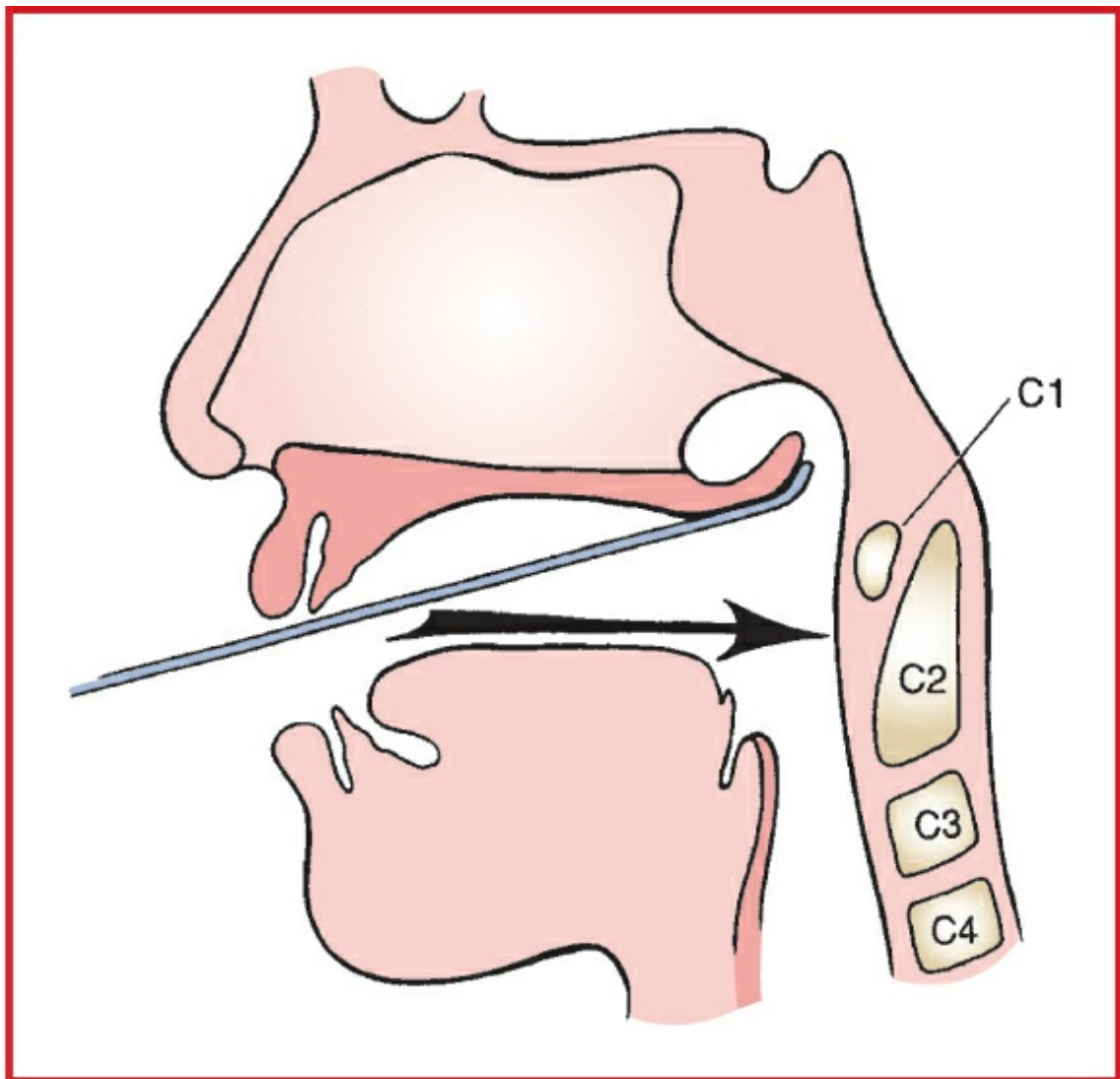


Figure 1.3-1 The transoral approach. (Reproduced with permission from An HS, Cotler JMSpinal Instrumentation. Williams & Wilkins, Baltimore: 1992.)

Variants of the transoral procedure

Transpalatal exposure with removal of hard palate or a **tongue-splitting transmandibular approach** may be required for adequate exposure of the clivus or upper C-spine, respectively. A **high cervical anterior retropharyngeal approach** is rarely used to approach C1-C3 without traversing the oral cavity and with less destabilization. A transcervical approach to the C2 body can be performed using an endoscope and a highly beveled tubular retractor.

Transodontoid screw fixation

Fractures of the odontoid process of C2 account for 10–20% of all C-spine fractures and are classified (**Anderson and D'Alonzo**) into three types, based on anatomical location. **Type I fractures**, which occur at the tip of dens, are treated conservatively. **Type II fractures**, which occur through the waist of the odontoid process, are the most

common and are often inherently unstable, requiring surgical treatment. Internal instrumented fixation is the ideal treatment for type II fractures because it provides immediate stability while preserving C1-C2 rotation. **Type III fractures** occur through the body of C2 and can be treated surgically or with external immobilization.

The anesthetized patient is placed in a supine position with traction of 5–10 lb. The mouth may be kept open using a radiolucent jaw distractor. A horizontal skin incision is made at approximately C5 level, and the platysma is cut along the skin incision. The anterior C-spine is exposed by opening the natural plane between the trachea and esophagus medially and carotid sheath laterally. Using a guide tube, a hole is drilled under fluoroscopy through the body of C2, the odontoid process, and its apex, through the fracture. The drilled hole is tapped, and an appropriately sized screw is placed (Fig. 1.3-2). At this point, fluoroscopy of the patient's neck in flexion and extension is done to exclude C1-C2 instability.

Usual preop diagnosis

Basilar impression (telescoping of the C-spine into the posterior fossa); odontoid fracture; rheumatoid arthritis with atlantoaxial instability; traumatic irreducible atlantoaxial instability; type II odontoid fractures (recent or remote)

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

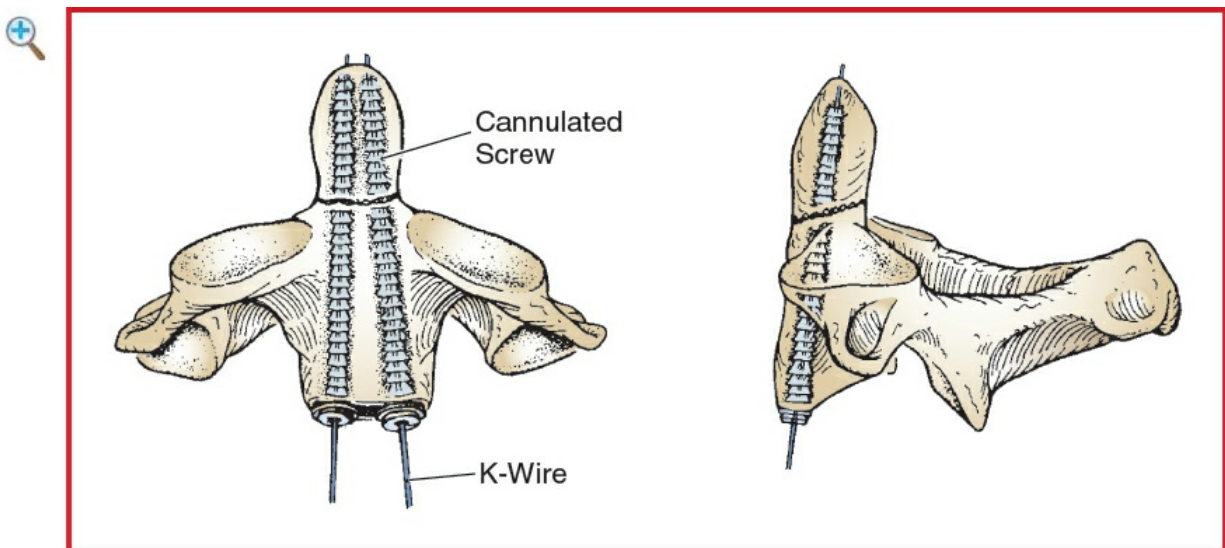


Figure 1.3-2 Screw fixation technique, using cannulated screws. K-wire provides provisional stabilization and allows guided tapping and screw placement. (Reproduced with permission from An HS, Cotler JM: Spinal Instrumentation. Williams & Wilkins, Baltimore: 1992.)

■ SUMMARY OF PROCEDURES

	Transoral	Transodontoid Screw Fixation
Position	Supine; head in mild extension with traction	Supine; head in mild extension with traction
Incision	Posterior oropharynx	Transverse skin incision in the anterolateral neck at C5 level
Special instrumentation	Operating microscope, I.I., Dingman retractor	K-wire, drill, odontoid screw instrumentation
Unique considerations	Requires ≥ 3 cm mouth opening; armored ETT must be taped securely away from the surgical field. Tracheostomy is needed occasionally. Fiberoptic intubation may be required.	Fiberoptic intubation in patients with C1-C2 instability; biplanar fluoroscopy is mandatory.
Antibiotics	Cefazolin 2 g iv (30–60 min before incision)	⇐
Surgical time	2–4 h	2–3 h
Closing considerations	Extubation may be delayed due to lingual swelling	Routine wound closure
EBL	25–200 mL	25–200 mL
Postop care	ICU or close observation unit. Monitor airway for respiratory obstruction. $\pm$ postop immobilization. Will require leaving NG tube in postop	PACU $\rightarrow$ room; cervical collar
Mortality	0–3%	1%

Morbidity	All: <20%: CSF leak/dural tear Infection Neural injury Pharyngeal wound dehiscence	All: <15% Hardware failure (screw pullout, screw fracture): 10% Nonunion: 5–10% Superficial wound infection: 2%
Pain score	3–5	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–85 yr (usually 20–60 yr)	15–90 yr (usually 20–60 yr)
Male:Female	1:2	1.5:1
Incidence	Rare	Common (10–20% of C-spine fractures)
Etiology	Neoplastic; traumatic; congenital; degenerative	Traumatic
Associated conditions	Rheumatoid arthritis	—

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Cervical Neurosurgical Procedures.

Suggested Readings

1. Apfelbaum RI, Lonser RR, Veres R, et al: Direct anterior screw fixation for recent and remote odontoid fractures. *J Neurosurgery* 2000; 93:227–36.
2. Fountas KN, Kapsalaki EZ, Nikolakakos LG, et al: Anterior cervical discectomy and fusion associated complications. *Spine* 2007; 32(21):2310–7.
3. Shriver MF, Kshetry VR, Sindwani R, et al: Transoral and transnasal odontoidectomy complications: a systematic review and meta-analysis. *Clin Neurol Neurosurg* 2016; 148:121–9.
4. Vender JR, Harrison SJ, McDonnell DE: Fusion and instrumentation at C1–3 via the high anterior cervical approach. *J Neurosurgery* 2000; 92:24–9.

POSTERIOR FUSION/FIXATION OF THE

UPPER CERVICAL SPINE

SURGICAL CONSIDERATIONS

Description

Craniocervical (occipitocervical, craniovertebral) fusion/fixation or instrumentation involves stabilization of the occiput and upper three or four cervical vertebrae, whereas **atlantoaxial fusion/fixation** involves stabilization of the atlas (C1) and axis (C2). Instability may be caused by congenital, traumatic, degenerative, neoplastic, or infectious conditions resulting in compression of the lower brain stem or cervical spinal cord. Symptoms may include paresthesias and/or weakness of the upper and lower extremities.

Atlantoaxial techniques

Atlantoaxial (C1-C2) fusion is performed in the prone position, with or without traction. Through a posterior midline incision, the occiput and upper C-spine are exposed. A posterior iliac or rib graft may be harvested and fashioned appropriately. The bone graft can be secured with wires to the decorticated segments to be fused. Traditionally, the fixation has been performed with sublaminar wires alone; however, other fixation techniques—such as **C1-C2 transarticular screw fixation** and more recently the **C1-C2 lateral mass fusion** techniques are being used more often because they are biomechanically stronger and permit early ambulation with minimal orthotic support.

In **C1-C2 posterior wiring techniques**, the posterior arches of C1 and C2 laminae are exposed through a midline incision. Of the various wiring techniques used, **Gallie's** and **Brooks'** are the most widely accepted. In **Gallie fusion** ([Fig. 1.3-3A](#)), a wire loop or cable is passed underneath the C1 arch and brought over a bone graft wedged between C1 and C2 and then tightened over the C2 spinous process. In **Brooks technique** ([Fig. 1.3-3B](#)), wires are passed beneath the C2 lamina and C1 posterior arch on each side and tightened over a bone graft placed between C1 and C2. The posterior aspects of C1 and C2 are decorticated to facilitate the bony fusion. Wiring techniques are simpler, but carry the risk of cord injury during wire placement.

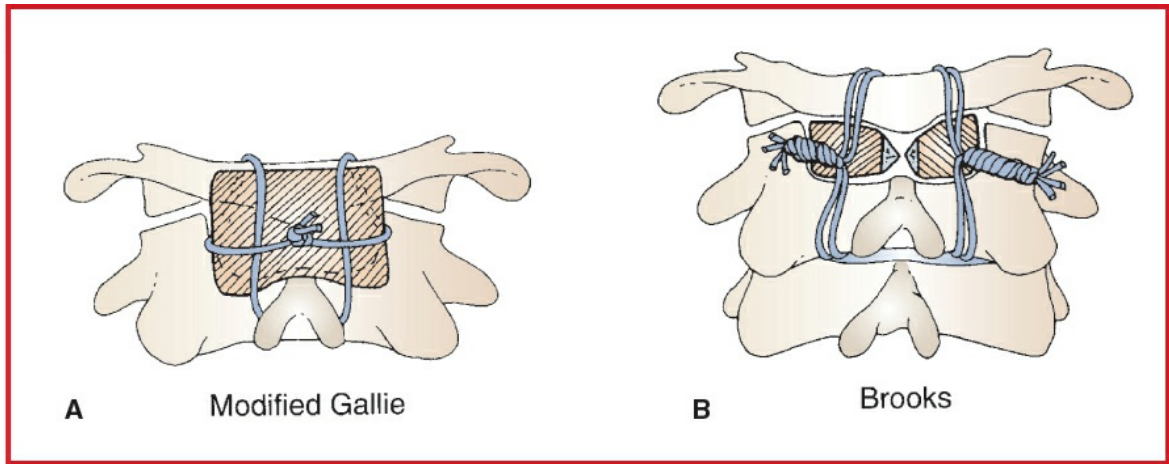


Figure 1.3-3 Posterior wiring techniques. **A:** Modified Gallie using an H-shaped bone graft from the iliac crest, contoured to fit over the posterior arches of C1 and C2. A double U-shaped 18- or 20-ga wire is passed under the arch of C1 from inferior to superior. **B:** Brooks-type fusion with doubled-twisted 24-ga wires passed under the arch of C1 and then under the lamina of C2. Rectangular iliac crest bone grafts are fitted in the intervals between the arch of C1 and each lamina of the axis. (Reproduced with permission from An HS, Cotler JM: Spinal Instrumentation. Williams & Wilkins, Baltimore: 1992.)

C1-C2 transarticular screw fixation was initially popular because of its greater biomechanical stability, higher fusion rates (87–100%), and superior fixation of atlantoaxial rotation. The occiput and C1-C3 vertebrae are exposed by a conventional posterior approach. Screws are placed from each lateral mass toward the anterior tubercle of C1 under fluoroscopic guidance. The major risks of this technique include injury to the vertebral artery (4.1%), malposition of screws, or instrumentation failure. Twenty percent of patients will have an anomalous vertebral artery, demonstrated by radiographic studies, precluding use of this technique.

In **C1-C2 lateral mass fusion**, the C-spine is exposed subperiosteally from occiput to C3-C4 vertebrae by a conventional posterior approach. In this technique, polyaxial screws are passed into the lateral mass of C1 ([Fig. 1.3-4](#)). In addition, C2 pedicle screws are also placed and then attached to the C1 screws by connecting rods ([Fig. 1.3-5](#)). If required, a reduction maneuver is carried out by repositioning the head or by direct manipulation of the C1 and C2 vertebrae. C1-C2 interfacetal fusion or posterior interlaminar fusion may be performed with wiring. The C1-C2 construct can be combined with cervical decompression. Because of the superior and medial placement of C2 pedicle screws, the risk of injuring the vertebral artery is less than with a transarticular fusion.

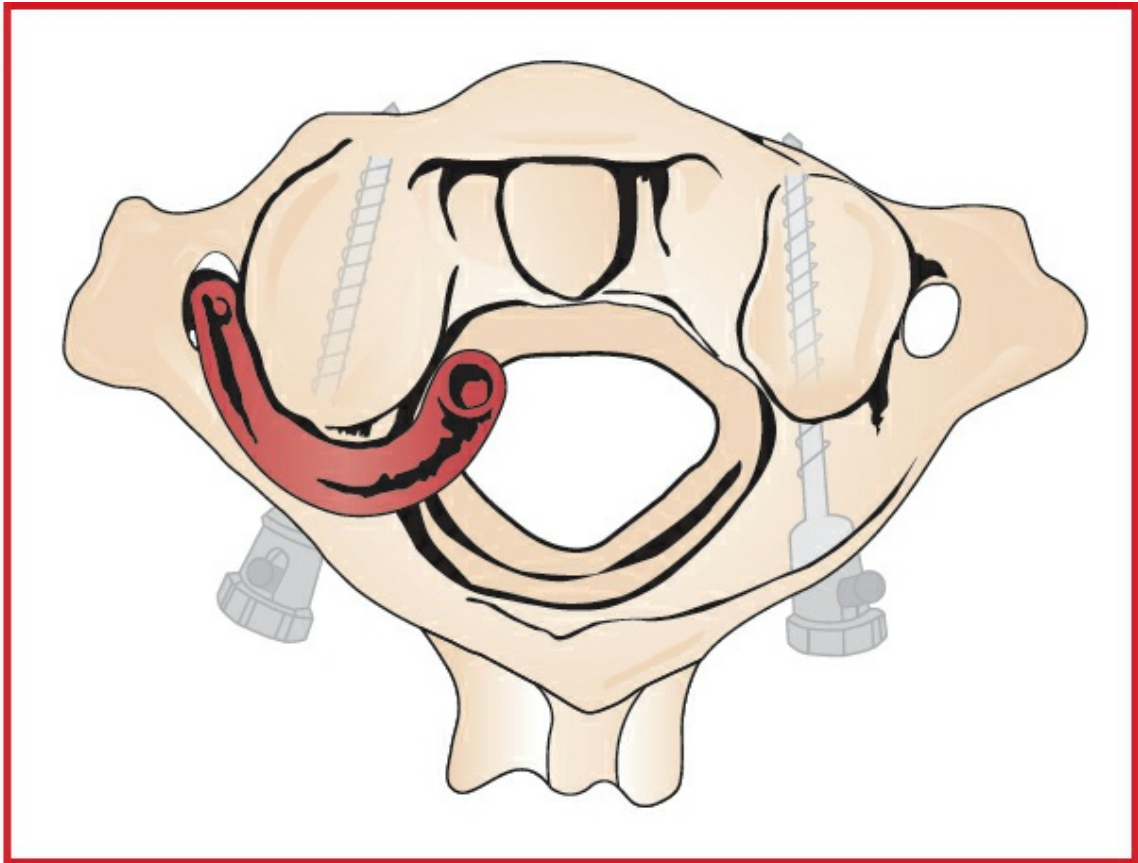


Figure 1.3-4 Axial view of C1 lateral mass screws. (Reproduced with permission from Harms J, Melcher RP: Posterior C1-C2 fusion with polyaxial screw and rod fixation. Spine 2001; 26(22):2467–71.)

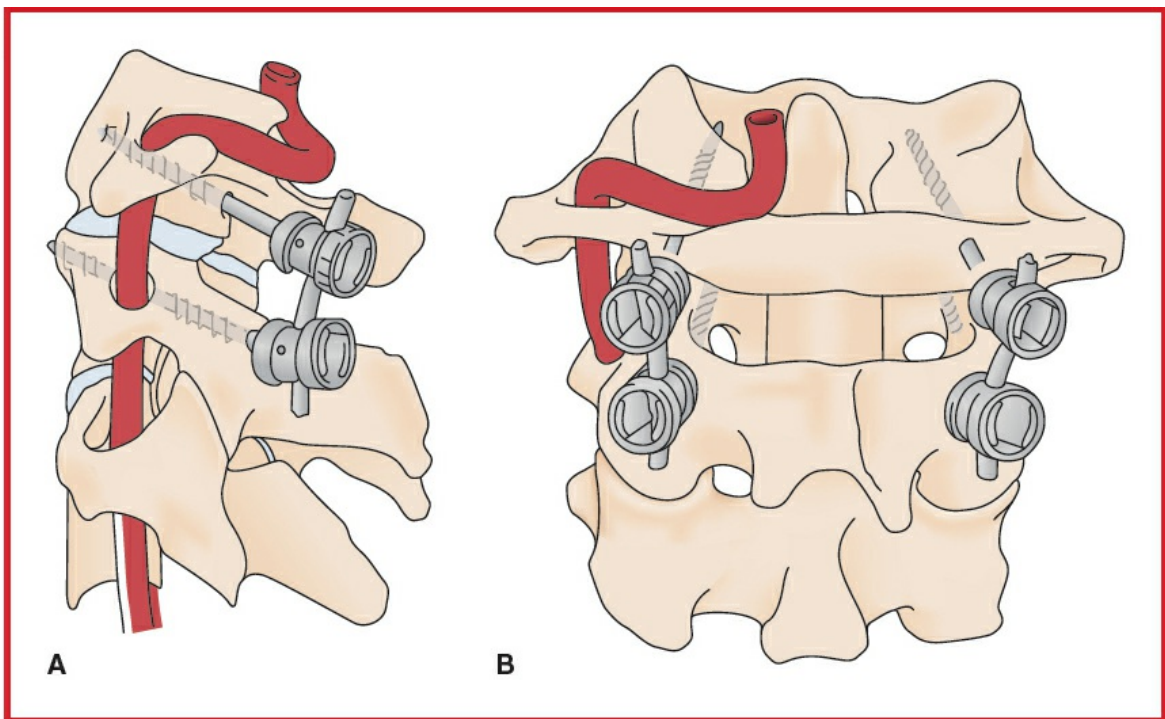


Figure 1.3-5 Lateral (A) and AP view (B) of C1-C2 fusion. (Reproduced with permission from Harms J, Melcher RP: Posterior C1-C2 fusion with polyaxial screw and rod fixation. Spine 2001; 26(22):2467–71.)

Craniocervical techniques

Occipitocervical fusion involves a surgical exposure similar to that of atlantoaxial fusion, except that a more extensive exposure of the occipital bone is required. In the past, fixation was performed with a **Luque rectangle/contoured rod and wiring** or plate and screws. An appropriately fashioned rib or iliac crest graft was then secured in place.

In **occipitocervical contoured rod fixation**, the occiput and posterior C-spine are exposed through a posterior incision, and trephines are made 2.5 cm to either side of the midline and about 2 cm above the foramen magnum. Wires or cables are passed from these occipital holes through the foramen magnum on both sides. Sublaminar wires are passed beneath laminae of the atlas, axis, and C3 vertebrae on each side and are tightened over a rod. Other cervical vertebrae may be included in the fixation as required. A tricorticate iliac or rib graft is fixed with wires over the occipitocervical region. Decortication of occipital bone and laminae of the atlas, axis, or C3 vertebrae is essential for bony fusion.

Occipitocervical plate fixation can be performed by using a T- or Y-shaped plate fixed by screws to the occiput and lateral masses of the cervical vertebrae. C1-C2 transarticular screws, lateral mass screws, or wiring techniques can be added for additional stability. Occipitocervical plating techniques are biomechanically stable, often obviating the need for postop halo immobilization; however, they can be technically challenging. The major concerns include possible dural penetration by occipital screws and obtaining adequate contouring of the construct.

Usual preop diagnosis

Transoral odontoid resection; occipitoatlantal instability; atlantoaxial instability; odontoid fractures; spinal fractures; cervical instability; previous failed fusions

■ SUMMARY OF PROCEDURES		
	Atlantoaxial (C1-C2) Fusion/Fixation	Occipitocervical Fusion/Fixation
Position	Prone, head in tong traction or pins	⇐
Incision	Posterior midline incision	⇐
Special instrumentation	Fluoroscopy; drills; wires/cables; lateral mass	Fluoroscopy; drills; rods, wires/cables; plates and

	plate and/or screws; transarticular screws	screws
Unique considerations	Iliac/rib autograft ± posterior ilium for graft. Fiberoptic intubation and SSEPs	⇐
Surgical time	2–3 h	3–4 h
Closing considerations	Routine wound closure	Halo vest may be needed in selected cases.
EBL	100–500 mL	⇐
Postop care	PACU, then → room	⇐
Mortality	<1%	⇐
Morbidity	Vertebral artery injury: 4.1% CSF leak Instrumentation failure Nonunion	Neural injury ⇐ ⇐ ⇐
Pain score	5–9	5–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–85 yr (usually 20–60 yr)
Male:Female	1:2
Incidence	Rare
Etiology	Neoplastic; traumatic; congenital; degenerative; infections; rheumatoid arthritis

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Cervical Neurosurgical Procedures.

Suggested Readings

1. Menendez JA, Wright NM: Techniques of posterior C1-C2 stabilization Neurosurgery 2007; 60(Suppl 1): S103–11.
2. Wright N, Lauryssen C: Vertebral artery injury in C1–2 transarticular screw fixation:

results of a survey of the AANS/CNS section on disorders of the spine and peripheral nerves. J Neurosurg 1998; 88:634–40.

ANTERIOR FUSION/FIXATION OF THE MID AND LOWER CERVICAL SPINE

SURGICAL CONSIDERATIONS

Description

The first description of the anterior approach for excision of a cervical disc was made by Smith and Robinson in 1958. This approach permits easy and safe access to the entire C-spine below C2. **Anterior cervical discectomy** is commonly indicated for the removal of herniated discs or osteophytes compressing the spinal cord or nerve roots. Multisegmental cervical spondylosis (narrowing of spinal canal) may require single- or multilevel corpectomy (removal of a vertebral body). During anterior cervical discectomy, an approach from the left side of the neck is often preferred because it minimizes the chances of injury to the recurrent laryngeal nerve. The dissection is carried along the avascular plane between the trachea and esophagus medially and the carotid sheath laterally (Figs. 1.3-6 and 1.3-7). The fascia is incised to expose the longus colli muscles and anterior C-spine. The disc level is confirmed using fluoroscopy. The annulus is incised, and the disc is removed in piecemeal fashion with the use of an operating microscope. **Fusion and instrumentation** are often performed after discectomy to maintain disc space height, restore normal cervical lordosis, prevent graft extrusion, facilitate early ambulation, and possibly prevent delayed deformity and pain due to collapse of the disc space. After the discectomy, osteophytes are removed from the vertebral bodies, and an appropriately sized bone graft or prosthesis is placed in the intervertebral space. PEEK and carbon fiber cages are radiolucent and allow good assessment of bony fusion. Fusion with instrumentation is often essential for immediate stability and early ambulation.

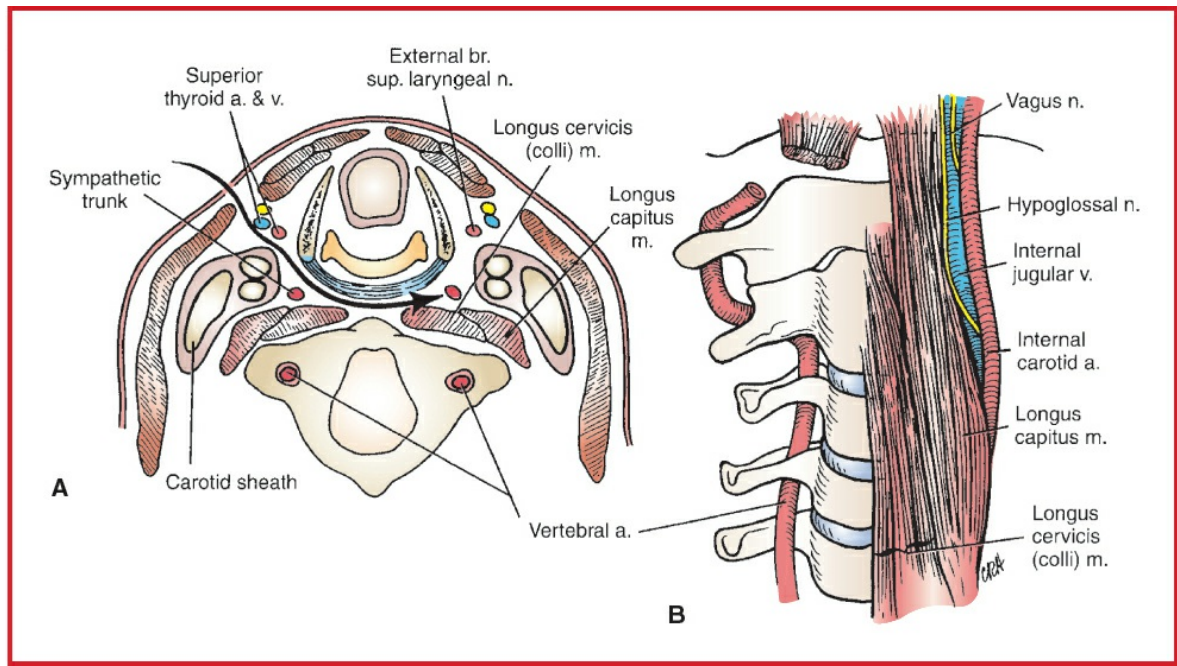


Figure 1.3-6 Upper C-spine. **A:** Cross-section showing the anteromedial approach. **B:** Anterior aspect, after stripping the longus colli muscle. (Reproduced with permission from An HS, Cotler JMSpinal Instrumentation. Williams & Wilkins, Baltimore: 1992.)

Anterior screw-plate fixation (with MRI-compatible titanium) is the preferred method of fixation for C2-C7. It provides stable fixation after discectomy or corpectomy, prevents bone graft migration, improves fusion rate, corrects spinal deformities, and may restore anterior and middle column function following cervical trauma. Plates and screws are placed under fluoroscopic guidance to prevent dural penetration or malposition. Hemodynamic changes should be monitored closely during the procedure, because ↓ HR or ↓ BP during the instrumentation may suggest cord compression.

Usual preop diagnosis

Cervical radiculopathy (nerve root compression due to disc herniation or osteophytic compression); cervical myelopathy (spinal cord compression by disc/osteophytes); cervical instability (ligamentous laxity or disruption)

Variant procedures

Cervical (vertebral) corpectomy and fusion are used to treat conditions in which there is anterior impingement of the spinal cord or narrowing of the spinal canal at the level of the vertebral body, including multisegmental cervical spondylitic compression, ossification of the posterior longitudinal ligament (OPLL), tumors, infections (e.g., TB osteomyelitis), or C-spine injury. The surgical exposure is similar to that for anterior cervical discectomy. A transverse neck incision is preferred for corpectomy involving two or three vertebrae; however, a vertical skin incision along the anterior border of the

sternomastoid may be used if more than three vertebrae are involved. Before the removal of a vertebral body, adjacent discs are resected. The posterior part of the vertebra and osteophytes at the posterior margins are excised. Reconstruction is accomplished with an autograft, allograft, or cages (metal or carbon fiber spaces filled with bone fragments). Some of the newer cages are expandable, facilitating a good fit. Supplemental fixation with plates and screws is essential to prevent graft extrusion, to facilitate fusion, and to permit early ambulation.

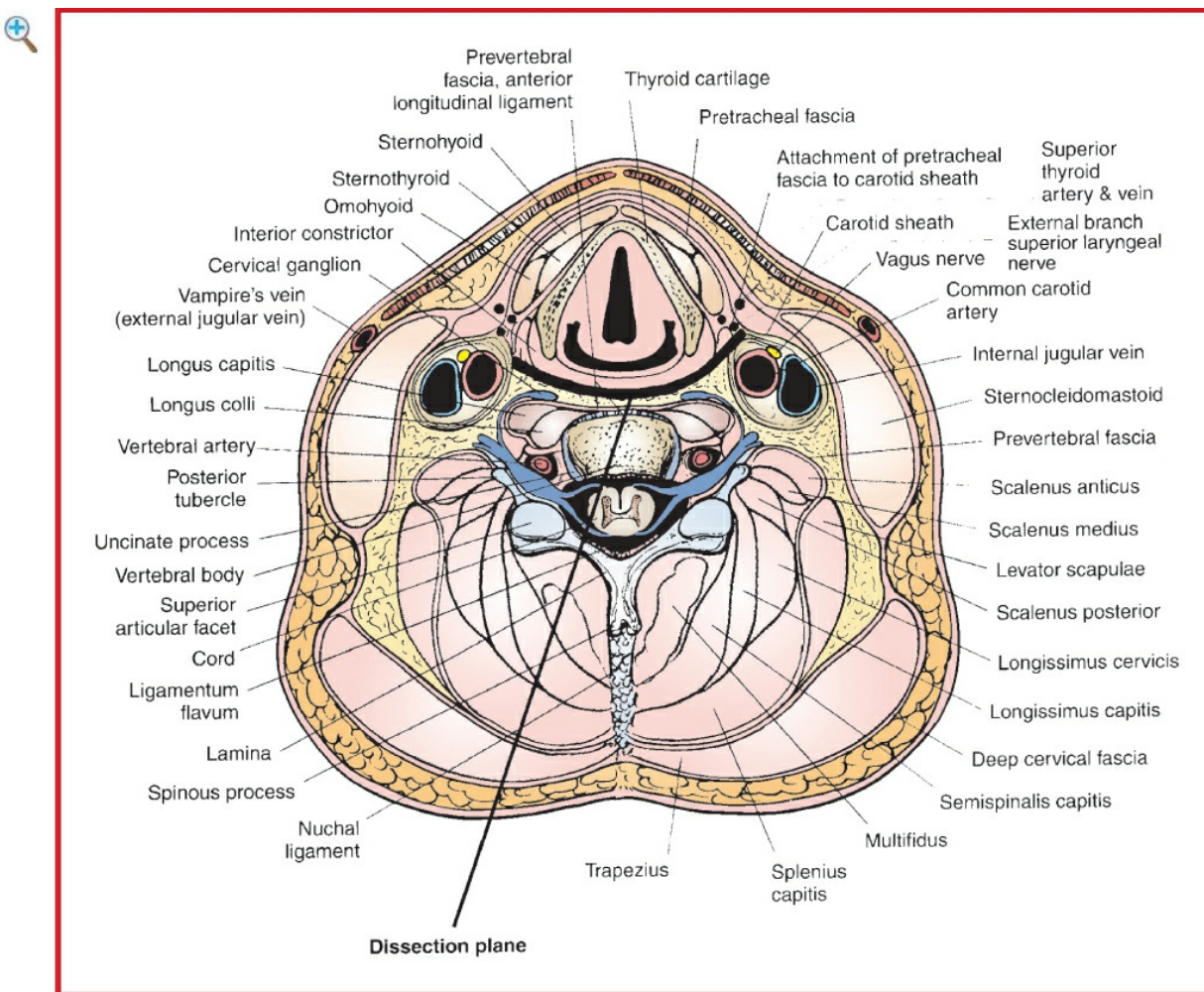


Figure 1.3-7 Cross-section of the C-spine at C5 level. Note the deep cervical fascia, the pretracheal fascia, and the prevertebral fascia. Note the relationship of the pretracheal fascia to the carotid sheath. Dissection plane is shown. (Modified with permission from Hoppenfeld S, deBoer P Surgical Exposures in Orthopaedics: The Anatomic Approach, 2nd edition. J.B. Lippincott, Philadelphia: 1994.)

A **cervical artificial disc** can be placed via an approach that is similar to that of an ACDF but instead of placing a bone graft or cage, a mobile prosthetic disc is placed according to manufacturer's instruction. No additional stabilization or bone grafting is required. Accurate placement of this disc is absolutely critical to the success of this procedure. Fluoroscopy is used extensively to ensure this. There is immediate stability

and because motion is preserved, early mobilization is recommended.

Usual preop diagnosis

Cervical myelopathy (spinal cord compression) 2° fracture of the C-spine (traumatic or pathologic); narrowing of the spinal canal due to congenital conditions; degenerative conditions, such as severe disc disease with osteophyte formation; OPLL; cervical instability (ligamentous laxity or disruption or destruction of bone due to tumor or infection); failed previous spinal fusion

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, and postop pain management options. See [Appendix K](#), p. K-1.

■ SUMMARY OF PROCEDURES		
	Cervical Discectomy ± Fusion and Plating; Artificial Disc Placement	Cervical Corpectomy ± Fusion and Plating
Position	Supine, head extended on headrest	⇐
Incision	Transverse anterolateral	⇐ or vertical anterolateral
Special instrumentation	Operating microscope; anterior cervical plates	⇐
Unique considerations	Fiberoptic intubation may be necessary; ± cervical traction	⇐
Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	1–2 h (+1 h for fusion/plating)	1.5–3 h for single level; add 20–30 min for each additional level
Closing considerations	Cervical collar (soft/hard)	⇐
EBL	25–250 mL	50–1000 mL
Postop care	PACU → room	⇐

Mortality Morbidity	<1%	0–5%
	Esophageal perforation:	⇐
	<1%	⇐
	Infection: <1%	⇐
	Massive blood loss—	⇐
	carotid or jugular injury:	⇐
	<1%	
	Myelopathy: <1%	⇐
	Nerve injury:	⇐
	Recurrent laryngeal nerve:	⇐
	5%	⇐
	Root: <1%	1–15%
	Sympathetic chain: <1%	⇐
	Postop instability: <1%	
	Instrument failure: <1%	
	Slipped graft: <1%	
Pain score	3–5	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–85 yr (usually 20–60 yr)	⇐
Male:Female	1:2	⇐
Incidence	Rare	⇐
Etiology	Neoplastic; traumatic; congenital; degenerative; OPLL	⇐

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Cervical Neurosurgical Procedures.

Suggested Readings

1. Brislin BT, Hilibrand AS: Avoidance of complications in anterior cervical spine revision surgery. *Curr Opin Orthop* 2001; 12(3):257–64.
2. Tasiou A, Giannis T, Brotis AG, et al: Anterior cervical spine surgery-associated complications in a retrospective case-control study. *J Spine Surg* 2017; 3(3):444–59.

POSTERIOR FUSION/FIXATION OF THE MID AND LOWER CERVICAL SPINE

SURGICAL CONSIDERATIONS

Description

Posterior cervical laminectomy (removal of lamina), **foraminotomy** (opening of the neural foramina), and **laminotomy** (removal of a portion of the lamina) are posterior procedures for decompression of the neural elements in the C-spine. These procedures are used to treat cervical radiculopathy 2° degenerative disc disease (e.g., herniated discs, osteophytes). The major advantage of **foraminotomy** over an anterior approach is that it does not require fusion and, thus, preserves the motion of the involved vertebral segments and obviates the need for immobilization for fusion. It also permits decompression of multiple levels, if required. Disadvantages of foraminotomy include an increased incidence of neck pain and the fact that it is not an effective approach to midline disc herniation. **Decompressive laminectomy** can be used to treat cervical canal stenosis (congenital or degenerative) and for removal of intraspinal masses (tumors, AVMs, infective granulomas), which may be extradural, intradural, extramedullary, or intramedullary. Depending on the location of the tumor, the surgeon may need to open the dura and/or spinal cord. Obviously, the intradural intramedullary tumors involve more risk and are more delicate to remove. Many laminae may be removed to expose and excise the tumor. Surgical adjunctive tools (e.g., CUSA, laser, surgical microscope) may be used to aid in removal of the tumor. Intraop evoked potential monitoring may be used during these procedures to test the integrity of the dorsal columns. After the tumor has been removed, the wound is closed in layers, as in a simple laminectomy.

Surgery is performed in the prone or sitting position through a posterior midline incision over the involved vertebrae. The paraspinal muscles are dissected off the spinous processes, and lamina and bone are removed piecemeal. The extent of the procedure depends on the indications for treatment. Hemostasis is achieved with bipolar cautery, and raw bone surfaces are sealed with bone wax. Topical hemostatic agents are used to aid in hemostasis in the epidural gutters. If the patient has an intradural tumor or process, such as syringomyelia, the dura is opened, and the operating microscope is used for this portion of the procedure. After the intradural procedure is complete, the dura is closed, and the surgeon may wish to test the integrity of the closure with a Valsalva-like maneuver (sustained inspiration to 30–40 cm H₂O). The wound is closed in layers, and a drain may be left in the epidural space. Multilevel laminectomies with

foraminotomies (involving partial removal of cervical facet joints) can result in late-onset cervical kyphosis, an extremely difficult condition to treat. These patients are usually considered for concomitant posterior fusion and instrumentation, especially in the presence of cervical segmental instability.

Newer less traumatic techniques can be used to perform foraminotomies and discectomies in the cervical spine in a **minimally invasive (MIS)** fashion. These use one of many tubular retractor systems (e.g., **METRx** [Medtronic, Memphis, TN, USA]) These afford similar exposure but minimize blood loss, scar, and pain by spreading the muscles. The disadvantage is unfamiliar exposure, difficulty with retractor placement, and potential of neurologic injury by inadvertent penetration of the interlaminar space.

Posterior cervical wiring techniques include (a) **interspinous wiring** (Fig. 1.3-8) (wires are passed through drilled holes in the base of adjacent spinous processes and then tightened), (b) **sublaminar wiring** with Luque rods or rectangles (sublaminar wires are passed at each level on both sides and are tightened over the rods or rectangles), and (c) **a triple-wire technique** with the first wire being passed through drill holes at the base of each spinous process and the second and third wires passed through the same holes and then through drill holes in the previously placed bone grafts. This latter technique is biomechanically sound because it places the bone grafts in compression. Wiring techniques, although stable in flexion, however, are less stable in extension and rotation, and they cannot be performed in patients with prior laminectomy or requiring laminectomy.

In the **posterior cervical lateral mass screw fixation** technique, the C-spine is exposed through a midline incision over the involved vertebral segments. The lateral mass (bony column between facet joints) is identified, drilled, and tapped. Cortical screws are passed into the lateral mass and fixed with plates or rods. The trajectory of the screws in the lateral mass is to the upper outer corner in the classic Magerl technique. The entry point should be ~1 mm medial to the midpoint of the lateral mass (Fig. 1.3-9). The axial trajectory angulation should be 25°, and the sagittal angulation should be 45°, which is in-line with the facet joints (Figs. 1.3-10 and 1.3-11). Adjacent facet joints are decorticated, and bone grafts are placed. Lateral mass plating provides a rigid multisegmental fixation and can be performed in patients with prior laminectomy. The major risks involved with this procedure are nerve root and vertebral artery injuries.

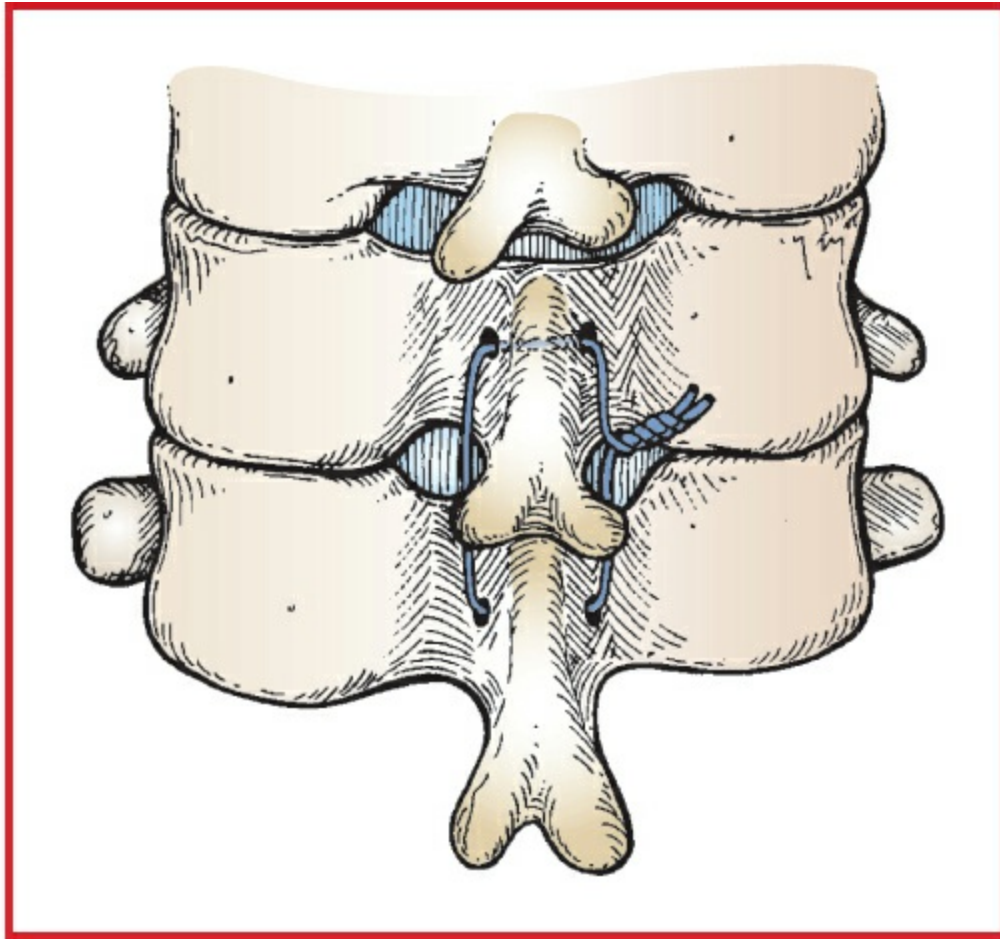


Figure 1.3-8 Intraspinous wiring. Wires passed through drilled holes in base of adjacent spinous processes and tightened. (Reproduced with permission from An HS, Cotler JMSpinal Instrumentation, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 1999.)

Cervical pedicle screw plate fixation is an effective alternative to lateral mass fixation. In this technique, screws are passed under fluoroscopic guidance into the cervical pedicles and secured to plates or rods. This procedure is technically demanding, as the cervical pedicles are narrow and in close proximity to nerve roots, vertebral artery, and spinal cord, and their trajectories are typically unfamiliar. This technique is biomechanically stable and permits the correction of deformity by application of compression or distraction forces. It is most commonly performed at C2 where the pedicles are relatively larger.

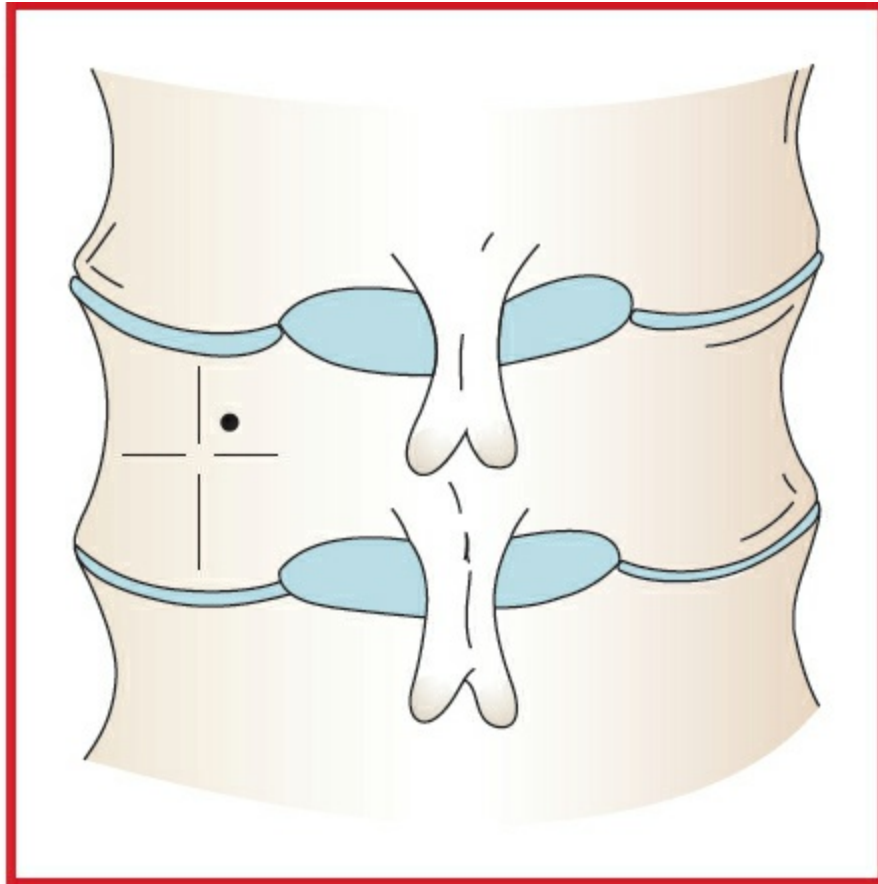


Figure 1.3-9 Magerl technique lateral mass screw entry point. (Reproduced with permission from Barrey C, Mertens P, et al: Quantitative anatomic evaluation of cervical lateral mass fixation with a comparison of the Roy-Camille and the Magerl screw techniques. Spine 2005; 30(6):E140–7.)

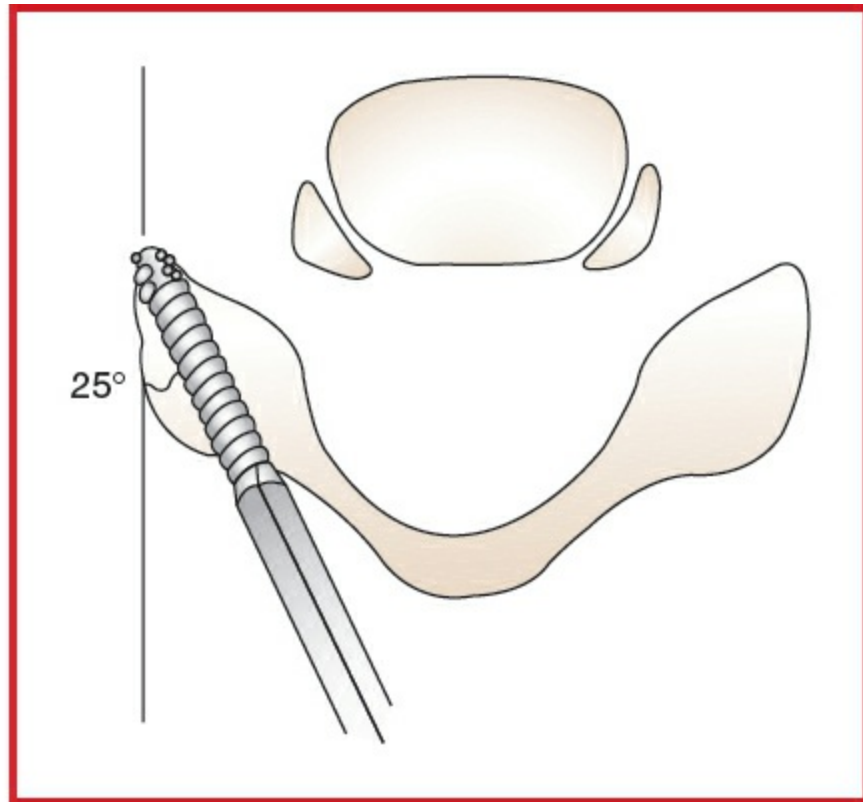


Figure 1.3-10 Magerl technique lateral mass screw axial trajectory angulation. (Reproduced with permission from Barrey C, Mertens P, et al: Quantitative anatomic evaluation of cervical lateral mass fixation with a comparison of the Roy-Camille and the Magerl screw techniques. *Spine* 2005; 30(6):E140–7.)

Usual preop diagnosis

Cervical radiculopathy (nerve root compression); cervical myelopathy (spinal cord compression); cervical disc disease (herniation or degeneration of one or more cervical discs); C-spine injury

Variant procedures

Patients with panvertebral disease (involving anterior and posterior elements of the spine) and three-column spinal instability often require **combined anterior and posterior decompression, reconstruction, and instrumentation**. Anterior screw plates provide a strong tension band to resist vertical/horizontal translation and neck extension; however, they are less able to resist flexion or rotation. By contrast, posterior cervical plates strongly resist flexion or rotation but are less able to resist extension. Thus, in the presence of three-column spinal instability, combined anterior and posterior instrumentation often is required. This technique provides rigid fixation of spinal segments and avoids the need for rigid external orthotic devices.

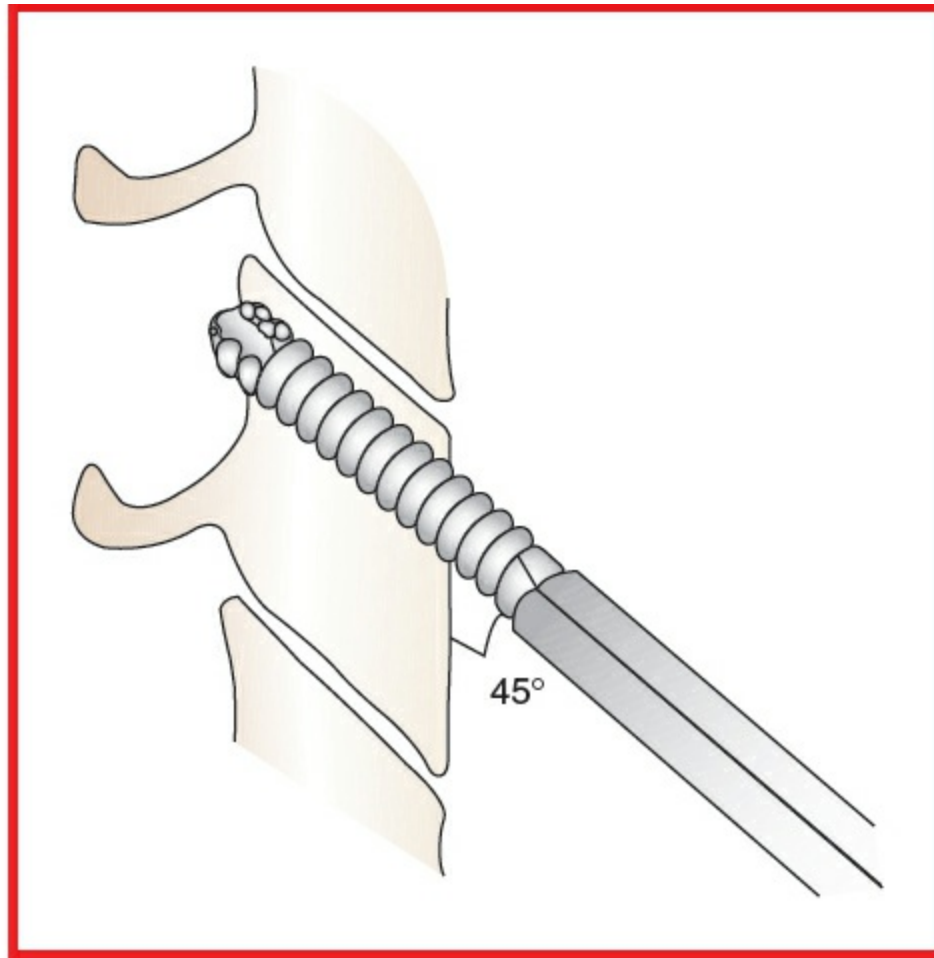


Figure 1.3-11 Magerl technique lateral mass screw sagittal trajectory angulation. (Reproduced with permission from Barrey C, Mertens P, et al: Quantitative anatomic evaluation of cervical lateral mass fixation with a comparison of the Roy-Camille and the Magerl screw techniques. *Spine* 2005; 30(6):E140–7.)

Combined instrumentation techniques are challenging and require several special considerations. Patients with any unstable C-spine may require fiberoptic intubation, intraop cervical traction, and electrophysiologic monitoring. **Anterior and posterior cervical instrumentation** is usually carried out in a single surgical session although may be staged. The transition between the anterior and posterior approaches requires a specialized operating table (e.g., Jackson spinal table or Stryker frame) and careful coordination among the entire OR team. The long duration of surgery may be associated with increased incidence of respiratory complications, blood loss, and prolonged ICU stays. In rare instances, an additional anterior or posterior approach may be performed (“540-degree procedure”).

Usual preop diagnosis

C-spine injury causing three-column (severe, unstable) injuries; cervicothoracic junctional pathologies; correction of kyphotic deformities; panvertebral disorders

involving C-spine (neoplasms, infection, spondylitic myelopathy); failed symptomatic anterior cervical fusions

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Cervical Laminotomy/Foraminotomy	Cervical Laminectomy with Instrumentation	Combined Anterior/Posterior Cervical Instrumentation
Position	Prone or sitting; pin fixation or horseshoe headrest	Prone; pin fixation or horseshoe headrest	⇐ + supine
Incision	Posterior midline neck	⇐	⇐ + transverse longitudinal anterolateral
Special instrumentation	Operating microscope for intradural procedures	⇐ + Implants, pedicle screws with plates	Implants; and posterior plates with
Unique considerations	Fiberoptic intubation; I.I. to assess correct level	⇐; I.I. for screw placement	Implants; and posterior plates with
Antibiotics	Cefazolin 2 g	⇐	⇐
Surgical time	1.5–2 h	⇐ Add 30–45 min/level of instrumentation	4–5 h
EBL	25–500 mL	⇐	250–600 mL
Postop care	PACU → room	⇐	⇐; sometimes needs short stay
Mortality	0–3%	⇐	⇐

Morbidity	All <5%: Neurologic deterioration Myelopathy Nerve root injury CSF leak Postop instability Infection	⇐ ⇐ Additionally: Instrumentation failure Vascular injury Nerve root injury 2° screws	5–10% ⇐ Addition Bone graf dislodgem Respirato problems
Pain score	2–6	5–10	7–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–85 yr (usually 20–60 yr)
Male:Female	1:2
Incidence	Rare
Etiology	Neoplastic; traumatic; congenital; degenerative; infections; rheumatoid arthritis

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Cervical Neurosurgical Procedures.

Suggested Readings

1. Cason GW, Herkowitz HN: Cervical intervertebral disc replacement. *J Bone Joint Surg Am* 2013; 95(3):279–85.
2. Leven D, Meaie J, Radcliff K, et al: Cervical disc replacement surgery: indications, technique, and technical pearls. *Curr Rev Musculoskelet Med* 2017; 10(2):160–9.

ANTERIOR CERVICOTHORACIC SPINE SURGERY

■ SURGICAL CONSIDERATIONS

Description

The **anterior approach** to the **cervicothoracic junction** (CTJ, C7-T3) is performed for discectomy, stabilization of spinal fractures, tumor resection, spinal reconstruction, and instrumentation. Anterior approaches to the CTJ often are challenging, as this area represents a rapid transition from cervical lordosis to thoracic kyphosis, resulting in

abrupt increase in the depth of the wound. The confluence of great vessels and visceral (trachea, esophagus) and neural structures at the thoracic inlet makes them susceptible to injury. The modified anterior approach utilizes a “hockey-stick” incision to allow greater access to the lower cervical and upper thoracic spine. This technique does not require any additional bone resection during the approach. The **transsternal** approach involves a longitudinal incision along the anterior border of the sternomastoid, extended over the midline of the sternum to the xiphisternum. The sternum is divided using an oscillating saw and retracted to expose the anterior aspect of the CTJ. The wide exposure available with this approach permits vertebral resection, reconstruction, and instrumentation. The **transclavicular** approach involves a T-shaped incision over the clavicles, with a vertical limb extending down the midline of the sternum. Subplatysmal flaps are elevated, and the sternal and clavicular heads of the sternomastoid are detached from their origin and retracted superolaterally. The medial third of the clavicle and manubrium are resected to provide an excellent direct anterior approach to the CTJ for vertebral decompression, reconstruction, and stabilization.

Variant procedure and approaches

Axillary thoracotomy and high transthoracic thoracotomy approaches permit an anterolateral exposure to CTJ. As these procedures involve entering the thoracic cavity, they typically require one-lung ventilation (OLV) during the procedure.

Usual preop diagnosis

C7-T3 disc disease, fracture, tumor, and deformity

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Transsternal Approach	Transclavicular Approach	Axillary/High Thoracotomy
Position	Supine	⇐	Lateral
Incision	Longitudinal incision along anterior border of sternomastoid	T-shaped incision over clavicles with a vertical limb down the midline of sternum	Lateral chest wall

Special instrumentation	Anterior cervical plates Special oscillating saw for sternal opening	⇐	⇐ Special blades for clavicular resection
Unique considerations	OLV not necessary	⇐	OLV necessary
Antibiotics	Cefazolin 2 g iv	⇐	⇐
Surgical time	3–4 h	⇐	⇐
Closing considerations	Soft cervical collar	⇐	⇐
EBL	200–500 mL	⇐	⇐
Postop care	PACU → room	⇐	⇐
Mortality	1–2%	⇐	⇐
Morbidity	Wound infection/breakdown Injury to great vessels at thoracic inlet	⇐ ⇐	⇐ Cosmetic deformity Respiratory problems
Pain score	7–10	7–10	7–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	30–50 yr
Male:Female	1:1
Incidence	Uncommon
Etiology	Disc herniations; trauma; tumor; infections

ANESTHETIC CONSIDERATIONS FOR CERVICAL NEUROSURGICAL PROCEDURES

(Procedures covered: anterior/posterior fusion/fixation of upper and mid/lower C-spine and anterior cervicothoracic spine surgery)

PREOPERATIVE

Surgery of the C-spine is common primarily because of the frequency of herniation of a cervical intervertebral disc causing compression of the adjacent spinal nerve roots. Other less frequent indications for cervical surgery include acute or chronic instability of the neck requiring fusion, removal of a tumor of the spinal cord, or craniocervical decompression for Arnold-Chiari malformation.

Respiratory

Trauma to the head and high C-spine may be associated with loss of consciousness and/or acute respiratory failure resulting in severe hypoxemia and hypercapnia. These cases require immediate airway control. While efforts should be made to minimize neck movement, these concerns should not be allowed to delay the restoration of adequate ventilation and oxygenation. Direct laryngoscopy with manual in-line stabilization is a rapid and reasonable approach to emergently securing the airway without producing excessive neck motion. If a video laryngoscope or a fiberoptic bronchoscope is immediately available, these techniques may allow airway control with less motion of the neck but may also take more time to perform. C-spine trauma patients should be assessed for the presence of other injuries such as pulmonary contusion, pneumo- or hemothorax, tracheal injury, or facial injury that may affect pulmonary function, impede ventilation, or make airway control difficult.

Tests: Consider ABG to quantify the degree of respiratory impairment, if present.

Cardiovascular

Acute fractures of the C-spine and associated spinal cord trauma may → ↓ sympathetic tone → ↓ BP and ↓ HR (neurogenic or spinal shock). Generally, this condition can be treated effectively with crystalloid and/or colloid infusion and atropine to ↑ HR. Rarely is it necessary to use vasopressors to maintain BP or HR. If spinal cord injury is suspected, an A-line must be placed, and MAP should be >80 mm Hg at all

	times to avoid vascular insufficiency of the spinal cord. Severe cervical spine disease may compromise blood flow through the vertebral arteries. If present, careful positioning and maintenance of adequate MAP are essential. Tests: As indicated from H&P
Neurologic	Patients with herniation of a cervical disc generally complain first of pain in the neck, particularly with lateral rotation of the head. The pain may radiate down one or, rarely, both arms. As nerve compression continues, patients begin to develop weakness and atrophy of specific muscle groups in the arm. These Sx, however, are not specific to herniation of a disc and may be caused by a spinal cord tumor or cyst. Patients with acute fractures of the neck and attendant spinal cord trauma at the T1 level may be paraplegic to some degree, whereas fractures above C5 may result in quadriplegia and loss of phrenic nerve function. Injuries between these two levels result in variable loss of motor and sensory functions in the upper extremities. A careful documentation of preop sensory and motor deficits is important. Be aware that changes in the head or neck position may alter patient signs and symptoms. Tests: MRI is the primary diagnostic test to evaluate disease affecting the spinal cord. Patients with suspected acute cervical trauma are usually evaluated with multiple imaging techniques.
Hematologic	Antiplatelet agents should be stopped 5 d before elective surgery. At least 2 U of PRBCs should be cross-matched if large blood loss is anticipated. Expect more blood loss during surgeries involving multiple levels, tumor resection (especially metastases), or at C1 and C2. Tests: Hct; T&C; others as indicated from H&P
Laboratory	Other tests as indicated from H&P

Premedication

Elective cases for chronic disease should be premedicated as usual. Acute trauma victims demand extreme caution with premedication since the possibility of intoxication, brain, chest, and/or spinal cord trauma may predispose the patient to serious adverse effects. Proper topicalization and cautious sedation (consider dexmedetomidine: load 0.5–1 mcg/kg over 10 min) are critical when attempting awake intubation to prevent coughing and gagging-induced motion of the neck.

ERAS

Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water.

INTRAOPERATIVE

ANESTHETIC TECHNIQUE: GETA

Induction

It is important to discuss the severity of the surgical lesion and the planned intubation with the surgeon before proceeding. The ideal technique for airway management in patients with cervical spine disease remains a topic of debate. All airway management techniques (including mask ventilation) will likely result in some C-spine movement, but direct laryngoscopy seems to result in more motion than other intubation techniques. Minimizing neck motion during airway management is an important goal. For patients with a stable neck, orotracheal intubation using standard laryngoscopy is acceptable. Manual in-line stabilization will likely reduce neck motion but will also make direct laryngoscopy more difficult. Alternative techniques should be employed in patients with an unstable C-spine, if a difficult airway is anticipated or if neck manipulation exacerbates symptoms. Depending on the clinical situation,

fiberoptic intubation before or after induction (with or without an LMA), a lighted stylet device, or a video laryngoscope (e.g., a GlideScope®) might be most appropriate (with care being taken to minimize neck motion). An awake intubation has the advantage of permitting a neurologic assessment after intubation and positioning.

Whether it is done before or after intubation, induction is typically accomplished with propofol 0.5–2.5 mg/kg. If MEP monitoring will be used, consider succinylcholine or low-dose rocuronium (0.5 mg/kg) vs. deep anesthesia and remifentanyl bolus (1.5 mcg/kg) to facilitate intubation yet permit the rapid return of neuromuscular transmission.

ERAS

Verify preincision antibiotics. Assess volume status (see [Appendix K](#)). Maintain normovolemia and normothermia. Avoid hypothermia. Dexamethasone (4–8 mg iv) may ↓ PONV. Sequential pneumatic compression devices should be used to ↓ VTE risk. Consider ketamine (0.25 mg/kg/h) and/or lidocaine (1–2 mg/kg/h) infusions as components of MMPM especially in chronic pain patients.

Maintenance

If no neurophysiologic monitoring is used, then standard maintenance is sufficient. Neuromuscular blockade with rocuronium 0.6 mg/kg or vecuronium 0.1 mg/kg is helpful for positioning the patient and insertion of the Dingman retractor. After muscle dissection is complete and the retractors are in place, further use of relaxants is usually not necessary.

When EP monitoring is used the anesthetic plan must be adapted to ensure optimal monitoring conditions. Standard NMB doses will completely abolish MEPs and must be avoided although small NMB doses may be needed to suppress interfering EMG activity. Doses >0.5 MAC each of a volatile anesthetic and N₂O interfere with MEP and, to a lesser extent, SSEP monitoring. An anesthetic consisting of 50% N₂O in

O₂ with up to 0.6% isoflurane in combination with a remifentanyl infusion (0.05–2 mcg/kg/min) will provide anesthesia while permitting both MEP and SSEP monitoring. TIVA may be preferred by local custom when monitoring EPs but is not necessary. The neurophysiology monitoring team must be updated when conditions that affect the evoked potentials change (depth of anesthesia, anesthetic agent, blood pressure, PCO₂, PO₂). As soon as neuromonitoring is complete, switching to an inhaled anesthetic and stopping the propofol will allow for a timely emergence. In general, keep MAP > 80 mm Hg to ensure adequate perfusion pressure to the cord.

Emergence

If a cervical fusion has been performed and the patient has been returned to a halo device or body jacket, it is desirable to leave ETT in place until the patient is fully awake, responding to commands, and able to manage his/her own airway as reintubation is likely to be difficult. **NB:** Immediate airway obstruction 2° soft tissue occlusion or superior laryngeal nerve damage may occur on extubation. A useful way to test for airway patency is to deflate the cuff of the tracheal tube and determine that the patient is able to breathe around the tube as well as through it. If there is any question about adequacy of the airway, it is prudent to leave the ETT in place and spray lidocaine 4% 4 mL down the trachea using a laryngotracheal anesthesia device (LTA) before emergence. This technique will usually prevent or minimize coughing or bucking on the ETT for about 15–30 min. One also should consider inserting an airway exchange catheter (AEC) through the ETT before its removal. AECs are well tolerated and can be left in place until one is confident that no further airway compromise will occur. So long as the AEC is not touching the carina, it will not induce coughing or bucking, and the patient can talk without difficulty.

	This catheter will provide a conduit for immediate reinsertion of an ETT if airway obstruction from early or delayed swelling, bleeding, or hematoma formation should occur. Remifentanyl will not provide postop analgesia and a longer-acting opioid (e.g., hydromorphone 0.2–0.6 mg iv) must be administered prior to emergence. Suggest wound infiltration with LA. Consider iv acetaminophen 1 g if po dose given >4–6 h ago and hydromorphone (0.2–0.6 mg iv). Discuss possible use of ketorolac or celecoxib.	
Blood and fluid requirements	16–18 ga × 1 BSS/LR @ 1–3 mL/kg/h T&C 1–2 units depending on type of surgery	While blood transfusion is rarely needed for operations on the C-spine, expect more blood loss during surgeries involving multiple levels, tumor resection (especially metastases), or C1 and C2. Suggest goal-directed fluid management (see Appendix K, p. K-4). Zero balance or goal-directed fluid management is recommended.
Monitoring	Standard monitors ± Arterial line ± Urinary catheter ± BIS monitor MEP/SSEP	If a standard BP cuff is to be used, consider placing it on a leg at the ankle. If placed on an arm without rigid protection (e.g., sleds), the surgeons tend to lean on the cuff, making it difficult to take consistent, reliable measurements. If the patient has a CV, respiratory, or metabolic

	± CVP line ± Precordial Doppler	disorder (e.g., insulin-dependent diabetes), consider inserting an arterial catheter. For patients with spinal cord injury or severe stenosis, an arterial line should be placed prior to induction when hypotension is most likely. If a posterior surgical approach with patient in seated position is planned, arterial and CVP catheters should be placed. If a CVC is placed for the purpose of aspirating air from the RA, then its location must be confirmed before the start of surgery by x-ray, transcatheter ECG, or echocardiography. A precordial Doppler flow probe also should be placed on the anterior chest wall; confirm position by injecting 1 mL of agitated NS into CVP line and listening for the characteristic change in Doppler sound, alternatively by transcatheter ECG or by x-ray.
Positioning	Supine: ✓ and pad pressure points	For anterior cervical discectomy and/or fusion, patient is

✓ eyes
Shoulder roll
± Cervical traction

positioned supine with a roll under the shoulders or neck, and the head is moderately extended. A cervical strap may be placed below the chin and behind the occiput and attached to a weight of 5–10 lb hung over the head of the bed. If patient is in a halo or tongs, 5–10 lb of weight is attached to the device. Alternatively, the surgeon may request traction intermittently during insertion of bone plugs for fusion. The surgical incision can be made on either side of the neck at the discretion of the surgeon.

Prone:
✓ and pad pressure points
✓ eyes
✓ genitalia

The head will usually be placed in a Mayfield three-point attachment. Ensure that there is no posterior pressure on the ETT, as it can injure the uvula or soft palate. Ensure a neutral C-spine position. Malpositioning, when maintained for several hours, may result in catastrophic damage to the spinal cord. If there are signs of cord dysfunction on evoked potential monitoring with

no obvious physiologic or surgical explanation, consider adjustments to the position of the head to try to relieve any occult malpositioning.

Sitting:

✓ and pad pressure points

✓ eyes

VAE monitoring:

✓ ETT position

Although seldom used today, there are some advantages to the sitting position compared to the prone position. For the surgeon: (a) easier access to the lesion, (b) less blood loss, (c) less interference from CSF, and (d) lower incidence of neurologic injury. For the anesthesiologist: (a) less chance of ETT and line dislodgement, (b) less chance for inadvertent pressure injury, (c) easier assessment and management of ventilation, and (d) easier access to patient for insertion of additional catheters, if necessary.

NB: measure (or calculate) blood pressure at the level of the head to avoid cerebral hypoperfusion. The previous notes of caution about malpositioning of the neck apply in the sitting position as well.

Complications

VAE

The major disadvantage of the sitting position is the risk of VAE, particularly paradoxical air embolism through a PFO (or, rarely, through the pulmonary circulation) → CNS or coronary emboli. Incidence of VAE is up to 76% in patients operated on in the seated position. VAE is easily detected using a combination of Doppler, ETCO_2 , and ETN_2 analysis and complications are rare. If VAE is suspected ($\text{Sx} = \downarrow \text{ETCO}_2$, $\uparrow \text{ETN}_2$, $\downarrow \text{BP}$, dysrhythmias), notify the surgeon to flood the field, d/c N_2O , lower the head if possible, and aspirate the right atrial catheter using a syringe. This generally will confirm the Dx as well as provide treatment. If VAE continues, and the surgeon has difficulty identifying the site of air entrainment, consider using $\text{PEEP} \leq 10 \text{ cm H}_2\text{O}$ or bilateral jugular compression to increase cerebral venous pressure. PEEP may promote paradoxical air embolism.

Esophagus perforation	In cases of massive systemic air embolism, hypothermia and hyperbaric oxygen therapy should be started ASAP. A possible complication of anterior cervical surgery is esophageal perforation. One way to assess for this type of injury is to flood the operative field with saline and then force air through the oropharynx and look for any signs of bubbling. These can be occult and manifest postop as fistulas as well.
Retraction nerve injury	Another potential complication of anterior surgery is retraction injury to various nerves including the recurrent laryngeal. Pressure can be taken off the nerve by reducing the ETT cuff pressure once the retractors are in place to mitigate any compression due to the nerve being trapped between the retractor and the ET cuff.
Hypotension (calculate MAP at head level)	↓ BP caused by venous pooling, inadequate venous return → ↓ CO can be treated by wrapping lower extremities while patient

is supine, infusing adequate fluid volume to maintain right heart filling pressure, and avoiding excessive depth of anesthesia. Any acute drop in MAP may be due to a vascular injury, either obvious (vertebral artery injury) or occult (deep great vessel injury) until proven otherwise. The source of bleeding should be identified. Again, be sure to measure (or calculate) blood pressure at the level of the head to avoid cerebral hypoperfusion.

POSTOPERATIVE

Complications

Airway obstruction,
edema
Hematoma
Neurologic deficit

Airway obstruction is usually the result of soft tissue falling back against the posterior pharyngeal wall, which may be difficult to correct by forward displacement of the mandible. Tissue edema from prone positioning and surgical manipulation, vocal cord paresis from recurrent laryngeal nerve injury, or hematoma formation may also contribute.

Obstruction may require oral or nasal airway or reintubation. Consider inserting an airway exchange catheter before removing the ET tube. If a hematoma is suspected as the cause of respiratory distress, the wound must be opened immediately for evacuation. This should be done before any attempts at intubation, because direct laryngoscopy may not be possible if tracheal compression distorts the view. Emergent tracheostomy or cricothyrotomy can be lifesaving in patients with extreme soft tissue swelling.

Tension pneumothorax

Delayed respiratory insufficiency usually is caused by either development of a tension pneumothorax or from bleeding into the neck at the surgical site, with progressive compression of the airway. If a tension pneumothorax is suspected and circulatory signs are stable, immediate CXR should confirm the Dx. If

circulation is failing, an 18- or 20-ga needle catheter should be inserted immediately anteriorly at the 2nd intercostal space on the suspected side to relieve the pneumothorax. If the Dx is airway obstruction from bleeding into the neck, the wound should be opened immediately and clots and blood removed. This should be done before any attempts at intubation because direct laryngoscopy may not be possible if tracheal deviation distorts the view. Emergent tracheostomy or cricothyrotomy can be lifesaving. It is sometimes difficult to distinguish airway obstruction from tension pneumothorax by physical signs. One useful way is to check for the "puff sign." With airway obstruction from any cause, what gas moves in and out of the airway does so very slowly because of an obstruction. In contrast, with tension pneumothorax, gas moves out of the airway

with great speed because of high intrapleural pressure. By applying positive pressure to the airway and listening at the patient's mouth for the sound of gas escaping as airway pressure is released, one hears either a puff or jet of air escaping (tension pneumothorax) or slow, gradual exit of air (airway obstruction).

Multimodal pain management	IV hydromorphone IV acetaminophen Consider PCA ± Fentanyl 25–50 mcg prn ± Ketorolac (30 mg iv) or celecoxib (200 mg po)	0.2–0.6 mg iv before emergence 1 g q6h for adults, 15 mg/kg q6h if ≤50 kg or ≤12 yr. If not given preop or if 4–6 h has elapsed since previous dose. Consult with surgeon regarding possible hemostasis concern.
ERAS	Ondansetron 4 mg iv ± Metoclopramide 10–20 mg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization with physiotherapy.
Tests	CXR (as indicated) Hct	Repeat neurologic exam prior to discharge from PACU.

Suggested Readings

1. Aziz M: Airway management in neuroanesthesiology. *Anesthesiol Clin* 2012; 30(2):229–40.
2. Cason GW, Herkowitz HN: Cervical intervertebral disc replacement. *J Bone Joint Surg Am* 2013; 95(3):279–85.

3. Cavallone LF, Vannucci A: Review article: extubation of the difficult airway and extubation failure. *Anesth Analg* 2013; 116(2):368–83.
4. Chen CJ, Saulle D, Fu KM, et al: Dysphagia following combined anterior-posterior cervical spine surgeries. *J Neurosurg Spine* 2013; 19(3):279–87.
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ANTERIOR THORACIC SPINE SURGERY

SURGICAL CONSIDERATIONS

Description

The **anterior approach** to the mid and lower thoracic spine (T4-T12/L1) is performed for spinal fracture, scoliosis/kyphosis, tumor, and infection. The **anterior transthoracic approach** provides a wide and easy exposure of the thoracic spine from T4-T10. The patient is placed in a lateral decubitus position (right or left, based on spinal pathology). An incision is made over the involved vertebrae and extended rostrally one or two intercostal spaces. The muscles and ribs are retracted, the pleura are opened, and lungs retracted to expose the vertebral bodies. Discectomy, corpectomy, bony reconstruction, and stabilization can be performed as required under radiographic guidance. The risk of spinal cord injury depends on the extent of surgery and reconstruction. The complex vasculature can be challenging. Hemostasis is vital to prevent hemothorax. Chylothorax and lymphatic injury are rare.

The **transition zone** at the thoracolumbar (TL) spine (T11-L1) is somewhat difficult to

expose and requires a combined transthoracic and retroperitoneal approach through the diaphragm (**thoracolumbar transdiaphragmatic approach**). A left-sided approach is preferred because it is easier to retract the spleen and stomach than the liver. The skin incision is made over the 10th rib, down to thoracic muscles, and the rib is resected subperiosteally to provide wide exposure. Blunt dissection separates the peritoneum from the undersurface of the diaphragm and lateral and posterior abdominal walls. With gentle retraction of the lung and abdominal contents, the diaphragm is well visualized and is sectioned circumferentially from the chest wall. This provides an excellent exposure of the anterior aspect of the TL junction. Most procedures can be performed with minimal retraction of lung tissue at this level; thus, OLV is not typically required. Vertebral resection, reconstruction, and stabilization are performed with radiographic guidance. Reconstruction of the diaphragm is important to prevent herniation. This is rare with pure retroperitoneal approaches.

Variant procedure or approach

An 11th rib extrapleural-retroperitoneal approach offers an alternative approach to the TL junction. It provides excellent exposure without the need to incise the diaphragm, resulting in less morbidity and reduced risk of pulmonary complications. Because the pleural cavity is not entered, chest drains and OLV are not needed.

Thoracoscopic spine surgery

This technically challenging procedure is performed in the lateral position with GA, using OLV. Four or more 10–15-mm portals are made, with the working portal centered over the target vertebra. The optical (scope) portal is placed two or three intercostal spaces cranial to the target vertebra. Separate portals anterior to the working channel allow suction/irrigation and retraction. When using **thoracoscopic instrumentation**, hardware is placed through the portals in the chest wall under fluoroscopic guidance. Blood loss must be minimized as hemostasis is difficult with this approach. Because instrumentation requires a wide exposure of the spine, OLV is essential. The major advantages of thoracoscopic surgery include minimal rib retraction; minimal blood loss, with consequent early removal of chest drain; reduced wound pain; early ambulation; and low morbidity. These factors combine to reduce hospital stay.

Usual preop diagnosis

Fractures (usually at the TL junction); scoliosis; primary and metastatic tumors of the spine; pyogenic and tuberculous osteomyelitis; Scheuermann kyphosis

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, and postop pain

management options. See [Appendix K](#), p. K-1.

■ SUMMARY OF PROCEDURES			
	Transthoracic (T4-T10)	Transdiaphragmatic (T11-L1), 10th Rib	Thoracoscopic
Position	Lateral decubitus + axillary roll	⇐	⇐
Incision	Over involved vertebrae	Along 10th rib	3–4 portals (10–15 mm)
Special instrumentation	Anterior thoracic instrumentation/cages	⇐	Thoracoscopic spinal instrumentation
Unique considerations	DLT ± OLV	⇐	DLT; OLV mandatory
Antibiotics	Cefazolin 2 g iv	⇐	⇐
Surgical time	2–6 h	⇐	⇐
Closing considerations	Transfer to bed before emergence. NB: sudden movement may dislodge grafts or implants	⇐ + May need closure of diaphragm	⇐
EBL	200–5000 mL; nontumor cases: 200–400 mL	⇐	50–200 mL
Postop care	Chest tube to water seal	⇐	⇐
	Chest physiotherapy/incentive spirometer Short ICU stay is usual. Bleeding >200 mL/h → re-exploration		Shorter period of chest drain/ICU stay

Mortality	3–5%	⇐	⇐
Morbidity	Overall: 5–15%	⇐	⇐ + Problem with OLV
	DVT	⇐	—
	—	—	Conversion open surgery 4%
	Neurologic: 1–2%	⇐	—
	Infection	⇐	—
	Vascular injury	⇐	—
	Sepsis	⇐	—
	Atelectasis	⇐	⇐
	Pneumonia	⇐	⇐
Pain score	7–10	7–10	5–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	12–30 yr (scoliosis surgery); >40 yr (tumor and infection surgery)
Male:Female	1:1; except for > scoliosis surgery in females
Incidence	20,000/yr
Etiology	Scoliosis, idiopathic (50%); trauma (20%); scoliosis, neuromuscular (15%); infections, tumors (10%); scoliosis, congenital (5%)
Associated conditions	See Etiology (above)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Thoracolumbar Neurosurgical Procedures.

Suggested Readings

1. Foreman PM, Naftel RP, Moore TA II, et al: The lateral extracavitary approach to the

- thoracolumbar spine: a case series and systematic review. *J Neurosurg Spine* 2016; 24(4):570–9.
2. Sundaresan N, Shah J, Feghali JG: A transsternal approach to the upper thoracic vertebrae. *Am J Surg* 1984; 148:473–7.
 3. Sundaresan N, Shah J, Foley KM, et al: An anterior approach to upper thoracic vertebrae. *J Neurosurg* 1984; 61:686–90.

POSTERIOR THORACIC SPINE SURGERY

SURGICAL CONSIDERATIONS

Description

Thoracic laminectomy (midline removal of the lamina) and **costotransversectomy** (off midline removal of the rib head and transverse process) are procedures for decompressing the neural elements of the thoracic spine via a posterior approach. (Commonly used approaches are shown in [Fig. 1.3-12](#).)

Thoracic laminectomy is used to treat spinal cord compression due to disc herniation, neoplasm, or trauma. It also is used to gain access to the spinal canal or spinal cord for various intradural mass lesions, including syringomyelia. Thoracic laminectomy is done through a posterior midline incision centered over the involved vertebrae. The paraspinal muscles are retracted subperiosteally from the spinous processes and laminae on both sides. Laminae are removed piecemeal with rongeurs or drills. Extensive laminectomy involving several segments or requiring removal of facet joints may require stabilization with transpedicular screws or hooks. Intradural procedures are performed with an operating microscope and microneurosurgical instruments. After completion of the intradural procedure, watertight closure of the dura is obtained and tested with Valsalva maneuver (sustained inspiration at 30–40 cm H₂O).

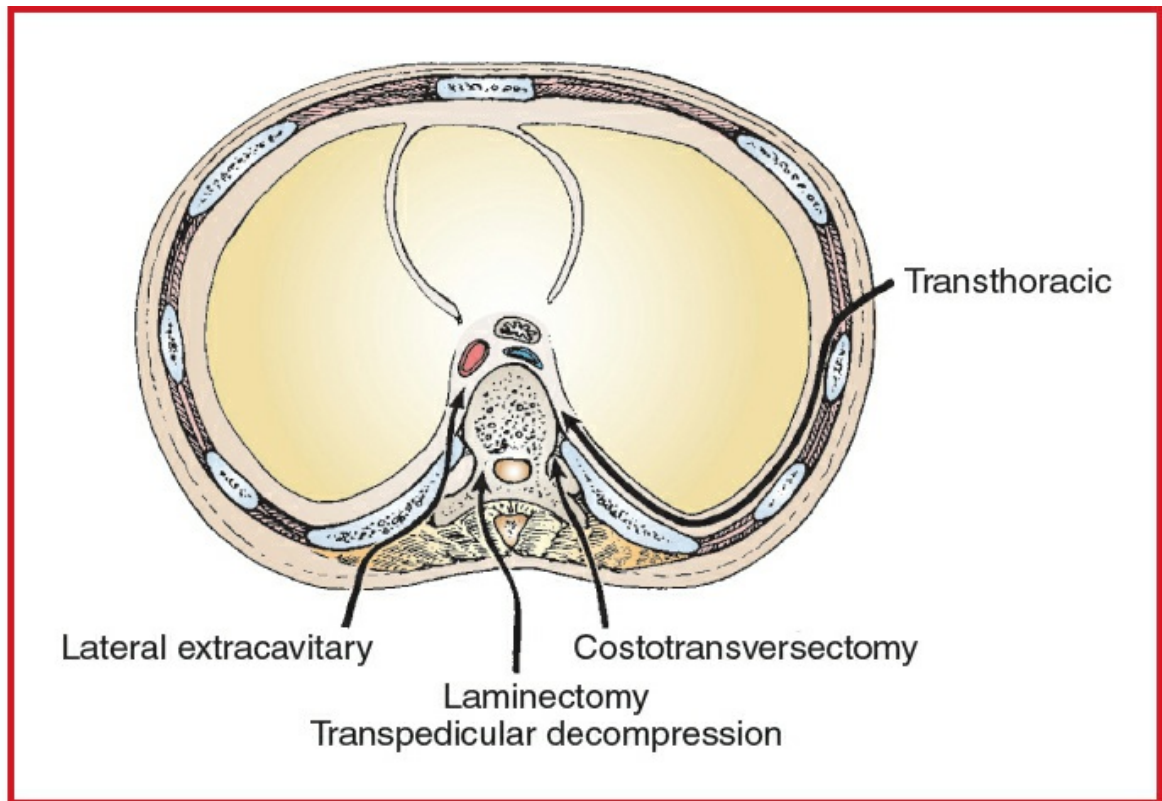


Figure 1.3-12 Surgical approaches to the thoracic spine. (Reproduced with permission from Tindall GT, Cooper PR, Barrow DL: The Practice of Neurosurgery, Vol. II. Williams & Wilkins, Baltimore: 1996.)

A **transpedicular approach** may be indicated for removal of herniated discs, excision or decompression of tumors, and the treatment of infection involving vertebral bodies. This procedure is performed in the prone position through a posterior midline incision centered over the affected vertebrae. The paraspinal muscles are retracted subperiosteally on both sides to expose laminae and facet joints. Facet joints and the superior half of the involved pedicle are drilled out to expose the lateral limits of the thecal sac and the nerve roots. A laterally herniated disc can be removed in a piecemeal fashion. If required, total removal of the pedicle is done to facilitate adequate bony decompression. Posterior instrumentation by pedicle screws, sublaminar wiring with rods, or a hook-rod construct may be performed.

The **lateral extracavitary approach** is a modification of a **costotransversectomy** and provides access to the anterior and posterior elements of the spine, thereby avoiding the need for a thoracotomy. This approach is performed with the patient in the prone position. A midline skin incision is made three levels above and below the involved vertebrae. The lower part of the incision may be curved over the involved side, if needed. A myocutaneous flap is developed by dissecting the scapular muscles (trapezius, rhomboids, etc.) laterally. Paraspinal muscles are freed from the spinous processes and dorsal spinal elements to enable retraction, which exposes the entire rib

cage and dorsal vertebral elements. Subperiosteal resection from its costovertebral tip to the posterior bend of the appropriate rib is done. The parietal pleura is gently separated from the ribs and the vertebrae to expose the posterolateral aspect of the vertebral bodies. The transverse process, pedicle, and laminae are removed, as required, to permit direct visualization of the cord during decompression of the vertebral body. Discectomy/corpectomy, vertebral reconstruction, and instrumentation are performed as required. Complete spondylectomy and anterior reconstruction are possible with this approach with minimal retraction to the cord. Combined anterior-posterior instrumentation can be used to leverage off each other for deformity correction. At the end of the procedure, the operative field is filled with saline to check for any evidence of air leak. A small chest tube can be placed if an air leak is present. A layered wound closure with drain is performed.

Thoracic pedicle screws have largely replaced the older hook-rod construct and **Harrington** rods as the fixation of choice. These can be technically challenging because the thoracic pedicles tend to be rather narrow and variable. Potential complications include injury to surrounding nerves, spinal cord, blood vessels (both local and great vessel), and lung parenchyma.

Usual preop diagnosis

Thoracic radiculopathy (nerve root compression); thoracic disc disease (herniation or degeneration of thoracic disc); thoracic myelopathy (spinal cord compression); thoracic canal stenosis (degenerative); infections (TB, pyogenic osteomyelitis) and tumors to spine (primary bone tumors or metastatic); intraspinal tumors; syringomyelia

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Thoracic Laminectomy	Transpedicular Approach	Lateral Extracavitary Approach
Position	Prone	⇐	⇐
Incision	Posterior midline	⇐	⇐; with/without hockey-stick extension
Special instrumentation	Operating microscope	⇐	⇐ ± anterior and posterior spinal

	± EP monitoring ± Pedicle screws/hooks		instrumentation
Unique considerations	I.I. localization	⇐	⇐
Antibiotics	Cefazolin 2 g iv	⇐	⇐
Surgical time	2–3 h	⇐ Add 30–45 min for bony/tumor decompression	3–7 h, based on need for anterior bony resection/reconstruction
Closing considerations	Often no cast	TL support/orthosis	⇐
EBL	100–1000 mL	200–1000 mL	300–3000 mL
Postop care	PACU → room	⇐	⇐ ± ICU
Mortality	0–5%	⇐	⇐
Morbidity	All <5% Neurologic: Myelopathy Nerve root injury Massive blood loss CSF leak	⇐ ⇐	⇐ ⇐
Pain score	6–10	6–10	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–85 yr (usually 20–60 yr)
Male:Female	1:2
Incidence	Rare
Etiology	Neoplastic; traumatic; congenital; degenerative

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Thoracolumbar Neurosurgical Procedures.

Suggested Readings

1. Sairyo K, Chikawa T, Nagamachi A: State-of-the-art transforaminal percutaneous endoscopic lumbar surgery under local anesthesia: discectomy, foraminoplasty, and ventral facetectomy. *J Orthop Sci* 2018; 23(2):229–36.
2. Yu L, Chen X, Margalit A, et al: Robot-assisted vs freehand pedicle screw fixation in spine surgery—a systematic review and a meta-analysis of comparative studies. *Int J Med Robot* 2018; 14(3):e1892.

ANTERIOR LUMBAR/LUMBOSACRAL SPINE SURGERY

SURGICAL CONSIDERATIONS

Description

The use of anterior procedures is steadily increasing among spine surgeons. Most procedures permit short-segment instrumentation of the spine, which often obviates the need for subsequent posterior fixation. The most significant disadvantage of these procedures involves the risk of injury to the great vessels; thus, these procedures are commonly done in association with a vascular or general surgeon. There is also risk of injury to the peritoneal contents and the neural plexus around the lumbosacral spine. Anterior instrumentation systems generally fall into three categories: (a) **plating systems** (e.g., Z plating, anterior locking plates, and MACS-TL), (b) **rod systems** (e.g., Kostuik-Harrington, Kaneda, and Moss-Miami), and (c) **interbody devices** (e.g., cages and allografts).

An **anterior lumbar interbody fusion** (ALIF) can be done with artificial cages or bone. After exposure of the disc space, the exact midline of the space is marked and verified with fluoroscopy. A spacing guide determines the exact position for pilot holes, and a partial discectomy is performed through these pilot holes, which are distracted and later reamed. Bone or cage is then attached to a specialized implant driver for insertion under fluoroscopic guidance. Harvested bone chips and other fusion enhancers are placed into the cages or around the bone dowel. ALIF provides immediate mechanical stability and long-term load support, with the ability to heal through the disc space. Its primary disadvantages include bleeding and possible major vessel injury.

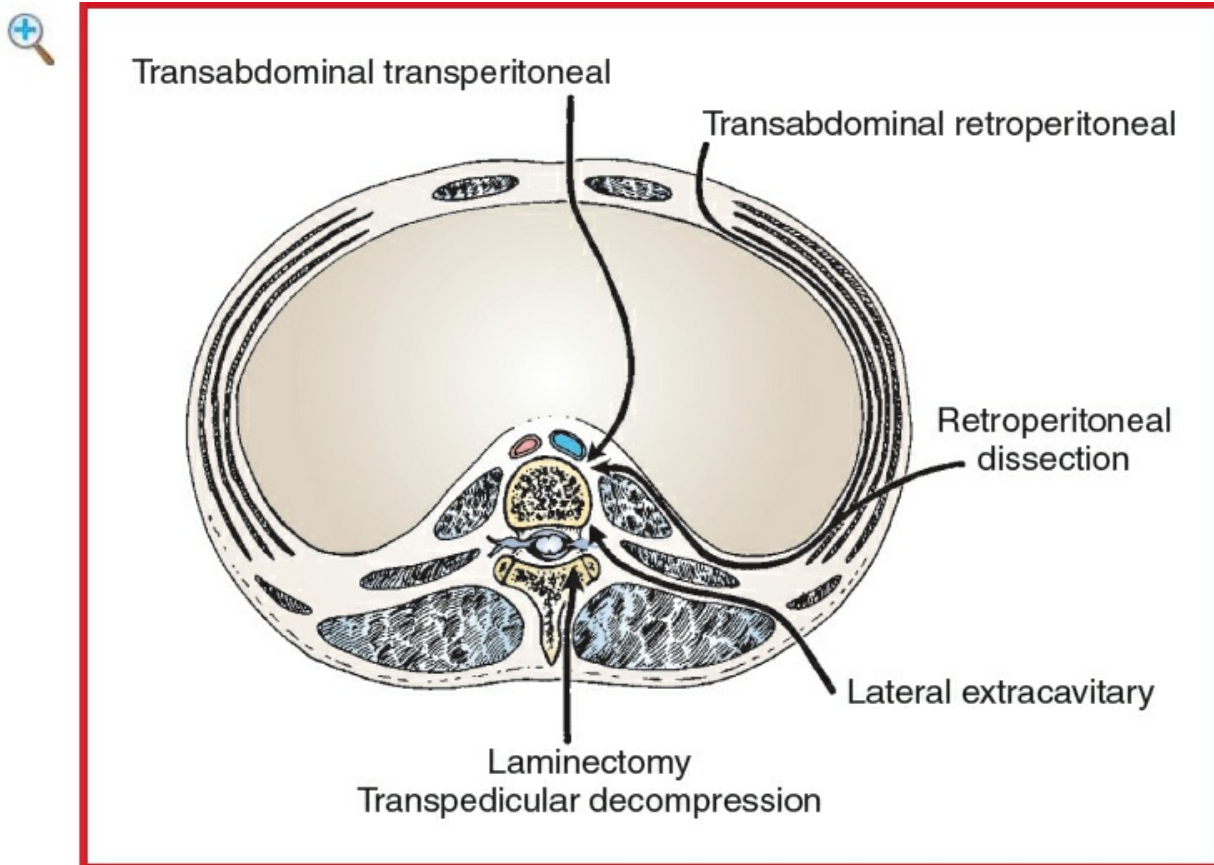


Figure 1.3-13 Surgical approaches to the lumbar spine. (Reproduced with permission from Tindall GT, Cooper PR, Barrow DL: The Practice of Neurosurgery, Vol. II. Williams & Wilkins, Baltimore: 1996.)

A **lumbar artificial disc** can be placed via this approach. The approach is similar to that of the ALIF but instead of placing a bone graft or cage, a mobile prosthetic disc is placed according to manufacturer's instruction. No additional stabilization or bone grafting is required. Accurate placement of this disc is absolutely critical to the success of this procedure. Fluoroscopy is used extensively to ensure this. There is immediate stability, and because motion is preserved, early mobilization is recommended.

Anterior instrumentation after ALIF can be done with a variety of screws and/or plates. These include a screw and screw/plate combination to buttress the graft from falling out. Other devices provide plate stabilization across interspaces. These can be used alone or with a posterior stabilization surgery. These add time to the anterior case and run the risk of hardware failure, but have the advantage of potentially improving the overall fusion rate and decreasing the graft related complications.

A **transperitoneal approach** involves laparotomy through a Pfannenstiel or subumbilical vertical midline incision. After opening the peritoneum, intestines are retracted to expose the anterior aspect of the lower lumbar and lumbosacral spine, an exposure that is often difficult to achieve with the retroperitoneal approach. Exposure of

L4-L5 disc spaces requires mobilization of the aorta and inferior vena cava, along with its bifurcations.

Variants of the transperitoneal approach

A **laparoscopic transperitoneal approach** often is used at the L5-S1 level. With the patient supine, Trendelenburg position is used to move the small intestine away from the operative field. The procedure is performed through one 10-mm portal for a 30° endoscope, two 5-mm portals for retraction, and one 20-mm working portal for instruments. For access to the L5-S1 level, the posterior peritoneum is incised at the base of the sigmoid mesocolon with endoscopic scissors. Laparoscopic interbody fusion and instrumentation are performed as required, using specially designed long-alignment tubes, distraction plugs, and a reamer, as in the open procedure. The peritoneum is closed with endoscopic sutures or clips. The major advantages of this technique are related to the minimal manipulation of abdominal viscera required and minimal trauma to the abdominal wall. In addition, postop pain, recovery time, and length of hospitalization are often less, permitting an early return to the patient's normal activities.

Variants of the retroperitoneal approach

A lateral retroperitoneal approach provides an excellent exposure of the lumbar spine from L1-S1 through a flank incision. The procedure is performed in the lateral decubitus position. The skin incision is made from the lateral border of the paravertebral muscles at the midlumbar level to the lateral border of the rectus abdominis. The incision is angulated below the umbilicus for exposure of the lower lumbar and lumbosacral junction and is carried down to the peritoneum.

The **supine retroperitoneal approach** is accomplished through a left paramedian incision, and the peritoneum and abdominal contents are retracted. This is one of the most common approaches. Ligation of lumbar intersegmental arteries and tributaries of the iliac vein may be required to allow a direct anterior exposure from L3-S1. An approach surgeon trained in vascular or general surgery can often expose even more. With blunt dissection, the peritoneum is peeled off the lateral and posterior abdominal walls, diaphragm, and iliopsoas, exposing the anterior aspect of the lumbar spine. Vertebral resection and reconstruction are carried out in the routine manner. During this procedure, the great vessels, ureter, and sympathetic trunk need to be protected. Monopolar cautery is avoided because it can cause injury to the presacral plexus, which can result in retrograde ejaculation.

A **laparoscopic retroperitoneal approach** can be used for performing an ALIF following discectomy in patients with lumbar segmental instability. This procedure is

performed in the right lateral decubitus position. A 10- to 12-mm port is made in the posterior axillary line midway between the 12th rib and iliac crest, and a trochar is advanced into the peritoneum. Retroperitoneal dissection is accomplished by balloon inflation, with 1000 mL of air or saline, through a trochar. This procedure is carried out under direct vision through the laparoscope. Discectomy, fusion, or instrumentation requires two additional working portals. Retroperitoneal insufflation with CO₂ may be required during the procedure.

Usual preop diagnosis

Degenerative disc disease; segmental instability; vertebral fractures; benign and neoplastic diseases of the lumbar spine; vertebral osteomyelitis; TB

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Transperitoneal Instrumentation	Retroperitoneal Instrumentation	Endoscopic Approach
Position	Supine	Supine/lateral decubitus	Right lateral decubitus
Incision	Pfannenstiel's/vertical subumbilical	Flank incision	3–4 ports
Special instrumentation	Z plates, Kaneda instrumentation, MACS-TL, femoral rings, cages; intervertebral disc replacements	⇐	Endoscopic instrumentation; balloon dissector
Unique considerations	NG tube, preop bowel prep	⇐	⇐
Antibiotics	Cefazolin 2 g iv	⇐	⇐
Surgical time	3–4 h	⇐	⇐
Closing considerations	Transfer to bed while anesthetized. Smooth	⇐	Rapid closure of ports

	emergence necessary to avoid graft/implant disruption		
EBL	200–600 mL	200–5000 mL; nontumor cases: 200–400 mL	100–500 mL
Postop care	PACU → ward; patients with infections/tumors → ICU. Postop ileus common	⇐	⇐
Mortality	Malignancy and sepsis: 1–2% Elective: <1%	⇐ ⇐	—
Morbidity	Neurologic injury: 2– 6% Vascular injury: 2– 15% Retrograde ejaculation: 5–35%	⇐ ⇐ <5%	— — —
Pain score	6–7	6–7	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Variable, infant–adult (usually 20–60 yr)
Male:Female	1:1 (usually)
Incidence	Uncommon
Etiology	Degenerative; lumbar segmental instability; neoplastic; traumatic; infectious

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Thoracolumbar Neurosurgical Procedures.

Suggested Reading

1. Kozak JA, O'Brien JP: Simultaneous combined anterior and posterior fusion. A independent analysis of a treatment for the disabled low-back pain patient. *Spine* 1990; 15(4):322–8.

POSTERIOR LUMBAR SPINE SURGERY

SURGICAL CONSIDERATIONS

Description

Lumbar laminotomy (partial removal of lamina) and **laminectomy** (complete removal of lamina) are procedures for decompressing the neural elements of the lumbar spine via a posterior approach. They can be used to treat lumbar radiculopathy 2° degenerative disc disease (e.g., herniated discs or osteophytes). **Decompressive laminectomy** can be used to treat compression of the cauda equina, usually 2° degenerative disease, congenital stenosis, neoplasm, and, occasionally, trauma. Lumbar laminectomy is also used to gain access to the spinal canal for dealing with intradural tumors, arteriovenous malformations (AVMs), and other spinal cord lesions.

Through a vertical midline incision, the lumbodorsal fascia is exposed, and then the paraspinal muscles are dissected off the spinous process and lamina of the segments intended for decompression. The level may need to be checked by intraop x-ray or fluoroscopy if the surgeon is not able to identify location based on visual confirmation of anatomic level. The bone landmarks are identified and ligamentous attachments are cut. The bone is removed piecemeal with rongeurs, gouges, or power drills. Care is taken not to injure the underlying dura. If a dural tear is found, it usually must be repaired. The surgeon may want a **Valsalva-like maneuver** (sustained inspiration at 30–40 cm H₂O) performed to test the integrity of the repair. If disc is to be removed, the dura is retracted and the annulus incised. The disc is removed piecemeal with a series of curettes and disc-biting rongeurs. There is a risk of damage to retroperitoneal structures (e.g., great vessels or intestines) during this portion of the procedure. More commonly, there may be troublesome epidural bleeding, which may be difficult to control and will necessitate transfusion. Hemostasis is obtained prior to closure. The wound is closed in layers, and a drain may be left in the epidural space. The patient is rolled supine onto a bed at the completion of the procedure.

Minimally invasive surgery (MIS) lumbar surgery is performed through a short

paramedian incision at the level of the affected disc. Under radiographic guidance, a series of soft tissue dilators are inserted over a previously placed guide wire to create an operative corridor through the paraspinous musculature. A tubular retractor is inserted over the dilators and connected to a flexible support arm assembly. Endoscope or microscope is used to perform laminotomy and discectomy and/or decompressive laminectomy. As the retractor is withdrawn at the end of the procedure, the paraspinal muscles resume their normal anatomic position, obliterating the dead space. Skin margins are closed with subcuticular sutures. Although all the major risks of surgery are still present, the blood loss, postop pain, and hospital stay are reduced. In theory, less trauma to the paraspinous muscles should improve the postop function of the spine.

Usual preop diagnosis

Lumbar radiculopathy (nerve root compression); lumbar disc disease (herniation or degeneration of the lumbar discs); lumbar canal stenosis; lateral recess stenosis; neurogenic claudication; herniated disc; metastatic tumor to the spine; lumbar spine tumor; lumbar spondylosis (degeneration of the lumbar spine); spondylolysis (structural defect in the pars interarticularis of the vertebra); spondylolisthesis (slipping of one vertebra over another)

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Lumbar Laminectomy	Lumbar Laminotomy	MIS
Position	Prone	⇐	⇐
Incision	Posterior midline	⇐	Paramedian port over disc
Special instrumentation	± Operative microscope	⇐	Endoscopic instrumentation (optional)
Unique considerations	I.I. localization to assess the correct level	⇐	⇐
Antibiotics	Cefazolin 2 g iv	⇐	⇐

Surgical time	1–2 h for single level; add 0.5–1 h/additional level	2 h for single level; add 0.5 h/additional level	⇐
EBL	25–500 mL	50–1000 mL	10–50 mL
Postop care	PACU → room	⇐	⇐
Mortality	0.5%	⇐	⇐
Morbidity	All <5%:	⇐	⇐
	CSF leak	⇐	⇐
	Nerve root injury	⇐	⇐
	Infection	⇐	—
	Postop instability	—	—
	Massive blood loss	—	—
	Injury to retroperitoneal structures	⇐	⇐
Pain score	4–7	4–9	3–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–85 yr (usually 30–60 yr)
Male:Female	3:2
Incidence	Common
Etiology	Degenerative; traumatic; neoplastic; infectious

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Thoracolumbar Neurosurgical Procedures.

Suggested Readings

1. Blamoutier A: Surgical discectomy for lumbar disc herniation: surgical techniques. *Orthop Traumatol Surg Res* 2013; 99(1 Suppl):S187–96.
2. Overley SC, Cho SK, Mehta AI, et al: Navigation and robotics in spinal surgery where are we now? *Neurosurgery* 2017; 80(3S):S86–99.
3. Wu Y, Vázquez-Barquero A: Stimulus-evoked electromyographic monitoring during minimally invasive transpedicular implantation of screws in lumbosacral spine: threshold value, methodology and clinical effectiveness. *World Neurosurg* 2017;

POSTERIOR LUMBAR FUSION AND INSTRUMENTATION

SURGICAL CONSIDERATIONS

Description

Posterior lumbar spinal fusion may relieve low back pain resulting from intervertebral movement. This surgery is often indicated for segmental lumbar instability, spondylolisthesis, or iatrogenic instability due to extensive laminectomy or facetectomy.

The **pedicle screw stabilization** technique provides rigid three-column spinal fixation and is the preferred mode of instrumentation in lumbar spinal surgery ([Fig. 1.3-14](#)). Pedicle screws are passed after tapping the entry site and are fixed with rods or plates on each side of each vertebral segment. The major risks with pedicle screw fixation include screw malposition and nerve root injury. Pedicle screws may be combined with hooks to provide fixation of the lumbar/thoracolumbar spine, an approach that improves the stability of the construct and minimizes the risk of instrumentation failure. Facet screw stabilization can also be performed to fix levels together. This is usually not used in a stand-alone fashion but in combination with anterior fixation. Instrumentation can be placed via percutaneous techniques that decrease blood loss and patient pain; however, complications often go undetected and unseen.

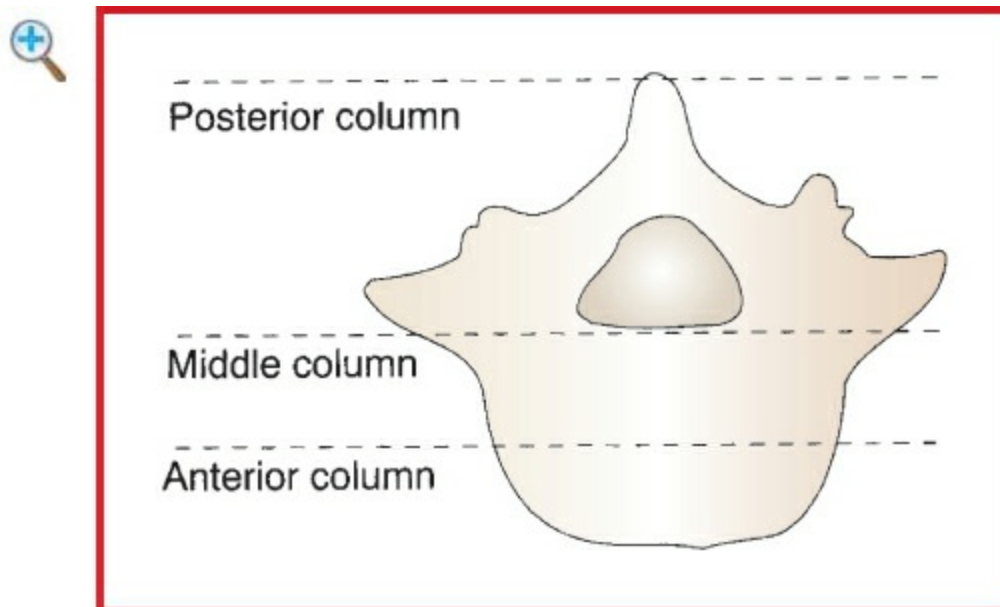


Figure 1.3-14 Spinal three-column model developed by F. Denis. Disruption of elements of two or more columns renders the spine unstable. (Reproduced with permission from Tindall GT, Cooper PR, Barrow DL: *The Practice of Neurosurgery*, Vol. II. Williams & Wilkins, Baltimore: 1996.)

Posterolateral fusion (PLF) is performed through a posterior midline incision, with the paraspinal muscles retracted subperiosteally from the affected lumbar vertebrae. Decompressive laminectomy and discectomy are performed as needed. PLF is performed by decorticating the facet joints and transverse processes. Bone graft is then placed over the decorticated bone. Instrumentation with pedicle screws and plate/rod constructs often is done for stability and to facilitate fusion.

Posterior lumbar interbody fusion (PLIF) consists of a bilateral laminectomy and removal of the inferior facet and the medial portion of the superior facet. The dural sac is retracted, and a total discectomy, together with the removal of cartilaginous end plates, is performed. The anterior disc space is packed with bone graft. Appropriately sized rectangular bone grafts or cages are inserted into the posterior half of the disc space on both sides to provide structural support close to the center of rotation. The nerve roots above and below the disc space should be visualized during the procedure to avoid excessive retraction. Instrumentation with pedicle screws and a rod/plate construct is often added to facilitate early fusion and ambulation while preventing the extrusion of the graft. The major advantage of this procedure is that it provides the ability to achieve combined anterior and posterior spinal fusion while avoiding the significant morbidity often associated with anterior lumbar surgery. Its major disadvantages include the potential risk of nerve root injury and compromise of the structural integrity of both facet joints.

Transforaminal lumbar interbody fusion (TLIF) is a modification of the PLIF, using a unilateral posterolateral approach. A **hemilaminectomy and total facetectomy** is performed on one side. A near total discectomy is performed and the first bone graft or cage is inserted across the disc space to the contralateral side. A second bone graft may be inserted into the ipsilateral posterior disc space, and satisfactory placement of the bone grafts is confirmed by fluoroscopy. Supplemental pedicle screw stabilization is indicated as in PLIF. This approach has been adopted to be performed with MIS techniques to afford circumferential lumbar decompression and stabilization via an all posterior approach.

Direct/extreme lateral interbody fusion (DLIF/XLIF) is an MIS procedure performed by having the patient positioned on his/her side where the surgeon's approach to the spine will be directly laterally through the retroperitoneal cavity. A specialized retractor is used with multiple long blades that allow for visualization of the spine

through the deep layers of tissue. **Neuromonitoring** (EMG nerve root mapping) is essential in the procedure to help avoid nerve injury. Therefore, minimal or no paralytic should be used as it may confound monitoring.

Transaxial lumbosacral fusion (Trans1) is an **MIS** procedure performed by having the patient positioned prone. A complete L5-S1 discectomy and fusion with screw instrumentation is performed. This is done via a 1-cm incision at the base of the sacrum. Specialized tubular dilators and shims protect the visceral contents while a reamer and disc remover tools are used to remove the disc. There is minimal blood loss, but occult injury to peritoneal contents including the viscera and blood vessels can occur acutely or present in a delayed fashion.

Usual preop diagnosis

Lumbar segmental instability; spondylolisthesis; iatrogenic lumbar instability; lumbar disc disease; spondylolysis; mechanical back pain syndrome

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Posterolateral Fusion	PLIF	TLIF (DLIF/XLIF)
Position	Prone	⇐	⇐ (DLIF/XLIF: Lateral)
Incision	Posterior midline	⇐	⇐ (DLIF/XLIF: Lateral)
Special instrumentation	Drills, pedicle screws, or hooks	⇐; osteotomes, curettes, bone plugs, or cages	⇐; angled curettes, angled impactors
Unique considerations	I.I. localization and guidance	⇐	⇐ (DLIF/XLIF: EMG)
	Cefazolin 2 g iv	⇐	⇐

Antibiotics			
Surgical time	2 h for single level + 0.5 h/additional level	3–4 h for single level + 1 h/additional level	2–3 h for single level + 1 h/additional level
Closing considerations	No brace/lumbosacral corset	⇐	⇐
EBL	250–500 mL	250–1000 mL	250–750 mL (less with MIS)
Postop care	PACU → room	⇐	⇐
Mortality	0–5%	⇐	⇐
Morbidity	Nonunion: 20–30%	10–20%	⇐
	Nerve root injury	⇐	⇐
	CSF leak	⇐	⇐
	Infection	⇐	⇐
	Massive blood loss		
Pain score	6–10	6–10	6–10 (4–7 with MIS)

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–85 yr (usually 30–60 yr)
Male:Female	3:2
Incidence	Common
Etiology	Degenerative; traumatic; neoplastic; infectious

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Thoracolumbar Neurosurgical Procedures.

COMBINED ANTERIOR AND POSTERIOR INSTRUMENTATION OF THE THORACIC AND

LUMBAR SPINE

SURGICAL CONSIDERATIONS

Description

Patients with multilevel vertebral collapse, unstable three-column injuries, severe kyphosis or scoliosis, and/or neoplastic or infective conditions involving multiple spinal levels often require **combined anterior and posterior instrumentation**. This approach provides (a) complete circumferential neural decompression, which facilitates maximal neuronal recovery; (b) rigid short-segment spinal fixation, which facilitates early ambulation with minimal orthotic support; and (c) maximal correction of deformities with low instrumentation failure and high fusion rates. The combined approach maximizes the possibility of complete resection of the neoplastic or infective process. Patients with major systemic disease or poor marrow reserve may require staged procedures. Combined instrumentation procedures are often lengthy, requiring 5–10 h of surgery. Major related morbidities include infection, wound breakdown, respiratory complications, and significant blood loss. The transition between anterior and posterior procedures should be performed carefully to minimize disruption of the instrumentation.

Usual preop diagnosis

Lumbar segmental instability; spondylolisthesis; iatrogenic lumbar instability; spondylolysis; mechanical back pain syndrome

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol and smoking cessation, review of nutritional status, and postop pain management options. See [Appendix K](#), p. K-1.

■ SUMMARY OF PROCEDURE	
Position	Prone + supine/lateral
Incision	Posterior midline skin incision + transverse
Special instrumentation	Anterior and posterior spinal implants and instrumentation sets
Unique considerations	OLV for anterior thoracic approaches; radiologic localization; intraop EP monitoring (optional)
Antibiotics	Cefazolin 2 g iv

Surgical time	4–10 h
EBL	500–5000 mL
Postop care	PACU → room; sometimes needs short ICU stay
Mortality	0–3%
Morbidity	Respiratory problems: atelectasis, pneumonia Infection Bone graft dislodgement
Pain score	7–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–85 yr (usually 30–60 yr)
Male:Female	3:2
Incidence	Common
Etiology	Congenital; degenerative; traumatic; neoplastic; infectious

Suggested Readings

1. Denis FF: The three column spine and its significance in the classification of acute thoracolumbar spinal injuries. *Spine* 1983; 8:817–31.
2. Lowe TG, Tahernia D: Unilateral transforaminal posterior lumbar interbody fusion *Clin Ortho* 2002; 394:227–36.

ANESTHETIC CONSIDERATIONS FOR THORACOLUMBAR NEUROSURGICAL PROCEDURES

(Procedures covered: anterior/posterior thoracic spine surgery, anterior lumbar/lumbosacral spine surgery, posterior lumbar spine surgery, posterior lumbar fusion and instrumentation, and combined anterior/posterior instrumentation of the thoracic and lumbar spine)

PREOPERATIVE

Surgery of the lumbar or thoracic spine is common as a result of a variety of spinal disorders, including herniation of lumbar or thoracic intervertebral discs causing

compression of the adjacent spinal cord or nerve roots, spinal stenosis from bony overgrowth causing compression of the nerve roots or spinal cord, spondylolisthesis, traumatic injury to the spine, and removal of a spinal tumor or AVM. Because these diseases span a wide age range, the patients may be healthy, or they may have severe cardiovascular and/or respiratory disorders. Generally, the surgical incision is made in the posterior thoracic or lumbar region, but the surgeon may elect to use an abdominal (transperitoneal) or flank (retroperitoneal) approach. Sometimes, both anterior and posterior approaches are used sequentially. Finally, some anterior abdominal or thoracic procedures are performed endoscopically.

Neurologic	Patients with a herniated disc or spinal stenosis generally complain of pain, often in the pelvis or radiating down one or both legs. As nerve or spinal cord compression continues, patients develop motor weakness and atrophy of muscle groups in the legs as well as sensory deficits and/or bowel/bladder dysfunction. These changes can also result from a spinal cord tumor, vascular malformation, or cyst. MRI helps delineate the etiology of the neural compression. Tests: MRI
Hematologic	Blood loss may be substantial if the surgeon plans both an anterior and posterior approach in the same patient, or if the operation includes both spinal cord decompression and posterior spinal instrumentation. For these operations, 2 U of blood should be available at all times. Consideration should also be given to using a red cell salvage device. Generally, the anticipated blood loss will be proportional to the number of levels involved in the surgery. Prepare for significant blood loss in tumor resection cases.
Laboratory	Preop Hct, platelet count, and coagulation studies—to assess baseline. Other tests as indicated from H&P.
Premedication	Anxiolysis (e.g., midazolam 2–4 mg iv) and analgesia (e.g., meperidine 25 mg iv) as necessary. These patients may be experiencing an exacerbation of

	their chronic pain because they have likely been off NSAIDs and may have skipped their normal AM
ERAS	meds. Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water.

INTRAOPERATIVE

Anesthetic technique

GA is almost invariably used for these operations because it maximizes patient comfort, provides airway control, and permits use of controlled hypotension. Spinal and epidural anesthesia are, in principle, excellent techniques for lumbar surgery, particularly for removal of a lumbar intervertebral disc, but they are seldom used because of the medicolegal concern that the regional anesthetic may be blamed for a new neurologic deficit, if one should occur as a result of the surgery. Regional anesthesia is generally not suitable for lumbar fusion or removal of a spinal cord tumor or cyst because of the potentially prolonged duration of the surgery.

Induction	Standard induction unless MEP monitoring will be used. If the surgeon selects a transthoracic approach, one-lung ventilation will be necessary to deflate the lung on the operative side, and a compatible airway technique must be employed (e.g., DLT, bronchial blocker). If MEP monitoring will be used, consider succinylcholine vs. low-dose rocuronium (0.5 mg/kg) vs. deep anesthesia and remifentanyl bolus (1.5 mcg/kg) to facilitate intubation yet permit the rapid return of neuromuscular transmission.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Avoid hypothermia. Dexamethasone (4–8 mg iv) may ↓ PONV. Sequential pneumatic compression devices should be used to ↓ VTE risk. Consider ketamine (0.25 mg/kg/h) and/or lidocaine (1–2 mg/kg/h) infusions as components of MMPM

especially in chronic pain patients.

Maintenance

Standard maintenance will suffice if no neurophysiologic monitoring is used. Once exposure is completed with the posterior approach, muscle relaxation is no longer needed. For thoracoscopic or anterior abdominal approaches, neuromuscular blockade should be maintained until spine surgery is complete and closure is under way. When evoked potential monitoring is used, the anesthetic plan must be adapted to ensure optimal monitoring condition. Standard NMB doses will completely abolish MEPs and must be avoided although small NMB doses may be needed to suppress interfering EMG activity. More than 0.5 MAC each of volatile anesthetics and N₂O interferes with MEP and, to a lesser extent, SSEP monitoring. An anesthetic consisting of 50% N₂O in O₂ with up to 0.6% isoflurane in combination with a remifentanil infusion (0.05–2 mcg/kg/min) will provide anesthesia while permitting MEP and SSEP monitoring. TIVA (see [Appendix B, p. B-3](#)) may be preferred by local custom when monitoring MEPs but is not necessary. The propofol infusion should be tapered as the drug saturates the various compartments of distribution. The neurophysiology monitoring team must be updated as conditions that affect the evoked potentials change (depth of anesthesia, anesthetic agent, blood pressure, PCO₂, PO₂). After the baseline SSEP and motor-evoked potentials have been established, some surgeons will request paralysis during dissection of the muscles from the bones. This can be accomplished safely with rocuronium (20–40 mg).

Emergence

Standard emergence, after the patient has been returned to the supine position on a bed or gurney. If the intubation was difficult or the operation was prolonged and airway edema or respiratory

	depression is possible, it may be advisable to leave the patient's trachea intubated overnight. Consider performing a leak test if there is concern for airway edema and extubate over an airway exchange catheter. Keep in mind that remifentanyl will not provide postop analgesia and that another longer-acting opioid (e.g., hydromorphone 0.2–0.6 mg iv) must be administered prior to emergence. Suggest wound infiltration with LA. Consider iv acetaminophen 1 g if po dose given >4–6 h ago and hydromorphone (0.2–0.6 mg iv). Discuss possible use of ketorolac or celecoxib.	
Blood and fluid requirements	14–18 g/dL × 1–2 BSS/LR @ 3–5 mL/kg/h up to a maximum of 40 mL/kg. Use goal-directed fluid management (see Appendix K, p. K-4)	To ↓ the chance of ischemic optic neuropathy in the prone position, do not administer >40 mL/kg of crystalloid regardless of EBL or duration of surgery. If additional fluid is needed, administer albumin 5% or blood. Blood transfusion is often necessary when there has been extensive bony decompression and fusion. Consider the use of a red cell salvage device if large blood loss is anticipated in noncancer surgery. Some tumors, particularly metastases, can bleed profusely. Epidural venous bleeding, particularly during an anterior approach, can also be severe and difficult to control. Zero balance or goal-directed fluid management is recommended.
Control of blood loss	Controlled	In the absence of cord

	hypotension	compression, modest ↓ in BP may be helpful to ↓ blood loss when extensive surgery is anticipated. This may be accomplished with anesthetics and opiates or by the use of α- and β-blockers (e.g., labetalol, esmolol) and clevidipine infusion. BP should not be <60 mm Hg MAP in young adults or <80 mm Hg in elderly patients. After the bony dissection is complete, any possible benefit of deliberate ↓ BP wanes, and a more normal BP is advisable.
Monitoring	Standard monitors ± Arterial line ± CVP line	For simple back surgery, standard monitors are sufficient. If controlled ↓ BP or if an extensive posterior or anterior and posterior approach are planned, an arterial catheter is essential to monitor BP, and a CVC is recommended for infusion of vasoactive drugs and monitoring of CVP.
	± Urinary catheter ± BIS monitor (for TIVA)	A urinary drainage catheter is also desirable if surgery is expected to last several hours or substantial fluid shifts are anticipated. Do not administer excessive crystalloid volume to promote urine output. Urine output is often less than expected when patients are positioned prone. Increased intra-

	± Neuromonitoring	abdominal pressure may → ↓ GFR. SSEPs + MEPs should be used to monitor posterior and anterior cord pathways if there is a risk of cord injury during surgery.
Positioning	✓ and pad pressure points ✓ eyes and ears frequently ✓ breasts and genitals ✓ free abdominal movement Neutral C-spine	Usually, patients are positioned prone on a Jackson table, Wilson frame, or bolsters. Less commonly, the knee-chest position on an Andrews table is necessary. For anterior approaches, the patient will be supine; for retroperitoneal or transthoracic approaches, the patient will be positioned lateral or semilateral with some flexion in the table. When prone, the head is placed in a neutral position using a foam pillow with cutouts for the eyes, nose, and chin. Whenever possible, maintain the patient's head at or above the level of the heart to lessen the development of facial or intracranial edema. If the patient also has cervical spine disease, it is advisable to place a cervical collar before turning the patient prone. If an anterior approach and then posterior approach are to be used in the same patient, either move the

patient to a gurney after the anterior portion is completed, and then roll the patient back onto the same operating table, or transfer the patient from one OR table to another. Alternatively, perform the operation on a Jackson table, which allows the patient to be turned from supine to prone without having to move them. If this table is used, the anesthesiologist must pay careful attention to all lines and wires so they are not inadvertently pulled out during the 180° rotation. Other options are to use a horseshoe headrest or place the head in Gardner-Wells tongs or in a Mayfield head clamp. When placed in a headrest, the breathing circuit should be supported so that the ET tube is strain-free to prevent gradual extubation due to gravity or pressure injury to the posterior oropharynx. Typically, the arms are abducted; they should be limited to <90° at the shoulder and extended <90° at the elbow. When the arms are tucked, take special care to ensure that all stopcocks under the drapes are secured in the desired position and

		capped to prevent bleeding or leakage and padded to prevent pressure injury. Elbows, knees, feet, and arm pressure points need padding.
Complications	↓ BP Bowel or ureteral injury Hemorrhage	↓ BP may be 2° abdominal compression and ↓ venous return. ↑ blood loss may occur 2° epidural vein engorgement, abdominal compression, or vascular injury. Any acute ↓ BP may be due to a vascular injury, either obvious (segmental artery or epidural vein) or occult (deep great vessel injury) until proven otherwise. The source of bleeding should be identified. If ↓ BP persists despite vigorous blood and fluid administration, suspect bleeding into the retroperitoneal space or abdomen. Alert the surgeon of this possibility and prepare for immediate exploration. Tumor resection, especially of metastases, can result in massive blood loss.

POSTOPERATIVE

Complications	↓ BP Hemorrhage Nerve injury	Postop bleeding may persist. Drain output and vital signs should be monitored closely. As above, acute hypotension
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	Blindness	may be the result of ongoing hemorrhage. Rarely, an epidural hematoma can form, resulting in evolving neurologic symptoms. In either case, emergent surgical re-exploration should be considered. Intubation may be more difficult if airway edema has accumulated during the initial operation. Postop blindness is a rare but devastating complication of surgery in the prone position. Risk factors include prolonged duration of surgery, large blood loss, and large volume crystalloid administration.
Multimodal pain management	Epidural opiate IV acetaminophen (1 g) PCA Acetaminophen 1 g iv ± Fentanyl 25–50 mcg prn ± ketorolac (30 mg iv) or celecoxib (200 mg po)	Some surgeons may place a mixture consisting of Duramorph® ± fentanyl, ± steroid, in Avitene® hemostat into the epidural space before closing. This provides excellent postop analgesia for 12+ h. If not given preop or if 4–6 h has elapsed since previous dose. Consult with surgeon regarding possible hemostasis concern.
ERAS	Ondansetron 4 mg iv ± Metoclopramide	Encourage early enteral nutrition, Foley catheter

	10–20 mg iv	removal, and mobilization with physiotherapy.
Tests	Hct; document neurologic status	If postop bleeding is suspected, serial Hct determinations are useful.

Suggested Readings

1. Aziz M: Airway management in neuroanesthesiology. *Anesthesiol Clin* 2012; 30(2):229–40.
2. Elgafy H, O'Brien P, Blessinger B, et al: Challenges of spine surgery in obese patients. *Am J Orthop (Belle Mead NJ)* 2012; 41(3):E46–50.
3. Kurd MF, Kreitz T, Schroeder G, et al: The role of multimodal analgesia in spine surgery. *J Am Acad Orthop Surg* 2017; 25(4):260–8.
4. Lachiewicz PF: The role of intravenous acetaminophen in multimodal pain protocols for perioperative orthopedic patients. *Orthopedics* 2013; 36(2 Suppl):15–9.
5. Lee LA: Perioperative visual loss and anesthetic management. *Curr Opin Anaesthesiol* 2013; 26(3):375–81.
6. Mergeay M, Verster A, Van Aken D, et al: Regional versus general anesthesia for spine surgery. A comprehensive review. *Acta Anaesthesiol Belg* 2015; 66(1):1–9.
7. Pendi A, Acosta FL, Tuchman A, et al: Intrathecal morphine in spine surgery: a meta-analysis of randomized controlled trials. *Spine (Phila Pa 1976)* 2017; 42(12):E740–7.
8. Rabai F, Sessions R, Seubert CN: Neurophysiological monitoring and spinal cord integrity. *Best Pract Res Clin Anaesthesiol* 2016; 30(1):53–68.
9. Sugrue PA, Halpin RJ, Koski TR: Treatment algorithms and protocol practice in high-risk spine surgery. *Neurosurg Clin N Am* 2013; 24(2):219–30.
10. Willner D, Spennati V, Stohl S, et al: Spine surgery and blood loss: systematic review of clinical evidence. *Anesth Analg* 2016; 123(5):1307–15.

Suggested Viewing

Links are available online to the following videos:

Anterior Cervical Decompression and Fusion (ACDF): https://www.youtube.com/watch?v=yfSkOF_DAfA

Dr. Ali Bydon Performs a Minimally Invasive Lumbar Discectomy: <https://www.youtube.com/watch?v=2j9QyiOxVF4>

Minimally Invasive Tubular Microdiscectomy: 3-D Video: https://www.youtube.com/watch?v=Cpkr_oy5Enc

Scoliosis Spinal Fusion Animation: <https://www.youtube.com/watch?v=WBIf4AQj5s0>

C1 Lateral Mass Fixation: A Comparison of Constructs: <https://www.youtube.com/watch?v=ZRcafQ3xLSs>

Dr. Ali Bydon Removes an Intramedullary Spinal Cord Tumor: <https://www.youtube.com/watch?v=HoWWRQfWYTo>

Posterior Spinal Instrumentation & Fusion for Idiopathic Scoliosis: <https://www.youtube.com/watch?>

[v=TbQaT7z54E4](#)

Transforaminal Lumbar Interbody Fusion (TLIF) Surgery: <https://www.youtube.com/watch?v=1I5j4EPomW4>

L5 S1 Fusion TLIF Back Surgery Video: <https://www.youtube.com/watch?v=I2IHCdX28D0>

Posterior Cervical Foraminotomy Herniated Disc Neck Surgery: <https://www.youtube.com/watch?v=0edUnIJpg3o>

CHAPTER 1.4

Carotid Endarterectomy

SURGEONS

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ANESTHESIOLOGIST

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CAROTID ENDARTERECTOMY

SURGICAL CONSIDERATIONS

Gary K. Steinberg and Robert L. Dodd

Description

Carotid endarterectomy (CEA) is frequently used to treat severe atherosclerotic occlusive disease involving internal carotid arteries at the common carotid artery bifurcation. Atherosclerotic carotid artery disease commonly causes thromboembolic or hemodynamic stroke and transient ischemic attacks (TIAs). Recent studies proved the efficacy of this operation, compared with medical treatment for symptomatic high-grade stenosis (70–99%), symptomatic moderate stenosis (50–69%), and asymptomatic high-grade stenosis ($\geq 60\%$).

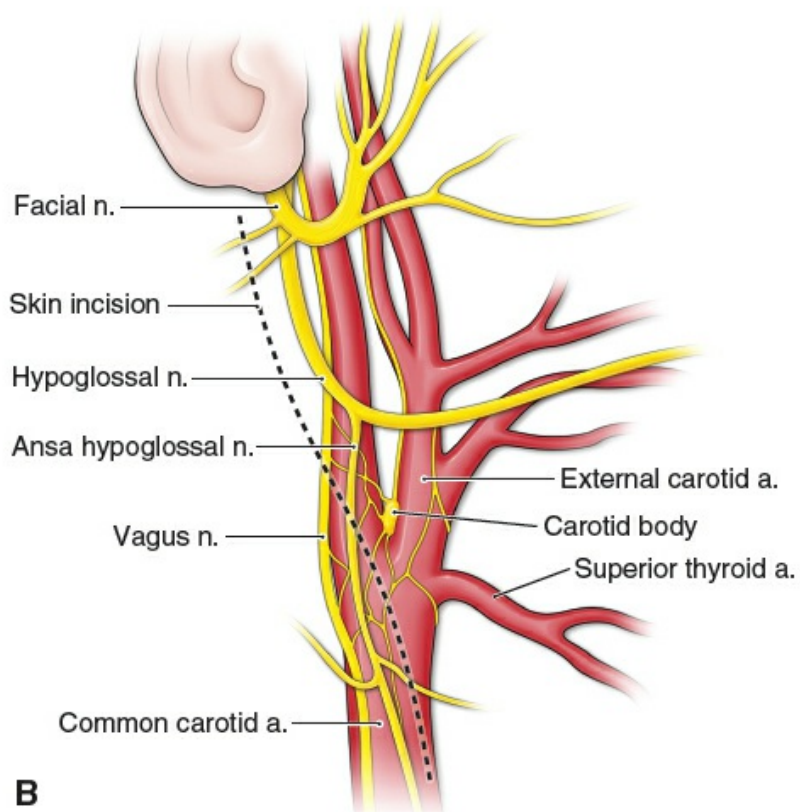
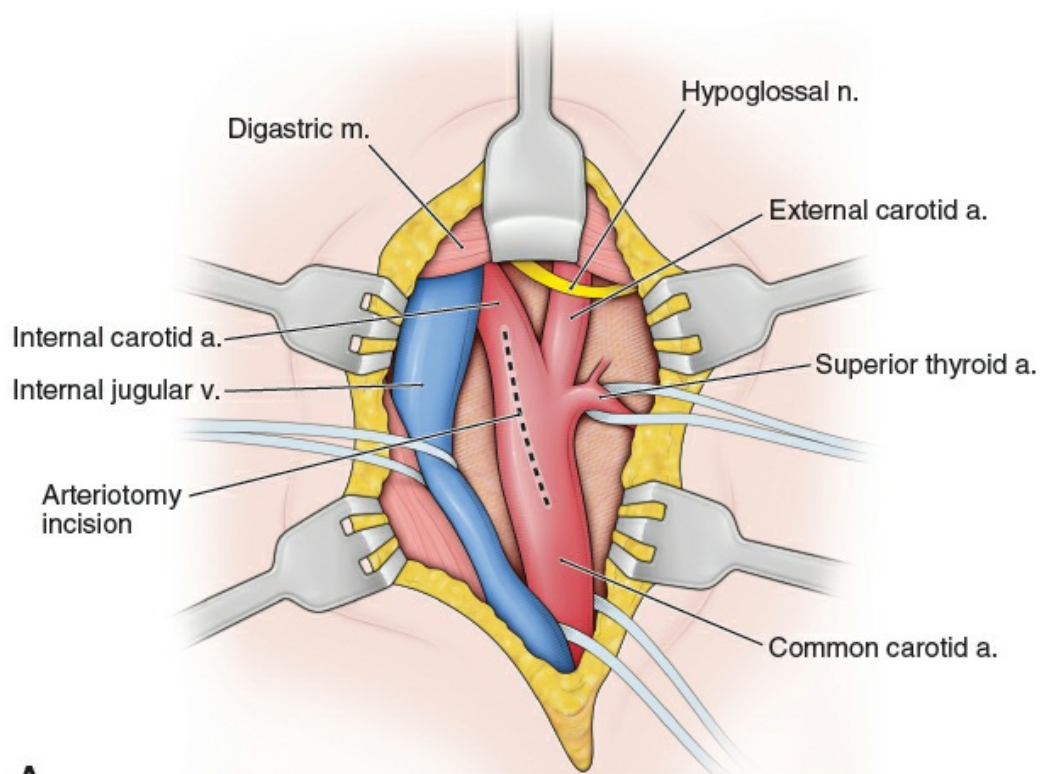


Figure 1.4-1 **A:** Exposure and control of the carotid artery. **B:** Nerves at risk during carotid endarterectomy.

The operation involves opening the common carotid artery and the proximal internal carotid artery in the neck ([Fig. 1.4-1](#)), removing atherosclerotic plaque from the inside of the artery, and repairing the wall of the arteries (media and adventitia). Opening the carotid artery (**arteriotomy**) requires temporary occlusion of the proximal common carotid artery, distal internal carotid artery, external carotid artery, and, usually, its first branch, the superior thyroid artery. The entire procedure can be achieved under continued occlusion of these vessels if the collateral blood flow to the territory supplied by the occluded internal carotid is deemed adequate (on the basis of intraop EEG monitoring, internal carotid artery back-bleeding, stump pressures, testing in an awake patient, CBF studies, or angiography). Alternatively, an internal shunt between the proximal common carotid artery and distal internal carotid artery can be placed in an appropriately heparinized patient after the arteriotomy for use during the endarterectomy. Often a synthetic graft (e.g., Dacron) or, occasionally, a vein graft is used to reconstruct (“patch”) the arteriotomy site and increase the luminal diameter.

Variant procedure or approaches: The vascular surgeon’s approach to CEA is described in [Chapter 6.3](#). Carotid stenting is described in [Interventional Neuroradiology, Section 13.1](#). Stenting appears to be inferior to CEA in outcome studies (↑ stroke, ↑ death) but is associated with ↓ MI. EC-IC bypass for treatment of carotid occlusion is described in [Section 1.1](#).

Usual preop diagnosis

Stroke; TIAs; carotid artery stenosis; carotid artery dissection

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possibility of lines, and urinary catheter postop. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Parallel and anterior to the sternocleidomastoid centered at the level of the thyroid cartilage; occasionally, if “patching” arteriotomy, may have to harvest portion of greater saphenous vein from the leg.
Special	Magnification loupes; vascular instruments ± shunt

instrumentation	(Bard, Javid, Pruitt-Inahara)
Unique considerations	Techniques for monitoring cerebral perfusion: EEG-spectral analysis or raw EEG, somatosensory- and motor-evoked potentials, back-bleeding, internal carotid artery stump pressure (>50 mm Hg), or transcranial Doppler. CEA under regional anesthesia permits direct patient testing. Full anticoagulation with heparin (typically 100 U/kg iv) during arterial occlusion ± reversal with protamine (typically 0.5 mg/kg iv) 10 min after repair and reopening of carotid arteries. Maintaining ↑ BP during internal carotid artery occlusion (MAP 90–110). The use of neuroprotective agents just before internal carotid artery occlusion (e.g., iv propofol 1–2 mg/kg as necessary to produce EEG burst suppression).
Antibiotics	Cefazolin (2 g iv q6h)
Surgical time	~3 h
Closing considerations	Avoid ↑ BP or ↓ BP (typical MAP 80–100), meticulous hemostasis
EBL	50–150 mL
Postop care	Control of BP (MAP 80–100 mm Hg); start aspirin on postop d 1; PACU, ICU, or other monitored bed × 3–24 h
Mortality	0.3–1.1% (if combined CABG/CEA: up to 17.7%)
Morbidity	Postop MI: 0.5–4% Nerve injury: up to 39% (typically 8%) • Vagus, recurrent/superior laryngeal nerve (hoarseness): 4% • Hypoglossal nerve (the tongue deviates to the side of injury): 4% • Mandibular branch of the facial nerve (lower lip weakness): 2% • Glossopharyngeal nerve (loss of gag reflex, dysphagia): <1% • Transverse cervical and greater auricular nerves

(cutaneous anesthesia): 12%

Cerebral hyperperfusion syndrome (if postop BP is not well controlled)

Intraop MI: 1–2%

Postop bleeding: 1.7–2.7%

Wound infection: rare

Pain score

3

■ PATIENT POPULATION CHARACTERISTICS

Age range

50–90 yr

Male:Female

1:1

Incidence

~140,000 CEAs/yr in the United States (steadily declining)

Etiology

Atherosclerosis; occasionally, traumatic (dissection)

Associated conditions

HTN; CAD; PVD; smoking; obesity; alcohol abuse; hyperlipidemia

ANESTHETIC CONSIDERATIONS

(Procedure covered: carotid endarterectomy)

PREOPERATIVE

The incidence of occlusive or ulcerative lesions of the extracranial or intracranial vasculature increases with advancing years. Generally, these lesions are asymptomatic until the cross-sectional area of the vessel is decreased by at least 50%. This is because in most patients the cerebral vasculature has excellent collateral circulation, most importantly the circle of Willis (normal in only 50% of patients), but also persistent anastomoses from the basilar artery and extra- to intracranial collateral flow via the ophthalmic artery or branches of the vertebral artery. Patients presenting for CEA generally fall into one of three categories: (a) those with TIAs, presenting with symptoms that may be focal or generalized, (b) patients with completed stroke (If the stroke is recent (<2–4 wk), some surgeons will not operate on the patient for fear of converting an ischemic infarct into a hemorrhagic infarct; however, if the infarct is small and clinical deficit minor, early surgery may be beneficial. Most studies recommend immediate implementation of best medical therapy followed by CEA (or

stenting) within 2 wk of a related TIA/stroke.), and (c) patients with asymptomatic bruit, which usually is found during a routine physical examination of the neck. These bruits are of concern because they may signal the development of carotid stenosis that could benefit from surgical intervention.

Respiratory

Patients should be asked to stop smoking prior to anesthesia, even if only the night before. Although cessation of smoking for such a short time will not lessen the volume of secretions appreciably or make the airways less irritable to a foreign body, such as an ETT, it will provide sufficient time for the carbon monoxide levels in the blood to decrease, thereby enhancing O₂-carrying capacity. Any Hx of pulmonary disease should be evaluated with spirometry, ABGs, and CXR. If there is evidence of pulmonary infection, appropriate antibiotic therapy should be instituted. If secretions are excessive, preop pulmonary physiotherapy, including bronchodilator therapy, may be indicated.

Tests: ABGs; spirometry; CXR, if indicated from H&P

Cardiovascular

Careful evaluation of cardiovascular status is essential and should include a detailed Hx of cardiovascular function and serial determinations of BP in both arms to establish the range of pressures that normally occur and whether there are regional differences. If BP is different in the two arms, it should be measured intraop and postop in the arm with the higher values.

Also, a preop ECG is mandatory. The reason for concern about cardiovascular function is twofold: (a) it is often necessary to administer vasoactive drugs to regulate BP during CEA, either to maintain it at a normal value or sometimes to increase it as much as 20% above the highest resting pressure to maintain optimal collateral circulation during intraop carotid occlusion. (b) The incidence of perioperative MI in this surgical population is >1% and represents the most common major postop complication. Except in

the case of emergencies, anesthesia and surgery should not proceed in the face of severe, uncontrolled HTN, diabetes, or recent MI. Antihypertensive medications should be continued up to the time of anesthesia.

Tests: ECG; others as indicated from H&P

Neurologic

The Sx of cerebrovascular insufficiency are due to either critical stenosis or occlusion of cerebral vessels, combined with inadequate collateral circulation. The degenerative plaques or mural thrombi readily break off from the vessel wall and cause focal ischemic lesions. Manual occlusion of the carotid arteries is not an appropriate test of tolerance to temporary circulatory occlusion, as it may endanger the patient by precipitating embolization from an ulcerative lesion or by inducing bradycardia and ↓ BP from activation of the carotid sinus reflex. It is desirable, however, to position the patient's head in the operative position as a test of the effect of that position on CBF. It is well documented that hyperextension and lateral rotation of the head may occlude vertebral-basilar flow and, if sustained, contribute to postop cerebral ischemia. Sx of dizziness or diplopia will emerge with this maneuver if CBF is compromised.

Tests: Carotid and cerebral angiography will identify the type of lesion (ulcerative or stenotic), its location, and extent of collateral circulation. Other commonly used techniques include MR or CT angiography and duplex ultrasonography. The anesthesiologist should examine the angiograms of the patient or discuss the case with the surgeon to understand the type, location, and extent of the lesion.

Hematologic

Aspirin or other anti-Plt therapy usually is begun preop to ↓ the risk of periop thromboembolic complications. Aspirin may be continued up to the

	day of surgery. Tests: PT; PTT
Laboratory	Tests as indicated from H&P
Premedication	The use of premedication in patients undergoing CEA is controversial. Should a new TIA or stroke occur in the immediate preop period, its Sx may be difficult to distinguish from those associated with excessive responses to premedication. In this population, detailed discussion about the anesthetic and surgical plan, with appropriate reassurance, is usually sufficient. If medication is desired, midazolam 1–3 mg is preferable to opiates.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water and aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique (regional)

Historically, regional anesthesia—in the form of superficial and deep cervical plexus blocks, supplemented as needed by a local field block—was used for most CEAs. It provided the opportunity to evaluate cerebral function during a trial occlusion of 2–3 min. If the patient showed no adverse effects, the operation was completed under regional anesthesia. If the patient developed neurologic changes, a shunt was inserted or GA was induced and ET intubation performed, following which the operation was completed. Regional anesthesia, however, has several disadvantages: this may cause the absence of cerebral protection, patients may tolerate carotid occlusion for 10 min or more before suddenly losing consciousness or developing a Sz, and conversion to GETA may be technically difficult. Nevertheless, the technique still has some advocates among anesthesiologists and surgeons. It is claimed that (a) it decreases the need for a surgical shunt and, thereby, avoids the complications of shunt insertion; (b) it decreases the length of stay in the hospital; and (c) as an anesthetic technique, it is well accepted by some patients with fast surgeons.

Anesthetic technique (GETA)

GETA offers both direct and indirect advantages for patients undergoing CEA: cerebral

protection by decreasing CMRO_2 and redistributing flow toward the potentially ischemic area; greater patient comfort; and the ability to regulate PO_2 , PCO_2 , and MAP. Despite these arguments favoring GA, recent studies comparing regional and GA suggest that there is no clear outcome advantage of one technique over the other, although GA may be associated with a slightly higher risk for postop MI.

Induction	Propofol (1–2 mg/kg), when carefully titrated, is suitable for induction, as arterial BP will generally remain at acceptable levels ($\pm 20\%$ of baseline) in normovolemic patients. Propofol will $\downarrow$ CMRO_2 , constrict normally reactive cerebral vessels, and $\rightarrow$ a redistribution of CBF toward potentially ischemic areas. Etomidate (0.1–0.4 mg/kg) may be useful for induction of anesthesia in hemodynamically unstable patients. A muscle relaxant (e.g., vecuronium 0.15 mg/kg or rocuronium 0.6 mg/kg) is administered for tracheal intubation. An analgesic (e.g., fentanyl 2–5 mcg/kg or remifentanyl 1–3 mcg/kg) may be given to minimize the cardiovascular responses to ET intubation. These opiates have minimal effects on CBF or CMRO_2 .
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider dexamethasone (4 mg iv) to $\downarrow$ PONV. Avoid long-acting opioids or scopolamine patch, which may confound postop neuroexam. Sequential pneumatic compression devices should be used to $\downarrow$ VTE risk.
Maintenance	Isoflurane (up to 0.6%) or sevoflurane (up to 1%) inspired, in combination with N_2O 50% and remifentanyl (0.05–0.2 mcg/kg/min), is satisfactory and will not interfere with EEG or EP/MEP monitoring. TIVA offers no advantages but may be comforting to the neuromonitoring team. Just before cross-clamping of the carotid artery, an additional dose of propofol sufficient to produce burst suppression on the EEG (usually 0.5–1 mg/kg propofol) may be

	administered for cerebral protective effects. A low-dose NTG infusion may be beneficial for patients at high risk for MI. Before cross-clamp, the surgeon will request heparin, usually 100 mg/kg iv. Recheck ACT (goal: 2× baseline).	
Emergence	Total carotid occlusion time should be noted on the anesthetic record. After 10 min and once bleeding from the arteriotomy site has been controlled, protamine (typically 0.5 mg/100 U heparin) is administered iv slowly. If ↓ BP occurs, the rate of protamine administration should be slowed. If vasopressors were used during surgery, patient should be weaned from them during emergence because HTN is likely as patient awakens from anesthesia. The need to control BP with labetalol and/or a combination of esmolol and NTG is likely in the emergence phase. Antiemetic prophylaxis (e.g., ondansetron 4 mg iv 30 min before end of case) is usually appropriate. Prophylactic β-blockers may be useful in this patient population. Consider iv acetaminophen if preop po dose is not given.	
Blood and fluid requirements	IV: 16 ga × 2 NS/LR @ 5–10 mL/kg/h	Blood replacement is seldom an issue. Studies indicate that blood glucose > 180 mg/dL may ↓ tolerance of the brain to ischemia; thus, it is prudent to avoid glucose-containing solutions and treat levels >180 mg/dL. One iv is dedicated to drug infusions. Zero balance fluid management is recommended (see Appendix K, p. K-2). Albumin 5% If volume expansion is

		needed, albumin stays in the intravascular compartment longer than crystalloid solutions.
Monitoring	Standard monitors Arterial line (preinduction for symptomatic patients) UO PPV/PVI ±Stump pressure transducer	Marked fluctuations in BP may occur, necessitating vasoactive drug therapy. The arterial pressure transducer should be placed at the level of the head (tragus) to accurately assess CPP. A CVP catheter is seldom necessary. Vasoactive drugs usually can be given safely through a second iv placed in the proximal arm. A second pressure transducer setup may be needed for stump pressure measurements. Monitors incorporating EASI ECG analysis technology can emulate 12-lead ECGs using only five actual leads. ETT cuff pressure should be monitored and kept at the minimum required to prevent leak. This will minimize ischemic injury to the tracheal mucosa and to the recurrent laryngeal nerve, which may be trapped between the cuff and surgical retractor.
Cerebral perfusion	rCBF	Regional CBF (rCBF) ≥ 25

monitoring

EEG (multilead)

mL/min/100 g brain is satisfactory, and <20 mL/min/100 g suggests cerebral ischemia; however, rCBF measurements are not generally available in OR. Common BIS-type monitors are inadequate for detecting most ischemic events. Even multilead EEG may not identify small focal areas of ischemia. The depth of anesthesia and arterial CO_2 must be stable, or the EEG may not be interpretable. There is a significant incidence of false positives and negatives with EEG monitoring.

Stump pressure (back pressure)

Stump pressure (pressure distal to the carotid clamp) is used to evaluate the adequacy of cerebral perfusion. Cerebral ischemia rarely occurs at stump pressures >40 mm Hg. False positives are not uncommon and result in a shunt being placed when none is needed. The simplicity of the measurement and its relative validity when pressure is >40 mm Hg

still make it a useful clinical method for ensuring adequate perfusion.

Cerebral oximetry (rSO₂)

Cerebral oximetry (e.g., Somanetics Invos) is often used to evaluate superficial frontal cortex perfusion during CEA. With carotid occlusion, ipsilateral rSO₂ values may decrease suggesting inadequate cerebral perfusion. The method is nonquantitative, and an unknown portion of the signal may originate from structures superficial to the cortex perfused by branches of the external carotid artery. Postop increased rSO₂ may predict onset of cerebral hyperperfusion syndrome.

SSEP/MEP

SSEPs/MEPs are being used with increasing frequency as a means of determining the adequacy of cerebral perfusion during temporary occlusion of the carotid artery. This technique requires continuous monitoring by personnel with special training. Areas of focal ischemia may not be detected.

Transcranial Doppler
(TCD)

False-positive and false-negative results can occur. TCD scanning alone or combined with EEG monitoring is a useful method for detecting microemboli (air or particulate matter) during CEA. It also has been suggested that TCD can be used as a guide for regulating BP postop to minimize the occurrence of post-CEA cerebral hyperperfusion states. This technique requires continuous monitoring by personnel with special training and is technically difficult to implement intraop.

Control of BP

Keep MAP $\geq$ awake levels

Autoregulation

Vasopressors:
phenylephrine infusion

It is desirable to maintain MAP at or slightly above the patient's recorded resting pressure while awake. Volatile anesthetics impair autoregulation; therefore, the higher the pressure, the more likely it is that cerebral perfusion will be adequate during surgical occlusion. A pure α -adrenergic agonist (e.g., phenylephrine) is ideal to support BP because it has minimal dysrhythmogenic

	Coronary vasodilation NTG infusion	potential. A V-4 or V-5 lead usually will reveal myocardial ischemia induced by ↑ BP. A simultaneous infusion of NTG (0.2–0.4 mcg/kg/min) may improve coronary flow during periods of induced ↑ BP. If hypertensive episodes occur during surgery, infusions of esmolol ± NTG/clevidipine work well to control BP.
	Labetalol (pre-emergence)	Long duration effect (2–10 h) makes intraop titration difficult.
Positioning	✓ pad pressure points ✓ eyes	
Complications	Sudden ↓ HR or ↓ BP	Surgical stimulation of carotid baroreceptors. Rx: have surgeon apply 2–3 mL of 1% plain lidocaine to carotid bulb.

POSTOPERATIVE

Complications	Circulatory instability: ↓ BP	Circulatory instability is common. ↓ BP may be due to hypovolemia, residual anesthetic, dysrhythmias, or exposure of the carotid sinus baroreceptors to a new higher pressure, causing an exaggerated
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HTN
MI
Stroke

reflex response. Rx by volume expansion and/or pressors.

HTN may be due to loss of the normal carotid baroreceptor mechanism. It may → excessive bleeding, ↑ myocardial O₂ consumption, dysrhythmias, MI, intracerebral hemorrhage, and ↑ ICP from cerebral edema.

Loss of carotid body function

Ipsilateral chemoreceptor function is lost in most patients after CEA → ↓ ventilatory and circulatory responses to hypoxia and a modest increase in resting arterial PaCO₂. These patients should be given supplemental O₂ postop. Special attention must be directed toward preventing atelectasis or other pulmonary or circulatory abnormalities that might cause hypoxemia.

Respiratory insufficiency
Tracheal compression
Vocal cord paralysis
Tension pneumothorax

Acute respiratory insufficiency may occur 2° hematoma formation with tracheal deviation, vocal cord paralysis from surgical traction on laryngeal nerves, or tension pneumothorax from dissection of air

through the wound into the mediastinum and pleural space. A hematoma that causes respiratory distress should be evacuated before reintubation is attempted if possible.

Circulatory insufficiency
MI
Tension pneumothorax

If there is evidence of circulatory insufficiency, consider MI or tension pneumothorax and treat appropriately.

Intimal flap → stroke

Should a patient emerge from anesthesia with a new neurologic deficit, immediate surgical exploration of the operative site or immediate cerebral angiography should be performed to determine if an intimal flap or thrombosis has formed at the site of operation. These are surgically correctable lesions and, if corrected immediately, may lessen the severity of the subsequent neurologic deficit.

Multimodal pain management

Acetaminophen (1 g iv)
±Fentanyl 25–50 mcg prn

Complaints of “sore throat” are common and rarely require opiate treatment.
If not given preop or if 4–6 h has elapsed since

ERAS	Ondansetron 4 mg iv	previous dose. Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	As indicated by Sx	Consider urgent carotid ultrasound or cerebral angiography for delayed emergence.

Suggested Readings

1. Hye RJ, Voeks JH, Malas MB, et al: Anesthetic type and risk of myocardial infarction after carotid endarterectomy in the Carotid Revascularization Endarterectomy versus Stenting Trial (CREST). *J Vasc Surg* 2016; 64(1):3–8.e1.
2. Jones DW, Brott TG, Schermerhorn ML: Trials and frontiers in carotid endarterectomy and stenting. *Stroke* 2018; 49(7):1776–83.
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4. Lal BK, Meschia JF, Howard G, et al: Carotid stenting versus carotid endarterectomy: what did the carotid revascularization endarterectomy versus stenting trial show and where do we go from here? *Angiology* 2017; 68(8):675–82.
5. Leichter SW, Mouwad NJ, Welch K, et al: Outcomes of carotid endarterectomy under general and regional anesthesia from the American College of Surgeons' National Surgical Quality Improvement Program. *J Vasc Surg* 2012; 56(1):81–8.e3.
6. Lokuge K, de Waard DD, Halliday A, et al: Meta-analysis of the procedural risks of carotid endarterectomy and carotid artery stenting over time. *Br J Surg* 2018; 105(1):26–36.
7. Poi MJ, Echeverria A, Lin PH: Contemporary management of patients with concomitant coronary and carotid artery disease. *World J Surg* 2018; 42(1):272–82.
8. Sajid MS, Vijaynager B, Singh P, et al: Literature review of cranial nerve injuries during carotid endarterectomy. *Acta Chir Belg* 2007; 107(1):25–8.
9. Schechter MA, Shortell CK, Scarborough JE: Regional versus general anesthesia for carotid endarterectomy: The American College of Surgeons National Surgical Quality Improvement Program perspective. *Surgery* 2012; 152(3):309–14.
10. Sejersten M, Wagner GS, Pahlm O, et al: Detection of acute ischemia from the EASI derived 12-lead electrocardiogram acquired in clinical practice. *J Electrocardiol* 2007; 40:120–6.

Suggested Viewing

Links are available online to the following video:

Carotid Endarterectomy: https://www.youtube.com/watch?v=9lDf3sx_ZrE